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Master of Physics (Computational Physics)

Residual Hamiltonians for Quantum Differential-Equation Solvers: Descriptor Form, Conditioning and Verified State Preparation

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Declaration

This is to certify that:

1. This thesis comprises of my original work.
2. Due acknowledgment has been made in the text to all other materials used.
3. This thesis consists of less than 60 pages inclusive of tables and figures but exclusive of references and appendices.

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Summary of Student Achievement

Abstract

Brief summary.....

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Notation and Conventions

A quick reference for symbols used across the thesis; each is properly introduced at its first use in the text below.

Typeface. Fixed discretisation matrices are blackboard bold ($\mathbb{G}$, $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$, $\mathbb{M}$, $\mathbb{n}$, $\mathbb{B}$, $\mathbb{D}^{(0)}$, $\mathbb{S}$, $\mathbb{U}$, $\mathbb{A}$, $\mathbb{F}$); assembled objects are italic (A_{std}, A_{desc}, R_{std}, R_{desc}, H_{std}, H_{desc}, V, W, H'). The subscripts std and desc name the two forms (§ 4.2); bare A, H and R appear only in statements true of both forms and in § 3, where they are Wu et al.'s own and are printed in blackboard bold. Calligraphic letters are $\mathcal{T}$ (Gram map), $\mathcal{H}$ (Hilbert space) and $\mathcal{O}$ (order symbol).

Discretisation

Symbol	Meaning	Introduced
n	Qubits per Chebyshev axis (the resolution).	§ 3.1
$N = 2^n$	Truncation dimension of one Chebyshev axis.	§ 3.1
k	Order of an ordinary differential equation.	§ 3.2
p	Degree of a polynomial nonlinearity.	§ 6.5
n_f	Qubits in the field (block) register of the descriptor form.	§ 5.3
$\dim w$	Padded descriptor state dimension, $2^{\lceil \log_2 K \rceil} \prod_a N_a$ with K physical blocks ($2^{\lceil \log_2 (k+1) \rceil} N$ on one axis).	Appendix G.2

The Chebyshev model

Symbol	Meaning	Introduced
T_ℓ	Chebyshev polynomial of the first kind, degree ℓ.	§ 3.1
$\lvert\tau(x)\rangle$	The raw Chebyshev feature state.	§ 3.1
φ_k	Normalised Chebyshev basis functions, $\varphi_0 = T_0/\sqrt{N}$, $\varphi_k = T_k/\sqrt{N/2}$; $\lvert\tau(x)\rangle$ holds their values at x.	§ 4.1
ψ, η, η_e	Normalised coefficient state; its scale; $\eta_e = \lVert b \rVert^2$.	§ 3.1
ψ_e	The ideal state $b/\lVert b \rVert$ of a truncated solution f_N with coefficients b; the model represents f_N exactly at (ψ_e, η_e).	§ 4.1
$\hat{f}^{(j)}$	Coefficients of $f^{(j)}$: one block of the descriptor state.	§ 5.3
w, E	The descriptor state $(\hat{f}^{(0)}; \ldots; \hat{f}^{(k)})$; the map from a field's coefficients to the descriptor state that represents it.	§ 5.3

Operators and constraints

Symbol	Meaning	Introduced
$\mathbb{G}$	Differentiation matrix, $f \mapsto f'$, in coefficient space: dense, strictly upper triangular, $\|\mathbb{G}\|_2 = \Theta(N^2)$ (Wu et al. print it as $\mathbb{G}_n^\top$).	§ 3.2
$\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$	Banded factors of the differentiation matrix, $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$.	§ 5.2
$\mathbb{M}_{x^p}$, $\mathbb{M}_a$	Multiplication by x^p (the exact lift, $2N \times N$) and by a smooth $a(x)$.	§§ 3.2 and 4.2
$\mathtt{n}_1$, $\mathtt{n}_x$	Product-to-sum folds.	§ 3.4
$\mathbb{B}(x)$, $\mathbb{D}^{(0)}(x_s)$	Rank-one zero (invariant) and collapse (regular) constraint rows.	§ 3.2
x_z, x_m, x_s	Constraint points: a zero, an extremum, a source datum.	§ 3.2
R_0, U_{odd}, Λ	Factors of the structured exact encoding of $\mathbb{G}$.	§ 4.5
$\mathbb{S}_a(x^*)$, $\hat{g}$	Datum-slice row on axis a at x^*; the coefficients of the datum function.	§ 4.3
$\mathbb{U}$	The Chebyshev–Gauss transform used by the nodal fold.	§ 6.6
$\|\mathbb{G}\|_S$	Wu et al.'s norm $\max\big(\|\mathbb{G}\mathbb{G}^\top\|_1, \|\mathbb{G}^\top\mathbb{G}\|_1\big)$.	§ 3.3

Residuals and Hamiltonians

Symbol	Meaning	Introduced
A_{std}, A_{desc}	Residual of the standard form; of the descriptor form.	§ 4.2
H_{std}, H_{desc}	Effective Hamiltonian of each form; bare H only in regime-independent statements.	§ 4.2
$R = [R_1; \ldots; R_L]$	Stacked residual, with $H = R^\dagger R$.	§ 7.3
$H_\otimes$	The doubled-space Hamiltonian.	§ 6.4

Block encoding, filtering and state preparation

Symbol	Meaning	Introduced
α, α_R, $\alpha_H = \alpha_R^2$	Subnormalisation: generic, of R, of H.	§ 2.4.1
V, W	Stacked-residual reflection; qubitised walk.	§ 7.3
$H' = 2H/\alpha_R^2 - I$	The normalised Hamiltonian that V block-encodes.	§ 7.3
σ_0, σ_1, $\sigma_{\max}$	Smallest, second-smallest and largest singular values of R; $\lambda_j = \sigma_j^2$.	§ 7.2
Δ	Spectral gap $\lambda_1 - \lambda_0 = \sigma_1^2 - \sigma_0^2$.	§ 7.2
x_0, x_1; h_j	Ground and first excited values of H'; their heights $1 + x_j = 2\sigma_j^2/\alpha_R^2$.	§ 7.4
$w(\cdot)$, θ_0	The minimax filter's edge map of $[x_1, 1]$ onto $[-1, 1]$; its rate $\operatorname{arccosh}(-w(x_0))$.	§ 7.4
β	Imaginary time.	§ 7.4
d, q_k, p_G	GQSP degree; filter coefficients; success probability.	§ 7.4
ε	Target: the excited spectrum's suppression relative to the ground state, and the infidelity bar (10^{-6}).	§ 7.4
ψ_{init}, γ	Initial state; its overlap $\lvert\langle\psi_{\text{ref}}\rvert\psi_{\text{init}}\rangle\rvert$ with the reference.	§ 7.4
ζ, S	Block-rescaling constant; the diagonal of block weights, so that the rescaled residual is $R_{\text{desc}}S$.	§ 5.6
w_f	Weight of the field block in the descriptor's ground state; $\gamma^2 \le w_f$.	§ 5.5
α_A, Q_m	Subnormalisation of the descriptor's system rows; the reflection $I - 2\lvert m\rangle\langle m\rvert$.	§ 5.7
$\epsilon_{\mathbb{G}}$, μ	Entry error and per-bit coefficient of Wu et al.'s encoding of $\mathbb{G}$.	Appendix D.2

Nonlinearity

Symbol	Meaning	Introduced
K	Carleman truncation order.	§ 6.5
M	Dimension of the base state u.	§ 6.5
ρ	Nonlinearity-to-dissipation ratio; $\rho < 1$ is sufficient, not necessary, for the Carleman truncation error to fall as ρ^K.	§ 6.5
F_1, F_2, $\mathbb{A}_K$	Linear and quadratic terms; the lifted Carleman generator.	§ 6.5

Systems and complex coefficients

Symbol	Meaning	Introduced
C	Number of components (fields) of a system.	§ 6.1
z	A system's state: one block per component.	§ 6.1
$\mathbb{A}$	A system's coupling matrix, $u_i' = \sum_j \mathbb{A}_{ij} u_j$.	§ 6.1
$\mathbb{F}$	Ratio row: fixes the direction of a vector datum and leaves its scale free.	§ 6.2
y, y_a, $y_\perp$	The vector datum; its anchor component; its off-anchor part.	§ 6.2
w_0	A system's stated initial data; it weights the initial state.	§ 6.2
$c = \operatorname{Re} c + i \operatorname{Im} c$	A complex coefficient, in the realification example.	§ 6.3

Acronyms

DNS direct numerical simulation.

FOSLS first-order system least squares.

GQSP generalised quantum signal processing.

HHL Harrow–Hassidim–Lloyd.

IR intermediate representation.

IVP initial-value problem.

LCU linear combination of unitaries.

NISQ noisy intermediate-scale quantum.

ODE ordinary differential equation.

PDE partial differential equation.

PIHM physics-informed effective Hamiltonian method.

PINN physics-informed neural network.

QFT quantum Fourier transform.

QITE quantum imaginary-time evolution.

QSVT quantum singular value transformation.

SVD singular value decomposition.

1 Introduction

1.1 Differential Equations as a Target for Quantum Computation

1.2 Physics-Informed Hamiltonians and Where Their Cost Lies

1.3 Contributions and Scope

1.4 Outline

remove once previous section complete

2 Background and Literature Review

Every quantum algorithm for a differential equation pays a ratio that its discretisation fixes before any circuit is built: the condition number of a linear system, the norm of a generator times the evolution time, the ratio of nonlinearity to dissipation in a lift, or the norm of an encoded Hamiltonian against its gap. An accurate discretisation can still make that ratio large, and it typically grows with the resolution. The review follows the ratio through three bodies of work: the quantum routes to differential equations, judged by what each assumes, which ratio it pays, what it returns and where it fails; the spectral methods on which the physics-informed construction rests, for which classical numerical analysis has already shown how the choice of basis and of unknowns sets the ratio and how it can be lowered; and the quantum primitives that turn a ratio into a count of queries, the filter degree of Equation (2.2). The physics-informed effective Hamiltonian method itself is presented and judged separately, in § 3.

2.1 Quantum Computation Preliminaries

We use the circuit model of Nielsen and Chuang [1]; the gates of the thesis's circuits are listed in Appendix A. The state of n qubits is a unit vector in $\mathbb{C}^N$, $N = 2^n$, and a vector $v \in \mathbb{C}^N$ is *amplitude-encoded* as $|v\rangle = v/\|v\|$. The encoding stores only the direction of v, so its length must be carried separately (for a differential equation, the scale η of § 3.1), and it gives access to v only through measurement, which returns the index j with probability $|v_j|^2/\|v\|^2$; recovering all N entries takes at least of order N repetitions (§ 2.4.3). Circuits of unitary gates act on these states, and a non-unitary operation is applied by acting on the system together with *ancilla* qubits prepared in $|0\rangle$ and keeping only the outcomes in which they are measured in $|0\rangle$ again. This *post-selection* succeeds with a probability that enters the cost; every non-unitary matrix of the thesis, from a residual to a filter, acts on a state this way (§ 2.4.1).

An algorithm that accesses a matrix only through a circuit encoding it is costed in *queries*, the uses of that circuit, and its gate count is the number of queries times the gates per query, on a register of system and ancilla qubits. The asymptotic results reviewed below, and the filter degree of Equation (2.2), count queries; the gates per query belong to the encoding circuit. § 4 separates the two, and § 7.7 fixes how gates are counted. A tilde on $\mathcal{O}$, Ω or Θ suppresses factors logarithmic in N and in $1/\varepsilon$, and a cost polynomial in $n = \log_2 N$ is called poly-logarithmic.

Every cost in the thesis is that of a fault-tolerant machine, whose logical qubits and gates are protected by error correction. Current noisy intermediate-scale quantum (noisy intermediate-scale quantum (NISQ)) processors have no error correction, and noise limits the depth of a reliable circuit [2]; the variational algorithms of § 2.2.3 target them. A filter applies its block encoding as many times in sequence as its degree, which grows with the resolution, so its feasibility is judged against fault-tolerant resource estimates (§ 9.4) and NISQ hardware is out

of scope.

2.2 Quantum Algorithms for Differential Equations

Aaronson's caveats on the linear-systems algorithm apply to every route below: the input must load efficiently, the operator must be well conditioned and efficiently accessible, and the output is a quantum state, useful only where a few properties of the solution suffice [3]. Table 2.1 compares the routes under these headings.

2.2.1 Linear-Systems and Linear-Combination Routes

Harrow, Hassidim and Lloyd (Harrow–Hassidim–Lloyd (HHL)) prepare a state proportional to $A^{-1}b$ for an s-sparse $N \times N$ matrix of condition number κ in time $\tilde{\mathcal{O}}(s^2\kappa^2 \log N/\varepsilon)$, and show that time $\kappa^{1-\delta}$ for any $\delta > 0$ would imply BQP = PSPACE [4]. Discretising a differential equation into such a system gave the first solvers: Berry with a linear multistep method, at a cost polynomial in $1/\varepsilon$ [5]; Berry, Childs, Ostrander and Wang with a truncated Taylor series, at a cost polynomial in $\log(1/\varepsilon)$ [6]; and Childs and Liu with Chebyshev collocation in time, whose collocation matrix is dense and whose condition number they bound by a polynomial of degree 3.5 in the number of nodes [7]. Childs, Liu and Ostrander extended finite-difference and spectral discretisations to partial differential equations at the same dependence on precision [8], and Costa et al. brought the linear-systems solver to the optimal $\mathcal{O}(\kappa \log(1/\varepsilon))$ with a discrete adiabatic theorem [9]. The ordinary-equation solvers take the equation as a first-order system $x' = Ax + b$, reducing any higher order before discretising, so the conditioning they pay is that of a first-order operator.

The linear combination of Hamiltonian simulations writes the propagator of a dissipative linear system as a weighted integral of unitary evolutions, implemented as a linear combination of unitaries with optimal state-preparation cost [10]; Schrödingerisation maps a linear evolution to a Schrödinger equation in one more dimension by a warped phase transformation [11]. Neither solves a linear system. Jennings et al. give the first non-asymptotic costs, constant factors included, and prepare the history state of any stable linear system with complexity $\mathcal{O}(T^{1/2})$ in the evolution time, a fast-forwarding that classical stability permits [12].

These costs are proven, logarithmic in N and at best in $1/\varepsilon$. Learning all N amplitudes of the output takes at least of order N repetitions, and for equations in a fixed number of dimensions with smooth solutions the advantage over the best classical methods is at most polynomial [13]. Both κ and $\|A\|$ grow with the resolution, which makes the discretisation part of the cost (§ 2.3).

2.2.2 Nonlinear Equations: Carleman and Related Linearisations

Leyton and Osborne's algorithm, the first for nonlinear equations, transforms the amplitudes of many copies of the state nonlinearly, at resources exponential in the integration time [14].

The route developed most since is Carleman linearisation [15, 16]: a polynomial system such as $u' = F_1 u + F_2 u^{\otimes 2}$ is embedded in an infinite linear system for the monomials $u, u^{\otimes 2}, u^{\otimes 3}, \ldots$, truncated at order K and solved as a linear system. Liu et al. treat dissipative quadratic equations in M variables at cost $T^2 q\,\mathrm{poly}(\log T, \log M, \log 1/\varepsilon)/\varepsilon$, with q the decay of the solution, when the ratio R of nonlinearity and forcing to dissipation satisfies $R < 1$, and prove the problem intractable in the worst case for $R \geq \sqrt{2}$ [17]. Krovi makes the dependence on precision logarithmic and relaxes the normality of the linear part to a negative log-norm [18].

The condition $R < 1$ bounds the truncation error uniformly over states, so it is sufficient and not necessary: a lift may converge for $R > 1$, which § 6.5 (writing the ratio ρ) measures and § 8.4 reports. Dissipation must still dominate. The lift has dimension of order M^K, and its truncation is controlled only where the nonlinearity is weak against the damping.

2.2.3 Variational Routes

Lubasch et al. represent the solution by a parametrised circuit, evaluate a nonlinear cost on several copies of the state, and minimise it classically, as demonstrated on the nonlinear Schrödinger equation [19]. The differentiable quantum circuits of Kyriienko, Paine and Elfving represent the solution as the expectation value of a circuit with input x, differentiate it with respect to x by parameter shifts on the feature map, and train it on the residual of the equation and its data [20], the loss of a physics-informed neural network [21]. The variational linear solver finds a ground state the same way, minimising $\langle x|A^\dagger(I - |b\rangle\langle b|)A|x\rangle$, whose zero-energy ground state solves $Ax \propto b$ [22]; encoding time in a register, as the Feynman–Kitaev clock does, puts a whole trajectory in one ground state [23], and a space–time encoding of this kind has given the evolution of Burgers' equation from a single ground-state computation [24]. These circuits are shallow enough for NISQ devices, but nothing bounds the number of optimisation steps or guarantees the optimum: the gradients of a sufficiently random parametrised circuit vanish exponentially in the number of qubits, the barren plateau [25], and each evaluation of the cost is a sampled expectation value.

2.2.4 Ground-State and Physics-Informed Routes

Preparing the solution of a linear system as the null vector of a positive semi-definite Hamiltonian, without optimisation, predates the physics-informed work. Subaşı, Somma and Orsucci do so by randomised adiabatic evolution at cost $\mathcal{O}(\kappa \log \kappa/\varepsilon)$ [26], An and Lin by a time-optimal adiabatic schedule [27], and Costa et al. by a discrete adiabatic walk [9]. In Lin and Tong's form [28], for positive-definite A with $\|A\| = 1$ and $Q_b = I - |b\rangle\langle b|$, the Hamiltonian with off-diagonal blocks AQ_b and $Q_b A$ has $|0\rangle|x\rangle$ in its null space beside $|1\rangle|b\rangle$, which a measurement of the first qubit removes; the rest of its spectrum lies at a distance of order $1/\kappa$ from zero, and an optimal Chebyshev filter projects onto the null space. The gap, and with it the filter's cost, is set by κ (§ 2.4.3).

The physics-informed line arrived at the same object from machine learning. Physics-informed

neural networks are trained on the residual of the equation at sample points and on its data [21], and differentiable quantum circuits carried that loss to quantum models. Paine, Elfving and Kyriienko moved the model into a latent space in which the solution's Chebyshev coefficients are the amplitudes of a state, so that differentiation and multiplication by x act as fixed matrices and the loss needs no evaluation on a grid [29]. Wu, Paine, Philip, Gentile and Kyriienko removed the training: the summed Gram matrices of those operators form an effective Hamiltonian whose ground state is the solution, prepared by a filter [30]; § 3 presents and judges the method. In concurrent work, Paine compares Hamiltonian constructions and function encodings (grid values against spectral coefficients) on linear and nonlinear examples, and finds that both choices affect the recovery of the solution, the spectral gap, its degeneracy and the readout [31]. On the few small instances studied the normalised gap depends on both choices without a general ordering; no scaling with resolution or block-encoding circuit is given, and the lift for products is polynomial only, needing more registers as the degree grows.

2.2.5 Critical Assessment

Each route in Table 2.1 pays a ratio set by how the equation is discretised and encoded, which an accurate discretisation can still make large: the condition number of a linear system, the norm of a generator times the evolution time, the ratio of nonlinearity to dissipation in a lift, or the norm of an encoded Hamiltonian against its gap. Every route returns a quantum state, so its advantage holds only where a few properties of the solution suffice [3, 32], and is at most polynomial in a fixed, low number of dimensions (§ 2.2.1). The thesis therefore compares quantum methods with one another, at equal accuracy, in queries and gates, and claims no advantage over classical solvers. The guarantees differ: the linear-systems and ground-state routes have proven costs, the variational routes none, and the Carleman route worst-case sufficient conditions that a given problem may beat. Wu et al. leave the cost driver of their method open; its entry in Table 2.1 is this thesis's reading (§§ 3.5 and 4.6), and the descriptor form of the last row keeps that driver and changes the operator whose norm enters it.

2.3 Spectral Methods and Their Conditioning

The ratios of Table 2.1 are set when the equation is discretised. For the Chebyshev discretisation on which the physics-informed construction is built, classical numerical analysis locates the source of the ratio, the density of differentiation in coefficient space, and two ways of lowering it: a basis in which differentiation is banded, and derivatives carried as unknowns.

2.3.1 Chebyshev Discretisation and Differentiation

A function analytic on $[-1, 1]$ has a Chebyshev series $f = \sum_k b_k T_k$ with geometrically decaying coefficients, so the N-term truncation converges faster than any power of N [33]; the physics-informed construction, like the spectral methods it draws on, works with these coefficients for that reason. A differential equation becomes a finite system by Galerkin projection onto the

Table 2.1: Each quantum route's cost is set by a ratio its discretisation fixes. Rows: the families of § 2.2 and this work; columns: assumptions, cost driver, output, failure mode. A, the generator of a linear system; κ, its condition number; T, evolution time; R, Carleman ratio of nonlinearity and forcing to dissipation; K, Carleman truncation order; M, number of unknowns; N, number of Chebyshev terms; $\mathbb{G}$, Chebyshev differentiation matrix; α_H, Δ, subnormalisation and spectral gap of a block-encoded Hamiltonian H. The physics-informed row's cost driver is this thesis's reading of Wu et al.; the last row's degree holds for a regular one-axis ordinary differential equation (ODE) only.

Approach	Assumes	Cost depends on	Returns	Fails where
Linear systems [4, 6, 7, 9, 28]	linear equation in first-order form; efficient access to A and loading of the data	κ of the discretised system; $\|A\|T$	the solution, or its history, as a state	κ grows with the resolution; readout; at most polynomial gain at fixed dimension
Linear combination of Hamiltonian simulations; Schrödingerisation [10, 11]	linear, stable dynamics	$\|A\|T$; the decay of the solution's norm	the state at time T	nonlinear or unstable dynamics; readout
Carleman linearisation [17, 18]	polynomial nonlinearity; dissipation dominating ($R < 1$ sufficient)	the linear solver's cost on a lift truncated at order K	the lifted state, whose first level is the solution	intractable in the worst case for $R \geq \sqrt{2}$; lift dimension of order M^K
Variational [19, 20, 22, 24]	a trainable parametrised circuit	optimisation steps times samples, unbounded	a trained circuit	no guarantee of convergence; barren plateaus
Physics-informed effective Hamiltonian [30]	a solution well represented by N Chebyshev terms; data rows that isolate it	not analysed by its authors; here, $\sqrt{\alpha_H/\Delta}$, with $\|H\|$ set by powers of the dense $\mathbb{G}$	the solution's coefficients as a ground state; the field at any x	the norm grows with each derivative eliminated; the nonlinear route projects (§ 3.5)
This work: the descriptor form and the Carleman lift	as above; a regular one-axis problem for the best scaling; a known spectral edge	$\sqrt{\alpha_H/\Delta}$ with $\|H_{\text{desc}}\| = \Theta(N^2)$; degree $\tilde{\Theta}(N)$ on a regular one-axis ODE	as above; decoded classically	the gap falls on two axes and where the leading coefficient vanishes; not polylogarithmic

basis, by the tau method, which projects but replaces the highest-degree rows by the boundary conditions, or by collocation at a set of nodes [34, 35]. Wu et al.'s construction is closest to the tau method, with the equation acting on the coefficients and the data imposed as rows at points, but its system is overdetermined and solved by least squares (§ 3.2).

Differentiation in coefficient space is exact on polynomials of degree below N and dense, because the derivative of T_p contains every T_k of the opposite parity below it: the matrix is strictly upper triangular, about a quarter of its entries are non-zero, and its norm grows as N^2 (§ 4.2). An operator of order k built from its powers grows as N^{2k}, and so, typically, does the condition number of a tau or collocation discretisation [34, 36].

2.3.2 The Ultraspherical Method

Olver and Townsend make differentiation banded by letting the derivative live in another basis [36]. Since $T_k' = k\,C_{k-1}^{(1)}$, with $C_k^{(1)} = U_k$ the Chebyshev polynomials of the second kind, differentiation from Chebyshev to $C^{(1)}$ coefficients is a single superdiagonal, and conversion of a series' own coefficients to $C^{(1)}$ has two non-zero diagonals. Raising the basis one ultraspherical (Gegenbauer) level $C^{(\lambda)}$ per derivative gives, for an equation of order k with polynomial coefficients, an operator banded apart from its boundary rows, whose condition number grows linearly with the truncation and is bounded after a diagonal preconditioner, in the 2-norm for Dirichlet conditions [36]. The dense differentiation matrix of this thesis factorises as $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$, with $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ the method's first conversion and differentiation operators up to a factor of two (Appendix B.5); the descriptor form of § 5 keeps the two factors, and its block rescaling takes the place of the preconditioner.

Olver and Townsend eliminate the lower derivatives and solve one banded system for the solution's coefficients. The descriptor instead keeps every derivative as an unknown in the Chebyshev basis, tied to the next by the same pair $(\tilde{\mathbb{B}}, \tilde{\mathbb{D}})$ (§ 2.3.3). The full ultraspherical reduction would give a banded residual on N unknowns without the descriptor's block register; it is neither built nor compared here (Appendix B.5), and it is the natural next comparison (§ 9.5).

2.3.3 First-Order System Least Squares

The normal equations of a least-squares method square the residual operator, so high order compounds the conditioning: least squares applied directly to a second-order elliptic equation needs continuously differentiable finite elements and gives condition numbers of order h^{-4} in the mesh width h, against h^{-2} for a Galerkin method [37]. First-order system least squares (first-order system least squares (FOSLS)) introduces derivatives as unknowns, such as a flux $v = \nabla\phi$, and minimises the residuals of the resulting first-order system [38]. Where the functional is equivalent to a product of H^1 norms, the condition numbers return to order h^{-2} and the unknowns nearly decouple, so block-diagonal preconditioners are effective [37, 39].

The physics-informed Hamiltonian $H = R^\dagger R$ is such a functional, its energy the squared residual

of the equation and its data (§ 3.2). Eliminating every derivative through $\mathbb{G}$, as Wu et al. do, is the high-order least-squares method; carrying the derivatives as unknowns is FOSLS in a spectral basis. The quantum ordinary-equation solvers of § 2.2.1 already work with first-order systems, and the physics-informed construction departs from them in eliminating the derivatives into one dense operator. A filter applied through a block encoding of R pays the conditioning of R (§ 2.4.3); the squaring falls only on an eigensolver acting on H. A first-order formulation still lowers what the filter pays, because each block of R then carries one derivative instead of k (§ 4.6).

What this thesis adds. Both ideas are classical. The increment is their transfer to a block-encoded physics-informed Hamiltonian and its consequences for preparing the ground state: a proof that the reformulated residual has the standard form's kernel, so that the prepared state is the same solution (§ 5.3); an encoding of the banded residual as one flat linear combination of unitaries with $\mathcal{O}(\log n)$ ancilla qubits (§ 5.7); the finding that the carried derivatives erode the Hamiltonian's gap, with an a priori block rescaling, at no gate cost, that restores it on one axis and not on two (§§ 5.5 and 5.6); and a measurement of what the reformulation does to the filter's degree, as built and at an ideal encoding (§§ 8.2 and 8.3).

2.3.4 Fourier–Galerkin Truncation

On a periodic domain the Fourier basis diagonalises differentiation, $\partial_x e^{\mathrm{i}kx} = \mathrm{i}k\, e^{\mathrm{i}kx}$, so linear terms act mode by mode and a quadratic term becomes a convolution of coefficients [34]. A Galerkin truncation keeps exactly the convolution terms whose output lies in the retained set of modes; a pseudo-spectral evaluation on a grid aliases the discarded terms onto retained modes unless it is dealiased [34]. The fluids of § 6.7 use Galerkin truncations on real Fourier modes, so their quadratic coupling is a fixed matrix of exact coefficients, as the Carleman lift requires; a pseudo-spectral simulation serves only as the reference for the truncation's error.

2.4 Block Encodings, Signal Processing and Filtering

A block encoding applies a non-unitary matrix on a quantum computer, a polynomial transformation of the block encoding builds a filter, and the filter's degree counts the queries that a prepared ground state costs.

2.4.1 Block Encodings and Linear Combinations of Unitaries

Definition 1 (Block encoding [40]): A unitary U on $m+n$ qubits is an (α, m, ϵ) *block encoding* of an $N \times N$ matrix A if

$$\left\| A - \alpha \left(\langle 0^m | \otimes I \right) U \left(|0^m\rangle \otimes I \right) \right\| \leq \epsilon; \tag{2.1}$$

α is its *subnormalisation*, and the encoding is *exact* when $\epsilon = 0$.

Applied to $|0^m\rangle\,|\psi\rangle$ and post-selected on the ancillas' zero, an exact block encoding applies A/α with success probability $\|A\psi\|^2/\alpha^2$. The block of a unitary has norm at most one, so $\alpha \geq \|A\|$ for every exact block encoding, and the excess $\alpha/\|A\|$ multiplies the cost of every algorithm that uses it. The standard construction is the linear combination of unitaries (linear combination of unitaries (LCU)) [41]: for $A = \sum_{t<L} c_t U_t$ with unitaries U_t, PREPARE loads $\sum_t \sqrt{|c_t|/\alpha}\,|t\rangle$ on $\lceil \log_2 L \rceil$ ancilla qubits, SELECT applies U_t controlled on $|t\rangle$, with the phases of the c_t absorbed into the U_t, and PREPARE† returns the ancillas, giving an exact block encoding at $\alpha = \sum_t |c_t|$. An exact combination whose SELECT indexes L distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [42]. Explicit circuits exist for many structured and sparse matrices [43, 44]; each carries its own α, and the two forms of the residual compared in § 8 differ in both $\|A\|$ and $\alpha/\|A\|$.

If U is Hermitian and block-encodes H/α, the walk $W = (2\Pi - I)\,U$, with $\Pi = |0^m\rangle\,\langle 0^m| \otimes I$, satisfies $\Pi W^k \Pi = T_k(H/\alpha)$: its k-th power block-encodes the k-th Chebyshev polynomial of the encoded matrix at one query per power [45], so a polynomial of degree d in H/α costs d queries once its combination is formed.

2.4.2 Quantum Signal Processing

Quantum signal processing interleaves a single-qubit signal rotation with fixed phase rotations to realise a polynomial of the signal [46]. The quantum singular value transformation (quantum singular value transformation (QSVT)) lifts the construction to block encodings, applying a degree-d polynomial P to the singular values of the encoded matrix with d queries and one further ancilla [40], and places search, Hamiltonian simulation and linear systems in one framework [47]. Its polynomials are bounded by one on $[-1, 1]$ and of definite parity, so a function of indefinite parity, such as the imaginary-time filter $e^{-\beta x}$, is split into even and odd parts, each with its own phase sequence, and the two are summed by a further linear combination at a cost in subnormalisation; Wu et al. use this mixed-parity construction (§ 3.3). Generalised quantum signal processing (generalised quantum signal processing (GQSP)) drops the parity constraint [48]. SU(2) rotations on one signal qubit, with the encoded unitary applied controlled on it, realise any $P(z) = \sum_{k \leq d} q_k z^k$ with $|P| \leq 1$ on the unit circle at d queries, and applied to the walk W they give $\sum_k q_k T_k$ of the encoded matrix (§ 7.4). The phases follow from a complementary polynomial, computed by a fast-Fourier-transform formula with explicit error guarantees [49], through the inverse of the nonlinear Fourier transform to which GQSP is equivalent [50, 51].

2.4.3 Ground-State Preparation by Filtering

A filter is a function of H that keeps the ground eigencomponent of an initial state ψ_{init} and suppresses the rest. Imaginary-time evolution $e^{-\beta H}$ does so once β exceeds the inverse gap [52], and inverse iteration H^{-k} by powers [53]; expanded in polynomials, both are implementable by the transformations above. For an eigenvalue at zero with the rest of the spectrum in $[-1, -\Delta] \cup$

$[\Delta, 1]$, Lin and Tong's eigenstate filter, a Chebyshev polynomial in x^2, has the least maximum on that set among polynomials of its degree equal to one at zero, and needs degree $\mathcal{O}(\Delta^{-1}\log(1/\varepsilon))$ [28]. Their ground-state algorithm approximates a shifted sign function, given a bound that separates the ground energy from the rest of the spectrum, with $\mathcal{O}\big((\alpha/(\gamma\Delta))\log(1/\varepsilon)\big)$ queries for subnormalisation α, gap Δ and overlap $\gamma = |\langle\psi_0|\psi_{\text{init}}\rangle|$ with the ground state; they argue, by reduction from unstructured search, that its dependence on γ and Δ is essentially optimal [54]. Both assume the gap, or bounds that imply it, known in advance, a standard assumption that this thesis shares (§ 7.4).

Both place the ground eigenvalue inside the encoded spectrum; at its edge the degree falls quadratically. Let $H \succeq 0$ be block-encoded with its spectrum mapped into $[-1, 1]$ and the ground value at the lower edge, as by $2H/\alpha_H - I$, with gap $\Delta = \lambda_1 - \lambda_0$ and a ground value small against Δ. The excited spectrum lies in $[x_1, 1]$, at a distance of order Δ/α_H from the ground value. Among polynomials of degree d bounded by one on $[-1, 1]$, the Chebyshev polynomial of $[x_1, 1]$ has the least maximum there relative to its value at the ground state, by Chebyshev's classical inequality [55]; it is the edge counterpart of Lin and Tong's filter. Outside its interval it grows as the hyperbolic cosine of the degree times the square root of the distance, so suppressing the excited spectrum to ε relative to the ground state needs the degree

$$d = \Theta\left(\sqrt{\frac{\alpha_H}{\Delta}}\,\log\frac{1}{\varepsilon}\right), \qquad \alpha_H = \alpha_R^2. \tag{2.2}$$

The square root is the quadratic gap amplification available for frustration-free Hamiltonians, sums of positive terms that all vanish on the ground state [56]; $H = R^\dagger R$ is one when its kernel is exact and nearly one when its ground value is small. Encoded through R at subnormalisation α_R, the walk's eigenphases are set by the singular values of R/α_R, and the gap the filter resolves is of order $\sigma_1/\alpha_R = (\Delta/\alpha_H)^{1/2}$ (§ 7.3).

Each degree is one query, so d counts the queries of one filtered preparation before amplification. The circuit enters only through $\alpha_H \geq \|H\|$ (§ 2.4.1), so with this filter no encoding brings the degree below order $\sqrt{\|H\|/\Delta}\,\log(1/\varepsilon)$. With $H = R^\dagger R$ encoded through R, $\alpha_H = \alpha_R^2$ and the degree is linear in α_R; for an exact kernel, whose gap is the square of the smallest non-zero singular value of R, $\sqrt{\alpha_H/\Delta}$ is at least the ratio of the largest singular value of R to that one, the residual's effective condition number. The cost of a filtered ground state is therefore set by the norm of the operator it is built from, measured against its gap. The bound holds for polynomial filters given the edge and the gap, among which Chebyshev's inequality makes the degree least; over all algorithms the thesis claims no more than Lin and Tong's optimality supports (§ 9.2).

Post-selection succeeds with probability of order γ^2, and amplitude amplification raises it to near one with $\mathcal{O}(1/\gamma)$ repetitions [57], so a preparation costs of order d/γ queries; an answer-independent initial state with non-vanishing overlap is part of the method (§ 7.4).

Readout. An overlap or expectation value is estimated to additive error δ by a Hadamard or swap test with $\mathcal{O}(1/\delta^2)$ repetitions, or with $\mathcal{O}(1/\delta)$ by amplitude estimation, including variants without phase estimation [57–59]; a full field needs of order N such estimates, the readout problem of quantum differential-equation algorithms [32, 60]. Wu et al. read the field point by point with an interferometric overlap measurement (§ 3.3). This thesis decodes the field classically from the prepared state, which its classical computations make available (§ 7.6), and leaves readout, a limitation shared with every route of Table 2.1, unaddressed.

2.5 Synthesis

The physics-informed effective Hamiltonian sits where the three strands meet. It is a ground-state route, so it inherits the filters' guarantees and their cost law, Equation (2.2): the square root of an encoded Hamiltonian's norm over its gap. It is a least-squares functional on a Chebyshev discretisation, so it inherits spectral accuracy together with the conditioning of dense differentiation, the property that the ultraspherical method and first-order-system least squares each address. And it descends from physics-informed learning but trains nothing. Its authors name the gap's behaviour and the cost of differentiation as questions for future work; in the terms of Equation (2.2) the first is the degree's denominator and, as § 3.5 argues, the second sets its numerator. § 3 presents the method on the same attributed footing as the material above, judges it, and poses those questions in the form this thesis answers.

3 The Physics-Informed Effective Hamiltonian Method

This chapter presents the physics-informed effective Hamiltonian method (PIHM) of Wu, Paine, Philip, Gentile and Kyriienko [30] and evaluates it. The method turns a differential equation into a ground-state problem. The unknown function is expanded in Chebyshev polynomials whose coefficients are the amplitudes of an n-qubit state $|\psi\rangle$; the equation and each of its data conditions become matrix operators that annihilate $|\psi\rangle$ when it holds the solution's coefficients; and the Gram matrices of these operators are summed into an effective Hamiltonian H whose ground state is the solution. Preparing that ground state needs a block encoding of H, a filter that removes its excited spectrum, and a measurement that reads the field back. The construction descends from differentiable quantum circuits [20] through their Chebyshev latent space [29].

The method is claimed to need no variational training, prepare a continuous analytic function for $x \in [-1, 1]$, and "necessary terms [operator circuits] have $O(\mathrm{poly}(n))$ scaling and can be implemented efficiently" [30]. Wu et al. use QSVT-based quantum imaginary-time evolution (QITE) and expect a cost of order $\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon)$, but state that the gap is "not easy to bound" and leave its dependence to future work. We first present Wu et al.'s own terms, its model, Hamiltonian, circuits and nonlinear extension and evaluate them here, and present an improved construction in § 4.

3.1 The Chebyshev Latent-Space Model

On n qubits, with $N = 2^n$, the quantum Chebyshev state or feature map is

$$|\tau(x)\rangle_n = \frac{1}{2^{n/2}} T_0(x) |0\rangle^{\otimes n} + \frac{1}{2^{(n-1)/2}} \sum_{k=1}^{N-1} T_k(x) |k\rangle , \tag{3.1}$$

whose components are the first N Chebyshev polynomials of the first kind. The model is a scaled overlap

$$f_Q(x) = \sqrt{\eta}\, f_q(x), \qquad f_q(x) = \langle \tau(x)|\psi\rangle_n , \qquad \langle\psi|\psi\rangle = 1, \tag{3.2}$$

where the amplitudes of $|\psi\rangle$ are the series' weighted coefficients and $\eta > 0$ is a scaling factor [29, 30]. The N states $|\tau(\bar{x}_j)\rangle_n$ at the Chebyshev nodes $\bar{x}_j = \cos[\pi(j + \frac{1}{2})/N]$ are orthonormal, but elsewhere $|\tau(x)\rangle_n$ is not a unit vector; a circuit prepares its normalised version by post-selection, and Wu et al. discard the norm, "since it can be implicitly incorporated into the scaling factor", so their derivations use the unnormalised state. Every constraint below is linear in $|\psi\rangle$ and fixes only its direction. The scale is recovered after the ground state $|\psi_g\rangle$ has been prepared, from one known value of the solution, the *regular* datum $f(x_s) = y_s$: $\sqrt{\eta} = y_s / \langle \tau(x_s)|\psi_g\rangle_n$.

3.2 Hamiltonian Construction

Differentiation acts on the coefficients by a fixed matrix, $f'_q(x) = \langle \tau(x)|\mathbb{G}|\psi\rangle_n$, with $\mathbb{G}$ the Chebyshev differentiation matrix.[1] $\mathbb{G}$ is dense, because the derivative of T_p contains every T_k of the opposite parity below it [36]: one high coefficient of f reaches every lower coefficient of f'. Its entries are given in § 4.2. Higher derivatives are its powers, so a homogeneous linear ODE with constant coefficients, $\sum_j a_j f^{(j)} = 0$, is represented by

$$A |\psi\rangle = 0, \qquad A = \sum_j a_j \mathbb{G}^j. \tag{3.3}$$

For variable coefficients, $\mathbb{M}_{x^p}$ is introduced [29]. Using the Chebyshev product-to-sum identity, it can be shown that $x^p \langle \tau(x)|_n = \langle \tau(x)|_{n+1} \mathbb{M}_{x^p}$. For each power p, it is a $2N \times N$ matrix that carries x^p onto a register one qubit wider, so that every term of the equation shares the $(n + 1)$-qubit basis. A polynomial coefficient $a_j(x)$ enters as the sum of the lifts of its monomials. A source $r(x)$ is brought into the same form through the regular datum: the collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}_n(x_s)/f(x_s)$, with $\mathbb{B}_n$ defined below, maps the solution's coefficients to the constant $1/\sqrt{\eta}$, so a source expanded in a order $\bar{p}$-truncated Maclaurin series enters the equation's operator as $-\big(\sum_{p\le\bar{p}} c_p \mathbb{M}_{x^p}\big)\mathbb{D}^{(0)}(x_s)$, with c_p the Maclaurin coefficients of r.

Boundary and initial conditions enter as data constraints built from the rank-one row $\mathbb{B}_n(x) = \sqrt{2^n}\, |0\rangle^{\otimes n} \langle \tau(x)|_n$, which maps the model to the constant function equal to its value at x. An *invariant* constraint has a zero target and is unchanged by scaling: $f(x_z) = 0$ gives $\mathbb{B}_n(x_z) |\psi\rangle =$

[1] Wu et al. define the lower-triangular $\mathbb{G}_n$ and apply their transpose, $\mathbb{G}_n^\top$, to the coefficient vector. We write that upper-triangular matrix as $\mathbb{G}$ throughout.

0, and $f'(x_m) = 0$ gives $\mathbb{B}_n(x_m)\mathbb{G}\,|\psi\rangle = 0$. Wu et al. place these at boundaries where the solution vanishes or at interior zeros and extrema of the known solution, and uses one invariant constraint for each second-order ordinary example, with the regular datum fixing the scale.

A function of several variables is carried on one register per variable, with the product of the feature maps as its basis; Wu et al. treat two. For variables x and y the state is $\langle\tau(x)|_n \otimes \langle\tau(y)|_n$ on two n-qubit registers; derivatives act as $\mathbb{G} \otimes I$ and $I \otimes \mathbb{G}$, and a constraint on a boundary as $\mathbb{B}_n(x_z) \otimes I$. The invariant sets of the three partial differential equation (PDE) examples are chosen by hand: Laplace's equation uses its three homogeneous sides, so its non-zero side appears in no constraint, and the heat and wave examples place interior points at zeros of the known solutions (the sets are listed in Appendix K).

Each operator is turned into a Gram matrix, and the Gram matrices are summed:

$$H = \mathcal{T}(A) + \sum_i \mathcal{T}(\mathbb{B}_i), \qquad \mathcal{T}(X) = X^\top X, \qquad \langle\psi|H|\psi\rangle = \|A\psi\|^2 + \sum_i \|\mathbb{B}_i\psi\|^2, \tag{3.4}$$

where $\mathbb{B}_i$ runs over the invariant constraints. The energy of a state is its total squared violation of the equation and its data. Wu et al. describe H as "a real symmetric positive-definite matrix with distinct eigenvalues" and recast the equation as the eigenvalue problem $H\,|\psi_g\rangle = \lambda_{\min}\,|\psi_g\rangle$ with $\lambda_{\min} \approx 0$. It calls H physics-informed because $\mathcal{T}(A)$ embeds the differential constraint as the loss of a physics-informed neural network (PINN) [21] embeds it, while the $\mathcal{T}(\mathbb{B}_i)$ embed the data, but nothing is trained: the ground state is prepared. Wu et al.'s worked residual at $N = 4$ and the entries of the lifts are given in Appendix C.

3.3 Circuits, Filtering and Readout

Block encodings. Each term of H block-encoded separately and combined by a LCU (§ 2.4.1). The rank-one rows $\mathbb{B}_n(x)$ and $\mathbb{D}^{(0)}(x_s)$ pair the inverse Chebyshev feature map with a reflection about the all-zero state, on $n+2$ qubits at subnormalisation $\sqrt{2^{n+1}}$, and run "with high probability"[30]. $\mathbb{G}$ is encoded *approximately* by two conditioned subroutines that load its entries as rotation amplitudes. The circuit's top-left block is $\mathbb{G}$ divided by the prefactor

$$\alpha_{\mathbb{G}}^{\mathrm{Wu}} = \left(2^{n-1} + 2^n\right) \|\mathbb{G}\|_S, \qquad \|\mathbb{G}\|_S = \max\!\left(\|\mathbb{G}\mathbb{G}^\top\|_1,\ \|\mathbb{G}^\top\mathbb{G}\|_1\right), \tag{3.5}$$

on $2n + 3$ qubits, "with negligible errors in matrix entries that exponentially decrease as n increases", and with $O(n^3)$ gates once the multi-controlled gates are decomposed: "polynomial in n and thus efficient" [30]. The registers, subroutines and angles are given in Appendix D.1.

Ground-state preparation. The ground state is prepared by QITE [52], which multiplies each eigencomponent of an initial state by $e^{-\beta E}$, so that the lowest dominates once β exceeds the inverse gap. Wu et al. apply it to $\tilde{H} = H/\|H\|_F$ by an indefinite-parity QSVT sequence [40] whose phases are fitted to $e^{-\beta x}$ (§§ 2.4.2 and 2.4.3). The time is set by an "approximate

lower bound",

$$\beta \;\geq\; \frac{\lambda_{\max}}{\lambda_2\,(n-1)}, \tag{3.6}$$

with $\lambda_{\max}$ and λ_2 the largest and second-smallest eigenvalues of $\tilde{H}$ (or H). Wu et al. expect a preparation time of order $\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon)$ by eigenstate filtering [54], or $\Delta^{-1/2}$ by algorithmic thermalisation [61], and notes that the gaps "are problem-dependent and in general are not easy to bound". Its phase sets and the dilation of $\tilde{H}$ are given in Appendix K.5.

Readout. The field is read out point by point by an interferometric overlap measurement [53, 62]. Measuring the all-zero outcome after the inverse feature map returns only the squared overlap, so Wu et al. measure two probabilities, one for the prepared state and one for that state superposed with a reference, and combine them into the signed overlap $\langle\tau(x)|\psi_g\rangle$ and with η, the field value $f_Q(x)$. The identity is restated in Equation (F.6).

3.4 Doubled-Space Extension to Nonlinearity

A nonlinear term such as $(f')^2$ cannot be expressed as a linear operator on $|\psi\rangle$. Wu et al.'s treatment doubles the space: the unknown becomes $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$ on $2n$ qubits, a product of two model functions is the overlap of $|\Psi\rangle$ with a product of two feature states, and the product-to-sum rule folds that product back into one Chebyshev series on one extra qubit,

$$\langle\tau(x)|_n \otimes \langle\tau(x)|_n = \langle\tau(x)|_{n+1}\;\mathbb{m}_1, \qquad \mathbb{m}_1 \in \mathbb{R}^{2N\times N^2} \tag{3.7}$$

(their Eq. 8), so that $f^{(i)}f^{(j)}$ becomes $\mathbb{m}_1(\mathbb{G}^i \otimes \mathbb{G}^j)$ on $|\Psi\rangle$. Linear terms are carried into the same space by tensoring them with the collapse row $\mathbb{D}^{(0)}(x_s)$, which supplies the constant of one copy (their Eq. 30), and the Hamiltonian is built as in Equation (3.4). Wu et al. demonstrate the construction on two quadratic ordinary equations at $n = 2$ and reports that the Hamiltonian then has at least 2^{2n-1} degenerate zero-energy states, so that solving the equation is "converted to that of seeking a degenerate state" that "closely matches the analytical solution". The construction is quadratic: a term of degree p needs p copies, a register of dimension N^p (Appendix H.2), as Paine's concurrent work also observes [31].

3.5 Critical Assessment

Strengths. The construction is unified: ordinary and partial differential equations and their data constraints take one least-squares form, and nonlinear terms have a route into it. Its kernel is exact whenever the solution is a polynomial within the truncation. The Chebyshev basis needs no spatial grid, so the model and all its derivatives are available at every $x \in [-1, 1]$ from one prepared state, and the Hamiltonian has few terms, each with polynomially many gates.

Weaknesses in the cost. With this construction the spectrum of the Hamiltonian (not just its circuits) may fix the cost. A polynomial filter's degree grows as the square root of the encoding's subnormalisation over the gap, and no encoding brings that subnormalisation below $\|H\|$ (Equation (2.2)). Because H is the Gram matrix of the residual, the square root is the residual's own norm over the root of the gap. Every derivative enters the residual as a power of $\mathbb{G}$, whose norm grows as N^2 (§ 4.2), so an operator of order k grows as N^{2k}, the familiar ill-conditioning of dense Chebyshev differentiation [36]. For an equation whose leading coefficient does not vanish, and a gap that does not grow with N, the number of queries therefore grows with each derivative eliminated through $\mathbb{G}$, whatever the circuit. A least-squares functional built from operators of high order is the setting in which first-order-system least squares carries the derivatives as unknowns instead [38]. The expected time $O(\Delta^{-1}\,\mathrm{polylog}(1/\varepsilon))$ leaves its units open: if Δ is the gap of $\tilde{H}$, it is the gap of H divided by $\|H\|_F$, and the norm re-enters there. Wu et al.'s report of "a sizable gap for the studied problems" is therefore not yet evidence about the cost. Separately, the prefactor of Equation (3.5) is not compared with $\|\mathbb{G}\|_2$, which it exceeds by a factor growing with N, so the circuit for $\mathbb{G}$ is loose even against the matrix it encodes. § 4.5.1 and § 4.6 separate the two.

Weaknesses in the evidence. Each inhomogeneous example combines a rounded constraint point with a truncated Maclaurin source, and either alone sets an error floor that no increase in n removes: a zero imposed where the solution does not vanish contradicts the equation, and a truncated source changes the equation being solved (§§ 4.2 and 4.3, Appendix K). The PDE examples sometimes take their invariant points from the known solution, which a new problem does not supply, and their invariant sets omit the initial or boundary function that fixes the solution, so the kernel degenerates as n grows beyond the printed resolution. The heat equation's stated initial function sits at $t = 0$, inside the time domain, where it would pose a backward problem (§ 4.3). The description of H as positive definite with distinct eigenvalues is stronger than a sum of Gram matrices allows, and the case the method aims at, a solution represented exactly, is a zero eigenvalue (§ 4.4). The time rule (Equation (3.6)) contains no ε, so it cannot certify a target accuracy, and it needs $\lambda_{\max}$ and λ_2, which the method does not supply in advance (Appendix K.4).

Weaknesses in scope. The nonlinear route projects rather than prepares: any function of the Hamiltonian acts as a scalar on its zero-energy eigenspace, so a filter returns the projection of the initial state onto a kernel whose dimension, by Wu et al.'s own count, is at least 2^{2n-1}, and the direction it picks within that kernel is set by the initial state, not by the equation (§ 6.4). The readout identity treats the prepared state as the ground state with certainty, although a filter followed by post-selection succeeds with probability below one, and it returns one point of the field per measurement campaign (§ 7.6). The construction is stated for one real scalar field, and Wu et al. set complex-valued states aside, since with a Fourier basis "the resulting total energy function is of a complicated form, rendering it difficult to find the ground

state" (§ 6).

Open problems. Wu et al.'s Discussion leaves three questions open: "studying the spectral gap dependence is an important question for the future work"; the choice of basis, for which "one guiding principle here can be the cost of differentiation, where the goal is minimizing the depth of quantum circuits to implement derivatives, multiplication etc."; and the data constraints, whose choice it calls "crucial for regularizing solutions and growing the gap" [30]. Paine's concurrent work [31] finds, on a few small instances, that the normalised gap depends on the Hamiltonian's construction and the function encoding, but gives no scaling with resolution or circuits, so it raises the first question without answering it. The first weakness above gives the first two problems a common quantitative form. The gap must be set against the Hamiltonian's norm, and the cost of differentiation, which Wu et al. measure by the depth of a circuit, is also paid in that norm, a factor growing as N^2 in the residual's norm for each derivative eliminated through $\mathbb{G}$, at every query of the filter.

Read against the degree of Equation (2.2), the assessment leaves three questions, which the next three chapters answer in turn:

(i) *Is the query count set by the encoder or by the Hamiltonian?* (§ 4)

(ii) *Can the residual be reformulated so that its Hamiltonian is banded and well conditioned without changing its solution, and what does the reformulation cost?* (§ 5)

(iii) *Can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel, and does the reformulation's advantage survive there?* (§ 6, which first carries several fields and complex-valued fields as coupled linear systems, since the lift that answers it is one)

The questions follow the two terms of the degree. The first two concern its numerator: (i) asks whether the norm that sets the query count belongs to the circuit or to the operator, and (ii) whether changing what the residual carries can lower that norm without moving the kernel, and what the change costs elsewhere, in the state, the circuit or the gap. The third concerns the denominator at its limit. A degenerate kernel has no gap above its ground value, so a filter cannot single out one state in it; (iii) therefore asks first for a Hamiltonian whose ground state is unique, the condition under which the degree is defined at all, and only then whether the reformulation still lowers it.

4 The Standard Form and Its Cost

Wu et al.'s case for efficiency counts gates: every term of their Hamiltonian has a circuit of polynomially many, so one application of it is cheap (§ 3.3). The filter applies the Hamiltonian as many times as its degree, and the degree depends on the circuit only through its subnormalisation α_H, which no block encoding brings below $\|H\|$ (Equation (2.2)). Two quantities

are therefore folded into α_H. We call the ratio $\alpha_H/\|H\| \geq 1$ the *encoder's cost* and $\|H\|$ itself the *Hamiltonian's cost*: the first belongs to the circuit, the second to the operator. Since the degree is linear in $\alpha_R = \alpha_H^{1/2}$, the encoder's cost enters it as $\alpha_R/\|R\|$. The first question of the assessment asks which of the two sets the query count, and answering it needs an encoding tight enough that whatever growth remains cannot be the circuit's.

The construction under examination is that of Wu et al. [30], built on the Chebyshev latent space and lifts of Paine et al. [29], and it is stated here once, in the form every later chapter runs. We call it the *standard form*: the state holds the field alone, and every derivative is eliminated through the dense differentiation matrix $\mathbb{G}$, so that a term $f^{(j)}$ acts as $\mathbb{G}^j$ on the field's coefficients and the equation is one block of a stacked residual. Where a printed choice would make a problem inaccurate, answer-dependent or ill posed, the standard form replaces it at the point where the construction meets it, and Table 4.2 collects the departures.

4.1 The Model

We use Wu et al.'s model, Equations (3.1) and (3.2), unchanged, and write it in the orthonormal basis that makes its scale explicit. The prefactors of the feature map define the basis functions $\varphi_0 = T_0/\sqrt{N}$ and $\varphi_k = T_k/\sqrt{N/2}$ for $1 \leq k < N$, which are orthonormal under the discrete inner product on the N Chebyshev nodes, and $\langle\tau(x)| = (\varphi_0(x), \ldots, \varphi_{N-1}(x))$ is the row of their values at x. A truncated solution $f_N = \sum_{k<N} b_k \varphi_k$ is then $f_N(x) = \langle\tau(x)|b\rangle = \|b\| \, \langle\tau(x)|\psi_e\rangle$ with $|\psi_e\rangle = b/\|b\|$, so the model f_Q of Equation (3.2) represents f_N exactly at $|\psi\rangle = |\psi_e\rangle$ and $\eta = \eta_e$, where, by Parseval's identity under the Chebyshev weight (Appendix B.3),

$$\eta_e = \|b\|^2 = \frac{N}{\pi} \int_{-1}^{1} \frac{f_N(x)^2}{\sqrt{1-x^2}} \, dx. \tag{4.1}$$

The state carries the direction of b and the scale its length, and η_e is fixed by the solution and the resolution alone, growing with N for a smooth solution. It describes representability only: the ground state $|\psi_g\rangle$ is set by the residual's kernel and approaches $\pm\,|\psi_e\rangle$ only as far as f_N satisfies the constraints. The regular datum recovers the scale and its sign: $y_s/\,\langle\tau(x_s)|\psi_g\rangle$ is negative when the ground state, defined only up to sign, opposes the datum, so Wu et al.'s $\eta > 0$ is a convention and the decoded field carries the sign.

The feature vector is kept unnormalised, as Wu et al.'s derivation uses it. Its squared norm oscillates about one and reaches $(2N-1)/N$ at $x = \pm 1$ (Appendix B.3), and every constraint row of § 4.3 inherits that weight. Where the residual has no exact kernel the ground state is a least-squares compromise between its rows, and normalising $\tau(x)$ point by point, which gives Wu et al.'s $\tilde{\tau}(x)$, would change the rows' relative weights and with them the solution; it would also make collapse rows at different points disagree. The rows are therefore built, and block-encoded, from the unnormalised vector. A function of several variables is carried on one register per axis, each with its own n, with the feature vector $\langle\tau(x_1)| \otimes \langle\tau(x_2)| \otimes \ldots$ in Kronecker order. The running example of this chapter is Wu et al.'s Fig. 3b: the critically damped oscillator

$f'' + 4f' + 4f = 0$ with $f(-1) = 0$ and $f(0) = \frac{1}{2}$, whose solution is $f = \frac{1}{2}(1 + x)e^{-2x}$. It is R2a in the benchmark problems of Table 4.1, which the rest of the thesis uses. They are ordered so that each adds one capability to those above it, and a failure therefore localises to the capability that introduced it; each label names the problem with the data of § 4.3, not the printed panel.

Table 4.1: The benchmark problems R1–R10, each adding one capability to those above it. R2–R6 are the examples of Wu et al. [30] (their Figs 3–7) with this thesis's data; R7–R10 go beyond one real field. $\{a_2, a_1, a_0\}$, coefficients of $a_2 f'' + a_1 f' + a_0 f = 0$; l, m, Legendre degree and order; a/ν, initial amplitude over viscosity. Equations, data and references in full: Tables J.1 and J.2.

	Problem	New capability
R1	$f' + 2f = 0$, $f(-1) = 1$	the smallest end-to-end system
R2	three second-order equations, $\{1, 4, 4\}$, $\{1, -2, -3\}$, $\{1, 5, 400\}$	order two; Dirichlet and Neumann data; stiffness (R2c)
R3	Legendre, $m = 0$ ($l = 4, 5$), and associated Legendre, $m = 1$ ($l = 5, 6$)	variable coefficients and lifts; an exact kernel; the substitution of § 4.3 (R3c, R3d)
R4	four inhomogeneous equations	sources by their Chebyshev coefficients
R5	Laplace (two space axes), heat and wave (one space and one time axis)	two axes; initial and boundary functions as datum slices
R6	$4f'' + 2f'^2 + f = 0$; $f'' - 2f^2 + x = 0$	the doubled space (a degenerate kernel)
R7	$u' = v$, $v' = -\pi^2 u$	two coupled fields; ratio rows
R8	$y' = -y + y^2$, $y(0) = \frac{1}{2}$	the Carleman lift on a time axis, both forms
R9	viscous Burgers, periodic, $a/\nu = 0.03$–10 on 4–16 sine modes	a field-scale lift across its convergence boundary
R10	two-dimensional Navier–Stokes in vorticity form, periodic, 12–80 modes	incompressible flow; the tensor layout to 316 modes

4.2 Differentiation and Multiplication

In the normalised basis the differentiation matrix has the entries

$$[\mathbb{G}]_{k,p} = \begin{cases} 2p, & p > k \geq 1,\ p - k \text{ odd}, \\ \sqrt{2}\,p, & k = 0,\ p \text{ odd}, \\ 0, & \text{otherwise}, \end{cases} \tag{4.2}$$

and it is exact on the represented space, since the derivative of a polynomial of degree below N stays in it [36]. $\mathbb{G}$ is upper triangular, dense and nilpotent. $\ker \mathbb{G}^j$ is the space of polynomials of degree below j, and about half the positions above the diagonal are non-zero. Its norm grows as N^2: $\|\mathbb{G}\|_2/N^2$ converges to $1/(2j_{-3/4,1}) \approx 0.472$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and $\|\mathbb{G}^k\|_2/N^{2k}$ converges to a positive constant for every k. Because the basis is orthonormal at the Chebyshev nodes, $\mathbb{G}$ is orthogonally similar to the matrix that differentiates nodal values, so the growth is not an artefact of the normalisation, and its order is the classical one for dense Chebyshev differentiation [34, 36]. The structure, the similarity and the bounds $N^2/(4\sqrt{2}) \leq \|\mathbb{G}\|_2 \leq N^2/\sqrt{2}$ are proved in Appendix B.4, and the two limits in Appendix E.4.

A polynomial coefficient $a(x)$ acts by the lift of Paine et al. [29], $x^p \langle\tau(x)|_n = \langle\tau(x)|_{n+1} \mathbb{M}_{x^p}$ (Wu et al.'s Eq. 5), extended by linearity to $\mathbb{M}_a = \sum_p a_p \mathbb{M}_{x^p}$. It is exact whenever $\deg a \le N$, a constant enters as $a_0 \mathbb{M}_1 = a_0 \sqrt{2}\,[\,I;\,0\,]$, and $\|\mathbb{M}_a\|_2 \le \sqrt{2} \sup|a|$ (Proposition 12). The equation $\sum_j a_j(x) f^{(j)} = r(x)$ then has the equation block

$$A_{\text{std}} = \sum_j \mathbb{M}_{a_j} \mathbb{G}^j - \mathbb{M}_r\, \mathbb{D}^{(0)}(x_s), \qquad \mathbb{M}_r\, \mathbb{D}^{(0)}(x_s) = \frac{|\hat{r}\rangle \langle\tau(x_s)|}{f(x_s)}, \tag{4.3}$$

in which the source enters through the collapse row of § 4.3, with $\hat{r}$ the coefficients of r on the lifted register. The subscript marks the standard form, defined in Definition 2.

Wu et al.'s truncated Maclaurin source (§ 3.2) poses a different equation, whose solution the ground state then approximates, with an error floor that no increase in n removes (Appendix K.2). The standard form loads the Chebyshev coefficients of the source instead, through the same collapse row, so the source is as accurate as the series that represents the solution, geometrically convergent for a source analytic on the interval [33].

Wu et al. treat at most two variables. On any number of axes a term is the Kronecker product over the axes of $\mathbb{M}_a \mathbb{G}^{j_a}$, each axis lifted if any term on it needs the exact product, and a constant coefficient on a lifted axis enters through $\mathbb{M}_1$ (Appendix C.2). The benchmark problems use at most two.

4.3 Constraints and Data

The constraints are built from Wu et al.'s rank-one rows (their Eqs 9–11). The zero row $\mathbb{B}(x) = \sqrt{N}\,|0\rangle \langle\tau(x)|$ maps a series to the constant function $f_N(x)$, so $\mathbb{B}(x)\mathbb{G}^j$ annihilates exactly the series whose j-th derivative vanishes at x. The collapse row $\mathbb{D}^{(0)}(x_s) = \mathbb{B}(x_s)/f(x_s)$ maps the solution to the constant 1 whatever its scale, which is what lets a source enter Equation (4.3).

A zero row is exact only at a point where the solution's derivative of that order vanishes. Wu et al.'s rounded constraint points (§ 3.2) demand a zero the solution lacks, least squares trades it against the equation, and the error floors at every n; on R4b to R4d the rounded point and the truncated source each floor it on their own (Appendix K.2). The standard form takes the exact point, the root of the solution's derivative nearest Wu et al.'s value. The root depends on the answer: we compute it from the reference solution, outside the method, so a new problem must supply its constraint points with its data. We keep the invariant row, rather than impose the problem's stated data, so that the residual keeps Wu et al.'s rows and only the defect changes; the stated data could enter without the answer, as point datum rows of Definition 2, which no run here uses.

For an ordinary equation of order k whose leading coefficient has no zero on $[-1, 1]$, the invariant constraints fix the solution up to its scale, which the regular datum then fixes, exactly when they have rank $k - 1$ on the equation's k-dimensional solution space (Proposition 11). The

rank is that on the solutions, not of the functionals, so $k-1$ constraints chosen without regard to the equation can fail. Where the count does not apply, for an inhomogeneous equation or a vanishing leading coefficient, and for the gap at a given n, the computed kernel decides. We call it *trusted* when the residual's near-kernel is one-dimensional, its smallest singular value separated from the next by a ratio of at least 20 (§ 7.2). Trust certifies that the residual singles out one state, not that the field it decodes to has converged, which is measured separately.

A partial differential equation has an infinite-dimensional solution space and no such count. Wu et al.'s invariant sets (§ 3.2) omit the function that fixes the solution: on Laplace's equation (R5a) a whole family of modes satisfies the equation and its three homogeneous sides, only one of which matches the omitted top side, so the near-kernel grows once the truncation resolves the others (Appendix C.5); the sets of the heat equation (R5b) and the wave equation (R5c) fail in the same way. The standard form imposes the initial or boundary function g itself, on its slice $x_a = x^*$, by the *datum-slice* rows

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle \tau(x_s)|}{f(x_s)}, \tag{4.4}$$

with $\mathbb{S}_a(x^*)$ the zero row $\mathbb{B}(x^*)$ on axis a and the identity on the others, $\hat{g}$ the coefficients of g broadcast along axis a, and x_s a regular point. They vanish on the solution and its multiples and leave the scale free (Proposition 9), are built from the problem's statement, serve both forms, and on R5a leave a one-dimensional kernel that is trusted at every resolution computed from $n = (3, 3)$ to $(6, 6)$. The heat problem is posed forward from $t = -1$, with the solution evaluated there as its initial function, since data imposed at the printed $t = 0$ would pose a backward heat problem on $[-1, 0]$ (Appendix C.3); like R4's constraint points, these data are set up from the known solution.

Wu et al.'s associated Legendre examples R3c and R3d (their Figs 4c and 4d, $l = 5$ and 6 with $m = 1$) multiply the equation through by $1 - x^2$, which leaves a double zero of the leading coefficient at each end and a solution $P_l^1 = -\sqrt{1-x^2}\,P_l'$ [63, §14.6] with a branch point there, which a Chebyshev series approximates only algebraically, so that neither problem has a trusted ground state at the printed resolution (Appendix C.4). The standard form solves for $v = f/\sqrt{1-x^2}$ instead, whose equation has a leading coefficient with simple zeros and the polynomial solution $v = -P_l'$, represented exactly once $N > l - 1$ (Proposition 10): the error on R3c is 4.9×10^{-15} at $n = 3$. It is also the form the descriptor can run (§ 5.6).

Vector data, such as the initial values of a coupled system, enter as ratio rows, defined with systems in § 6.2. Every reference is the problem's analytic or Galerkin solution; the one printed solution that does not satisfy its own problem, Laplace's, is noted in Appendix K.3.

4.4 The Standard Form

The equation block and the constraint rows are stacked into one residual,

$$R_{\text{std}} = \begin{pmatrix} A_{\text{std}} \\ C_1 \\ \vdots \\ C_L \end{pmatrix}, \qquad H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}} = A_{\text{std}}^{\dagger} A_{\text{std}} + \sum_i C_i^{\dagger} C_i, \tag{4.5}$$

every block at weight one, so that H_{std} is a sum of Gram matrices of Wu et al.'s form, Equation (3.4), with the datum slices among the C_i. The ground state of H_{std} is the smallest right singular vector of R_{std}. The classical reference computes it from R_{std} by a singular value decomposition, or by Lanczos iteration beyond its reach (§ 7.2), and never forms H_{std}, which would square the residual's condition number [64] (§ 7.2). The circuit likewise encodes R_{std}, and H_{std} is never formed (§ 7.3).

H_{std} is positive semi-definite, and singular in the very case the method aims at, a solution the truncation represents exactly, contrary to Wu et al.'s description (§ 3.2). When the solution is a polynomial of degree below N, any source is represented exactly and every constraint holds exactly at its point, $\mathbb{G}$ differentiates the solution and the lifts multiply it exactly, so its coefficients are annihilated by R_{std} and give H_{std} an exact zero eigenvalue; Legendre's equation $(1-x^2)f'' - 2xf' + 20f = 0$ (R3a, Wu et al.'s Fig. 4a), whose solution is P_4, has one from $n = 3$. Otherwise the truncated solution leaves a residual, $\lambda_0 > 0$, and the ground state minimises the total squared residual: the construction is least squares. Nor need the bottom of the spectrum be distinct, as Wu et al.'s partial differential sets show (§ 4.3). The thesis relies on neither property; it relies on an isolated ground state, measured as a trusted kernel (§ 7.2).

Definition 2 (The standard form): For a linear problem on one or more Chebyshev axes, the *standard form* is the stacked residual R_{std} of Equation (4.5), whose equation block is Equation (4.3), lifted on every axis where a coefficient or a source needs the exact product, and whose constraint rows, each at weight one, are built from four kinds of row: zero rows $\mathbb{B}(x^*)\mathbb{G}^j$, at a point or on a slice, for data that vanish; the collapse row $\mathbb{D}^{(0)}(x_s)$, which carries a non-zero point datum $f^{(j)}(x^*) = y^*$ as the row $\mathbb{B}(x^*)\mathbb{G}^j - y^*\mathbb{D}^{(0)}(x_s)$, as it carries a source into the equation block; ratio rows for vector data (§ 6.2); and datum slices (Equation (4.4)) for initial and boundary functions. Its Hamiltonian is $H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}}$, and its solution is the smallest right singular vector of R_{std}, scaled by the regular datum $f(x_s) = y_s$.

Table J.3 lists the four kinds as model inputs. Table 4.2 collects every departure from the printed method: those above, and those of the pipeline, made where it is built. The departures in the residual change its data, never $\mathbb{G}$, the lifts or the Gram structure, so H_{std} has Wu et al.'s norm to leading order, the constraint rows growing more slowly than the equation block (Appendix E.4); R3c and R3d, whose substitution changes the equation, are costed in the form the standard form runs. Only the pipeline's departures change the cost, and they lower it: the

structured encoding of § 4.5 is far cheaper than Wu et al.'s. Against the published method the baseline is therefore not weakened: it has the same Hamiltonian, the cheapest exact encoding built, and better accuracy. How every comparison with the descriptor corrects for the difference between their encodings is declared with the design of the experiment (§ 7.7).

Table 4.2: Every departure from the printed method removes a defect of the printed choice or extends what the residual can carry. The standard form and pipeline of this thesis against the method of Wu et al. [30]. *Data* rows leave $\mathbb{G}$, the lifts and the Gram structure untouched, so H_{std} keeps Wu et al.'s norm to leading order (the R3c/R3d substitution, which changes the equation, excepted); *reach* rows extend the residual beyond one real field, one axis and linearity; *pipeline* rows alone change the cost. Constraint points and the heat equation's initial function are set up from the reference solution, so a new problem supplies its own. $\mathbb{G}$, Chebyshev differentiation matrix; R_{std} and $H_{\text{std}} = R_{\text{std}}^{\dagger} R_{\text{std}}$, the standard form's stacked residual and Hamiltonian; α_R, the residual's subnormalisation; $\|H\|_F$, Frobenius norm; ε, target suppression; p_G, post-selection success probability; Where, the section that argues the row. Evidence for every row: Appendix K.

Choice	This thesis	Defect of the printed choice	Where
Data: what the residual carries			
Constraint points	the exact point, set up from the reference solution	a rounded zero the solution lacks floors the error	§ 4.3
Sources	Chebyshev coefficients on the lifted register	a truncated Maclaurin series poses a different equation	§ 4.2
PDE data	datum slices for initial and boundary functions; the heat data at $t = -1$	hand-picked invariant sets let the near-kernel grow with n; data at $t = 0$ pose a backward heat problem	§ 4.3
Double zero	R3c and R3d solved for $v = f/\sqrt{1-x^2}$	a branch point limits convergence to algebraic	§ 4.3
Reach: what the residual can express			
Axes	any number (two run)	a limit of scope: two axes	Appendix C.2
Fields	several fields on one state, ratio rows, realification	a limit of scope: one real scalar field	§ 6
Nonlinearity	the Carleman lift, in both forms; the doubled space kept in the standard form	the doubled space projects onto a kernel of dimension at least 2^{2n-1}	§§ 6.4 and 6.5
Pipeline: how the residual is encoded, filtered, solved and read			
Hamiltonian	the stacked residual, solved and encoded from R_{std}; H_{std} never formed	forming H squares the residual's conditioning	§§ 4.4 and 7.2
Encoding of $\mathbb{G}$	an exact factorisation, α within a factor 2.12 of $\|\mathbb{G}\|_2$	an approximate encoding, α looser than $\|\mathbb{G}\|_2$ by about N^3	§ 4.5
Filter	the minimax edge filter, its degree set a priori	the time rule does not reach the suppression ε	§ 7.4
Phases and scale	GQSP phases from the target alone; scaled by α_R^2	fitted phases carry no guarantee; $\|H\|_F$ is not a subnormalisation	§§ 7.3 and 7.4
Initial state	the geometric product state; its overlap γ reported	not specified	§ 7.4
Readout	the field decoded classically from the prepared state	the interferometric identity assumes $p_G = 1$	§ 7.6

The baseline is validated against Wu et al.'s printed results. With every printed choice restored, the rounded points, the Maclaurin sources and the hand-picked invariant sets, the same construction reproduces the matrices printed in Wu et al.'s supplementary material at $n = 2$ entry by entry, which checks the operators. Its basis gives, for the exact solution, the printed scale factor of each of Wu et al.'s eleven ordinary examples to within one unit of the last printed digit, which checks the normalisation: for Fig. 3b, as printed at $n = 3$, the exact solution gives

$\eta_e = 1.29009$ against the printed 1.29. The printed scale factors of their three partial differential and two nonlinear examples are not reproduced, by the exact solution's η_e or by any ground state we computed, and the cause is not settled. The record of every example, the triage of the printed choices and the verdict on each of Wu et al.'s claims are given in Appendix K.

4.5 Encoding the Derivative

Every term of the standard form is a product of lifts, rank-one rows and powers of $\mathbb{G}$. The rank-one rows have exact circuits whose subnormalisation is fixed by their own norm, so the term whose encoding decides the encoder's cost is $\mathbb{G}$.

4.5.1 Wu et al.'s Encoding

Wu et al.'s circuit for $\mathbb{G}$ (their Fig. 9; § 3.3) selects each entry (k, p) by a register subtraction over a superposition of offsets and loads the column value $2p$ as an amplitude through a ladder of small controlled rotations. Its claims about gates and entry errors hold, the first as a bound and the second to leading order (Appendix D.1). The claim that decides its cost is the prefactor of Equation (3.5), which is valid but not compared with $\|\mathbb{G}\|_2$. Since $\|\mathbb{G}\|_S = \|\mathbb{G}^\top \mathbb{G}\|_1$ grows as $N^4/3$, the prefactor grows as $N^5/2$, against $\|\mathbb{G}\|_2 \sim N^2/(2j_{-3/4,1})$, so its looseness tends to $j_{-3/4,1}N^3$ with $j_{-3/4,1} \approx 1.06$; at $n = 7$ it is 2.2×10^6 (exact classical computation, Appendix E.5). To cost it beside the others we rebuild its mechanism. The rebuilt circuit's own subnormalisation is a third of the printed prefactor, so every cost quoted for Wu et al.'s encoding is the kinder reading; its entries are approximate, with an error of order N^{-5} (Proposition 13), so it enters resource estimates only, composed into the same stacked residual as the exact encoding below. The rebuild, its subnormalisation, its error bound and the verdict on each of Wu et al.'s circuit claims are given in Appendix D.1.

4.5.2 An Exact Factorisation

Wu et al.'s circuit is approximate, and its prefactor large, because it loads the column value of each entry as a rotation amplitude. Every non-zero entry of Equation (4.2) is a product of three factors, each depending on one index alone: a row weight, equal to $1/\sqrt{2}$ in row 0 because of the normalisation of T_0 and to one elsewhere; a selector depending only on the offset $p - k$, non-zero when $p > k$ and $p - k$ is odd; and the column value $2p$, the weight with which differentiation passes T_p to the lower polynomials. Each factor has an exact block encoding with no amplitude loaded by rotation.

Proposition 1 (Exact factorisation of the differentiation matrix)**.** *For every $n \geq 1$ the differentiation matrix on the normalised Chebyshev coefficients factorises as*

$$\mathbb{G} = R_0 \left(2U_{\text{odd}}\right) \Lambda,$$
$$R_0 = I - \left(1 - \tfrac{1}{\sqrt{2}}\right) |0\rangle \langle 0|, \qquad \left(U_{\text{odd}}\right)_{kp} = \left[p > k,\ p - k\ \textit{odd}\right], \qquad \Lambda = \operatorname{diag}(p), \tag{4.6}$$

where U_{odd} is the sum of the $N/2$ truncated shifts by odd offsets.

The proof compares the two sides entry by entry with Equation (4.2) and is given in Appendix E.1. As a check on our implementation, not on the proposition, the block of the composed circuit matches the independently assembled $\mathbb{G}$ entrywise to 3.4×10^{-13} (absolute) for $n \leq 7$.

4.5.3 Circuits and Cost

Each factor is encoded as a linear combination of unitaries [41] whose terms are unitary already, so no part of the encoding is approximate. The column index Λ is a sum of $n+1$ Pauli-Z strings, encoded at $\alpha_\Lambda = N - 1 = \|\Lambda\|_2$, and the row weight R_0 is a positive combination of the identity and the reflection about $|0\rangle$, encoded at $\alpha_{R_0} = 1 = \|R_0\|_2$ (Appendix E.1). The sum of shifts is the one factor that needs arithmetic (Figure 4.1). Its PREPARE puts an offset register in the uniform superposition over the $N/2$ odd offsets ℓ, and its SELECT is the in-place subtraction $p \leftarrow p - \ell$ of a ripple-carry circuit [65], which sets a borrow flag when $p < \ell$. Post-selecting the flag on zero removes exactly the terms that would wrap around, so the block is $\frac{2}{N}\big[p > k,\ p-k \text{ odd}\big]$: an exact encoding of U_{odd} at $\alpha = N/2$ with $O(n)$ Toffoli gates, in which one PREPARE serves all $N/2$ terms because SELECT is a permutation (Appendix E.2).

Quantikz circuit: offset register ℓ (X on the lowest qubit, $H^{\otimes(n-1)}$ on the rest), system register p, carry and borrow ancilla; a ripple-carry subtractor $p \leftarrow p - \ell$ spanning all five wires; PREPARE†; post-selection of ℓ and the borrow flag on $\langle 0|$.

Figure 4.1: The odd-offset shifts are one subtraction. PREPARE (left) puts the offset register ℓ in a uniform superposition over the $N/2$ odd offsets, by an X on its lowest qubit and Hadamards on the rest. SELECT is a ripple-carry subtractor [65] that replaces the system register p by $p - \ell$ and sets the borrow flag when $p < \ell$. PREPARE† (right), with post-selection of ℓ and the borrow flag on $|0\rangle$, keeps $\frac{2}{N}|k\rangle\langle p|$ for every odd $\ell = p - k \geq 1$, and the wrapped terms vanish exactly. The block is $U_{\text{odd}}/(N/2)$, so $\alpha = N/2$, on n system and $n+2$ ancilla qubits.

The three encodings multiply, and so do their subnormalisations. Their product is an exact block encoding of $\mathbb{G}$ at $\alpha_{\mathbb{G}} = 1 \cdot N \cdot (N-1)$.

Proposition 2 (Cost of the structured encoding)**.** *The factorisation of Proposition 1 compiles to an exact block encoding of $\mathbb{G}$ on n system qubits with*

$$\alpha_{\mathbb{G}} = N(N-1), \qquad \frac{\alpha_{\mathbb{G}}}{\|\mathbb{G}\|_2} \longrightarrow 2j_{-3/4,1} \approx 2.117, \qquad O(n^2) \text{ gates}, \qquad n+4 \text{ ancilla}, \tag{4.7}$$

of which $n-1$ ancilla are the selector of the $N/2$-term combination U_{odd}.

The limit is that of $\|\mathbb{G}\|_2/N^2$ (§ 4.2). The ancilla are the $n+2$ of U_{odd}'s offset register, carry and borrow, and one flag for each of the two products, the other factors' selectors reusing those qubits; the count is exact in the circuit for $n = 2$ to 8. The gate count assumes an ancilla-free synthesis of multi-controlled gates and falls to $O(n \log n)$ with a linear-cost synthesis that borrows idle qubits [66]. The proofs, and the few transpiled counts available, are given in Appendix E.3. At the sizes reached, the ratio $\alpha_{\mathbb{G}}/\|\mathbb{G}\|_2$ is 1.62 at $n = 2$ and rises to 2.11 at $n = 10$ (Table 4.3), against a ratio for Wu et al.'s encoding that grows as N^3. R_0 and Λ are encoded at their own norms, so the limiting factor is $\pi/2$ from U_{odd} and $4j_{-3/4,1}/\pi \approx 1.35$ from the product (Lemma 5).

The $n-1$ qubits that hold the offsets' superposition are the price of writing U_{odd} as one combination of its $N/2$ shifts; what that implies for the growth of the ancilla, against the descriptor's, is argued in § 5.8.

Table 4.3: The structured encoding of $\mathbb{G}$ stays within a constant of its norm; Wu et al.'s prefactor grows as N^3. Subnormalisation α of each encoding of the Chebyshev differentiation matrix $\mathbb{G}$, over its spectral norm $\|\mathbb{G}\|_2$, at n qubits ($N = 2^n$): the structured exact encoding (Proposition 2), which rises towards $2j_{-3/4,1} \approx 2.117$, and Wu et al.'s printed prefactor (Equation (3.5)). $j_{-3/4,1}$, first positive zero of the Bessel function $J_{-3/4}$. Full table to $n = 10$: Table E.1. Exact classical computation.

n	$\|\mathbb{G}\|_2$	Structured (Proposition 2)	Wu et al.'s (Equation (3.5))
2	7.39	1.623	49.9
4	121	1.986	4.05×10^3
7	7739	2.100	2.20×10^6
10	495 308	2.115	1.14×10^9

The remaining terms of the residual are built from the same pieces: a polynomial coefficient through its lift, a combination of truncated shifts built by the same post-selected subtraction (Appendix C.7); the rank-one rows by Wu et al.'s exact circuits; a source by one rotation tree over its Chebyshev coefficients; and a power $\mathbb{G}^k$ as a product of encodings that share one ancilla register with one flag per product, at subnormalisation $\alpha_{\mathbb{G}}^k$ while $\|\mathbb{G}^k\|_2 < \|\mathbb{G}\|_2^k$. The stacked residual is assembled into one reflection (§ 7.3) and verified against the independently assembled Hamiltonian (§ 7.5).

4.6 The Standard-Form Hamiltonian

For constant coefficients the structured encoding holds the encoder's cost of the whole residual, as well as that of $\mathbb{G}$, to a constant. On R2a at $n = 8$ the stacked residual's encoding has $\alpha_R = 4.26 \times 10^9$ against $\|R_{\text{std}}\|_2 = 5.51 \times 10^8$, both computed exactly from the built circuit and the assembled residual, a ratio of 7.74. From 4.72 at $n = 2$ the ratio rises to 7.57 at $n = 6$, levelling off towards the looseness of the residual's dominant term, $\mathbb{G}^2$, whose ratio $\alpha_{\mathbb{G}}^2/\|\mathbb{G}^2\|_2$ tends to a constant (Appendix E.4); no limit is proved for the whole residual. Since the degree is linear in α_R, this ratio is the whole of the encoder's cost in the degree; in $\alpha_H = \alpha_R^2$ it is squared, 59.9 at $n = 8$. With Wu et al.'s encoding the same ratio is 9.4×10^{11} at $n = 7$ and

6.1×10^{13} at $n = 8$: it grows as N^{3k}, a factor N^3 for each of the k factors of $\mathbb{G}$ in the leading term.

For an equation of order k with constant coefficients and $a_k \neq 0$, $\|\mathbb{G}^k\|_2/N^{2k}$ tends to a positive constant, so $\|R_{\text{std}}\|_2 = \Theta(N^{2k})$ and $\|H_{\text{std}}\| = \|R_{\text{std}}\|_2^2 = \Theta(N^{4k})$, the constraint rows growing more slowly (Appendix E.4). The gap is measured; no proof is given. On R2a it is 7.04 at $n = 3$ and 7.04 at $n = 10$, and on $f'' - 2f' - 3f = 0$ (R2b, Wu et al.'s Fig. 3c) it is 0.415 at $n = 4$ and 0.415 at $n = 10$, unchanged to the figures shown at every n between, while $\|H_{\text{std}}\|$ on R2a grows from 2.8×10^5 to 2.0×10^{22}. Because no block encoding has α_H below the norm, Equation (2.2) gives

$$\|H_{\text{std}}\| = \Theta\big(N^{4k}\big), \quad \Delta = \Theta(1) \text{ (measured)} \quad \Longrightarrow \quad d = \tilde{\Omega}\big(N^{2k}\big) \text{ for every block encoding,} \tag{4.8}$$

and $d = \tilde{\Theta}(N^{2k})$ for the structured encoding, whose $\alpha_R/\|R_{\text{std}}\|_2$ stays bounded. At second order, the order of Wu et al.'s examples, $\|H_{\text{std}}\| = \Theta(N^8)$ and no block encoding brings the degree below $\tilde{\Omega}(N^4)$. This is the Hamiltonian's cost, and its growth can be measured without any encoding. Every block encoding of R_{std} has $\alpha_R \geq \|R_{\text{std}}\|_2$, and the minimax filter's degree does not increase as α_R falls, so the degree at $\alpha_R = \|R_{\text{std}}\|_2$, which we call the *ideal-encoding bound*, is the least this filter needs on this residual through any encoding of it (Proposition 18). The floor is the minimax edge filter's, which prepares every state in this thesis; for polynomial filters in general it holds only as far as the optimality of Lin and Tong's construction supports [54].

The polynomial gate count that Wu et al. report for their operators [30], and that the rebuilt circuits bear out (Appendix D.1), is the cost of one query. The filter makes $\tilde{\Omega}(N^{2k})$ queries, exponentially many in n, however few gates each takes. Wu et al.'s claim of efficiency therefore holds for the circuit and says nothing about the method, consistent with its leaving the gap to future work.

The two costs predict different growth for the degree. With Wu et al.'s encoding the looseness, growing as N^{3k}, multiplies a degree that already grows as N^{2k}, so the degree grows as about N^{5k}; with the structured encoding, and at the ideal-encoding bound, it grows as N^{2k}. The difference between the first two is the encoder's cost; the common exponent is the Hamiltonian's, and § 8.2 measures all three. For a ground value near zero $\Delta \approx \sigma_1^2$, so $\sqrt{\|H_{\text{std}}\|/\Delta} \approx \|R_{\text{std}}\|_2/\sigma_1$: the filter pays the conditioning of the residual, and only an eigensolver acting on $H_{\text{std}} = R_{\text{std}}^\dagger R_{\text{std}}$ pays its square [64] (§ 4.4). That conditioning grows as N^{2k} because each of the k derivatives is eliminated through $\mathbb{G}$. A residual whose blocks each carry one derivative would grow only as the norm of one derivative operator, whatever the order, and less again if that operator were banded. These are the classical remedies: first-order-system least squares carries the derivatives as unknowns so that no block is of high order [38], and the ultraspherical method makes differentiation banded [36] (§§ 2.3.2 and 2.3.3).

The conclusion holds for constant coefficients only: there the structured encoding bounds

$\alpha_H/\|H_{\text{std}}\|$, and for a lifted variable coefficient it is loose. On R3a the circuit pays for the product $\mathbb{M}_a\mathbb{G}^2$ as $\|\mathbb{M}_a\|\,\|\mathbb{G}^2\| \gg \|\mathbb{M}_a\mathbb{G}^2\|$, because the coefficient $1-x^2$ vanishes at the ends of the interval, where the derivatives of Chebyshev polynomials are largest. For the same reason the residual's norm is smaller than the constant-coefficient law gives: $\|H_{\text{std}}\|$ is 6.04×10^8 at $n=7$ and 9.70×10^9 at $n=8$, growing as N^4 where the constant-coefficient law gives $N^{4k}=N^8$. On such a problem the encoder's cost reappears, and only the ideal-encoding bound separates it from the Hamiltonian's; the convention of § 7.7 therefore quotes every cost both as built and at that bound, and the comparison with the descriptor on this problem is reported in § 8.2.

For constant-coefficient problems, then, the query count has two parts, and only one of them belongs to the circuit. The encoder's cost, a factor growing as N^{3k} in the residual's subnormalisation under Wu et al.'s encoding, is held to a constant by the structured encoding. The Hamiltonian's cost, $\tilde{\Omega}(N^{2k})$ in the filter's degree, remains under every encoding, because it is the residual's norm: a factor N^2 for each derivative eliminated through $\mathbb{G}$. This is the cost of differentiation that Wu et al.'s Discussion raises, paid in the number of queries where they looked for it in the depth of one, and the gap, measured flat, does nothing to offset it. With the circuit already tight, the quantity left to change is the operator. § 5 changes what the residual carries, keeping the derivatives as unknowns instead of eliminating them into the dense $\mathbb{G}$.

5 The Descriptor Reformulation

In the standard form the state holds the field alone, and every derivative is eliminated through the dense differentiation matrix $\mathbb{G}$, so that a term $f^{(j)}$ acts as $\mathbb{G}^j$ on the field's coefficients. § 4.6 found the cost of that form in its operator: with the encoding already tight, $\|\mathbb{G}^k\|_2=\Theta(N^{2k})$ makes $\|H_{\text{std}}\|=\Theta(N^{4k})$ for constant coefficients, and the filter's degree grows as N^{2k} against a gap that stays flat. A lower degree therefore needs a different residual, one whose Hamiltonian is banded and well conditioned, whose kernel is still the solution, and whose price, in the circuit, the state and the gap, can be stated. Whether such a residual exists, and what it costs, is the second question of § 3.5. The descriptor form answers it: its norm falls to $\Theta(N^2)$ at fixed block weights, and its gap, flat in the standard form, becomes the quantity the reformulation must defend.

5.1 The Idea

The density of $\mathbb{G}$ is a property of the representation, familiar from the dense, ill-conditioned matrices of the tau and collocation methods [36]: it arises because the derivative is required to be expanded in the same basis as the function. If the derivative may instead live in a neighbouring polynomial basis, differentiation is banded, and $\mathbb{G}$ factors into two banded matrices whose product is dense only because one of them is inverted. We therefore change the form in which the equation is written and leave the equation, its data and its solution as they were.

The descriptor form keeps the factors instead of the product. Where the standard form sub-

stitutes $\mathbb{G}^j$ for each derivative, the descriptor form carries $f', \ldots, f^{(k)}$ as unknowns beside f, ties consecutive derivatives by banded recurrences, and writes the equation over all of them. A first-order equation or system needs no extra unknowns, because the equation itself gives the derivative: multiplying the standard rows through by one banded factor is enough. We call the first layout the *chain* and the second the *mass form* (§ 5.3).

Neither ingredient is new to numerical analysis. Banded differentiation is the ultraspherical spectral method of Olver and Townsend [36], of which the factors of § 5.2 are the discrete shadow (§ 2.3.2). Carrying derivatives as unknowns so as not to square the condition number of a least-squares functional is the idea of first-order-system least squares [38] (§ 2.3.3). The physics-informed Hamiltonian is a least-squares functional, so both apply to it. The contribution of this thesis is their transfer to a block-encoded physics-informed Hamiltonian: a proof that the kernel is unchanged, an encoding of the residual as one flat linear combination of unitaries, an a priori rescaling that restores the gap the new unknowns erode, and a measurement of what all of this does to ground-state preparation.

Figure 5.1 shows the effect on the running example $f'' + 4f' + 4f = 0$ (R2a). At $n = 7$ the standard form's stacked residual holds 8384 non-zero entries on 128 unknowns, and the descriptor's holds 1402 on 512: four times as many unknowns, because of the block register that § 5.3 counts, and six times fewer entries. Apart from the rank-one constraint rows, which both forms share, the descriptor's residual is banded. Carrying the intermediate derivatives instead of eliminating them turns the residual from dense to banded with its kernel unchanged, so that the Hamiltonian's norm falls from $\Theta(N^{4k})$, which is $\Theta(N^8)$ at second order, to $\Theta(N^2)$ at fixed block weights; the price is a block register and a gap that the derivative blocks erode, which an a priori rescaling restores on one axis at no gate cost.

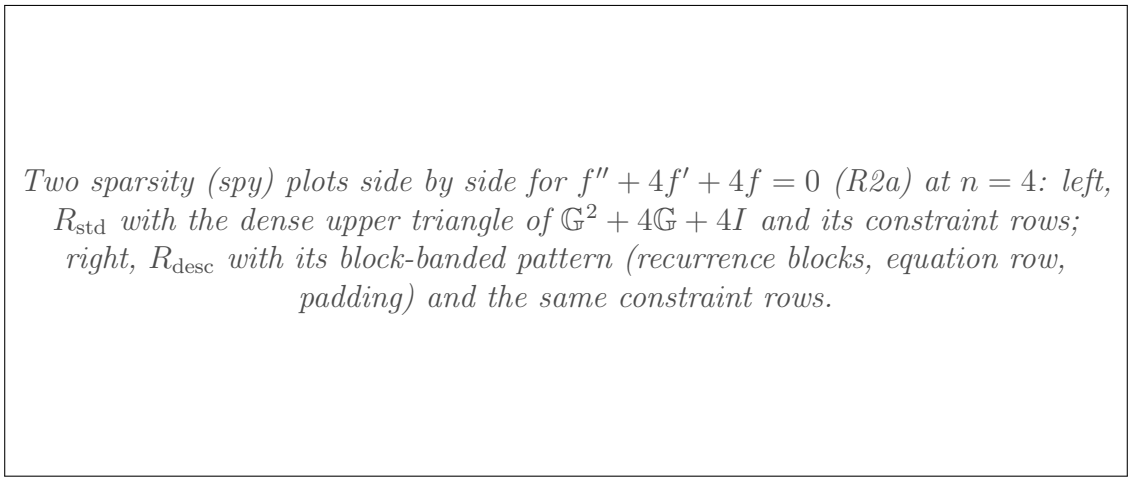


Figure 5.1: Dense against banded: the two residuals of the same equation. Sparsity of the stacked residuals of $f'' + 4f' + 4f = 0$ (R2a) at $n = 4$, constraint rows included. Left, the standard form's R_{std}, whose equation block $A_{\text{std}} = \mathbb{G}^2 + 4\mathbb{G} + 4I$ has a full upper triangle. Right, the descriptor's R_{desc}, whose recurrence blocks, equation row and padding are banded; the only dense rows are the rank-one constraint rows, which both forms share. The descriptor's residual has four times as many columns and, at $n = 7$, six times fewer non-zero entries (§ 5.1).

5.2 The Banded Factorisation

The Chebyshev coefficients of a function and of its derivative are tied by a three-term recurrence, each row of which ties one coefficient of the function to two of the derivative; the recurrence and its derivation are given in Appendix B. Truncated to N coefficients and written in the normalised basis of § 4.1, with $\hat{f}^{(0)}$ and $\hat{f}^{(1)}$ the coefficient vectors of f and f', it is the matrix identity

$$\tilde{\mathbb{B}}\,\hat{f}^{(1)} = \tilde{\mathbb{D}}\,\hat{f}^{(0)}. \tag{5.1}$$

$\tilde{\mathbb{D}}$ has a single non-zero diagonal, the first superdiagonal, with entries $(2\sqrt{2}, 4, 6, \dots, 2(N-1))$; $\tilde{\mathbb{B}}$ has the diagonal $(2, 1, 1, \dots)$ and the second superdiagonal $(-\sqrt{2}, -1, -1, \dots)$, the irregular first entries coming from the normalisation of T_0. Both hold $\mathcal{O}(N)$ non-zero entries against the $N^2/4$ of $\mathbb{G}$. Two properties are used throughout: $\tilde{\mathbb{B}}$ is upper triangular with a non-zero diagonal, so it is invertible; and since the singular values of a single-diagonal matrix are its entries' magnitudes, $\|\tilde{\mathbb{D}}\|_2 = 2(N-1)$, while $\|\tilde{\mathbb{B}}\|_2 \le 2 + \sqrt{2}$.

The dense differentiation matrix is therefore the quotient

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}, \tag{5.2}$$

exactly at every N (Appendix B). Away from its first row, $\tilde{\mathbb{B}}^{-1}$ is a triangle of ones on each parity class, and the density of $\mathbb{G}$ is that inverse; Equation (5.1) is the same linear map before the inverse is taken. Up to the basis normalisation, $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ are twice the ultraspherical method's operators for conversion from first-kind to second-kind Chebyshev polynomials and for differentiation into the second kind [36] (Appendix B.5). The ultraspherical method uses such operators to eliminate the derivatives, raising the basis once per derivative, and solves a banded system in f alone. We instead keep Equation (5.1) as a row of the system and each derivative as an unknown, which lets one pair of factors serve every order, every axis and every coupled field. The method's own reduction of an equation of order $k \ge 2$ to a banded operator on the N coefficients of f would also keep the standard form's kernel; it is not built here, and what it would change is stated in Appendix B.5.

5.3 Two Layouts and Kernel Equivalence

Consider the linear ODE of order k

$$\sum_{j=0}^{k} a_j(x)\, f^{(j)}(x) = r(x), \qquad x \in [-1, 1], \tag{5.3}$$

with the data of § 4.3, every function in it represented by its first N coefficients in the normalised Chebyshev basis.

The chain ($k \geq 2$). Let $\hat{f}^{(j)} \in \mathbb{C}^N$ be the coefficient vector of $f^{(j)}$ for $j = 0, \ldots, k$, stacked into the state

$$w = (\hat{f}^{(0)};\ \hat{f}^{(1)};\ \ldots;\ \hat{f}^{(k)}). \tag{5.4}$$

Two kinds of row couple the blocks. The *recurrence rows*

$$-\tilde{\mathbb{D}}\,\hat{f}^{(j)} + \tilde{\mathbb{B}}\,\hat{f}^{(j+1)} = 0, \qquad j = 0, \ldots, k-1, \tag{5.5}$$

apply Equation (5.1) block by block. They contain no derivative operator, and on the kernel they are equivalent to $\hat{f}^{(j+1)} = \mathbb{G}\hat{f}^{(j)}$. The *equation row* is the equation itself, $\sum_j \mathbb{M}_{a_j}\hat{f}^{(j)} - \mathbb{M}_r\mathbb{D}^{(0)}(x_s)\hat{f}^{(0)} = 0$, with $\mathbb{M}_{a_j}$ the multiplication operator of § 4.2, lifted exactly where the standard form lifts it, and the source entering through the collapse row as in Equation (4.3). The derivatives are unknowns of w and are never replaced by $\mathbb{G}^j\hat{f}^{(0)}$, so the dense quotient Equation (5.2) never appears. The name is borrowed from descriptor systems in control theory [67, 68], which likewise keep intermediate quantities as unknowns beside the state and tie them by algebraic rows.

Definition 3 (Descriptor residual operator): Collecting the recurrence rows and the equation row, the descriptor residual operator A_{desc} is the block matrix

$$A_{\text{desc}} = \begin{pmatrix} -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & & \\ & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{\mathbb{D}} & \tilde{\mathbb{B}} \\ \mathbb{M}_{a_0} - \mathbb{M}_r\mathbb{D}^{(0)} & \mathbb{M}_{a_1} & \cdots & \cdots & \mathbb{M}_{a_k} \end{pmatrix} \tag{5.6}$$

with $(k+1)N$ columns before padding, so that $A_{\text{desc}}\,w = 0$ is the equation.

The padded block register. On a quantum register the block index is held by $n_f = \lceil \log_2(k+1) \rceil$ qubits, which address 2^{n_f} blocks rather than $k+1$. The state is therefore padded to $2^{n_f}N$ amplitudes by blocks $b > k$, each pinned to zero by its own row $c_{\text{pad}}\hat{f}^{(b)} = 0$ with $c_{\text{pad}} = 1$. No other row reads a padding block, so the padding decouples: the Hamiltonian is block diagonal, and the spectrum of the physical blocks is unaffected. The padding still costs state: the chain's state has $2^{n_f}N$ amplitudes against the standard form's N, which is n_f more qubits and up to twice the unpadded $(k+1)N$ when $k+1$ is not a power of two: a second-order equation carries f, f' and f'' in four blocks. The constraint rows of § 5.4 are added to these block rows, and every row block, with the several-axis case and the padded count, is given in full in Appendix G.1.

The mass form ($k = 1$). A first-order system $z' = \mathbb{A}z$ with C components and a constant coupling matrix $\mathbb{A}$, a single first-order equation being the case $C = 1$, has in the standard form the rows $I_C \otimes \mathbb{G} - \mathbb{A} \otimes I_N$, one block per component. The equation already determines each

derivative, so a derivative block would carry nothing new; the mass form instead multiplies these rows on the left by $I_C \otimes \tilde{\mathbb{B}}$ and uses $\tilde{\mathbb{B}}\mathbb{G} = \tilde{\mathbb{D}}$,

$$(I_C \otimes \tilde{\mathbb{B}})\big(I_C \otimes \mathbb{G} - \mathbb{A} \otimes I_N\big) = I_C \otimes \tilde{\mathbb{D}} - \mathbb{A} \otimes \tilde{\mathbb{B}}. \tag{5.7}$$

Both factors are banded and neither contains $\mathbb{G}$. The name is taken from the ODE literature [69], $\tilde{\mathbb{B}}$ playing the mass matrix of $\tilde{\mathbb{B}}\dot{z} = \dots$. For $f' + 2f = 0$ with $f(-1) = 1$ (R1) the mass form is $\tilde{\mathbb{D}} + 2\tilde{\mathbb{B}}$ on N unknowns, where a chain would need $2N$ unknowns and elimination through $\mathbb{G}$ gives the dense $\mathbb{G} + 2I$; all three are written out at $N = 4$ in Appendix B. For a scalar equation with a constant coefficient the mass form is the ultraspherical method's own first-order operator. It has the standard form's state, initial state and readout, adds no qubits and has no derivative block to rescale (§ 5.6). Coupled fields and the Carleman lifts of § 6.5 are first-order systems with a constant coupling matrix, so in the descriptor form they take the mass form, which § 6.1 applies.

Theorem 1 (Kernel equivalence)**.** *Let R_{std} be the standard form's stacked residual (Definition 2) of a linear problem on one axis. Let R_{desc} stack the descriptor's recurrence, equation and padding rows with the same constraint rows, each reading the block of the derivative it constrains (§ 5.4), the equation row applying the standard form's multiplication operators. Let R_{mass} be the mass form's stacked residual of a first-order system with a constant coupling matrix. Then*

$$\ker R_{\text{desc}} = \big\{\, (p,\ \mathbb{G}p,\ \dots,\ \mathbb{G}^k p,\ 0) \ :\ p \in \ker R_{\text{std}} \,\big\}, \qquad \ker R_{\text{mass}} = \ker R_{\text{std}}. \tag{5.8}$$

Proof. Write $Ep = (p; \mathbb{G}p; \dots; \mathbb{G}^k p; 0)$. Chain: the padding rows force the padding blocks of a kernel vector w to zero, and since $\tilde{\mathbb{B}}$ is invertible each recurrence row gives $w_{j+1} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}\, w_j = \mathbb{G}w_j$, so $w = Ew_0$. The recurrence rows vanish on Ep for every p, because $(\tilde{\mathbb{B}}\mathbb{G} - \tilde{\mathbb{D}})\mathbb{G}^j p = 0$, and every other row of R_{desc} acts on Ep as the standard form's row of the same name acts on p: the equation row gives $\sum_j \mathbb{M}_{a_j}\mathbb{G}^j p - \mathbb{M}_r \mathbb{D}^{(0)}(x_s)p$, the standard form's equation block, and a constraint row reads the block that holds $\mathbb{G}^j p$. Hence $R_{\text{desc}}Ep$ vanishes exactly when $R_{\text{std}}p$ does, and E is injective because its first block is the identity. Mass form: R_{mass} is R_{std} with its equation rows multiplied on the left by the invertible $I_C \otimes \tilde{\mathbb{B}}$ (Equation (5.7)) and its constraint rows unchanged. □

The theorem holds on several axes, with E carrying $\mathbb{G}_a^j$ along each axis's chain, and for the block-rescaled residual of § 5.6, whose kernel is the image of this one under a diagonal map; both extensions are proved in Appendix G.3.

An exact kernel exists only where the solution lies in the span of the basis and meets every constraint exactly, as Legendre's P_4 does on R3a; otherwise both residuals have full rank, and their ground states are the smallest right singular vectors of different matrices. The descriptor measures a state's residual against a norm that carries its derivatives (§ 5.5), and the mass

form weights the equation rows by $\tilde{\mathbb{B}}$. We therefore claim equal kernels only. Both decoded fields converge to the solution, at rates measured in § 8.1 and not proved; at high resolution the computed singular vector's rounding error grows with the residual's norm relative to its gap, $\Theta(N^{2k})$ in the standard form against $\Theta(N)$ here (§ 5.9), which favours the descriptor's reference there.

5.4 Constraints and Partial Differential Equations

A data constraint sits on the block that holds the quantity it constrains. A Dirichlet condition $f(x_z) = 0$ reads block 0 with the zero row $\mathbb{B}(x_z)$, as in the standard form. A Neumann condition $f'(x_m) = 0$ reads block 1 with the same row, $\mathbb{B}(x_m)\hat{f}^{(1)} = 0$, where the standard form must write $\mathbb{B}(x_m)\mathbb{G}$. The regular datum reads block 0, so the field is decoded and scaled exactly as in the standard form (§ 4.1). A variable coefficient and a source enter the equation row as they enter the standard form's equation block, so the two forms differ only in how the derivatives are carried.

On several axes the field is block 0, and each axis a owns a chain $\partial_a f, \ldots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it, tied by that axis's factors $\tilde{\mathbb{B}}_a$ and $\tilde{\mathbb{D}}_a$ tensored with the identity on the other axes. A mixed derivative would need a block of its own; the construction does not build one, which limits it to equations without mixed derivatives. Initial and boundary functions enter both forms as the datum slices of Equation (4.4), read on the field block. The descriptor pays a larger state here: the heat equation ($k_t = 1$, $k_x = 2$) carries four blocks, and Laplace's and the wave equation ($k = 2$ on each axis) five, padded to eight, so their states are four and eight times the standard form's, two and three more qubits.

5.5 The Descriptor's Gap

The kernel is the standard form's; the spectrum above it is not, because the descriptor's state carries every derivative and its norm counts them all. The gap, which the standard form keeps flat on the problems of § 4.6 while its norm grows, is therefore where the reformulation can lose what its norm has gained. With E the map of Theorem 1, the descriptor's Rayleigh quotient on the states that satisfy the recurrences is

$$\frac{\|R_{\mathrm{desc}} E p\|^2}{\|E p\|^2} = \frac{\|R_{\mathrm{std}}\, p\|^2}{\sum_{m=0}^{k} \|\mathbb{G}^m p\|^2}, \tag{5.9}$$

because the recurrence and padding rows vanish on Ep and the remaining rows act as the standard form's. By the min–max principle, restricting the trial states to the range of E can only raise the quotient, so the descriptor's singular values are bounded above by the generalised singular values of the pair (R_{std}, E) [70]. The denominator carries every derivative, so the gap can be flat in N only where the equation controls each carried derivative: where no state p independent of the solution has a residual small against its largest carried derivative. The bound is an upper one in two senses. It does not prove flatness, and states that break the recurrences

can lie below it, since the least singular value of $\tilde{\mathbb{B}}$ falls with N, its vector at the ends of the interval (Appendix G.4). The gap is therefore measured. The standard form measures the same states by $\|p\|$, against which their residuals are large, so its gap stays flat; its difficulty is the other side of the quotient, the norm of Equation (4.8).

On a one-axis equation whose leading coefficient has no zero on $[-1, 1]$, the continuum a priori estimate $\|f^{(k)}\| \le (\|\sum_j a_j f^{(j)}\| + \|\sum_{j<k} a_j f^{(j)}\|)/\min|a_k|$ suggests that the quotient stays bounded away from zero on the complement of the kernel, and the measured gap, without rescaling, is flat: on R2a it is 0.458 at $n = 3$ and 0.458 at $n = 10$. The norm weighs block m by the size of $f^{(m)}$, so where the solution's derivatives are large the derivative blocks dominate it, and the ground state's weight on the field block, $w_f = \|\hat{f}^{(0)}\|^2/\|w\|^2$, is small; for an exact kernel it is $\|b\|^2/\sum_m \|\mathbb{G}^m b\|^2$, with b the solution's coefficients. An initial state prepared on the field block then overlaps the ground state by $\gamma \le w_f^{1/2}$, and amplification costs $1/\gamma$. On the stiff equation $f'' + 5f' + 400f = 0$ (R2c, Wu et al.'s Fig. 3d), whose solution oscillates at an angular frequency near 20, the gap is 7.7×10^{-6} at $n = 5$ and 7.5×10^{-6} at $n = 10$, flat and small, and w_f is 5.4×10^{-6}, against 4.3×10^{-3} on R2a.

The gap falls with N in two cases.

1. *A leading coefficient that vanishes.* A polynomial of degree N can concentrate within $\mathcal{O}(N^{-2})$ of an endpoint, where the Chebyshev points cluster, and each of its derivatives grows by a factor $\mathcal{O}(N^2)$ there, the extremal behaviour of Markov's inequality [55]. Where a_k vanishes at that endpoint, the leading term $a_k p^{(k)}$ of such a state is small, the lower-order terms are small as well, and the quotient is small against the top block. The measured gap falls as about N^{-4}: on Legendre's equation $(1 - x^2)f'' - 2xf' + 20f = 0$ (R3a), whose leading coefficient vanishes at both ends, it is 7.6×10^{-5} at $n = 5$ and 1.1×10^{-10} at $n = 10$, and on $(x - 1)f'' - xf' + f = (x - 1)^2$ (R4a, Wu et al.'s Fig. 5a), whose leading coefficient vanishes at $x = 1$, 4.9×10^{-5} and 4.3×10^{-11}. At an interior zero, where the Chebyshev resolution is $\mathcal{O}(N^{-1})$, the same argument, with Bernstein's inequality in place of Markov's [55], gives N^{-2}; no benchmark problem has one, and the rate is measured on a controlled equation in Appendix G.4.

2. *Two axes.* The equation constrains only a sum of top derivatives, $f_{xx} + f_{yy}$ or $f_t - \kappa f_{xx}$. A high-frequency state with $\mathbb{G}_x^2 p \approx -\mathbb{G}_y^2 p$, or with its weight in the block the equation weights least, has a small residual against large blocks, and only the datum rows penalise it. On the heat equation $\partial_t f - \frac{1}{25}\partial_x^2 f = 0$ (R5b, Wu et al.'s Fig. 6d) the gap is 3.2×10^{-5} at $n = 3$ per axis and 2.5×10^{-7} at $n = 5$, where the standard form's is 0.264 and 0.433.

The collapse is not an artefact of the lift, the padding, the arithmetic precision, the constraints, the basis or the classical reference: each was tested and ruled out, and a descriptor built in the ultraspherical basis collapses at the same rates where the leading coefficient vanishes. The tests and the full mechanism are given in Appendix G.4.

Try to prove that the gap is flat for a regular one-axis equation (the lower bound on the quotient of Equation (5.9 uniformly in N). If proved, put the proof in Appendix G (Appendix C and reword this paragraph, Equation (5.1

5.6 Block Rescaling

The remedy is to rescale the carried derivatives so that no block outweighs the equation's own scale. The block of order m is carried divided by ζ^m,

$$\hat{f}^{(m)} \longmapsto \zeta^{-m}\,\hat{f}^{(m)}, \qquad m = 0, \ldots, k, \tag{5.10}$$

which replaces R_{desc} by $R_{\text{desc}}S$, with S diagonal, equal to ζ^m on block m and to one on the padding blocks. Because S is invertible, the kernel becomes $S^{-1}\ker R_{\text{desc}}$ and Theorem 1 survives. The field block has weight one, so the readout is unchanged, and the padding keeps its weight and still decouples. What changes is the Hamiltonian, $SR_{\text{desc}}^{\dagger}R_{\text{desc}}S$. This is a congruence, so the spectrum moves: by Equation (5.9) the denominator becomes $\sum_m \zeta^{-2m}\|\mathbb{G}^m p\|^2$, and ζ sets how much each carried derivative weighs against the residual. From here on R_{desc} and H_{desc} denote the rescaled residual and its Hamiltonian, and the residual of § 5.5 is called *unscaled*.

In the circuit the rescaling adds nothing. Scaling the columns of block m by ζ^m multiplies the coefficient of every term of the block encoding of § 5.7 by ζ raised to the order of its column block, which adds no gate and no ancilla and changes only the subnormalisation. A general diagonal equilibration [71], with a different weight for each amplitude, would need a diagonal block encoding of its own at $\Theta(\dim w)$ gates; that is why the weights are constant on each block, and general equilibration appears in no circuit in this thesis.

The a priori rule. ζ is computed from the equation's coefficients and the resolution, never from a solution or a spectrum. For a one-axis equation whose leading coefficient a_k has no zero on $[-1, 1]$,

$$\zeta = 2\left(\bar{a}_j/\bar{a}_k\right)^{1/(k-j)}, \tag{5.11}$$

with j the lowest order present and $\bar{a}$ the mean of $|a|$ over the interval. For constant coefficients and $j = 0$, $(\bar{a}_0/\bar{a}_k)^{1/k}$ is the geometric mean of the magnitudes of the characteristic roots: the scale at which a solution $e^{\lambda x}$ has $\|f^{(m)}\| \approx |\lambda|^m\|f\|$, so that dividing block m by its m-th power balances the blocks at the equation's own scale and returns the field block's weight to order one. Where a_k has a zero on the interval, $\zeta = N/2$, which raises the quotient of the concentrated states of § 5.5 to order one. On several axes $\zeta_a = N_a$, weighted on an axis of second order so that the top-order terms balance across axes: the wave equation $f_{tt} = c^2 f_{xx}$, with wave speed c, takes (cN, N). Whether a_k vanishes is read from the equation.

The rule's two constants, the 2 in Equation (5.11) and the $\frac{1}{2}$ in $N/2$, were set once from a sweep of ζ over the benchmark problems and are never refitted to a problem. The check of the rule against the cheapest ζ on a grid is therefore in-sample: it shows that one rule serves the whole benchmark set, within a small factor of the best weight, and says nothing about a class of equation the set does not contain. The cost is flat over a window of ζ, so ζ need only be right to about a factor of two; the sweep, the window and the check are given in Appendix G.5.

What it restores. On every one-axis problem the descriptor runs, the rescaled gap no longer falls with N, and the field block regains its weight. The gap meant here is that of the physical subspace, the one the filter meets, since the padding alone would cap the gap of the whole spectrum at one. On R2c the gap rises from 7.5×10^{-6} to 0.715 and w_f from 5.4×10^{-6} to 0.774 at $n = 7$. On R3a the rescaled gap is 2.27 at $n = 5$ and 3.65 at $n = 10$, against 1.1×10^{-10} unscaled: it rises with n and levels off.

What it does not remove. Three limits remain, each a finding.

1. *The norm grows with ζ.* Block m carries ζ^m, so $\|R_{\text{desc}}\| \gtrsim \zeta^k \bar{a}_k$ and $\|H_{\text{desc}}\| \gtrsim \zeta^{2k}$. With ζ constant this costs nothing. With $\zeta \propto N$, which a simple zero requires, $\|H_{\text{desc}}\| = \Theta(N^{2k})$, and at a flat gap the degree at second order is about $\tilde{\Theta}(N^2)$. Whether this is an advantage over the standard form depends on where the coefficient vanishes. A coefficient that vanishes at both ends of the interval, where the derivatives of the Chebyshev polynomials are largest, lowers the standard form's norm too: on R3a, whose coefficient is $1 - x^2$, $\|H_{\text{std}}\|$ grows as N^4 where N^8 is the constant-coefficient law (§ 4.6). One that vanishes at one end only does not: on R4a, whose coefficient is $x - 1$, $\|H_{\text{std}}\|$ keeps the N^8 law. On R3, then, both norms grow as N^4 and the comparison at the ideal-encoding bound can go either way, while as built the structured encoding pays for the lifted multiplier in full; on R4a the descriptor's lower norm should hold at the bound as well. § 8.2 measures both, and reports the reversal it finds on R3 as a finding.

2. *Two axes.* The rescaled gap still falls with N, though far more slowly than unscaled: on R5b it is 0.253 at $n = 3$ per axis and 0.0942 at $n = 5$, and on Laplace's equation (R5a) 0.372 and 1.1×10^{-2}, against 1.1×10^{-4} and 1.1×10^{-8} unscaled.

3. *A double zero is beyond any block rescaling.* If a_k vanishes to second order at an endpoint, the concentrated states have quotient $\mathcal{O}(N^{-8}\zeta^{2k})$, which is the unscaled N^{-8} at $\zeta = 1$, so balancing them needs $\zeta \propto N^{4/k}$. At $k = 2$ that is $\zeta \propto N^2$, at which $\|H_{\text{desc}}\| = \Theta(N^8)$: the norm of a second-order standard form with constant coefficients, and more than the standard form's own norm on such a problem, which the vanishing coefficient lowers. Wu et al.'s associated Legendre examples R3c and R3d (their Figs 4c and 4d) multiply the operator by $1 - x^2$ and so have exactly this double zero. Both forms instead run the equivalent problem for $v = f/\sqrt{1 - x^2}$ of § 4.3, whose leading coefficient has simple zeros, and we make no claim for the double zero beyond the measurement on Wu et al.'s form in Appendix G.5.

Block rescaling is the descriptor's counterpart of the diagonal preconditioner of the ultraspherical method [36], which bounds the condition number of its banded system, and preconditioning a block-encoded system is established quantum practice [72, 73].

5.7 The Block Encoding

A norm of $\Theta(N^2)$ lowers the degree only as far as the circuit's subnormalisation follows it, so the descriptor's encoding must keep α_R within a small factor of $\|R_{\text{desc}}\|_2$, or the encoder's cost of § 4 returns. Every block of A_{desc} is a short sum of *bands*, each a diagonal followed by a truncated shift, the device used for the explicit block encodings of banded circulant matrices [43]. Write $\Lambda = \operatorname{diag}(0, 1, \ldots, N-1)$ for the index diagonal of Proposition 1. The diagonal of $\tilde{\mathbb{D}}$ is 2Λ except at column 1, where the normalisation of T_0 makes it $2\sqrt{2}$; the diagonals of $\tilde{\mathbb{B}}$ and of a constant multiplication are constant except at their ends; and a variable coefficient's multiplication and its truncation at the edge of the register contribute a few further exceptions. An affine diagonal $aI + b\Lambda$ is a sum of $n+1$ Pauli-Z terms, and each exception is one reflection Q_m:

$$\Lambda = \frac{N-1}{2}\, I - \sum_{j=0}^{n-1} 2^{j-1}\, Z_j, \qquad Q_m = I - 2\,|m\rangle\,\langle m|\,. \tag{5.12}$$

A_{desc} is therefore one flat LCU of $\mathcal{O}(n)$ terms,

$$A_{\text{desc}} = \sum_{t=1}^{L} c_t\, U_t, \qquad U_t = (\text{field word}) \otimes \text{SHIFT}^{\ell_t}\, P_t, \qquad \alpha_A = \sum_{t=1}^{L} |c_t|, \tag{5.13}$$

where the field word, a Pauli word on the block register, places the term's row and column blocks, SHIFT^{ℓ_t} is the truncated shift by ℓ_t, and P_t is the identity, a Z_j or a reflection. An affine diagonal whose end values a and $a + b(N-1)$ share a sign is encoded tightly, at $\alpha = \max(|a|, |a + b(N-1)|) = \|aI + b\Lambda\|_2$, so the dominant term $\tilde{\mathbb{D}}$ is encoded at its own norm $2(N-1)$ and its exception costs $\mathcal{O}(1)$. A truncated shift of the N-dimensional register is not unitary; it is applied on the register widened by one overflow qubit, so that rows falling outside the interval land in the half that the final projection discards. That qubit is the shift's only ancilla.

Algorithm 1 compiles the terms into the circuit of Figure 5.2.

Input: the blocks of A_{desc} as bands, the block weights ζ
Output: PREPARE_R, SELECT and $\text{PREPARE}_L^\dagger$; the subnormalisation α_A
Write every block as a sum of bands: a diagonal and a truncated shift;
Fit each diagonal as affine in the column index plus a few exceptions, Equation (5.12);
Merge terms with equal axis factors across blocks; expand the resulting block-register pattern in Pauli words;
Multiply the coefficient of each term by ζ^m for its column block m;
$\alpha_A \leftarrow \sum_t |c_t|$; on the joint selector index of term t, PREPARE_R loads $\operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ and PREPARE_L loads $\sqrt{|c_t|/\alpha_A}$;
SELECT: controlled on each selector register, apply the field word, then on each axis the diagonal and the shift;
return *PREPARE$_R$, SELECT, PREPARE$_L^\dagger$, α_A*;
Algorithm 1: Compiling the rows of A_{desc} into two preparations and one SELECT.

Two choices in it set the cost. Line 3 merges terms whose axis factors agree across the rows

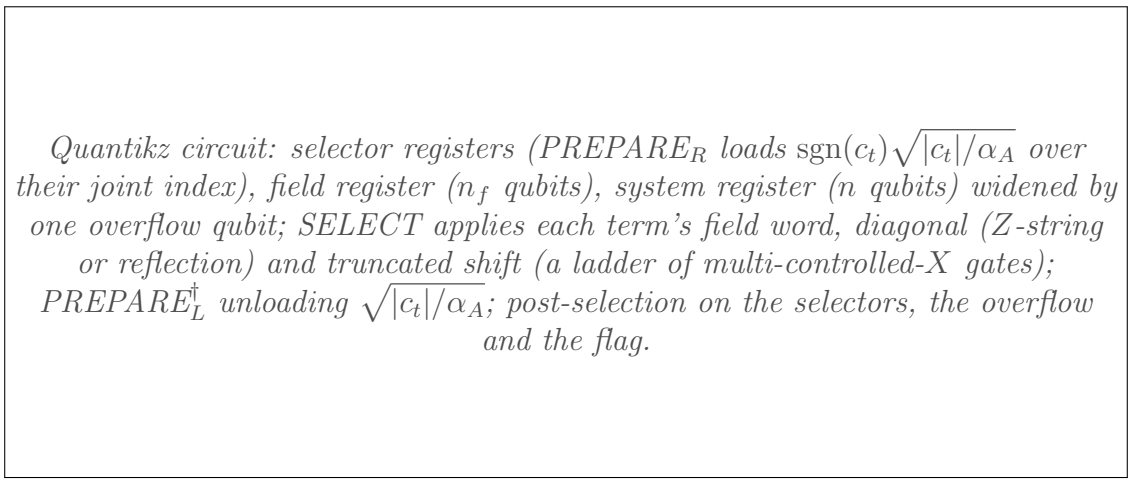


Figure 5.2: The whole descriptor residual is one preparation, one SELECT and one unpreparation. PREPARE$_R$ loads the signed amplitudes $\mathrm{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ over the joint index of the factored selector registers, and PREPARE$_L^\dagger$ unloads the unsigned $\sqrt{|c_t|/\alpha_A}$, so that the signs of the terms survive. SELECT applies each term's field word on the block register (n_f qubits), then on the system register (n qubits, widened by one overflow qubit) its diagonal, a Z-string or a reflection, and its truncated shift. Post-selecting the selectors, the overflow qubit and the flag on $|0\rangle$ leaves the system rows of A_{desc} divided by α_A, with the block rescaling carried entirely in the coefficients c_t.

and columns they join, so the recurrences' common $\tilde{\mathbb{D}}$ is paid for once and not k times. And the selectors are factored, one register for the field operation and, on each axis, one for the diagonal and one for the shift, so that SELECT applies each distinct operation once rather than once per term, which keeps the ancilla at $\mathcal{O}(\log n)$ (§ 5.8). The rank-one rows are encoded through the feature map, with a rotation tree loading a source's coefficients, so no dense state preparation appears anywhere. The compilation line by line, and the check of the terms against the assembled rows, are given in Appendix G.6.

5.8 Cost of the Descriptor Encoding

Proposition 3 (Cost of the descriptor encoding)**.** *For a descriptor residual of a fixed problem, the flat LCU of Equation* (5.13) *is an exact block encoding of the system rows (the recurrence, equation and padding rows) at $\alpha_A = \sum_t |c_t|$, with $\mathcal{O}(\log n)$ ancilla and $\mathcal{O}(n^2)$ gates. For the second-order family R2,*

$$\lceil \log_2(n+4) \rceil + 8 \text{ ancilla}, \qquad \mathcal{O}(n^2) \text{ gates}. \tag{5.14}$$

For the mass form of a scalar first-order equation, and for a second-order chain with polynomial coefficients whose leading coefficient has no zero on $[-1,1]$, at a fixed $\zeta \geq 1$, $\alpha_A/\|A_{\mathrm{desc}}\|_2 = 1 + \mathcal{O}(N^{-1})$ without a source and $1 + \mathcal{O}(N^{-1/2})$ with one.

The ancilla are the selector registers, the overflow qubit and a flag. Only the diagonal selector's width depends on n: for R2 it indexes the identity, the n gates Z_j and three reflections ($\tilde{\mathbb{B}}$'s diagonal and second superdiagonal and the first entry of $\tilde{\mathbb{D}}$), so it has $\lceil \log_2(n+4) \rceil$ qubits, while the shift selector holds the offsets 0, 1 and 2 on two qubits and the field selector has four.

The gate count follows because each distinct operation is applied once and the shift's multi-controlled ladder costs $\mathcal{O}(n^2)$. The subnormalisation tends to the norm because the dominant terms, the recurrences' $\zeta^j\tilde{\mathbb{D}}$, share one pattern on the block register that its Pauli expansion encodes at its largest entry, so their contribution to α_A is $2\zeta^{k-1}(N-1)$, which is also a lower bound on $\|A_{\text{desc}}\|_2$; every other term is $\mathcal{O}(1)$ in N at fixed ζ, except a source's rank-one term, whose coefficient vector grows as $N^{1/2}$. The proofs are given in Appendix G.7, and the ancilla count is verified for R2's system rows at $n = 2$, 4, 5 and 12.

The ancilla count is what expanding the diagonals in Pauli strings forces, and no more. A combination whose SELECT indexes L distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [42], and the index diagonal Λ has exactly $n+1$ non-zero Walsh coefficients, so an LCU that expands it in Pauli-Z strings has at least $n+1$ terms and $\Omega(\log n)$ selector qubits. The count is therefore of the least order among exact LCU encodings that expand their diagonals in Pauli strings; it is not claimed against encodings built otherwise, by arithmetic on the index for instance. The structured encoding of the standard form needs $n-1$ selector qubits for $\mathbb{G}$ alone, because it writes U_{odd} as one combination of its $N/2$ shifts (§ 4.5.3), so its ancilla grow as n where the descriptor's grow as $\log n$; the descriptor never forms that sum, though an encoding of U_{odd} that is not such a combination is not excluded. The better scaling is not yet visible at the sizes that circuits reach, where the whole reflection's ancilla are counted in § 8.2.

The proposition concerns the system rows; for the whole stacked residual, constraints included, the ratio is computed exactly from the built circuit's α_R and the assembled residual's norm. It approaches one on regular one-axis problems, $\alpha_R/\|R_{\text{desc}}\|_2 = 1.04$ on R2a at $n = 8$. It stays near two where the leading coefficient vanishes, 2.12 on R3a at $n = 7$, because $\zeta\tilde{\mathbb{D}}$, $\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2 a_2$ are then comparable and comparable terms add in ℓ^1 even when each is tight. On two axes it is 2.29 on R5b at $n = 4$ per axis. The structured standard-form encoding's ratio on R2a is given in § 4.6, and what the difference does to an as-built comparison in § 7.7.

The gates per query need not follow the subnormalisation. One query of the descriptor applies more terms than one of the structured standard form, through a block register, so on constant-coefficient problems its advantage is expected in the number of queries, the filter's degree, rather than in the gates per query; on a lifted variable coefficient, which the structured encoding pays for in full, its query can be the cheaper as well. § 8.2 counts both.

5.9 Consequences for the Spectrum and the Filter

Every block of the rescaled descriptor is $\mathcal{O}(N)$ at fixed ζ, the recurrences' $\zeta^j\tilde{\mathbb{D}}$ being the largest, and the block $\zeta^{k-1}\tilde{\mathbb{D}}$ alone has norm $2\zeta^{k-1}(N-1)$, so $\|R_{\text{desc}}\|_2 = \Theta(N)$ and $\|H_{\text{desc}}\| = \Theta(N^2)$ whatever the order k: the derivative is banded, and its norm is that of $\tilde{\mathbb{D}}$ rather than of $\mathbb{G}^k$. Through Equation (2.2), with an encoding whose subnormalisation stays within a constant of that norm, the rescaled gap, which is flat on a one-axis ODE whose leading coefficient does not

vanish, gives for that class of problems

$$\|H_{\text{desc}}\| = \Theta(N^2), \qquad \Delta = \Theta(1) \text{ (measured)} \quad \Longrightarrow \quad d = \tilde{\Theta}(N), \tag{5.15}$$

against $\tilde{\Theta}(N^{2k})$ for the standard form with the best exact encoding (Equation (4.8)). Where the leading coefficient vanishes, or on two axes, the rescaling's weights raise the norm, and the degree is about $\tilde{\Theta}(N^2)$ or grows more steeply; for a double zero no claim is made (Table 5.1). Neither form is polylogarithmic in N: $\tilde{\Theta}(N)$ is exponential in n, and the claim is a lower exponent only. The degrees are the minimax edge filter's, with the hedge of § 4.6 for polynomial filters in general.

Table 5.1: The descriptor lowers the degree's exponent where the equation controls every carried derivative. Hamiltonian norm $\|H\|$, spectral gap and minimax filter degree of the standard form and the descriptor, by class of problem, the descriptor's block weight ζ set by the a priori rule. k, equation order; a_k, leading coefficient; N_a, resolution of axis a; $\tilde{\Theta}$, up to logarithmic factors; flat, falls: the gap's behaviour as N grows; –, not applicable. Norms are proved; gaps are exact classical computations. A leading coefficient vanishing at both ends (R3) lowers the standard form's norm too, so there the simple-zero row is no advantage at the ideal-encoding bound $\alpha_R = \|R\|_2$; one vanishing at one end (R4a) does not.

Case	ζ	$\|H\|$	Gap	Degree
Standard, order k, constant coefficients	–	$\Theta(N^{4k})$	flat	$\tilde{\Theta}(N^{2k})$
Descriptor, one axis, $a_k \neq 0$ on $[-1, 1]$	fixed	$\Theta(N^2)$	flat	$\tilde{\Theta}(N)$
Descriptor, one axis, simple zero, $k = 2$	$N/2$	$\Theta(N^4)$	flat	$\tilde{\Theta}(N^2)$
Descriptor, two axes	N_a per axis	$\Theta(N^4)$	falls	steeper than $\tilde{\Theta}(N^2)$
Descriptor, double zero	$\propto N^2$ to balance	$\Theta(N^8)$	–	not claimed

The encoding sits at that norm. In the terms of § 4, the encoder's cost $\alpha_H/\|H\|$ on R2a at $n = 8$ is 1.09 for the descriptor against 59.9 for the structured standard form, so the descriptor lowers the Hamiltonian's cost and keeps the encoder's near one. § 8.2 measures the degrees that result, as built and at the ideal-encoding bound, and § 8.3 the gap on which Equation (5.15) rests.

The relative gap $\Delta/\|H\|$ is $\Theta(N^{-2})$ here and $\Theta(N^{-4k})$ in the standard form: on R2a at $n = 8$ the descriptor's is 6.9×10^{-7}, where the standard form's has fallen below double precision (§ 7.2). The classical reference forms neither Hamiltonian, so a small relative gap costs the filter's degree, not the resolvability of the ground state. Per query the two encodings also differ in how often one post-selected application succeeds, which compares the encodings and not the preparation (Appendix G.8).

5.10 Summary

Carrying the derivatives instead of eliminating them makes the residual banded with its solution unchanged, and has a price.

1. Keeping the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ of $\mathbb{G}$ and carrying the derivatives as unknowns, as a chain for equations of order two and above or in mass form for first-order equations and

systems, makes the residual banded, apart from the rank-one constraint rows both forms share, with its kernel exactly the standard form's (Theorem 1).

2. Its block encoding is one flat LCU with $\lceil \log_2(n+4) \rceil + 8$ ancilla for R2's system rows, the least order for an exact LCU that expands its diagonals in Pauli strings, and its subnormalisation tends to the norm on regular one-axis problems (Proposition 3). Its price is a larger state, by the chain's block register, and, on the constant-coefficient problems, more gates per query at the sizes reached (§ 8.2).

3. The Hamiltonian's norm is $\Theta(N^2)$, and for a one-axis equation whose leading coefficient does not vanish the filter's degree is $\tilde{\Theta}(N)$, against the standard form's $\tilde{\Theta}(N^{2k})$ (Equation (5.15)). Neither is polylogarithmic.

4. *The limit.* The carried derivatives shrink the gap where the solution's derivatives are large, and make it fall with N where the leading coefficient vanishes and on two axes. An a priori block rescaling restores it on one axis at no gate cost, at the price of a larger norm where the leading coefficient vanishes (on R3, which vanishes at both ends, the standard form's norm falls too); on two axes the gap still falls, and a double zero is beyond any block rescaling.

All of this concerns one real scalar field and a linear equation, the scope of Wu et al.'s construction, and the gain comes from replacing the dense $\mathbb{G}$ by banded factors wherever the residual differentiates in Chebyshev coefficients. Coupled fields, complex-valued fields and nonlinear terms each change what the residual must carry, and the nonlinear route Wu et al. propose gives the filter no unique ground state to find. § 6 takes the reformulation there; §§ 8.2 and 8.3 test the cost and the gap on the benchmark problems.

6 Systems, Complex-Valued Fields and Nonlinearity

Every residual so far has carried one real scalar field, the scope Wu et al. set themselves. They set complex-valued states aside, because with a Fourier basis, whose differentiation matrix is imaginary, the ground state of the resulting energy would be hard to find [30] (§ 3.5), and their route to nonlinearity, the doubled space of § 3.4, multiplies the register instead of adding to it. The doubled space also changes what the filter is given to find. Its kernel is degenerate, so there is no gap above the ground value, Equation (2.2) gives no finite degree, and a filter of any degree returns the initial state's projection onto the kernel. Before the reformulation's advantage can be asked about on a nonlinear equation, the equation needs a Hamiltonian with a unique ground state. That is the third question of the assessment: *can the solution of a nonlinear equation be prepared as the unique ground state of a Hamiltonian, rather than selected by projection from a degenerate kernel, and does the reformulation's advantage survive there?*

Each extension that prepares is the same object, a coupled linear system on one state with one block per component: several fields directly, a complex field by realification as two real ones, and

a polynomial nonlinearity by the Carleman lift, whose components are the field's monomials. The system is therefore built first, with its data and its complex-valued case, because the lift that answers the question is an instance of it; the doubled space, the one route that is not such a system, only projects. The constructions and proofs in full are in Appendix H.

6.1 Several Fields on One State

Consider a first-order system of C fields with a constant coupling matrix $\mathbb{A}$, $z_i' = \sum_j \mathbb{A}_{ij} z_j$, each field holding its own first derivative. A time interval $t \in [0, T]$ is carried on the Chebyshev axis $\xi = 2t/T - 1$, so $dz/d\xi = \frac{T}{2}\mathbb{A}z$. The state holds one block of N Chebyshev coefficients per field, the field index on a register of $\lceil \log_2 C \rceil$ qubits as the most significant index, padded where C is not a power of two with blocks pinned to zero by rows that no other row reads, so the padding decouples as the chain's does (§ 5.3). In the standard form (Definition 2) the state holds the fields alone and every derivative is eliminated through the dense differentiation matrix $\mathbb{G}$; the descriptor carries a system in the mass form of Equation (5.7). The equation rows are

$$\underbrace{I_C \otimes \mathbb{G}_\xi - \tfrac{T}{2}\,\mathbb{A} \otimes I_N}_{\text{standard form}}, \qquad \underbrace{I_C \otimes \tilde{\mathbb{D}}_\xi - \tfrac{T}{2}\,\mathbb{A} \otimes \tilde{\mathbb{B}}_\xi}_{\text{descriptor (mass form)}}, \tag{6.1}$$

the second being Equation (5.7) with $\mathbb{A} \to \frac{T}{2}\mathbb{A}$, so by Theorem 1 their kernels coincide. With the data rows of § 6.2, an initial-value problem has a one-dimensional near-kernel in either form, for the reason given in § 6.5. R7's rows are written out in Appendix H.1.

The descriptor's chain could instead carry each field with its derivative as a block, as it carries a field of higher order. The equation already gives each derivative as $\mathbb{A}z$, so such a block holds nothing the rows do not determine; it doubles the state and adds a block for the rescaling of § 5.6 to balance. It also costs gap: on the harmonic oscillator $u' = v$, $v' = -\pi^2 u$ (R7) at $n = 8$, the mass form's gap is 0.132 and the block-rescaled chain's 0.0628, a ratio of about 2.1 that holds from $n = 3$ upwards. The chain is kept only as a labelled ablation.

Every system and lift of §§ 8.4 and 8.5 is therefore compared on the footing of the mass form (§ 5.3): the same unknowns, data rows, initial state and prepared object, and equation rows that differ by one banded factor along the time axis. A system of higher order enters by the usual reduction to first order, as R7 is $u'' = -\pi^2 u$ written as two fields; none of higher order than R7's was run.

6.2 Data for Several Fields: Ratio Rows

A single field's initial or boundary value is imposed by a homogeneous zero row, which leaves the field's scale free, and a regular datum, which fixes it (§ 4.3). Vector data $y = (y_1, \ldots, y_C)$ at a point x^* need the same separation. With an anchor component a, $y_a \neq 0$, and $y_\perp = \sum_{c \neq a} y_c \, |c\rangle$

the rest of the data, the *ratio row* on the field register is

$$\mathbb{F} = y_a\big(I - |a\rangle\langle a|\big) - y_\perp\,\langle a|\,, \qquad (\mathbb{F}v)_c = y_a v_c - y_c v_a \;\; (c \neq a), \quad (\mathbb{F}v)_a = 0, \tag{6.2}$$

and the data rows are $(\mathbb{F}/y_a)\otimes\mathbb{B}(x^*)$. The kernel of $\mathbb{F}$ is exactly the line through y, so the rows impose $z_c(x^*) = (y_c/y_a)\,z_a(x^*)$: the direction of the data, with their scale free. The regular datum on the anchor fixes the scale, a component whose datum is zero is held to zero exactly, and the rows' singular values, $\|y\|$, $|y_a|$ and 0, weigh the data against the equation at the data's own scale (Appendix H.1, with their three-term circuit).

The initial state of a system is the geometric product state of § 7.4 on each field's block, weighted by the problem's stated data $w_0/\|w_0\|$: the problem's statement, never its solution. Equal weights would not do, since they cancel exactly on the Taylor–Green vortex of § 6.7, whose two modes carry data of opposite sign.

6.3 Complex-Valued Fields

Wu et al.'s remark locates the difficulty in the energy function, but the construction does not meet it. If an equation has complex coefficients, data or sources, its stacked residual R is complex, yet $H = R^\dagger R$ is still Hermitian and positive semi-definite, its ground state is still the smallest right singular vector of R, and the stacked reflection and the filter of § 7 act on $H' = 2H/\alpha_R^2 - I$ exactly as before. The non-Hermitian operator is R, which the filter never sees, so no non-Hermitian filter is needed. This is a statement about the formulation: no complex residual was prepared in this thesis.

The route that was run is realification. Writing $z = u + \mathrm{i}v$, a complex linear equation becomes a real system of twice as many fields; for one mode $z' = cz$ with a complex coefficient c,

$$\begin{pmatrix} u \\ v \end{pmatrix}' = \begin{pmatrix} \operatorname{Re} c & -\operatorname{Im} c \\ \operatorname{Im} c & \operatorname{Re} c \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \tag{6.3}$$

a system of § 6.1 with one more field-register qubit. The map $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$, a real-linear isomorphism, carries the complex problem's solutions onto the real system's, and real ratio rows on the realified data fix the solution's phase as well as its direction, so an initial-value problem keeps a one-dimensional near-kernel (§ H.1.2). R7 is an instance: with $z = \pi u + \mathrm{i}v$, its equations are the real and imaginary parts of $z' = -\mathrm{i}\pi z$, a single-mode Schrödinger-type equation, its state being the realified one with the first block divided by π. The fluids of § 6.7 meet Wu et al.'s remark directly, since they are Fourier–Galerkin systems; they are carried on real bases instead. Burgers' solution from odd data stays odd and lives on sine modes, so it needs no realification, and the vorticity is carried on $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$, an isometric real embedding of the complex Fourier variables (§ H.1.2).

One encoding is natively complex: the tensor layout of the Carleman coupling (§ 6.6) acts on

complex Fourier variables, and its LCU weights carry the phases $\mathrm{i}^{\sum_j a_j}$ of the Fourier symbols $\prod_j k_j^{a_j}$ it applies, with k_j the components of a wavevector. It is verified, exactly at the smallest size and by intermediate representation (IR) to field scale, and prepares no state. Native complex preparation is not built: the implementation accepts real coefficients, data and sources only, and no complex case is among R1–R10. The limit is the implementation's: the classical reference and the GQSP and multiplication stages already carry complex amplitudes, but the extension is untested. The research proposal's "complex-valued, non-Hermitian equations" are therefore answered at the level of the formulation, with realification as the route run (Table M.1); native complex preparation is future work (§ 10.2).

6.4 The Doubled Space

Wu et al.'s doubled space is not a linear system on one state. We run it in the standard form only, since its failure is a property of its kernel, which neither form changes. Its encoding is exact: with $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, each product is folded to one Chebyshev series by the nodal fold of § 6.6 and each linear term is carried by a copy of the collapse row (Appendix H.2). Its kernel is large. The equation block maps the N^2 amplitudes of Ψ to the $2N$ coefficients of the folded register, so it has rank at most $2N$. Let m be the number of invariant constraints; each adds rank at most one, so

$$\dim \ker H_{\otimes} \;\geq\; N^2 - 2N - m, \tag{6.4}$$

above Wu et al.'s count of $2^{2n-1} = N^2/2$ from $n = 3$ for $m \leq 2$. On $4f'' + 2(f')^2 + f = 0$ (R6a) at $n = 5$ the kernel has dimension 962 of 1024.

Proposition 4 (A filter projects onto the doubled kernel)**.** *Let $H_{\otimes} \succeq 0$ act on $\mathbb{R}^{N^2}$, $N \geq 2$, with Π the projector onto its kernel.*

1. *The symmetric product states $\{u \otimes u\}$ do not form a linear subspace, and a kernel does. A kernel that contains two independent product states therefore contains states that are not products, and no Hamiltonian has the product states as its kernel.*

2. *Let P_d be filters with $P_d(0) = 1$ whose largest magnitude on the rest of the spectrum tends to zero, as for the minimax filter as $d \to \infty$ or for $e^{-\beta H_{\otimes}}$ as $\beta \to \infty$. For any ψ_{init} with $\Pi\psi_{\text{init}} \neq 0$,*

$$\lim_{d\to\infty} \frac{P_d(H_{\otimes})\,\psi_{\text{init}}}{\left\|P_d(H_{\otimes})\,\psi_{\text{init}}\right\|} = \frac{\Pi\,\psi_{\text{init}}}{\left\|\Pi\,\psi_{\text{init}}\right\|}. \tag{6.5}$$

3. *A penalty $P \succeq 0$ with $\ker(H_{\otimes} + P) = \operatorname{span}\{\psi_* \otimes \psi_*\}$ must vanish on $\psi_* \otimes \psi_*$ and on no other vector of $\ker H_{\otimes}$: a condition stated in terms of the solution ψ_*.*

The proof is given in Appendix H.3. A filter returns the initial state's projection onto a kernel that holds the solution's product state among exponentially many others, chosen by ψ_{init}, not by the equation. A penalty that singles out the solution must be built from it and is excluded as answer-dependent; Wu et al.'s selection of "a degenerate state that closely matches the

analytical solution" is such a choice. We report the doubled space as a projection, never as a preparation.

6.5 The Carleman Lift in Space–Time

The Carleman lift of a polynomial system is a linear system on the system's monomials [15–18] (§ 2.2.2), so everything in §§ 6.1 and 6.2 carries over: the components become monomials, the ratio rows carry the data's moments, and $\mathbb{A}$ becomes the lifted generator $\mathbb{A}_K$. For $u \in \mathbb{R}^M$,

$$\frac{du}{dt} = \sum_{p=1}^{P} F_p\, u^{\otimes p} \quad \longrightarrow \quad \frac{dz}{dt} = \mathbb{A}_K\, z, \qquad z = \big(u_\alpha\big)_{1 \le |\alpha| \le K}, \quad u_\alpha = \textstyle\prod_i u_i^{\alpha_i}, \tag{6.6}$$

where the state holds every monomial of degree 1 to K once, level by level (the symmetric layout), and each row of $\mathbb{A}_K$ comes from differentiating a monomial and substituting u', the terms beyond degree K being dropped. A damped linear part gives a damped lift, since the eigenvalues of $\mathbb{A}_K$ are sums of those of F_1 [17]. The lift is carried in the scaled variable u/s, $s = \max(\|u_0\|, 1)$, a diagonal similarity of $\mathbb{A}_K$ that leaves the truncated solution unchanged and keeps the data's moments on level ℓ, the monomials of degree ℓ, from growing as $\|u_0\|^\ell$ and pulling the ground state's weight into the top level. Both facts are proved in Appendix H.4.

With λ_1 the eigenvalue of F_1 of largest real part and $\|F_2\|$ taken on symmetric tensors,

$$\rho = \frac{\|u_0\|\,\|F_2\|}{\big|\operatorname{Re}\lambda_1(F_1)\big|} \tag{6.7}$$

is the ratio of Liu et al., who prove for a dissipative quadratic system that $\rho < 1$ suffices for the truncation error at order K to fall as $\mathcal{O}(\rho^K)$, uniformly in time [17]. The condition is sufficient; nothing in the bound says the lift diverges at $\rho \ge 1$, and other analyses give conditions of a different form: a time-dependent radius of convergence [74], a weaker ratio in the ℓ^∞ norm for reaction–diffusion systems [75], and exponential convergence over all time for a class of systems [76]. It is a worst case over all data of norm $\|u_0\|$, since $\|F_2\|$ is a supremum over every pair of inputs and $|\operatorname{Re}\lambda_1|$ the slowest decay of any mode. It says nothing about the residual. For $y' = -y + y^2$ from $y(0) = \frac{1}{2}$ (R8), $\rho = \frac{1}{2}$; on the fluids it exceeds one on most builds. The convergence is therefore measured on every build, order by order against the exact solution of the truncated lift (§ 8.4).

What the residual guarantees is uniqueness. On the time axis the lifted system is Equation (6.1) with $\mathbb{A} \to \mathbb{A}_K$, in either form, with its data as ratio rows at $t = 0$ on the scaled moments of u_0. The truncated lift is a linear initial-value problem (IVP) whose initial value is fixed up to scale, so its solutions form a one-dimensional space. The residual, restricted to polynomials of degree below N_t, then has a one-dimensional near-kernel whose singular value falls spectrally in N_t, in both forms and for every $\mathbb{A}_K$, including a lift whose truncation diverges (Appendix H.5). The uniqueness is the initial-value problem's, which the lift carries; neither form contributes it.

The decoded field's error splits accordingly,

$$\|\hat{u} - u\|_\infty \;\le\; \underbrace{\left\|z_1^{(K)} - u\right\|_\infty}_{\text{the lift's}} + \underbrace{\left\|\hat{u} - z_1^{(K)}\right\|_\infty}_{\text{the time axis's}}, \tag{6.8}$$

with $z_1^{(K)}$ the first level of the truncated lift's exact solution. The lift's part does not depend on n_t and the time axis's falls spectrally to rounding, so the field's error falls with n_t until it meets the lift's floor and then holds; every build records both parts.

6.6 Couplings and Products on a Circuit

Everything in a system's or a lift's rows except its coupling is built from the bands and rank-one rows of § 5.7; the ratio rows are three LCU terms (Appendix H.1).

The coupling as a sum of matchings. A system's coupling $\mathbb{A}$, or a lift's $\mathbb{A}_K$ in the symmetric layout, is sparse and irregular, and its expansion in Pauli words has many terms even when it has few entries. We split it instead into *matchings*, each with at most one entry in every row and column: a permutation of the field register's basis states followed by a diagonal, whose entries are rotations on one flag qubit (§ H.1.1). Each matching is encoded at its largest entry, so the coupling's subnormalisation is the sum of the matchings' largest entries. The split is an edge colouring of $\mathbb{A}$'s bipartite graph of rows and columns. Let δ be the largest number of entries in a row or column: a greedy colouring uses at most $2\delta - 1$ matchings, δ is the least possible by König's theorem [77], and on every lift built the greedy colouring reached δ. The field-register factor is the same in both forms, so they pay the same for the coupling. On Burgers' lift on sixteen modes at $K = 3$, $\mathbb{A}_K$ has 968 components and splits into 31 matchings at $\alpha/\|\mathbb{A}_K\| = 1.34$; on the Navier–Stokes lifts the ratio is larger, 7.34 on eighty modes at $K = 2$ (Table H.1).

The tensor layout. The matchings grow with $\mathbb{A}_K$'s entries, which grow with the number of modes. For a Fourier–Galerkin equation, whose linear operators are Fourier multipliers and whose products are convolutions, a second layout's term count does not: level ℓ is carried as the whole tensor $u^{\otimes \ell}$ in ℓ slots of wavevectors, and the coupling is a sum of terms "F_p on slots $j, \dots, j + p - 1$", $2K - 1$ of them for a quadratic field whatever the number of modes (Proposition 28). Its multipliers vanish off the truncation, so it maps physical states to physical states with no leakage, and its weights carry the complex phases of § 6.3. Per query it becomes the cheaper of the two as the lift grows (Table H.1). Its subnormalisation is looser in two dimensions, where the fold's α is N. It is verified exactly on Burgers' smallest lift and by IR to 316 Navier–Stokes modes, and it encodes the coupling alone: it prepares no state.

Products. A product is pointwise on a grid, so a fold transforms to node values, keeps the diagonal and transforms back. The doubled space's nodal fold does this with the Chebyshev–

Gauss transform $\mathbb{U}$: it is exact and block-encoded at $\alpha = 2$, within 1.21 of its norm at every n computed, so tight to a constant but not optimal (Proposition 31). $\mathbb{U}$ is the discrete cosine transform, budgeted as a cited primitive [78] and not built gate by gate. The tensor layout's Fourier fold uses the quantum Fourier transform (QFT), which is built, at $\alpha = N^{1/2}$ per axis, the convolution's own norm.

6.7 Burgers and Navier–Stokes

The lift is tested on two periodic flows whose quadratic terms are convolutions. Truncated to finitely many Fourier modes on a real basis (§ 6.3) they are polynomial systems of the form of Equation (6.6). A build's error has three parts, each measured on its own: the truncation's, against the equation itself; the lift's, the truncated lift's exact solution against the Galerkin system's IVP; and the time axis's (Equation (6.8)). The systems, their couplings and their parameters are given in full in Appendix H.7.

Viscous Burgers (R9). On $x \in [0, 2\pi)$,

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad u(x,0) = a\sin x, \tag{6.9}$$

whose solution stays odd and lives on the sine modes $\sin kx$, $k \le M$, with $M = 4$, 8 or 16. The Galerkin system has $F_1 = \mathrm{diag}(-\nu k^2)$ and F_2 the projection of $-\frac{1}{2}(u^2)_x$. The least damped mode is $k = 1$, so $\rho = a\|F_2\|/\nu$, and $\|F_2\|$ grows with M. The strength runs from $a/\nu = 0.03$ to 10, across the lift's convergence boundary, and the Cole–Hopf solution [79, 80] measures the truncation's error.

Two-dimensional Navier–Stokes (R10). In vorticity form on the periodic square,

$$\omega_t + J(\psi,\omega) = \nu\,\nabla^2\omega, \qquad \nabla^2\psi = -\omega \ \text{ on } [0,2\pi)^2, \qquad J(\psi,\omega) = \psi_x\omega_y - \psi_y\omega_x, \tag{6.10}$$

with $\nu = 1$, the vorticity on the real modes $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$ of the half-plane with $0 < |\mathbf{k}|^2 \le k_c^2$ (12 to 80 modes), the stream function eliminated as $\hat{\psi}_{\mathbf{k}} = \hat{\omega}_{\mathbf{k}}/|\mathbf{k}|^2$, and every coupling rational and exact. Without viscosity the truncation conserves the energy and the enstrophy, the two invariants of two-dimensional flow, and modes of equal $|\mathbf{k}|$ do not interact (Appendix H.7); a pseudo-spectral direct numerical simulation (DNS) measures the truncation's error. The Taylor–Green vortex $\omega_0 = 2\sin x \sin y$ [81] occupies two modes of equal $|\mathbf{k}|$, so $\omega = \omega_0 e^{-2\nu t}$ exactly and its lift is exact at every order: R10a validates the whole chain at a known answer and does not test the lift's convergence. A counter-rotating vortex pair, scaled to $\|w_0\| = a$ and run at $a/\nu = 3$, 10 and 30, spans several $|\mathbf{k}|$, so its triads are active and its lift is tested. The tensor layout's coupling alone is built further, to 148 and 316 modes.

6.8 Preparation Versus Projection

Table 6.1 places the two nonlinear routes: the doubled space's kernel is a subspace that the solution's product structure is not, so a filter projects; the Carleman lift is a linear system on one state with its data fixed up to scale, so a filter prepares it, in either form.

Table 6.1: The Carleman lift prepares; the doubled space projects. Each nonlinear route by the forms it is run in, its kernel, what a filter returns from an initial state ψ_{init}, and what a trusted kernel (one-dimensional, gap ratio above 20) certifies. $N = 2^n$; m, number of invariant constraints; IVP, initial-value problem. For Carleman ratio $\rho < 1$ the lift's truncation error falls as ρ^K in the order K; the condition is sufficient, not necessary, and convergence is measured on every build.

Route	Forms run	Kernel	A filter returns	A trusted kernel certifies
Doubled space (Wu et al.; R6)	standard only	degenerate, dimension $\geq N^2 - 2N - m$ (Equation (6.4))	the projection of ψ_{init} onto the kernel	nothing: the kernel is never one-dimensional, so never trusted
Carleman lift (R8–R10)	both	one-dimensional near-kernel, from the lifted IVP's uniqueness	the truncated lift's solution	the lifted *linear* problem, not the lift's convergence

The last column carries the caveat. A trusted kernel certifies that the residual singles out the truncated lift's solution; how close that solution lies to the nonlinear one is the first term of Equation (6.8), measured separately. The two can part: past the convergence boundary of Burgers' lift the kernel stays one-dimensional and trusted while the lift diverges (§ 8.4).

What the descriptor adds is the time axis. At one Carleman order the two forms differ only by $I \otimes \tilde{\mathbb{B}}_\xi$ (Equations (5.7) and (6.1)), so the time axis contributes $\|\tilde{\mathbb{D}}_\xi\| = 2(N_t - 1)$ to the mass form's norm against $\|\mathbb{G}_\xi\| = \Theta(N_t^2)$ to the standard form's. Where the gap is flat, as on R7, the mass form's degree therefore grows more slowly with N_t; the coupling's part, $\frac{T}{2}\|\mathbb{A}_K\|$, is common to both and grows with the modes, so at fixed n_t the advantage should narrow as the truncation widens.

For polynomial nonlinearities, then, a Hamiltonian with a unique ground state exists in either form: the Carleman lift turns the equation into a linear initial-value problem on one state, which a filter prepares, where the doubled space can only project. What is prepared is the truncated lift's solution, and its distance from the nonlinear solution is measured, not certified, so on a lift a preparation has two separate tests, the kernel's trust and the lift's convergence. The reformulation's advantage survives on the lift through the time axis alone, and is expected to narrow as the truncation widens. Each of §§ 4 and 5 and this chapter thus ends on a residual and a predicted degree, which § 7 turns into prepared states on equal terms; §§ 8.4 and 8.5 test both halves of this answer, from the scalar R8 to two-dimensional Navier–Stokes.

7 Preparing and Verifying the Ground State

§§ 4 to 6 each end with a residual and a prediction: the filter's degree grows as $\sqrt{\alpha_H/\Delta}$ (Equation (2.2)), with a norm and a gap that each chapter derived for its own form. A prediction about a ratio becomes a comparison only if both of its terms are measured the same way for both forms: against the same classical reference, through the same filter to the same suppression, from the same answer-independent initial state, and with the encoder's share of α_H kept apart from the Hamiltonian's. One pipeline therefore serves every residual (Figure 7.1). A problem enters as data, its classical reference is computed from the stacked residual, the residual is block-encoded and made into a quantum walk, a polynomial filter of the walk is applied at a degree fixed in advance, the result is verified and the field is decoded. The two forms enter only through the residual they hand to this pipeline, so a difference measured between them in § 8 is a difference between their Hamiltonians $H = R^\dagger R$ or between their encodings, which the design of the experiment separates (§ 7.7). The statements hold for any stacked residual, so R, H and α_R carry no form subscript here.

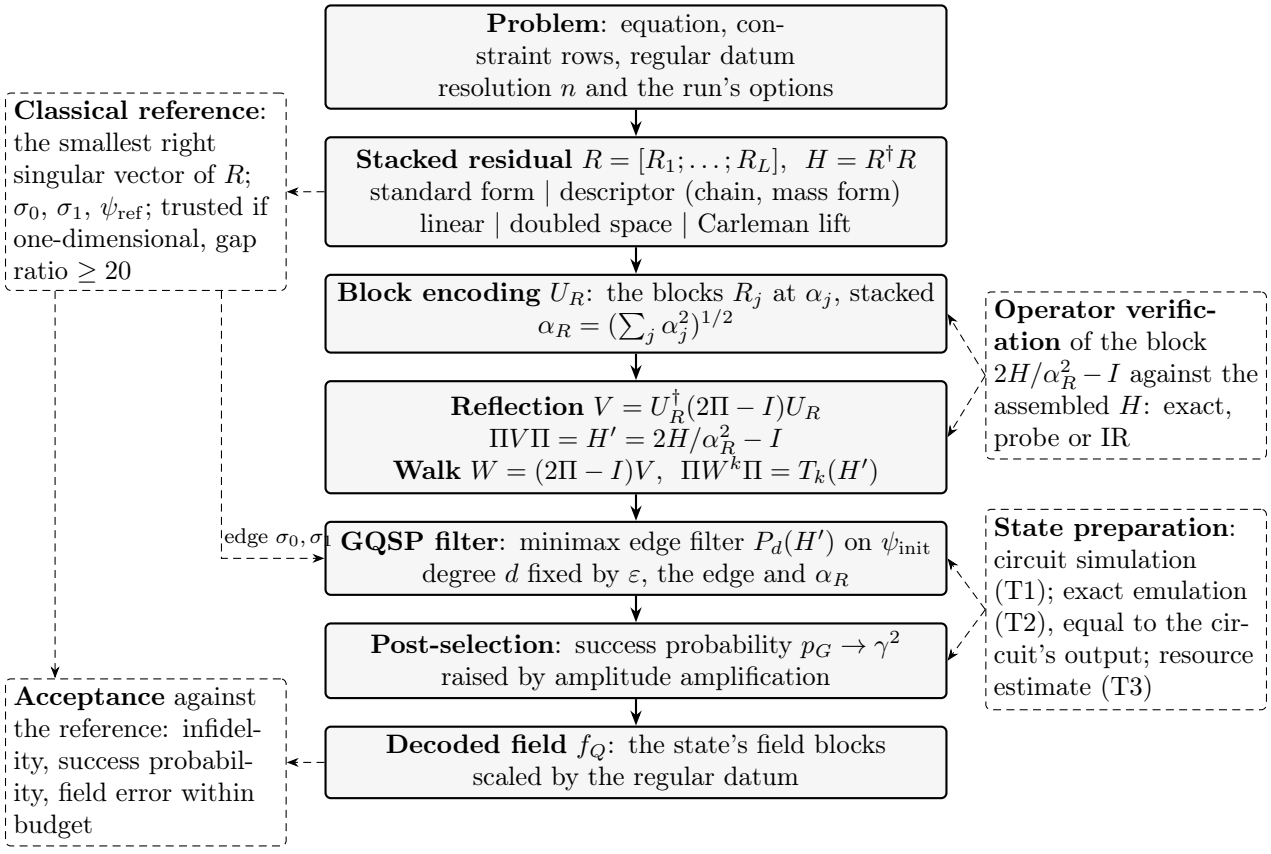


Figure 7.1: One pipeline serves both forms, from a problem's data to a decoded field. Centre (solid arrows), the stages: the forms differ only in the stacked residual R; its block encoding U_R has subnormalisation α_R; the reflection V encodes $H' = 2H/\alpha_R^2 - I$ and the walk W its Chebyshev polynomials $T_k(H')$; a GQSP circuit applies the minimax edge filter P_d, of degree d fixed before the run, to an answer-independent ψ_{init}; post-selection succeeds with probability $p_G \to \gamma^2$; the field f_Q is decoded classically. Left (dashed), the classical reference: the smallest right singular vector ψ_{ref} of R, whose two least singular values σ_0, σ_1 fix the degree, judges acceptance. Right (dashed), operator verification and the three tiers of state preparation. Π, projector onto the ancilla's zero state; ε, target suppression; γ, overlap of ψ_{init} with the ground state; IR, comparison of compiled terms. Every result is a classical computation; nothing ran on hardware.

7.1 Problems and Model Inputs

A problem is data: its fields and axes; its equation, as a sum of terms, each a coefficient function, a derivative order on an axis, a field and possibly a source; its constraint rows; its regular datum; and its reference solution. Nothing downstream depends on a problem except through these. The benchmark problems R1–R10 of Table 4.1 (§ 4.1) are given in this way; the panels as printed are run only to test the published method (Appendix K). The equations, references, constraint rows, resolutions and parameters of every problem are listed in Appendix J.

The data enter as rows of the four kinds of Definition 2, the model's boundary and initial conditions, each the same in both forms and listed with the problems that use them in Table J.3. Two kinds of datum are set up from the reference solution when the problem is built, for the reasons of § 4.3: the exact constraint points of R4 and R6b, and the heat equation's initial function at $t = -1$. Every other datum is the problem's statement.

A run is a problem at a resolution with a set of options: the form and, for the descriptor, its layout and block rescaling; the encoding of $\mathbb{G}$; the treatment of a nonlinearity; the filter; the target $\varepsilon = 10^{-6}$; and the initial state. Their values and defaults are listed in Table J.4 (Appendix J.3). The defaults are the configuration every comparison of § 8 uses; the alternatives are controls, labelled ablations, or the printed choices kept to test the published method.

7.2 The Classical Reference

Every result is judged against a classical reference: the ground state of H, computed from R and never from H, a departure from Wu et al.'s eigenvector of the formed Hamiltonian (Table 4.2). With the singular value decomposition $R = \sum_j \sigma_j \, u_j v_j^\dagger$ and $\sigma_0 \le \sigma_1 \le \cdots \le \sigma_{\max}$,

$$|\psi_{\mathrm{ref}}\rangle = v_0 = \operatorname*{arg\,min}_{\|\psi\|=1} \|R\psi\|_2, \qquad \lambda_j = \sigma_j^2, \qquad \Delta = \sigma_1^2 - \sigma_0^2. \tag{7.1}$$

The reason for working with R is accuracy. A backward-stable singular value decomposition (SVD) of R bounds the computed ground state's error by about $\epsilon_{\mathrm{mach}}\sigma_{\max}/(\sigma_1 - \sigma_0)$, by Wedin's theorem [82], while an eigensolver applied to the formed H is bounded only by about $\epsilon_{\mathrm{mach}}\sigma_{\max}^2/(\sigma_1^2 - \sigma_0^2)$, by the Davis–Kahan theorem [83], larger by about the residual's condition number (Proposition 34). On R2a at $n = 8$ the standard form's relative gap $\Delta/\|H\|$ is 2.3×10^{-17}, below $\epsilon_{\mathrm{mach}} = 2^{-52}$, so the eigensolver's bound exceeds one while Wedin's does not (Appendix I.4).

The dense SVD is used up to 8,192 physical columns, the descriptor's decoupled padding excluded (Proposition 20). Beyond it, the smallest singular triplets come from shift-invert Lanczos iteration on an augmented system that never forms $R^\dagger R$, with an a posteriori estimate of the ground state's error; it reaches residuals of 524 288 columns on the Navier–Stokes lifts (Appendix I.5).

A computed spectrum always has a smallest value, and the reference must decide whether it is

a kernel. The kernel is placed at the largest ratio σ_k/σ_{k-1} among the lowest seventeen singular values, and the reference is trusted (§ 4.3) when that kernel is one-dimensional with gap ratio at least 20, a threshold set above every ordinary spacing in the benchmark problems' spectra: a property of this set of spectra, not a theorem (Appendix I.6). When the truncated solution is close to the residual's minimiser, the gap ratio bounds its angle to the ground state by about σ_0/σ_1; the field's accuracy is measured separately.

A preparation is judged only where its reference is trusted, at three levels: the operator (each encoding against the assembled operator, § 7.5), the state (the infidelity $1 - |\langle\psi_{\mathrm{ref}}|\psi\rangle|^2$ in the form's own space) and the field (the L^∞ and relative L^2 errors of the decoded field against the reference solution on a uniform grid, § 7.6).

7.3 Block Encoding and the Quantum Walk

Each block R_j of the residual, the equation's rows and each constraint's, is block-encoded by its form's circuits at a subnormalisation α_j (§ 4.5.3 and §§ 5.7 and 6.6). The blocks are stacked rather than summed: a row-select register of $\lceil \log_2 L \rceil$ qubits is prepared in $\sum_j (\alpha_j/\alpha_R)\,|j\rangle$ and U_j is applied when it holds j, and unlike the selector of a linear combination of unitaries the register is never returned by an inverse preparation, so that

$$\big(\langle j| \otimes \langle 0_a|\big)\, U_R\, \big(|0\rangle \otimes |0_a\rangle\big) = \frac{R_j}{\alpha_R}, \qquad \alpha_R = \Big(\sum_{j=1}^{L} \alpha_j^2\Big)^{1/2}, \tag{7.2}$$

with a the blocks' shared ancilla and the row select read as part of the output. The subnormalisations add in quadrature, $\alpha_R = \|(\alpha_j)\|_2$, the form of the bound $\|R\|_2 \le (\sum_j \|R_j\|_2^2)^{1/2}$ itself, so a stack of tight blocks is loose only by as much as $\|R\|$ falls below that bound. Wu et al. combine their terms by a linear combination of unitaries (§ 3.3); a combination of the Gram terms $R_j^\dagger R_j$, each encoded as a product of two encodings of its block, would reach the same $\alpha_H = \sum_j \alpha_j^2$, where the stack needs one encoding of each block and one reflection.

That reflection is the standard way to block-encode $A^\dagger A$ from a block encoding of A, the construction that underlies qubitisation and the quantum singular value transformation [40, 45]. What is particular here is its application to the stacked residual, whose blocks enter in quadrature, and the verification of the whole Hamiltonian it encodes (§ 7.5).

Proposition 5 (Stacked-residual reflection)**.** *Let U_R satisfy Equation (7.2), let Π_a project the shared ancilla onto $|0_a\rangle$, and let Π_{in} project the shared ancilla and the row select onto zero. Then*

$$V = U_R^\dagger \big(2\Pi_a - I\big)\, U_R \quad \textit{satisfies} \quad V^\dagger = V, \quad V^2 = I, \quad \Pi_{\mathrm{in}}\, V\, \Pi_{\mathrm{in}} = \frac{2H}{\alpha_R^2} - I \equiv H', \tag{7.3}$$

with $H = R^\dagger R$, and V controlled on further qubits is $U_R^\dagger\, C(2\Pi_a - I)\, U_R$: only the reflection carries the control.

The proof is given in Appendix F.1. V block-encodes H' at subnormalisation one and H at $\alpha_H = \alpha_R^2$, at the cost of the stack, its inverse and one multi-controlled Z gate. Its spectrum needs no rescaling: $H \succeq 0$ and $\|H\| = \|R\|^2 \leq \alpha_R^2$, so H' has its spectrum in $[-1, 1]$, its eigenvectors are the right singular vectors of R, and its ground value sits at $x_0 = -1 + 2\sigma_0^2/\alpha_R^2$, at the edge of the interval when the kernel is exact. Its gap is $2\Delta/\alpha_R^2$: the encoding's subnormalisation divides the gap the filter must resolve, which is how the encoder's cost of § 4 enters the degree. Qubitisation [45] turns the reflection into a walk whose powers are Chebyshev polynomials of H' (Appendix F.2),

$$W = \big(2\Pi_{\text{in}} - I\big)\, V, \qquad \Pi_{\text{in}}\, W^k\, \Pi_{\text{in}} = T_k(H'), \tag{7.4}$$

for one power of W followed by one projection. A controlled walk needs only its two reflections controlled, so it is no wider than U_R (Appendix F.3). The descriptor's padding blocks decouple from the physical subspace (Proposition 20), so a state on that subspace stays there under every polynomial in W.

7.4 The Filter and Its Parameters

A filter is a polynomial P of H' that keeps the ground state and suppresses the rest. GQSP [48] realises any polynomial with $|P(z)| \leq 1$ on the unit circle as a sequence of single-qubit rotations on one signal qubit interleaved with d applications of W controlled on it, with no parity constraint (§ 2.4.2). On the signal's zero, restricted once by Π_{in}, the circuit block-encodes

$$\Pi_{\text{in}}\, P(W)\, \Pi_{\text{in}} = \sum_{k=0}^{d} q_k\, T_k(H'), \qquad P(z) = \sum_{k=0}^{d} q_k z^k, \qquad \sup_{|z|=1} |P(z)| \leq 1, \tag{7.5}$$

by Equation (7.4), so a filter given by its Chebyshev coefficients q_k is realised by the analytic polynomial with the same coefficients. The phases are computed from P alone, by Weiss's complement and the inverse nonlinear Fourier transform [49, 50], with no root-finding, and the polynomial they realise is checked against P (Appendix F.7).

The minimax edge filter. The filter that prepares every comparison and extension in this thesis is the polynomial of degree d that is smallest on the excited spectrum relative to its value at the ground state. With x_1 the first excited value of H' and w the affine map of $[x_1, 1]$ onto $[-1, 1]$,

$$\begin{aligned} P_d(x) &= \frac{T_d\big(w(x)\big)}{T_d\big(w(-1)\big)}, \qquad w(x) = \frac{2x - (x_1 + 1)}{1 - x_1}, \\ \sup_{x \in [x_1, 1]} \frac{|P_d(x)|}{|P_d(x_0)|} &= \frac{1}{\cosh(d\theta_0)}, \qquad \theta_0 = \operatorname{arccosh}\big(-w(x_0)\big). \end{aligned} \tag{7.6}$$

$|P_d| \leq 1$ on $[-1, 1]$ with equality at -1, and by Chebyshev's classical inequality [55] no polynomial of degree d achieves a smaller ratio; it is the eigenstate-filtering polynomial of Lin and Tong [28, 54] anchored at the spectrum's edge. We take the least degree that suppresses the

excited spectrum to ε relative to the ground state,

$$d = \left\lceil \frac{\operatorname{arccosh}(1/\varepsilon)}{\theta_0} \right\rceil, \qquad \theta_0 = \operatorname{arccosh}\Big(1 + \frac{2(h_1 - h_0)}{2 - h_1}\Big), \qquad h_j = 1 + x_j = \frac{2\sigma_j^2}{\alpha_R^2}, \tag{7.7}$$

with θ_0 computed from the heights h_j above the spectrum's edge, which keep their relative precision where x_0 and x_1 would round to -1, as they do whenever $\alpha_R \gg \sigma_1$. Then $d \approx (\alpha_R/2\sqrt{\Delta})\ln(2/\varepsilon)$, an instance of Equation (2.2).[2] Its coefficients are obtained by exact Chebyshev interpolation, so nothing is truncated. For an initial state of overlap γ with a one-dimensional target the prepared state's infidelity is at most $(1-\gamma^2)\varepsilon^2/\gamma^2$, a bound that is finite whenever $\gamma > 0$ (Appendix F.4).

The parameters are a priori. The degree needs ε, the circuit's own α_R and the spectrum's edge, σ_0 and σ_1 from the classical reference; it never reads the solution. The edge is assumed known. This is the standard known-gap assumption: a solver without a classical reference would need a lower bound on the gap in its place, at a degree proportional to the bound's inverse square root, and estimating it on the circuit is outside this thesis (§ 9.5). Wu et al.'s time rule, Equation (3.6), contains no ε and is kept only as a tested variant, which falls short of it (Appendix K.4).

Imaginary time. Wu et al.'s filter is imaginary time, $e^{-\beta(1+x)}$ in the variable of H', whose Chebyshev coefficients are modified Bessel functions (Appendix F.5). Both filters' Chebyshev coefficients alternate in sign and sum in magnitude to one, so both are realised at full weight by Equation (7.5) (Proposition 16). They share the factor $\alpha_R/\sqrt{\Delta}$ and differ in the logarithms, which bring imaginary time's degree close to twice the minimax filter's at equal suppression of the excited spectrum. The minimax filter is every comparison's filter; imaginary time runs only in the filter study of § 8.1, which measures that ratio in both forms, and in the reproduction of Wu et al. (Appendix K).

The initial state. The filter acts on the geometric product state,

$$|\psi_{\text{init}}\rangle \propto \bigotimes_{j=0}^{n-1}\Big(|0\rangle + r^{2^j}\,|1\rangle\Big) = \sum_{k=0}^{N-1} r^k\,|k\rangle\,, \qquad r = \pm\tfrac{1}{2}, \qquad \gamma = \big|\langle\psi_{\text{ref}}|\psi_{\text{init}}\rangle\big|, \tag{7.8}$$

on the field block and zero on every other block: n single-qubit rotations and no entangling gate. Its rationale is the geometric decay of a smooth function's Chebyshev coefficients, with a rate and a sign pattern guessed, not fitted. Both signs of r are run and the one with the larger measured success probability is kept, a rule that needs only what a quantum computer observes. A system's fields each carry the state weighted by the problem's stated initial data $w_0/\|w_0\|$ (§ 6.2). The uniform state and a seeded random state are controls, and γ is reported

[2] From $\operatorname{arccosh}(1+e) \approx \sqrt{2e}$, valid for a small normalised gap with $h_1 \ll 1$, which gives $\theta_0 \approx 2\sqrt{\Delta}/\alpha_R$ (Appendix F.4).

for every run. Where the ground state has weight outside the field block, in the descriptor's derivative blocks or a lift's higher levels, γ is at most the square root of the field block's weight (§ 5.5).

Success and total cost. The post-selection succeeds with probability $p_G = \|P_d(H')\psi_{\text{init}}\|^2 \geq \gamma^2 P_d(x_0)^2$, which tends to γ^2 as the ground value approaches the edge. Amplitude amplification [57] with $\lfloor \pi/(4 \arcsin \gamma) \rfloor$ rounds raises it to at least one half, so with $g(W)$ the two-qubit gate count of one walk the cost of a preparation is[3]

$$\text{queries} = (2k_{\text{aa}} + 1)\, d, \qquad \text{gates} = (2k_{\text{aa}} + 1)\, d\, g(W), \qquad k_{\text{aa}} = \left\lfloor \frac{\pi}{4 \arcsin \gamma} \right\rfloor, \tag{7.9}$$

linear in $1/\gamma$ for small γ; the one-qubit rotations, the initial state and the amplification's reflections grow more slowly and are left out. The departures of this section from Wu et al.'s filter, phases, normalisation and initial state are collected with the others in Table 4.2.

7.5 Verification and the Error Budget

Two questions are kept apart: whether each circuit encodes the operator it claims to, and whether the filter prepares the ground state. The first is *operator verification*. Its target is the whole block $2H/\alpha_R^2 - I$ of the reflection, with H formed from the residual assembled from the problem, never from the compiler that produced the circuit, so wherever the exact and probe modes reach, the composed Hamiltonian is verified whole rather than by parts.

It runs in three modes, with their protocols and tolerances in Table I.1. The *exact* mode reconstructs the whole block, one statevector evolution per input column, and certifies the circuit completely at that size. The *probe* mode compares the circuit's action on twenty structured and Haar-random vectors with sparse products of the target, which certifies the root-mean-square error and is evidence, not proof; it covers every circuit of at most 24 qubits that the exact mode does not. The *IR* mode sums the compiled terms of a lift or coupling back into a sparse matrix and compares it with the assembled operator at any size; the two sides are independent constructions from the problem, so the comparison is not a tautology, and the step from terms to gates is closed by the other two modes where a circuit can be simulated. Wider circuits are recorded for their subnormalisation and gate counts alone. The errors reached are given in § 8.1.

The second question is *state-preparation verification*, at three tiers. *Circuit simulation* (T1) runs the GQSP circuit on a statevector simulator, one controlled walk at a time. *Exact emulation* (T2) computes $\sum_k q_k T_k(H')\psi_{\text{init}}$ by the Chebyshev three-term recurrence with $H'v = 2R^\dagger(Rv)/\alpha_R^2 - v$, two sparse products a step and H never formed, up to degree 2^{20}. A *resource*

[3]The count takes γ from the classical reference, as the degree takes the edge; without it, fixed-point amplification costs a constant factor more. Running both signs of r doubles the preparations, a factor the count leaves out (Appendix F.8).

estimate (T3) takes α_R and the walk's gate count from the circuit, the edge and γ from the reference and the degree from Equation (7.7), and prepares nothing.

By Proposition 5, Equation (7.4) and Equation (7.5), the circuit's post-selected output is $\sum_k q_k T_k(H')\psi_{\text{init}}$, which depends on the encoding only through $\Pi_{\text{in}} V \Pi_{\text{in}} = H'$; two encodings with the same H' give the same output for every filter. T2 therefore computes the circuit's output exactly, up to rounding, for the circuit's own α_R, and T1 tests only that the circuit as built implements it, wherever both run (§ 8.1).

Every circuit simulation is preceded by an estimate of its memory and time from the circuit, and a run whose estimate exceeds its budget is recorded with the estimate and not attempted (Appendix I.7), so circuit simulation covers the smallest resolutions only; § 8.1 gives the widest circuit simulated. The limit is the simulation budget's: by the argument above, a circuit that implements V outputs T2's state at any size.

Each preparation is accepted against bars at every level (Table I.3): the state's infidelity at most ε and within ten times its a priori bound, the circuit against its emulation within 10^{-10} before normalising, and the field within an error budget. The budget splits the prepared state's distance from the reference into the filter's term, the truncation's and the realised phases' (Appendix I.8), and carries each to the field by the exact readout bound B of Proposition 36:

$$\|f_Q - f\|_\infty \;\le\; \underbrace{\|f_{\text{ref}} - f\|_\infty}_{\varepsilon_{\text{disc}}} + B\big(d_{\text{total}}\big), \qquad d_{\text{total}} \le d_{\text{filter}} + d_{\text{trunc}} + d_{\text{phase}}, \tag{7.10}$$

where f_{ref} is the reference's own field, so $\varepsilon_{\text{disc}}$ is the discretisation's error, and $B(d)$ is linear in d for small d. Every encoding is exact and the field is decoded classically, so there is no block-encoding term and no sampling term.

7.6 Reading Out the Solution

The field is read out classically: it is decoded from the prepared state's field blocks and scaled by the regular datum. With $\ell_x(\psi) = \langle \tau(x) | D\psi \rangle$ the field's value functional, D the map from the state to the field's coefficients (the field block; for a lift, its first level times the scale s),

$$f_Q(x) = f(x_s)\,\frac{\ell_x(\psi)}{\ell_{x_s}(\psi)}, \qquad \eta = \Big(\frac{f(x_s)}{\ell_{x_s}(\psi)}\Big)^2, \tag{7.11}$$

so the state's global phase and norm cancel and the decoded field carries the sign of the datum; η is the linear model's scale of § 4.1. Wu et al. read the field out point by point instead, by an interferometric measurement (§ 3.3) whose identity assumes a deterministic preparation; after post-selection with probability $p_G < 1$ it carries a factor $\sqrt{p_G}$, which leaves the field, a ratio, unaffected and the scale wrong by p_G (Equation (F.6)). The protocol is built and verified, with this correction, on R1, R2a and R5a at small n, at two circuits and $\mathcal{O}(1/\delta^2)$ shots per point for an overlap to δ; its study at scale is further work (§ 10.2), not the method's readout.

7.7 Design of the Numerical Experiment

The study varies the resolution (n per axis, n_t, the modes M and the Carleman order K), the form, the encoding of $\mathbb{G}$, the filter, the target ε and the initial state, one at a time from the defaults of Table J.4. Within a comparison of the forms everything else is fixed: the problem and its constraint rows, $\varepsilon = 10^{-6}$, the minimax filter, the geometric initial state on the field block and the readout.

Cost conventions. A query is one application of the walk W; a preparation costs d queries and an amplified one $(2k_{\mathrm{aa}} + 1)d$ (Equation (7.9)). Gates per query are the two-qubit gates of one walk after transpiling to $\{\mathrm{CX}, U\}$ [84], with one-qubit gates counted separately and dense primitives that are cited rather than built, the doubled space's discrete cosine transform, counted apart. Ancilla are counted on the composed reflection V: the row select, every block's ancilla and, in the descriptor's chain, nothing for the block register, which is part of the state.

Fairness. The standard form is costed with its structured encoding, the cheapest exact encoding of it built (§ 4.4). It is not as tight as the descriptor's encoding, whose subnormalisation approaches the residual's norm on regular one-axis problems (§ 5.8), while the structured encoding's stays within a constant of it for constant coefficients only and is loose for a lifted coefficient (§ 4.6). The degree is linear in α_R (Equation (2.2)), so an as-built comparison favours the descriptor by about the ratio of the two encodings' looseness, a factor that belongs to the encodings, not to the Hamiltonians. Every comparison is therefore quoted twice: as built, and at the ideal-encoding bound $\alpha_R = \|R\|_2$ of § 4.6, the least degree the minimax filter needs on the residual through any encoding of it (Proposition 18). The difference between a form's two degrees is its encoder's cost; a difference between the forms at the ideal bound is a difference between their Hamiltonians, and only that difference answers the first two questions. Where the order of the forms reverses at the ideal bound, the reversal is reported as a finding (§ 8.2).

Tiers and fits. Every number states its tier. Degrees beyond the emulation's reach are resource estimates: exact for this filter, given the spectrum and the circuit's α_R, but with no state prepared, and compared with T2 where both exist. Every resource estimate is computed at its own resolution, from that resolution's circuit and reference, and none is extrapolated; the growth of a degree with N is quoted as a measured exponent between stated resolutions. Both forms are given the exact spectral edge, so the known-edge assumption favours neither, though it leaves out the cost of finding the edge (§ 9.5).

7.8 Software and Computation

The pipeline is implemented in `pihm`, a Python package written for this thesis, on NumPy [85] and SciPy [86] for the classical linear algebra and Qiskit and Qiskit Aer for the circuits and their simulation [84]. Each concept has one implementation, shared by both forms and every

problem, so a difference between the forms cannot come from different code for the same step. Each claim the thesis makes has a test that recomputes it. The production runs were made on the Setonix supercomputer of the Pawsey Supercomputing Research Centre [87] at one tagged version of the code, and every number and figure in this thesis is regenerated from the archived records by one command. The package's architecture, the record format, the campaigns and the commands are given in Appendix L.

8 Numerical Study

The formulation chapters made their claims about operators. For constant coefficients the standard form's norm grows as N^{4k} against a flat gap; the descriptor's falls to $\Theta(N^2)$ at fixed block weights, at the price of a gap its derivative blocks erode; and a Carleman lift prepares where the doubled space projects. Each claim predicts a filter degree through Equation (2.2), and none of those degrees has yet been measured. The measurements below, on the benchmark problems of Table 4.1 with the pipeline and conventions of § 7, follow the order in which the predictions depend on one another. The pipeline is shown to compute what it claims before any cost is compared; the standard form's two costs are separated before the descriptor is set against them; the gap is measured where the norm's advantage could be lost; and the lifts come last, since the question of advantage arises only once a nonlinear equation is prepared at all. Every result is a classical computation and names its tier, T1, T2 or T3 (§ 7.5); a gap, a norm or a reference's error is an exact classical computation on the assembled residual. The results for every problem in both forms are tabulated in Appendix J.4; the reproduction of Wu et al.'s own panels, which tests the published method rather than the three questions, is recorded in Appendix K.

8.1 Validation of the Pipeline

Solved classically through its reference (§ 7.2), the standard form converges spectrally on every linear benchmark problem: the smallest trusted error of the decoded field is 3.2×10^{-13} on R2a at $n = 5$, 8.5×10^{-12} on the inhomogeneous R4b at $n = 5$, and 4.6×10^{-12} on the heat equation at $n = 5$ per axis. The descriptor, whose kernel is the standard form's (Theorem 1), reaches 3.3×10^{-15} on R2a. Beyond its best resolution the standard form's error rises again, to 1.4×10^{-6} on R2a at $n = 10$, where the descriptor's is 3.7×10^{-13}. The rise is consistent with rounding error amplified by the conditioning of a residual whose norm grows as N^4, an early sign of the Hamiltonian's cost of § 8.2.

The circuits encode the operators. Every verified circuit meets its mode's tolerance (Table I.1 and fig. 8.1) by at least four orders of magnitude. The exact mode's worst entrywise error is 5.9×10^{-15}, on a reflection of the doubled space, and its widest verification is a whole reflection on 25 qubits, of Wu et al.'s own R4d panel, whose Maclaurin source widens the register, at 1.8×10^{-15}. The probe mode's worst is 9.1×10^{-14}, on a Navier–Stokes lift's reflection

of 24 qubits, and the IR mode verifies the tensor layout's coupling on 316 modes to 3.6×10^{-17}. Every composed Hamiltonian of at most 24 qubits is therefore verified whole, and every lift and coupling by its terms at any size. The wider reflections, those of the two-axis descriptor problems from 25 qubits and of the larger lifts, are built and counted but not independently verified (§ 7.5).

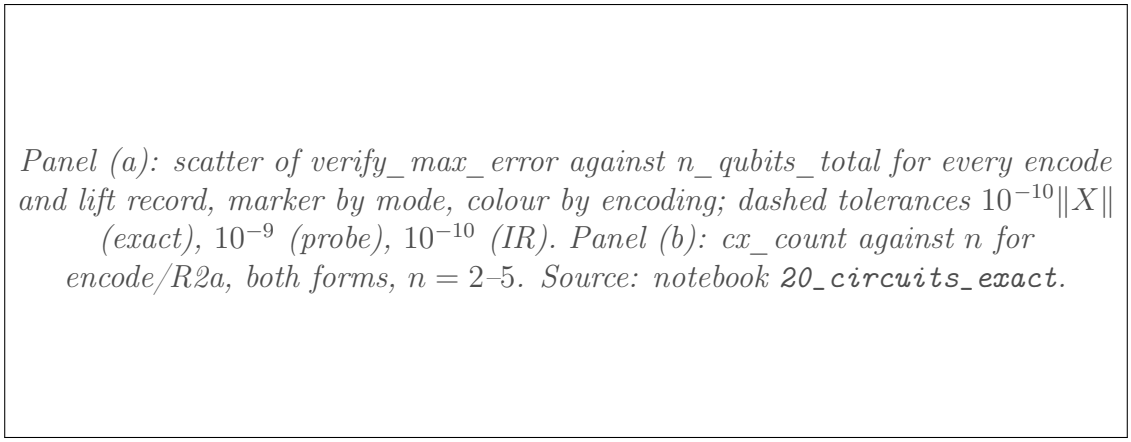


Figure 8.1: Every verified circuit encodes its operator to rounding, in every verification mode. (a) Largest error of each verified circuit against its independently assembled operator, against the circuit's width in qubits: exact mode (filled markers; the reflection's whole block $2H/\alpha_R^2 - I$, entrywise), probe mode (open markers; twenty structured and Haar-random inputs) and IR mode (crosses; a lift's or coupling's compiled terms), for Wu et al.'s, the structured and the descriptor encodings and the Carleman lifts, with each mode's tolerance dashed (Table I.1). (b) Transpiled two-qubit gates per query of the reflection against n on R2a, structured standard and descriptor forms. Every error lies at least four orders of magnitude below its tolerance; the widest exact verification, 25 qubits, is from the reproduction. Exact classical computation from the built circuits.

The circuit computes the emulation. On every circuit simulation with the minimax filter the simulated output agrees with the exact emulation's before normalising, within the bar of 10^{-10}. The worst difference is 1.9×10^{-12} in the descriptor (R4a at $n = 3$, degree 1126), 1.4×10^{-12} in the standard form (R4a at $n = 2$, degree 2093), and 5.8×10^{-13} on a Carleman lift (Burgers on four modes, 19 qubits), each the rounding of a statevector carried through hundreds to thousands of controlled walks. This confirms, where both run, the argument of § 7.5 that the exact emulation is the circuit's output, which is what licenses the emulation beyond the widths simulated. The widest circuit simulated for a problem as run has 20 qubits; the two-axis descriptor circuits, from 26 qubits, are beyond the simulation budget, the narrowest estimated at 3.7×10^5 s against a budget of 86 387 s, which is a limit of the classical simulation and not of the circuit. The imaginary-time filter's circuit simulations, run in the reproduction, meet the same bar on all but three records, at degrees of order 10^5; others at higher degrees meet it with room to spare, so the rounding depends on the problem as well as the degree. The three are reported in Appendix K.7.

Every prepared state meets its target. Every trusted preparation by exact emulation meets ε and its a priori bound, in both forms, on every linear problem and every Carleman lift, and its decoded field meets the error budget of Equation (7.10); the values per problem

are given in Table J.5. The bound is tight: on R2a in the descriptor at $n = 8$ the infidelity is 3.8×10^{-12} against the bound $(1-\gamma^2)\varepsilon^2/\gamma^2 = 4.2 \times 10^{-12}$ of § 7.4. Figure 8.2 carries the state's error to the field. The field's error falls spectrally with n until it meets the filter's contribution, and then holds: on R2a in the descriptor it is 1.8×10^{-6} at $n = 4$ and 1.7×10^{-6} at $n = 8$, while the discretisation's own error falls to 6.8×10^{-14}. From $n = 4$ the prepared field's accuracy is therefore set by ε: a more accurate field needs a smaller ε, at a degree logarithmic in $1/\varepsilon$ (Equation (7.7)), where a larger n gains nothing. The budget itself, 6.8×10^{-5} there, is conservative by more than an order of magnitude, consistent with a readout bound that takes the state's worst direction rather than its actual one.

field_err_linf, eps_disc, field_budget and its parts against n = 3–8 for stateprep/R2a/descriptor/T2. Source: notebook `32_field_and_budget`.

Figure 8.2: From moderate n the filter, not the resolution, sets the prepared field's accuracy, and the field stays within its error budget. Against n for R2a in the descriptor: the L^∞ error of the field decoded classically from the prepared state, the discretisation's error $\varepsilon_{\text{disc}}$ and the budget of Equation (7.10), split into its filter, truncation and phase terms carried to the field by the readout bound; minimax filter, $\varepsilon = 10^{-6}$. The error follows $\varepsilon_{\text{disc}}$ to $n = 3$ and then holds at the filter's contribution while $\varepsilon_{\text{disc}}$ falls to rounding. Exact emulation (T2).

What answer independence costs. The initial state and the spectral edge are the two inputs chosen without the solution. The geometric state of Equation (7.8) guesses that the solution's coefficients decay geometrically, and it is good where they do: on R1 at $n = 3$ its overlap is $\gamma^2 = 0.904$, against 6.4×10^{-3} for the uniform state and 0.27 for a random one. It is poor where the solution is a polynomial of low degree: on Legendre's equation R3a, whose solution is P_4, the uniform state's γ^2 is 0.321 against the geometric state's 0.166 (exact emulation, T2). The degree does not depend on the state, so every control meets ε at the same degree, and a poor guess costs the factor $1/\gamma$ of amplification in Equation (7.9). An underestimated edge costs degree, as an inverse square root of the gap (Equation (7.7)), and keeps the target, since it only widens the interval suppressed; a study on R2a bears this out, and in it Wu et al.'s own time rule falls short of ε (Appendix F.5).

The filter study. The comparison of the two filters, at three strengths of like error, is tabulated in Appendix F.5. At equal suppression with the ground state at the spectrum's edge, imaginary time needs 1.67–1.92 times the minimax filter's degree on synthetic spectra, whatever the gap, and every trusted reference has its ground state at the edge, so on the benchmark problems the certified ratio lies in that range in both forms. At equal measured infidelity the gain is smaller, because imaginary time suppresses the higher excited states more

strongly than the equiripple bound assumes, and at the loose $\varepsilon = 10^{-4}$ imaginary time needs fewer degrees on three of the eight cases. Because the ratio does not depend on the gap, the choice of filter changes the degree's constant and not its growth with N.

The pipeline is therefore faithful wherever it can be checked: each verified circuit is its operator, the simulated circuit is its emulation, and each emulated state meets its target and bound, so the differences measured in the rest of this chapter are differences between residuals.

8.2 The Cost of the Two Forms

Table 8.1 separates the two costs of § 4 on R2a ($f'' + 4f' + 4f = 0$) at $n = 8$, the largest resolution estimated for every encoding. The first two columns encode one Hamiltonian, H_{std}, so they share its norm, its gap and its ideal-bound degree, and differ only in how tightly they encode it: Wu et al.'s encoding exceeds the residual's norm by a factor of 6.1×10^{13} and the structured encoding by 7.74, and the degree falls between them by the same factor. That factor is the encoder's cost. The descriptor's column is a different Hamiltonian with the same kernel, encoded within a few per cent of its norm. The distance between the standard form's ideal-bound degree and the descriptor's is the Hamiltonian's cost, which no encoding of H_{std} recovers (Proposition 18).

Table 8.1: The encoder's cost and the Hamiltonian's cost are separate, and only the second is the reformulation's to remove. R2a ($f'' + 4f' + 4f = 0$) at $n = 8$, minimax filter at $\varepsilon = 10^{-6}$. The first two columns encode the same H_{std}, so their norm, gap and ideal-bound degree coincide; the third is the descriptor's H_{desc}, with the same kernel. $\|H\|$, spectral norm; Δ, spectral gap; $\alpha_R/\|R\|_2$, looseness, the encoding's subnormalisation over the residual's norm; ancilla, those of the reflection V; query, one walk step; growth, the exponent of $N = 2^n$ between $n = 7$ and 8; $\alpha_R = \|R\|_2$, the ideal-encoding bound. Norms, gaps and looseness: exact classical computation. Degrees, qubits and gate counts: resource estimates (T3), the descriptor's degree confirmed by exact emulation (T2).

	Standard form, H_{std}		Descriptor, H_{desc}
	Wu et al.'s encoding	structured encoding	
$\|H\|$, its law	$\Theta(N^8)$	$\Theta(N^8)$	$\Theta(N^2)$
$\|H\|$	3.0×10^{17}	3.0×10^{17}	4.2×10^{6}
Gap Δ	7.04	7.04	2.87
Looseness $\alpha_R/\|R\|_2$	6.1×10^{13}	7.74	1.04
Encoding	approximate, $\mathcal{O}(N^{-5})$	exact	exact (no $\mathbb{G}$)
Qubits	26	25	24
Ancilla	17	16	13
Two-qubit gates per query	38 744	46 324	46 260
Degree, as built	9.1×10^{22}	1.2×10^{10}	9128
Degree, $\alpha_R = \|R\|_2$	1.5×10^{9}	1.5×10^{9}	8736
Degree's growth, as built	$N^{10.0}$	$N^{4.01}$	$N^{0.945}$
Degree's growth, $\alpha_R = \|R\|_2$	$N^{4.00}$	$N^{4.00}$	$N^{1.01}$
Two-qubit gates, preparation	1.1×10^{28}	1.6×10^{15}	1.3×10^{9}

Figure 8.3 follows the same three encodings in N. The norms grow at their laws: between $n = 7$ and 8, $\|H_{\text{std}}\|$ rises from 1.2×10^{15} by a factor 2^8 and $\|H_{\text{desc}}\|$ from 1.0×10^{6} by a factor 2^2, while both gaps hold (§§ 4.6 and 5.9). The degree's exponents in Table 8.1 are slopes between

two resolutions of a degree that carries logarithmic factors, consistent with $\tilde{\Theta}(N^{10})$, $\tilde{\Theta}(N^4)$ and $\tilde{\Theta}(N)$ rather than fits to them. The two costs therefore appear as two separations on one plot. The separation between Wu et al.'s encoding and the structured one grows as $N^{3k} = N^6$, and an exact encoding of the same Hamiltonian removes it. The separation between the structured encoding and the descriptor grows as N^3, and only a different Hamiltonian removes it.

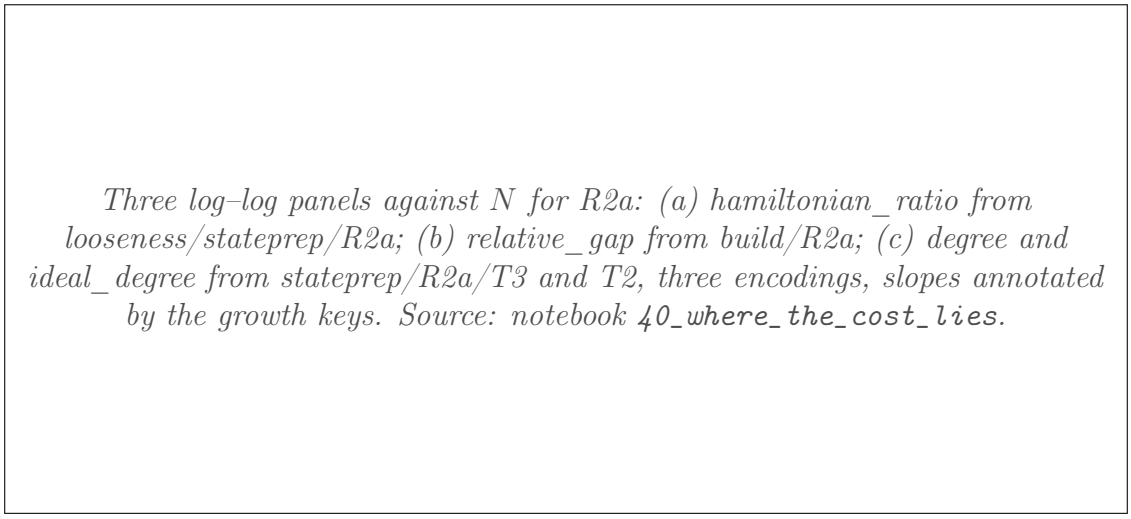


Figure 8.3: The encoder's cost and the Hamiltonian's cost are two separate gaps on one plot. R2a against $N = 2^n$, $n = 2$–8, log–log, for Wu et al.'s encoding, the structured encoding and the descriptor. (a) The encoding's looseness $\alpha_H/\|H\| = (\alpha_R/\|R\|_2)^2$: growing as N^{12} for Wu et al.'s encoding, bounded for the structured, falling towards one for the descriptor. (b) The relative gap $\Delta/\|H\|$: $\Theta(N^{-8})$ for H_{std}, $\Theta(N^{-2})$ for H_{desc}. (c) The filter's degree as built (solid) and at the ideal-encoding bound $\alpha_R = \|R\|_2$ (dashed); it is about $\frac{1}{2}\sqrt{\alpha_H/\Delta}\,\ln(2/\varepsilon)$, so (c) is the square root of (a) over (b). The vertical distance between Wu et al.'s and the structured curves is the encoder's cost; that between the structured and the descriptor curves at the ideal bound is the Hamiltonian's cost. Exact emulation (T2, filled) where it runs, the descriptor to $n = 8$ and the standard form to $n = 3$; resource estimates (T3, open) beyond. Norms and gaps are exact classical computations.

Table 8.2 extends the measurement to the classes of Table 5.1, with every linear problem and the coupled oscillator R7 in Table J.7. In the standard form on one axis, with constant coefficients, the degree grows as N^{2k} as built and at the ideal bound alike: the structured encoding holds the encoder's cost to a constant, and what remains is the Hamiltonian's. In the descriptor, on every one-axis problem whose leading coefficient does not vanish, it grows as about N, its exponents between 0.872 (R4d) and 0.983 (R1) as built and within a few per cent of one at the ideal bound. The stiff R2c is the exception as built, at $N^{0.656}$, because its degree is still rising towards the asymptotic regime at $n = 8$; its ideal-bound exponent is already 0.975. Where the leading coefficient vanishes (R3, R4a) the descriptor's degree grows as about N^2, the price of the rescaling's $\zeta \propto N$ (§ 5.6). On two axes it grows faster than N^2 per axis yet more slowly than the standard form's. These are slopes from $n = 4$ to 5 per axis, the largest resolutions estimated, and the least settled in the table; on the wave equation R5c, whose reference is trusted only from $n = 5$ per axis, they are indicative only.

The reversals. Two comparisons reverse, and both are findings. The first is R3 at the ideal bound, where the standard form needs the smaller degree on all four problems (Table J.6): on

Table 8.2: The descriptor's degree grows as about N where the equation controls every carried derivative; the standard form's grows as N^{2k} on one axis with constant coefficients. Exponent of $N = 2^n$ in the minimax filter's degree between the two largest resolutions estimated ($n = 7 \to 8$ on one axis, $n = 4 \to 5$ per axis on two), as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$, for representatives of each class of Table 5.1 (every problem: Table J.7). k, equation order; a_2, leading coefficient. The standard form's as-built exponent on R3a is its lifted multiplier's looseness; at the ideal bound both forms grow as N^2 there. †Reference not trusted at $n = 4$ per axis; slope indicative only. Two-point slopes of a degree with logarithmic factors: consistent with the laws, not fits of them. Resource estimates (T3).

		Standard		Descriptor	
Class	Problem	As built	Ideal	As built	Ideal
First order, $k = 1$	R1	2.01	2.00	0.983	1.01
	R7	2.00	2.00	0.954	0.994
Second order, $a_2 \neq 0$	R2a	4.01	4.00	0.945	1.01
	R2c	4.01	4.00	0.656	0.975
	R4d	4.01	4.00	0.872	1.00
Second order, a_2 vanishes	R3a	4.01	2.00	1.97	2.00
	R4a	4.01	4.01	1.99	2.02
Two axes, per axis	R5a	4.09	4.03	3.31	3.32
	R5b	4.01	3.90	2.23	2.30
	R5c†	2.93	2.86	2.75	2.76

R3a at $n = 8$, 1.1×10^5 against the descriptor's 2.8×10^5. As built the order is the other way, 1.7×10^{10} against 5.7×10^5, because the structured encoding pays for R3's lifted multiplier in full; the as-built advantage there is the encoder's, not the Hamiltonian's. The reversal belongs to R3 rather than to every vanishing coefficient. R3's coefficient $1 - x^2$ vanishes at both ends of the interval, where the derivatives of the Chebyshev polynomials are largest, and its standard-form norm grows only as N^4, from 6.0×10^8 at $n = 7$ to 9.7×10^9 at $n = 8$. The coefficient $x - 1$ of R4a vanishes at one end only; its $\|H_{\text{std}}\|$ grows as N^8, from 9.2×10^{15} to 2.4×10^{18}, and there the descriptor's degree is the lower at the ideal bound too, 4.0×10^5 against 6.7×10^9. That the endpoint concentration is what lowers R3's norm is an inference from these two problems, consistent with § 4.6, not a theorem. The second reversal is the heat equation at small n, taken up with its gap in § 8.3.

Queries and gates. The degree is not the whole cost. On R2a one query of the descriptor costs more two-qubit gates than one of the structured standard form at every resolution built and verified, 25 914 against 8094 at $n = 2$ and 34 098 against 22 856 at $n = 5$ (Figure 8.1), having more terms and a block register; its count grows more slowly, and the estimates at $n = 8$ put the two at parity (Table 8.1). Wu et al.'s encoding is the cheapest of the three per query and the loosest. On R3a, whose lifted multiplier the structured encoding pays for in full, the descriptor's query is the cheaper from $n = 4$: 38 864 against 43 108 at $n = 4$ and 48 542 against 52 038 at $n = 5$. The ancilla's growth, $\mathcal{O}(\log n)$ against n (§ 5.8), is not yet visible at these sizes: the whole reflection on R2a needs 12 ancilla in the descriptor at $n = 2$ and 13 at $n = 5$, against 10 and 13 in the standard form. On the constant-coefficient problems the

descriptor's advantage is therefore in the number of queries, and that number dominates the whole preparation's gate count (Table 8.1). No number at $n = 8$ is a simulation: the degrees are resource estimates (§ 7.7); the descriptor's agree with exact emulation on R2a through $n = 8$, while the standard form's lies far beyond the 2^{20} that exact emulation reaches (§ 7.5), and its size is itself the quantity under test.

The answer to the first question is therefore measured as well as argued: with Wu et al.'s encoding the cost is mostly the encoder's, an exact encoding removes that part, and what remains, $\tilde{\Theta}(N^{2k})$ in the degree for constant coefficients on one axis, is the Hamiltonian's. The descriptor removes the Hamiltonian's part on one-axis problems whose leading coefficient does not vanish; where it vanishes the descriptor's degree still grows as N^2, and on R3 at the ideal bound the standard form's is the lower.

8.3 The Descriptor's Gap and Its Rescaling

§ 5.5 predicted that the descriptor's gap would shrink where the solution's derivatives are large and close where the leading coefficient vanishes or a second axis enters, and that block rescaling would restore it on one axis only. Figure 8.4 measures this on every panel. Unscaled, the gap on the physical subspace at $n = 8$ is 0.458 on R2a but 7.5×10^{-6} on the stiff R2c and 2.7×10^{-8} on R3a, whose leading coefficient vanishes, and the field block holds as little as 5.4×10^{-6} of the ground state on R2c. Rescaled, the gaps are 2.87, 0.715 and 3.62 and the field block's weights 0.384, 0.774 and 0.993. Every reference here is trusted in both versions, so the rescaling changes the filter's work and not the prepared field. On two axes the Laplace equation's rescaled gap, its reference trusted from $n = 3$ per axis, falls from 0.372 to 6.9×10^{-2} and 1.1×10^{-2} at $n = 3$, 4 and 5 per axis, against 1.1×10^{-4}, 1.1×10^{-6} and 1.1×10^{-8} unscaled; at $n = 5$ per axis the heat and wave equations' rescaled gaps are 9.4×10^{-2} and 2.0×10^{-2}. That is a negative finding: on two axes block rescaling slows the gap's fall but does not stop it. The a priori rule of Equation (5.11) is never fitted, and its cost against the cheapest ζ on a grid is tabulated in Appendix G.5.

The overlap. Rescaling raises the ceiling $\gamma \le w_f^{1/2}$ on the initial state's overlap (§ 7.4); how close the geometric state comes to it is a separate matter of the guess. On R4b the ceiling is low: $w_f = 0.176$ and $\gamma^2 = 0.040$ at $n = 8$. On R3a, with w_f near one, γ^2 is still 0.16 at $n = 7$, because a guess of geometric decay is a poor one for a polynomial solution (§ 8.1); both by exact emulation (T2). The wave equation is the exception to trust rather than to overlap: below $n = 5$ per axis its near-kernel is several-dimensional, 5 at $n = 4$, consistent with a solution, $\cos 4\pi t \sin 2\pi x$, not yet resolved in t, and no preparation is claimed there (Table J.5).

Three problems in both forms. Figure 8.5 follows three problems through both forms, each chosen for what it tests.

- *R2c, the stiff $f'' + 5f' + 400f = 0$: where rescaling gains most on a regular problem.* Of

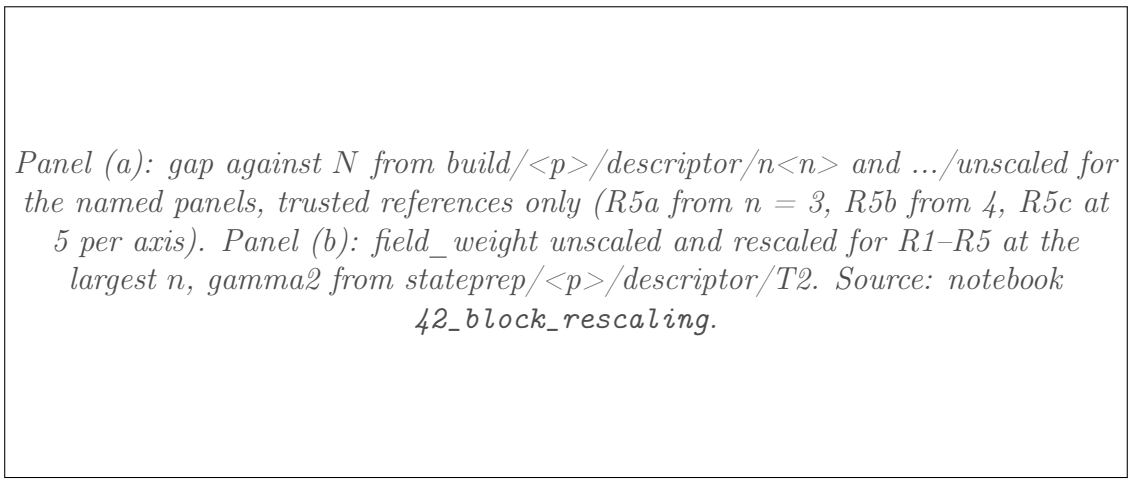


Figure 8.4: Block rescaling restores the descriptor's gap and the field block's weight on one axis; on two axes the gap still falls. (a) The gap Δ on the physical subspace against N, unscaled (open) and block-rescaled (filled), for R2a (regular), R2c (stiff), R3a and R4a (leading coefficient vanishing) and R5a–R5c (two axes, per axis), at the resolutions whose reference is trusted. (b) For every panel at $n = 8$ ($n = 5$ per axis on R5), the field block's weight w_f in the ground state, unscaled against rescaled, with the geometric initial state's overlap $\gamma^2 \leq w_f$ marked. Rescaling lifts the one-axis gaps to $\mathcal{O}(1)$ and w_f towards one; on two axes it gains orders of magnitude but the gap still falls with N. Exact classical computations (the reference's spectrum); γ^2 from exact emulation (T2).

the problems whose leading coefficient does not vanish, R2c has the smallest unscaled gap and field weight. Rescaled, the descriptor prepares it from $n = 5$, with infidelity 2.2×10^{-12} at $n = 8$ by exact emulation. Its degree at $n = 8$ is 2.4×10^5 against the standard form's 6.0×10^9 as built, and 1.8×10^5 against 7.8×10^8 at the ideal bound. Unscaled, the gap and the field block's weight are each about five orders of magnitude smaller, so an unscaled preparation would need more degrees and more amplification; how many more was not estimated, since rescaling also changes $\|H_{\text{desc}}\|$ and α_R.

- *R3a, Legendre's equation: the ideal-bound reversal.* Both forms prepare the exact kernel P_4 from $n = 3$. As built the descriptor's degree is the lower at every n from 3; at the ideal bound the standard form's is the lower at every n, and both grow as N^2 (Table 8.2). On this problem the descriptor's advantage is entirely the encoder's.

- *R5b, the heat equation: the gap still falls, and the descriptor gains nothing at the resolutions estimated.* Its reference is trusted from $n = 4$ per axis. There the descriptor's degree is the higher, 29 044 against 27 692 as built and 12 706 against 4015 at the ideal bound. At $n = 5$ it is the lower as built, 1.4×10^5 against 4.5×10^5, and at the ideal bound the two are at parity, 6.3×10^4 against 6.0×10^4. Its exponent per axis is the smaller (Table 8.2), so the order is expected to settle in the descriptor's favour beyond the resolutions estimated, but that is an extrapolation; the measured finding is the reversal at $n = 4$ and parity at $n = 5$.

The other problems, in both forms and at every tier, are given in Appendix J.4. The gap the carried derivatives erode is therefore restored by the a priori rescaling on every one-axis problem run and not on two axes, where on the heat equation the descriptor gains nothing at

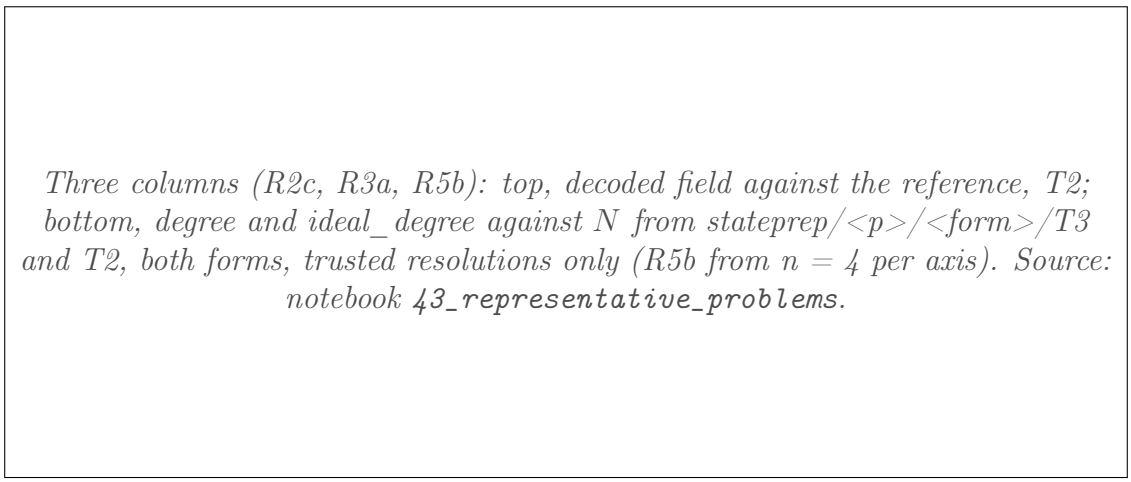


Figure 8.5: Three problems, three verdicts: rescaling's largest gain on a regular problem (R2c), a reversal at the ideal bound (R3a), and no gain at the resolutions estimated on two axes (R5b). Columns R2c, R3a and R5b. Top row: the field decoded from the prepared state against the analytic solution, both forms, by exact emulation (T2) at the largest trusted resolution. Bottom row: the minimax filter's degree against N (per axis on R5b), standard (squares) and descriptor (circles), as built (solid) and at the ideal-encoding bound $\alpha_R = \|R\|_2$ (dashed), at the resolutions whose reference is trusted; T2 filled where emulated, T3 open beyond. R2c: the descriptor lower by orders of magnitude in both comparisons. R3a: lower as built, higher at the ideal bound, both growing as N^2. R5b: higher at $n = 4$ per axis in both comparisons, at parity at the ideal bound at $n = 5$.

the resolutions estimated.

8.4 Systems and Nonlinear Problems

This section tests the third question's two halves: whether a nonlinear equation is prepared rather than projected (§ 6), and whether the descriptor's lower degree survives on the lifted systems, where it enters through the time axis alone (§ 6.8). The coupled oscillator R7 comes first, because every lift is a coupled linear system of its kind.

Two coupled fields. The harmonic oscillator R7, $u' = v$, $v' = -\pi^2 u$, is the realification of $z' = -\mathrm{i}\pi z$ (§ 6.3), carried as two fields on one state with its data as ratio rows. Both forms converge to rounding, the descriptor's field error 2.0×10^{-14} at $n = 5$, and both prepare it: from $n = 4$ the data-weighted geometric state has $\gamma^2 = 0.247$ in either form, and every trusted exact emulation meets ε. The descriptor carries the system in its mass form, whose gap is about twice the block-rescaled chain's (§ 6.1) and which keeps the whole ground state on the field blocks, where the chain's derivative blocks take weight from them, $w_f = 0.852$. The mass form's degree at $n = 8$ is 1.1×10^4 against the standard form's 1.4×10^6 (T3), and grows as N against N^2, the first-order case of Table 8.2. Simulated at $n = 2$ on 13 qubits, where the reference is not yet trusted, the circuit agrees with the emulation to 1.2×10^{-15}: a test of the circuit, not of a preparation. What is prepared is the realified real system, at the cost of one field-register qubit; no complex residual was prepared (§ 6.3).

Projection against preparation. The doubled space's kernel on R6a has dimension 10, 50, 226 and 962 at $n = 2$ to 5, in state spaces of 16 to 1024, so a filter returns the initial state's projection onto it (Proposition 4), which is never a preparation. The Carleman lift of $y' = -y + y^2$ (R8) has a one-dimensional, trusted kernel at every order from $K = 2$ to 6 in both forms, so the filter prepares it.

The lift on a time axis (R8). The field's error splits as Equation (6.8) predicts. At each order it is flat in n_t from $n_t = 3$, the time axis's part having fallen below the lift's, and it falls with the order, from 2.8×10^{-2} at $K = 2$ to 7.8×10^{-4} at $K = 6$, faster than the factor $\rho = \frac{1}{2}$ per order that the bound of § 6.5 allows. Both forms prepare every order to ε; the infidelity at $K = 4$ and $n_t = 8$ is 1.5×10^{-13} in the descriptor and 1.7×10^{-13} in the standard form. The descriptor's time axis lowers the degree, and the margin widens with n_t: at $K = 4$ it is 233 against 1228 at $n_t = 4$ and 1901 against 319 510 at $n_t = 8$, by exact emulation, and it is the lower at the ideal bound too (T3).

Burgers (R9): a trusted kernel can hide a diverging lift. Figure 8.6 measures the lift's convergence on viscous Burgers from the data $a \sin x$, against the strength a/ν and the number of sine modes M, with $\nu = 1$. The lift's error, its exact solution against the Galerkin system's, falls by a factor of about $a/(5\nu)$ an order: at $a/\nu = 1$ from 8.7×10^{-3} at $K = 2$ to 5.9×10^{-5} at $K = 5$, and at $a/\nu = 3$ from 0.20 to 0.034, still converging. It does not depend on M: at $a/\nu = 1$ and $K = 2$ it is 8.7×10^{-3} on sixteen modes, as on four, while the worst-case ratio of Equation (6.7) grows from $\rho = 1.86$ to 16.1; the data excite the higher modes only through the nonlinearity, and at this strength those modes plausibly carry too little of the solution to move the error at the digits shown. The lift converges at ρ far above one, and its convergence follows a/ν, not ρ. At $a/\nu = 10$ it diverges: the error grows from 4.56 at $K = 2$ to 24.4 at $K = 5$, against an amplitude of 10. Its kernel stays one-dimensional and trusted at every order, the gap ratio 5.3×10^{12} at $K = 5$, and both forms prepare it, with infidelity 1.2×10^{-12} in the descriptor and 2.2×10^{-12} in the standard form by exact emulation, while the decoded field is wrong by 24.4. This is a negative finding, analysed in § 6.5: trust measures the kernel's isolation, not the lift's accuracy. The boundary lies between $a/\nu = 3$ and 10; where the per-order factor $a/(5\nu)$ reaches one, near $a/\nu = 5$, is an extrapolation from these strengths, not a measurement. As on R8, the descriptor's time axis lowers the degree: at $a/\nu = 1$, $K = 2$ and $n_t = 5$ it is 747 against the standard form's 11 764 (T2).

The answer to the third question, measured: the Carleman lift is prepared in both forms on every problem run, from R8 to Burgers, where the doubled space only projects. What a preparation certifies is the lift; whether the lift is the nonlinear solution is measured order by order, and on Burgers it fails at $a/\nu = 10$ without any sign in the residual. The reformulation's advantage survives on these lifts, through the time axis: the descriptor's degree is the lower on R8 at $n_t = 4$ and 8 and on Burgers at the point compared, and § 8.5 tests whether it lasts as the coupling grows with the modes.

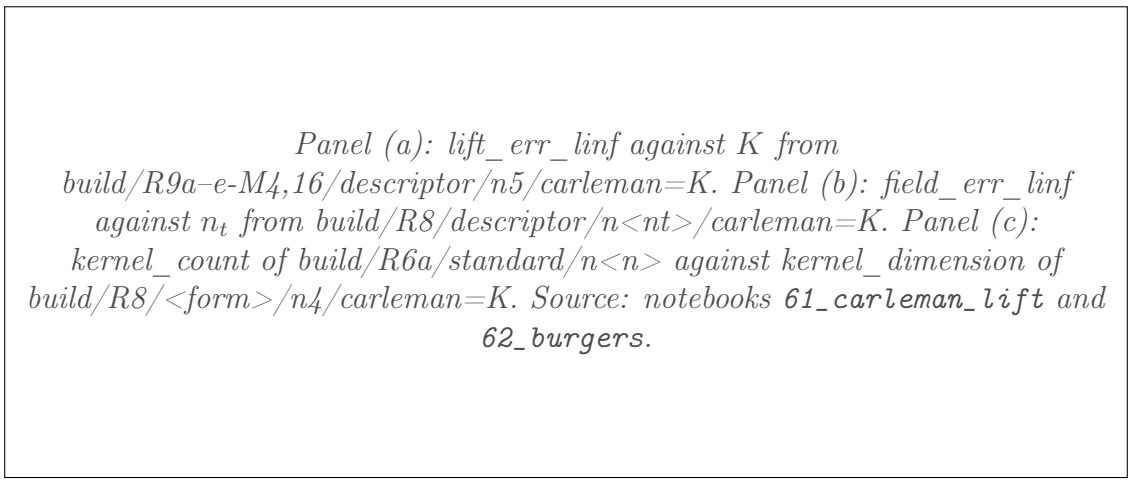


Figure 8.6: The Carleman lift's convergence follows a/ν, not ρ, and past it the kernel stays trusted while the lift diverges. (a) Viscous Burgers (R9): the lift's error against the Galerkin system against the Carleman order $K = 2$–5, at $a/\nu = 0.03$, 0.3, 1, 3 and 10, on $M = 4$ (filled) and 16 (open) sine modes; the curves for the two M coincide although ρ grows ninefold, and at $a/\nu = 10$ the error grows with K. (b) R8 ($y' = -y + y^2$, $\rho = \frac{1}{2}$): the field's error against n_t for $K = 2$–6, flat at the lift's floor from $n_t = 3$. (c) Kernel dimension: the doubled space on R6a ($n = 2$–5) against the Carleman lift on R8 ($K = 2$–6, both forms), which is one throughout. Exact classical computations from the references; every trusted preparation of these lifts meets ε by exact emulation (T2).

8.5 Two-Dimensional Navier–Stokes

Figure 8.7 carries the lift to two-dimensional incompressible flow, the vorticity equation of Equation (6.10) with $\nu = 1$, prepared on 12 to 48 Fourier modes and built classically to 80.

Validation. The Taylor–Green vortex (R10a), whose two modes do not interact, has an exact lift, and the pipeline returns it to rounding: the lift's error is 4.4×10^{-16} on twelve modes and 1.1×10^{-15} on eighty. This checks the Galerkin coefficients, the ratio rows on the stated data and the time axis against a known answer; it does not test the lift's convergence, which the vortex pair does.

The vortex pair. A counter-rotating pair of norm $\|w_0\| = a$ excites modes of several $|\mathbf{k}|$, so its triads are active. The truncation's own error against a direct simulation, about 4×10^{-3}, 4×10^{-2} and 0.2 on twelve modes at $a/\nu = 3$, 10 and 30, is a property of the Galerkin system, not of the lift. The lift's error is set by the strength, as on Burgers (Figure 8.7); on twelve modes at $a/\nu = 10$ it falls from 7.1×10^{-2} at $K = 2$ to 6.4×10^{-3} at $K = 3$. Unlike Burgers', it also grows with the modes: at $a/\nu = 10$ and $K = 2$ it is 0.36 on 48 modes against 0.071 on twelve, though it still converges there, to 0.18 at $K = 3$, at a ρ of 24. At $a/\nu = 30$ it converges on twelve modes, from 1.83 to 0.501, and diverges from 28: there the error grows from 5.28 to 6.07, and on 48 modes from 6.02 to 9.27, against $\|w_0\| = 30$, with every kernel trusted. This is the Burgers finding again, on a flow: trust does not detect it. A shorter horizon, $T = \frac{1}{8}$, restores convergence on 28 modes, from 3.65 to 2.36, but not on 48, where the error holds at 4.83, as the lift's time-dependent radius of convergence leads one to expect [74]. Every trusted preparation by exact emulation meets ε, the worst at infidelity 6.6×10^{-13}.

The time axis. At one Carleman order the two forms differ only in the time axis (§ 6.8), and the descriptor's degree is the lower in every pair estimated, on 12 to 48 modes at $K = 2$ to 4 and $n_t = 4$ and 5, as built and at the ideal bound. A qubit of n_t multiplies the standard form's degree by about four and the descriptor's by less than two: on the vortex pair at $a/\nu = 10$, twelve modes and $K = 2$, the degrees are 8604 and 1131 at $n_t = 4$, and 35 176 and 1970 at $n_t = 5$. At fixed n_t the margin narrows as the modes grow, because the coupling's part of the norm is common to both forms (§ 6.8) and the descriptor's encoding of it loosens with its matchings: on 48 modes the degrees are 11 512 against 4647, and the descriptor's looseness $\alpha_R/\|R\|_2$ has grown from 3.83 to 9.92. The margin narrows at the ideal bound too, from 1577 against 296 to 1688 against 469, so part of the narrowing is the Hamiltonian's and part the encoder's.

The tier, and the tensor layout. No circuit was simulated on Navier–Stokes. The narrowest prepared circuit, the descriptor at $n_t = 4$ on twelve modes, has 27 qubits, wider than any circuit simulated here; the probe-verified reflections at $n_t = 2$ are not prepared. The fluid results are therefore exact emulations, which equal the circuit's output (§ 7.5), and resource estimates. The tensor layout's coupling, the one native complex encoding built, keeps $2K - 1$ terms whatever the number of modes and is verified by IR to 316 modes, but prepares no state (§ 6.3); its gates per query against the matchings' are given in Table H.1.

Panel (a): vorticity snapshots, T2 against the Galerkin reference. Panel (b): degree and ideal_degree against M from stateprep/R10a–d-M<M>/<form>/n4/T3/carleman=2. Panel (c): per-order ratio of lift_err_linf against M from build/R10b–d-M<M>[-W8]/descriptor/n4/carleman=K. Source: notebook 70_navier_stokes.

Figure 8.7: Two-dimensional Navier–Stokes is prepared by exact emulation on up to 48 modes, and the descriptor's time axis lowers the degree in every pair. (a) Vorticity at $t = T$: the field decoded from the prepared state against the Galerkin reference, for the Taylor–Green vortex (R10a, exact lift) and the vortex pair at $a/\nu = 10$ (R10c), twelve modes, $K = 3$, $n_t = 4$; exact emulation (T2). (b) The minimax filter's degree against the number of modes M at $K = 2$, $n_t = 4$, standard (squares) and descriptor (circles), as built (solid) and at the ideal-encoding bound (dashed), for $a/\nu = 3$, 10 and 30; the ratio narrows as M grows. (c) The lift's per-order error ratio against M at $a/\nu = 3$, 10 and 30 over $T = 1$, and at $a/\nu = 30$ over the window $T = \frac{1}{8}$, with the line of unit ratio marked: above it the lift diverges while its kernel stays trusted. Resource estimates (T3) in (b); exact classical computations in (c). No circuit simulation (T1) was run on this problem.

The lift reaches two-dimensional incompressible flow: prepared by exact emulation on up to 48 modes, with the descriptor's time axis lowering the degree in every pair, and with the same caveat as on Burgers, now on a flow: at $a/\nu = 30$ the lift diverges from 28 modes with its kernel

trusted.

8.6 Summary of Findings

1. **The standard form's cost is its Hamiltonian's.** With Wu et al.'s encoding the filter's degree on R2a grows as about N^{10}, mostly the encoder's cost; an exact encoding removes that part, and on one-axis problems with constant coefficients the degree still grows as N^{2k}, both as built and at the ideal-encoding bound (Tables 8.1 and 8.2).

2. **The descriptor's degree advantage depends on the class of problem.** Its degree grows as about N on one-axis problems whose leading coefficient does not vanish, against N^{2k}, and as about N^2 where the coefficient vanishes; on two axes it grows faster than N^2 per axis. Per query its circuit costs more than the structured encoding's at small n and about the same at $n = 8$. *Negative:* the comparison reverses at the ideal bound on R3, whose coefficient vanishes at both ends, and on the heat equation the descriptor gains nothing at the resolutions estimated (§§ 8.2 and 8.3).

3. **The descriptor's gap erodes where a carried derivative is uncontrolled, and block rescaling restores it on one axis only.** On every one-axis problem run the rescaled gap is $\mathcal{O}(1)$ and the field block regains its weight, at no gate cost. *Negative:* on two axes the rescaled gap still falls with N (§ 8.3).

4. **Every verified circuit is its operator, and every trusted preparation meets its target.** Every composed Hamiltonian of at most 24 qubits is verified whole and every lift and coupling by its terms, the simulated circuit equals the exact emulation within 10^{-10} on every minimax record, and every trusted preparation meets ε, its a priori bound and its field budget. Wider reflections are built and counted but not independently verified (§ 8.1).

5. **Coupled systems and Carleman lifts are prepared in both forms, but a trusted kernel can hide a diverging lift.** The mass form's gap is about twice the chain's on R7, and the descriptor's time axis lowers the degree on every lift compared. *Negative:* on Burgers at $a/\nu = 10$, and on Navier–Stokes at $a/\nu = 30$ from 28 modes, the lift diverges while its kernel stays one-dimensional and trusted (§§ 8.4 and 8.5).

6. **Navier–Stokes is reached by exact emulation and resource estimate, not by circuit simulation.** Its narrowest prepared circuit is wider than any simulated; the tensor layout's coupling is verified to 316 modes and prepares nothing (§ 8.5).

Within its stated scope the central claim holds as measured: carrying the derivatives instead of eliminating them keeps the kernel and lowers the Hamiltonian's norm from $\Theta(N^8)$ to $\Theta(N^2)$ on a second-order equation, and with it the degree from about N^4 to about N on a one-axis equation whose leading coefficient does not vanish; the price is a block register and a gap the derivative blocks erode, which an a priori rescaling restores on one axis at no gate cost but

not on two. Where the leading coefficient vanishes the degree still grows as N^2, and on R3 at the ideal bound the standard form needs fewer queries. These are measurements on the problems run, between stated resolutions: the growth exponents are measured, not proved, and the descriptor's rescaled gap on one axis is measured, not derived. § 9 supplies the mechanism behind each finding and asks what the measured exponents imply for an end-to-end algorithm and for problems outside the benchmark set.

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9 Discussion

9.1 Why the Descriptor Form Wins, and What It Costs

9.2 End-to-End Complexity

9.3 Where the Construction Applies

9.4 Feasibility on Fault-Tolerant Hardware

9.5 Limitations

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10 Conclusions and Future Work

10.1 Conclusions

10.2 Future Work

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A Quantum Gate Reference

Table A.1: The gates used in this thesis.

Gate	Circuit symbol	Matrix
Hadamard		
Pauli X, Y, Z		
Rotation $R_Y(\theta)$		
Phase $P(\phi)$		
CNOT		
Toffoli		
Multi-controlled X		
Zero-state reflection $2\left\|0\right\rangle\left\langle 0\right\| - I$		
Quantum Fourier transform		

B Chebyshev and Ultraspherical Identities

This appendix records the recurrence that defines the banded factors $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ of § 5.2, their entries, the exact factorisation $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$, its connection with the ultraspherical method, and the fully populated $N = 4$ instance of the first-order descriptor operator.

B.1 The differentiation recurrence

For $f = \sum_{\ell\geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell\geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1}\, p_{\ell-1} - p_{\ell+1} = 2\ell\, a_\ell, \qquad \ell \geq 1, \tag{B.1}$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on p_0. Truncating to $N = 2^n$ coefficients and dropping the p_N term, Equation (B.1) is the matrix identity $\tilde{\mathbb{B}}_{\text{raw}}\, p = \tilde{\mathbb{D}}_{\text{raw}}\, a$ with $\tilde{\mathbb{D}}_{\text{raw}}$ a single superdiagonal, $(\tilde{\mathbb{D}}_{\text{raw}})_{\ell,\ell+1} = 2(\ell+1)$, and $\tilde{\mathbb{B}}_{\text{raw}}$ two diagonals, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell,\ell} = c_\ell$, $(\tilde{\mathbb{B}}_{\text{raw}})_{\ell,\ell+2} = -1$.

The recurrence follows from two trigonometric identities. With $x = \cos\theta$, $T_k' = k \sin k\theta / \sin\theta$, and $\sin(k+1)\theta - \sin(k-1)\theta = 2\cos k\theta \sin\theta$ gives $T'_{k+1}/(k+1) - T'_{k-1}/(k-1) = 2T_k$ for $k \geq 2$, with $T_1 = T_2'/4$ and $T_0 = T_1'$. Integrating $f' = \sum_k p_k T_k$ term by term with these, the coefficient of T_ℓ in f collects $c_{\ell-1}p_{\ell-1}/(2\ell)$ from $T_{\ell-1}$ and $-p_{\ell+1}/(2\ell)$ from $T_{\ell+1}$, which is Equation (B.1); the constant of integration affects a_0 alone, which no row of the recurrence reads. The truncation is exact: the derivative of a polynomial of degree below N has degree at most $N-2$, so the dropped coefficients p_N and p_{N+1} vanish, and the last row, $p_{N-1} = 0$, holds.

B.2 The normalised basis

The construction of § 5.2 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ $(\ell \geq 1)$, orthonormal under the discrete N-point Chebyshev inner product $\langle T_k, T_\ell\rangle_N = \frac{N}{2}\, c_\ell\, \delta_{k\ell}$ [88]:

since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \operatorname{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{\mathbb{B}} = S\,\tilde{\mathbb{B}}_{\text{raw}}\,S^{-1}$ and $\tilde{\mathbb{D}} = S\,\tilde{\mathbb{D}}_{\text{raw}}\,S^{-1}$, that is

$$\tilde{\mathbb{D}}_{\ell,\ell+1} = 2(\ell+1)\sqrt{c_\ell/c_{\ell+1}}, \qquad \tilde{\mathbb{B}}_{\ell,\ell} = c_\ell, \qquad \tilde{\mathbb{B}}_{\ell,\ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \tag{B.2}$$

so $\tilde{\mathbb{D}}$ has the single non-zero diagonal ($2\sqrt{2}$, 4, 6, 8, …) and $\tilde{\mathbb{B}}$ has main diagonal (2, 1, 1, …) and second superdiagonal ($-\sqrt{2}$, -1, -1, …). Both hold $\mathcal{O}(N)$ non-zero entries at any N. The dense Chebyshev derivative matrix is their exact quotient,

$$\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}, \tag{B.3}$$

an algebraic identity rather than a numerical approximation. $\tilde{\mathbb{B}}$ is upper triangular with the non-zero diagonal $(2, 1, \ldots, 1)$, so it is invertible. For any coefficients a the true derivative's coefficients satisfy the truncated rows of Equation (B.1), as shown above, and they are $\mathbb{G}a$; conjugating by S gives $\tilde{\mathbb{B}}\mathbb{G} = \tilde{\mathbb{D}}$, so the equality holds exactly at every N.

Proposition 6 (The inverse of the mass factor)**.** $\tilde{\mathbb{B}}^{-1}$ *is upper triangular and supported where the column index minus the row index is even. Its entries there are* 1 *in every row* $k \geq 1$*; in row* 0 *they are* $\frac{1}{2}$ *at column* 0 *and* $1/\sqrt{2}$ *at every even column* $q \geq 2$*.*

Proof. Away from row 0, $\tilde{\mathbb{B}} = I - P$ with P the shift by two, so on each parity class its inverse is $\sum_m P^m$, a triangle of ones. Row 0 of $\tilde{\mathbb{B}}x = e_q$ reads $2x_0 - \sqrt{2}x_2 = \delta_{0q}$, which gives $x_0 = \frac{1}{2}$ for $q = 0$ and, with $x_2 = 1$, $x_0 = 1/\sqrt{2}$ for even $q \geq 2$. □

Multiplying out $\tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$ with this inverse returns the entries of Equation (4.2): $\tilde{\mathbb{D}}$'s single diagonal forces the inner index to be $p-1$, the parity support of $\tilde{\mathbb{B}}^{-1}$ makes $p-k$ odd, and the entry is $2p$ times one, or times $1/\sqrt{2}$ in row 0. The density of $\mathbb{G}$ is the inverse of the banded $\tilde{\mathbb{B}}$, which is what the descriptor never forms. The triangle of ones has norm $\Theta(N)$ and $\tilde{\mathbb{D}}$ has norm $2(N-1)$; their product has the $\Theta(N^2)$ norm of $\mathbb{G}$.

B.3 The scale factor and the feature vector's norm

The basis functions $\varphi_0 = T_0/\sqrt{N}$ and $\varphi_k = T_k/\sqrt{N/2}$ $(1 \leq k < N)$ are orthonormal under the discrete inner product on the N Chebyshev nodes. Under the continuous Chebyshev weight $w(x) = (1-x^2)^{-1/2}$ they are orthogonal with a common norm: with $x = \cos\theta$, $\int_{-1}^{1} T_j T_k w\,dx = \int_0^\pi \cos j\theta \cos k\theta\,d\theta$, which is π for $j = k = 0$, $\pi/2$ for $j = k \geq 1$ and zero otherwise, so $\int \varphi_j\varphi_k w\,dx = (\pi/N)\,\delta_{jk}$ for every $j, k < N$. For $f_N = \sum_k b_k\varphi_k$ Parseval's identity is therefore $\int_{-1}^{1} f_N^2 w\,dx = (\pi/N)\|b\|^2$, which is Equation (4.1).

Proposition 7 (The feature vector's norm)**.** *With* $x = \cos\theta$*,*

$$\|\tau(x)\|^2 = 1 + \frac{1}{N}\Big(\frac{\sin(2N-1)\theta}{2\sin\theta} - \frac{1}{2}\Big), \qquad \|\tau(\pm 1)\|^2 = \frac{2N-1}{N}, \tag{B.4}$$

and $\|\tau(x)\|^2 \le (2N-1)/N < 2$ *on* $[-1,1]$.

Proof. Since $2T_k^2 = 1 + \cos 2k\theta$,

$$\|\tau(x)\|^2 = \frac{1}{N}\Big(T_0^2 + 2\sum_{k=1}^{N-1} T_k^2\Big) = \frac{1}{N}\Big(N + \sum_{k=1}^{N-1} \cos 2k\theta\Big).$$

The sum is the Dirichlet kernel $\sin((2N-1)\theta)/(2\sin\theta) - \frac{1}{2}$, at most $N-1$, which it attains at $\theta = 0$ and π. □

The norm oscillates about one and is largest at the ends, so a constraint row at an end of the interval weighs most. That weight is part of Wu et al.'s construction and is kept (§ 4.1). Normalising $\tau(x)$ point by point would also break the consistency of the collapse rows: two collapses $\mathbb{B}(x_s)/f(x_s)$ and $\mathbb{B}(x'_s)/f(x'_s)$ map the solution to the same constant, because $\mathbb{B}(x)b = f_N(x)\sqrt{N}\,|0\rangle$, whereas with $\tau(x)/\|\tau(x)\|$ they would map it to constants differing by the factor $\|\tau(x'_s)\|/\|\tau(x_s)\|$.

B.4 The structure and norm of the derivative matrix

Proposition 8 (Structure and norm of $\mathbb{G}$)**.** *For* $n \ge 2$, $\mathbb{G}$ *of Equation* (4.2) *is strictly upper triangular and nilpotent;* $\ker \mathbb{G}^j$ *is the space of polynomials of degree below* j*; it has* $N^2/4$ *non-zero entries; it is orthogonally similar to the matrix that maps the values of a polynomial of degree below* N *at the Chebyshev nodes to the values of its derivative there; and*

$$\frac{N^2}{4\sqrt{2}} \;\le\; \|\mathbb{G}\|_2 \;\le\; \frac{N^2}{\sqrt{2}}. \tag{B.5}$$

Proof. The entries are non-zero only for $p > k$, so $\mathbb{G}$ is strictly upper triangular and $\mathbb{G}^N = 0$. $\mathbb{G}$ represents differentiation exactly on polynomials of degree below N, so $\mathbb{G}^j b = 0$ exactly when the j-th derivative of the represented polynomial vanishes, that is when its degree is below j. Column p has a non-zero entry in each row $k < p$ of the other parity, $\lceil p/2 \rceil$ of them, and $\sum_{p<N} \lceil p/2 \rceil = N^2/4$.

For the similarity, let $\Phi_{jk} = \varphi_k(\bar{x}_j)$ at the nodes $\bar{x}_j$. The discrete orthonormality of the φ_k says that Φ is orthogonal, and Φb is the vector of nodal values of the polynomial with coefficients b, so $\Phi\mathbb{G}\Phi^\top$ maps nodal values to the nodal values of the derivative.

For the upper bound, $\|\mathbb{G}\|_2 \le \|\mathbb{G}\|_F$, and each of the $\lceil p/2 \rceil$ entries of column p is at most $2p$ in magnitude, so

$$\|\mathbb{G}\|_F^2 \le \sum_{p=0}^{N-1} 4p^2\Big(\frac{p}{2} + \frac{1}{2}\Big) = \frac{(N-1)^2N^2}{2} + \frac{(N-1)N(2N-1)}{3} = \frac{N^4}{2} - \frac{N^3}{3} - \frac{N^2}{2} + \frac{N}{3} \le \frac{N^4}{2}.$$

For the lower bound, let u be the unit vector uniform over the $N/4$ even rows $k < N/2$ and v

the unit vector uniform over the $N/4$ odd columns $p > N/2$. Every pair (k, p) has $p > k$ with $p - k$ odd, and its entry is $2p > N$ for $k \geq 2$ and $\sqrt{2}p > N/\sqrt{2}$ for $k = 0$, so

$$\|\mathbb{G}\|_2 \geq u^\top \mathbb{G} v \geq \frac{4}{N} \cdot \left(\frac{N}{4}\right)^2 \cdot \frac{N}{\sqrt{2}} = \frac{N^2}{4\sqrt{2}}. \qquad \square$$

The bounds give $\|\mathbb{G}\|_2 = \Theta(N^2)$. The exact limit, $\|\mathbb{G}\|_2/N^2 \to 1/(2j_{-3/4,1}) \approx 0.4724$, with $j_{-3/4,1}$ the first positive zero of the Bessel function $J_{-3/4}$, and the growth of every power, $\|\mathbb{G}^k\|_2 = \Theta(N^{2k})$, are proved in Appendix E.4.

B.5 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [36]. Stating the correspondence precisely is what fixes what this thesis adds to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis T_k but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1 - x^2)^{\lambda - 1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{\mathrm{d}T_k}{\mathrm{d}x} = k\, C_{k-1}^{(1)}, \qquad k \geq 1 \tag{B.6}$$

[36, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of u' in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because u''s coefficients now live in a different basis to u's own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ $(k = 0)$, $\frac{1}{2}$ $(k \geq 1)$, and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [36, Eq. 2.8]. Comparing entries, $\tilde{\mathbb{D}}_{\text{raw}} = 2\mathcal{D}_0$ and $\tilde{\mathbb{B}}_{\text{raw}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{\mathbb{B}}_{\text{raw}}\, p = \tilde{\mathbb{D}}_{\text{raw}}\, a$ of Equation (B.1) is literally $\mathcal{S}_0\, p = \mathcal{D}_0\, a$, doubled by the convention that folds the weight of T_0 into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow the framing of § 5.2 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that p's own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for p via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \geq 1$ in [36] raises the basis one level per derivative: a λ-th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $\mathrm{d}C_k^{(\lambda)}/\mathrm{d}x = 2\lambda\, C_{k-1}^{(\lambda+1)}$ ($k \geq 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \to C^{(\lambda+1)}$, still two diagonals [36, Eqs. 3.2–3.3]. To assemble one equation of order k, every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1} \cdots \mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$, giving one almost-banded operator $\mathrm{L} = \mathcal{M}_k[a^k]\mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1} \cdots \mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$ that is solved once for the coefficients of $u^{(k)}$ [36, §3.2] – so a k-th order equation needs k distinct conversion operators $\mathcal{S}_0, \ldots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of § 5.3 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $\hat{f}^{(j)}$ of w, and reuses the single $\lambda = 0$ pair $(\tilde{\mathbb{B}}, \tilde{\mathbb{D}})$ between every consecutive pair of blocks (Equation (5.5)). No elimination is performed, and no operator beyond $\tilde{\mathbb{B}}$, $\tilde{\mathbb{D}}$ and the multiplication operators of Appendix G.2 is required at any order k – the price is the larger state w that carries every intermediate derivative explicitly instead of eliminating it (Appendix G.2). What this thesis adds to [36] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [30], whose block structure is exactly what § 5.7 block-encodes.

For a first-order equation the two coincide. The descriptor's mass form multiplies the standard rows by $\tilde{\mathbb{B}}$ (Equation (5.7)), and for $f' = af$ with a a number, $\tilde{\mathbb{B}}(\mathbb{G} - aI) = \tilde{\mathbb{D}} - a\tilde{\mathbb{B}}$, which in the raw basis is $2(\mathcal{D}_0 - a\mathcal{S}_0)$: the ultraspherical method's own operator in the basis $C^{(1)}$, on N unknowns with no derivative block.

A reduction the descriptor does not build. For $k \geq 2$ the same left factor exists. In the raw basis $\mathcal{S}_1\mathcal{S}_0\mathbb{G}^2 = \mathcal{D}_1\mathcal{D}_0$ exactly, so $\mathcal{S}_{k-1} \cdots \mathcal{S}_0 \, A_{\text{std}}$ is the ultraspherical method's operator for a constant-coefficient equation: banded, with a bandwidth independent of N, on the N coefficients of f, and, since the conversions are invertible, with the standard form's kernel, as in Theorem 1. It would remove the chain's block register and its recurrence and padding rows. It would also change three things this thesis measures: the residual's rows are weighted by the conversions, so its least-squares ground state is that of a differently weighted problem; the conversions' entries would have to be encoded; and the gap mechanism of § 5.5 does not carry over without a measurement. This reduction was not built into the pipeline, and the thesis's descriptor is not compared against it.

B.6 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (B.2) gives

$$\tilde{\mathbb{B}} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \tilde{\mathbb{D}} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbb{G} = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad \text{(B.7)}$$

The top-right entry $3\sqrt{2}$ of $\mathbb{G}$ is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row ℓ of $\mathbb{G}$ is non-zero at columns $\ell + 1, \ell + 3, \ldots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{\mathbb{B}}$ or $\tilde{\mathbb{D}}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ (R1), the descriptor runs the mass form of Equation (5.7), on the field's own N coefficients and with no derivative block:

$$\tilde{\mathbb{D}} + 2\tilde{\mathbb{B}} = \begin{pmatrix} 4 & 2\sqrt{2} & -2\sqrt{2} & 0 \\ 0 & 2 & 4 & -2 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \tilde{\mathbb{B}}\,(\mathbb{G} + 2I), \qquad \mathbb{G} + 2I = \begin{pmatrix} 2 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \quad \text{(B.8)}$$

The standard form's equation block $\mathbb{G} + 2I$ carries the fill-in $3\sqrt{2}$; the mass form's has none, and every row touches at most three coefficients. Its kernel is that of $\mathbb{G} + 2I$, because $\tilde{\mathbb{B}}$ is invertible.

The same equation can be carried as a chain, with its derivative as a second block, $w = (\hat{f}^{(0)};\ \hat{f}^{(1)})$, the recurrence $-\tilde{\mathbb{D}}\,\hat{f}^{(0)} + \tilde{\mathbb{B}}\,\hat{f}^{(1)} = 0$ and the equation row $2\hat{f}^{(0)} + \hat{f}^{(1)} = 0$:

$$A_{\text{desc}} = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array}\right]. \quad \text{(B.9)}$$

The top four rows are the recurrence and the bottom four the equation. This chain doubles the state for a block the equation already determines, which is why first-order equations take the mass form (§ 5.3); it is kept only as the mass form's ablation, whose kernel and decoded field agree with the mass form's.

C The Standard Form of Wu et al. in Full

This appendix gives the $N = 4$ instance of the standard-form residual of § 4.2 for a constant-coefficient equation, the entries of the lifts $\mathbb{M}_{x^p}$ that the same section uses for a variable coefficient, and the constructions and proofs that § 4.3 points to.

Why a condition for all x is a finite linear system. A condition required at every $x \in [-1, 1]$, such as the differential equation, reaches the form $\langle\tau(x)|\,C\,|\psi\rangle = 0$ for all x. The components of $\langle\tau(x)|$ are linearly independent functions of x, so $\langle\tau(x)|\,\mathbf{v}$ vanishes for all x only if $\mathbf{v} = 0$. Taking $\mathbf{v} = C\,|\psi\rangle$, the condition for all x is therefore equivalent to the finite system $C\,|\psi\rangle = 0$, which no longer depends on x. This is the step that §§ 3.2 and 4.2 take without proof.

The standard-form residual at $N = 4$. For Wu et al.'s stiff pilot equation [30], $f'' + 5f' + 400f = 0$, the equation block of Equation (4.3) is $A_{\text{std}} = \mathbb{G}^2 + 5\mathbb{G} + 400I$. At $N = 4$, with the entries of Equation (4.2),

$$A_{\text{std}} = \begin{pmatrix} 400 & 5\sqrt{2} & 4\sqrt{2} & 15\sqrt{2} \\ 0 & 400 & 20 & 24 \\ 0 & 0 & 400 & 30 \\ 0 & 0 & 0 & 400 \end{pmatrix}. \tag{C.1}$$

It is upper triangular, as every polynomial in the strictly upper-triangular $\mathbb{G}$ is. At this size the largest off-diagonal entry, 30, is about one order of magnitude below the diagonal; the entries of $\mathbb{G}^2$ grow as N^4, so at larger N the derivative terms dominate the constant one.

The lifting operator $\mathbb{M}_{x^p}$. $\mathbb{M}_{x^p}$ is exact and *rectangular*: multiplying a degree-$(N-1)$ model by x^p can raise its degree by as much as p, so the result is represented one register wider, $2N \times N$, with no truncation. At $n = 1$ ($N = 2$, so $p \in \{1, 2\}$),

$$\mathbb{M}_{x^1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}, \qquad \mathbb{M}_{x^2} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{3}{2\sqrt{2}} \\ \frac{1}{2} & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}, \tag{C.2}$$

which reproduce Wu et al.'s printed matrices. The general entry follows from the product-to-sum identity conjugated into the normalised bases of $|\tau(x)\rangle_n$ and $|\tau(x)\rangle_{n+1}$ (Appendix C.7).

C.1 The faithful rebuild's conventions

C.2 The standard form on several axes

Stub: write this subsection out in full (derivation of the register count, a worked three-variable

§ 3.2 writes the two-variable case. For ω variables $\mathbf{x} = (x_1, \ldots, x_\omega)$, each carried on its own n-qubit Chebyshev axis u, the state lives on ωn qubits and the feature vector is $|\tau(\mathbf{x})\rangle = |\tau(x_1)\rangle_n \otimes \cdots \otimes |\tau(x_\omega)\rangle_n$, so that $f_q(\mathbf{x}) = \langle \tau(\mathbf{x}) | \psi \rangle$. The derivative along axis u is $\mathbb{G}_u = I^{\otimes(u-1)} \otimes \mathbb{G} \otimes I^{\otimes(\omega-u)}$; these act on different factors and commute, and a derivative of multi-index $\boldsymbol{\chi} = (\chi_1, \ldots, \chi_\omega)$ is $\partial^{\boldsymbol{\chi}} \mapsto \mathbb{G}^{\boldsymbol{\chi}} = \prod_u \mathbb{G}_u^{\chi_u} = \bigotimes_u \mathbb{G}^{\chi_u}$.

Each term t of the equation has a constant c_t, a multi-index $\boldsymbol{\chi}_t$ and a coefficient m_t that is a polynomial in a single variable x_{u_t} (the class the code builds; a coefficient that depends on several variables is not built). An axis is *lifted* when it carries a non-constant coefficient or when the equation has a source, in which case every axis is lifted. The term acts on axis u as $\mathbb{M}_{m_t} \mathbb{G}^{\chi_{t,u}}$ if $u = u_t$, as $\mathbb{M}_1 \mathbb{G}^{\chi_{t,u}}$ on a lifted axis without a coefficient, and as $\mathbb{G}^{\chi_{t,u}}$ on an unlifted axis, so that

$$A_{\text{std}} = \sum_t c_t \bigotimes_{u=1}^{\omega} \mathbb{M}_t^{(u)} \, \mathbb{G}^{\chi_{t,u}} \;-\; \mathbb{M}_r \, \mathbb{D}^{(0)}(\mathbf{x}_s), \tag{C.3}$$

with $\mathbb{M}_t^{(u)}$ as just described. The source term is the rank-one operator $\mathbb{M}_r\mathbb{D}^{(0)}(\mathbf{x}_s) = |\hat{r}\rangle \langle\tau(\mathbf{x}_s)| / y_s$, with $|\hat{r}\rangle$ the coefficients of r on the lifted register (the Kronecker product of the one-dimensional coefficient vectors where r is a product of single-variable polynomials). The equation block's output register has $\omega n + \omega_{\text{lift}}$ qubits, ω_{lift} being the number of lifted axes. For $\omega = 1$, Equation (C.3) is Equation (4.3).

An invariant constraint $\partial_u^j f = 0$ on the face $x_u = x_z$ is the row block $I^{\otimes(u-1)} \otimes \mathbb{B}_n(x_z)\mathbb{G}^j \otimes I^{\otimes(\omega-u)}$, or $\mathbb{B}_{n+1}(x_z)\mathbb{M}_1\mathbb{G}^j$ on its own axis if that axis is lifted. The identity factors are on n qubits whether or not their axes are lifted, so only the equation block, not each constraint, acts into the full lifted register. The regular datum at a point $\mathbf{x}_s$ is the rank-one row $\mathbb{D}^{(0)}(\mathbf{x}_s) = \bigotimes_u \mathbb{B}_n(x_{s,u})/y_s = 2^{\omega n/2} |0\rangle \langle\tau(\mathbf{x}_s)| / y_s$. Each $\mathbb{G}_u$ has the norm $\Theta(N^2)$ of $\mathbb{G}$.

C.3 Datum-slice rows

An inhomogeneous initial or boundary function g on the slice $x_a = x^*$ of a problem on several Chebyshev axes is imposed by the rows of Equation (4.4),

$$\mathbb{S}_a(x^*) - \frac{\hat{g}\ \langle\tau(x_s)|}{f(x_s)}, \qquad \mathbb{S}_a(x^*) = I \otimes \cdots \otimes \mathbb{B}(x^*) \otimes \cdots \otimes I \quad (\mathbb{B} \text{ on axis } a),$$

with $\hat{g}$ the coefficients of g on the other axes broadcast as a constant along axis a: its mode 0 on that axis, $\sqrt{N_a}$ times the coefficients of g, as $\mathbb{B}$ emits. $\langle\tau(x_s)|$ is the feature row at a regular point x_s of the full domain, where the value $f(x_s)$ is known.

Proposition 9 (Datum slices)**.** *On cf, with f the solution and c any scalar, the rows return $c\left(\hat{f}|_{x_a=x^*} - \hat{g}\right)$, with $\hat{f}|_{x_a=x^*}$ the broadcast slice of f. They are therefore homogeneous in the scale and vanish on the solution and its multiples.*

Proof. $\mathbb{S}_a(x^*)$ contracts axis a with $\tau(x^*)$, which gives the slice function's coefficients on the other axes times $\sqrt{N_a}\,|0\rangle$ on axis a: applied to cf, the broadcast slice $c\hat{f}|_{x_a=x^*}$. The second term applied to cf is $\hat{g}\, c\, f(x_s)/f(x_s) = c\,\hat{g}$. If $f|_{x_a=x^*} = g$ the difference is zero. □

The block is square on $\prod_a N_a$ coefficients but non-zero only on the $\prod_{b\neq a} N_b$ rows with mode 0 on axis a, which carry the slice; the rest are dropped from R without changing $R^\dagger R$. On one axis the slice is a point, and the rows reduce to the single row $\mathbb{B}(x^*) - y^*\mathbb{D}^{(0)}(x_s)$ of a point datum $f(x^*) = y^*$, or to $\mathbb{B}(x^*)\mathbb{G}^j - y^*\mathbb{D}^{(0)}(x_s)$ for a datum on a derivative (Definition 2). The rows are built from the problem's statement, the function g, the slice and one regular value, so they serve both forms and every resolution; whether the kernel is then one-dimensional is computed (§ 4.3).

Why the heat data sit at $t = -1$. Wu et al.'s heat problem, $\partial_t f = \frac{1}{25}\partial_x^2 f$ with $f(t, \pm 1) = 0$, states its initial function $\sin(2\pi x)$ at $t = 0$, inside the Chebyshev time domain $[-1, 1]$. Imposed there, it fixes the solution on $[-1, 0]$ by a backward heat problem: going back a time s, a

mode $\sin\frac{\ell\pi(x+1)}{2}$ of the data is multiplied by $\exp\bigl(\frac{1}{25}(\ell\pi/2)^2 s\bigr)$, a factor unbounded in ℓ, so the problem is ill posed in Hadamard's sense. The standard form states the problem forward from $t=-1$ instead, with the initial function $e^{4\pi^2/25}\sin(2\pi x)$, Wu et al.'s solution $e^{-4\pi^2 t/25}\sin(2\pi x)$ evaluated there. The solution is the same; the data are set up from it, as R4's constraint points are.

C.4 The substitution for R3c and R3d

Wu et al.'s R3c and R3d are the associated Legendre equation with $m=1$, multiplied through by $1-x^2$,

$$(1-x^2)^2 f'' - 2x(1-x^2)f' + \bigl(l(l+1)(1-x^2)-1\bigr)f = 0,$$

with $l=5$ and $l=6$. Its leading coefficient has a double zero at each end, and its solution $P_l^1 = -\sqrt{1-x^2}\,P_l'$ has a branch point there, so its Chebyshev coefficients decay only algebraically: $\sqrt{1-x^2} = |\sin\theta|$ has cosine coefficients of order k^{-2}. At the printed $n=5$ neither problem has a trusted ground state, and the error on R3c is 8.7×10^{-1}.

Proposition 10 (The substitution)**.** *With $f=\sqrt{1-x^2}\,v$ the equation becomes*

$$(1-x^2)\,v'' - 4x\,v' + \bigl(l(l+1)-2\bigr)v = 0, \tag{C.4}$$

whose leading coefficient has simple zeros at ±1 and whose solution $v=-P_l'$ is a polynomial of degree $l-1$.

Proof. Write $s=\sqrt{1-x^2}$, so that $s'=-x/s$ and $s''=-1/s^3$. Then $f'=sv'-xv/s$ and $f''=sv''-2xv'/s-v/s^3$. Substituting, $(1-x^2)^2f''=s^5v''-2xs^3v'-sv$, $-2x(1-x^2)f'=-2xs^3v'+2x^2sv$ and $\bigl(l(l+1)s^2-1\bigr)sv=l(l+1)s^3v-sv$. The sum is $s^5v''-4xs^3v'+\bigl(l(l+1)s^3-2s(1-x^2)\bigr)v = s^3\bigl[(1-x^2)v''-4xv'+(l(l+1)-2)v\bigr]$, and $s^3>0$ on $(-1,1)$. The solution is $v=f/s=-P_l'$. □

The data carry through: $f(\pm1)=0$ holds for every v and is dropped; R3c's $f'(0)=0$ is $v'(0)=0$, since $s(0)=1$ and $s'(0)=0$; R3d's $f(0)=0$ is $v(0)=0$; and the regular datum becomes $v(x_s)=f(x_s)/\sqrt{1-x_s^2}$, at $x_s=\frac{1}{2}$. The coefficients of Equation (C.4) are polynomials of degree at most two, inside the lift's range, and v is represented exactly once $N>l-1$, so the field error falls to rounding at the first such n (§ 4.3). The substitution removes this double zero because the branch point is a known factor of the solution; it is not a remedy for a double zero in general. The simple zeros it leaves are what the descriptor's block rescaling can treat (§ 5.6).

C.5 The uniqueness count for an ordinary equation

Proposition 11 (Uniqueness count)**.** *Let $\sum_{j=0}^{k} a_j f^{(j)} = 0$ on $[-1,1]$ with smooth coefficients and a_k nowhere zero, and let $c_1,\dots,c_m$ be invariant constraints (each sets a derivative of f at*

a point to zero). The solutions that satisfy all of them form a space of dimension $k - r$, where r is the rank of the c_i restricted to the k-dimensional solution space S. The ground state of the constraint energy is unique up to scale exactly when $r = k - 1$.

Proof. Since a_k has no zero, the equation is a regular linear system of order k and S has dimension k. The map $f \mapsto (c_i(f))_{i \le m}$ is linear on S with rank r, so its kernel, the solutions that satisfy every constraint, has dimension $k - r$. A ground state of zero energy is a nonzero element of that kernel, unique up to scale exactly when the kernel is one-dimensional. For $r = k$ the kernel is $\{0\}$ and no nonzero state has zero energy. For $r < k - 1$ it has dimension at least two. □

The rank is that of the constraints restricted to S, not of the functionals, so $k - 1$ constraints chosen without regard to the equation can fail. The regular datum fixes the scale of the selected solution and requires that it not vanish at the regular point. The proposition concerns the continuous problem: the assembled residual follows the count only as far as the truncation resolves S, and the dimension and gap of its kernel are computed, not inferred (§ 4.3). For an inhomogeneous equation, or one whose leading coefficient vanishes (R3a), no count is claimed here and uniqueness is judged by the computed kernel.

The Laplace invariant set. The modes $\sin \frac{\ell\pi(x+1)}{2} \sinh \frac{\ell\pi(y+1)}{2}$, $\ell \ge 1$, all satisfy the equation and its three homogeneous sides, and only $\ell = 1$ matches the top-side function, which the printed set does not impose. This is the continuous kernel that the truncation resolves as n grows. At the printed resolution the truncation hides the other modes and the printed set leaves a trusted one-dimensional kernel; as n grows it admits them, and at $n = (5, 5)$ the computed near-kernel has dimension 8. The datum slices of Appendix C.3 leave a one-dimensional kernel that is trusted at every resolution computed from $n = (3, 3)$ to $(6, 6)$, with gap ratio 1.7×10^9 at the last.

C.6 Wu et al.'s printed matrices at $n = 2$

C.7 Lifts and folds

For $a(x) = \sum_m a_m T_m(x)$ the identity $T_m T_k = \frac{1}{2}\big(T_{m+k} + T_{|m-k|}\big)$ gives the Chebyshev coefficients of $a\,T_k$, $X_{jk} = \frac{1}{2}\sum_m a_m\big([\,j = m + k\,] + [\,j = |m - k|\,]\big)$. In the normalised bases of n input and $n + 1$ output qubits, with σ_k the normalisation of T_k on each register ($\varphi_k = T_k/\sigma_k$), the lift is $(\mathbb{M}_a)_{jk} = (\sigma_j^{\text{out}}/\sigma_k^{\text{in}})\, X_{jk}$; a monomial coefficient is converted to its Chebyshev series first.

Proposition 12 (The lift)**.** $\mathbb{M}_a \in \mathbb{R}^{2N \times N}$ *represents the product* $a\,f_N$ *without truncation for every* f_N *of degree below* N *if and only if* $\deg a \le N$*. A constant enters as* $a_0 \mathbb{M}_1 = a_0 \sqrt{2}\,[\,I;\,0\,]$*, and* $\|\mathbb{M}_a\|_2 \le \sqrt{2}\,\sup_{[-1,1]} |a|$.

Proof. The product has degree at most $N - 1 + \deg a$, and $2N$ coefficients hold degrees up to $2N - 1$, so $\deg a \le N$ suffices; if $\deg a > N$, a series of degree $N - 1$ gives a product of degree

at least $2N$, which does not fit. Each normalisation σ_k on $n+1$ qubits is $\sqrt{2}$ times that on n, since $\varphi_k = T_k/\sqrt{N/2}$ on n qubits and $T_k/\sqrt{N}$ on $n+1$ for $k \geq 1$, and likewise for T_0, so the coefficients of a function are multiplied by $\sqrt{2}$ on the wider register, which gives $\mathbb{M}_1$. For the norm, Parseval's identity of Appendix B.3 on $n+1$ qubits, where the product is exact, gives $\|\mathbb{M}_a b\|^2 = (2N/\pi)\int a^2 f_N^2 w\,dx \leq 2\sup|a|^2\,(N/\pi)\int f_N^2 w\,dx = 2\sup|a|^2\,\|b\|^2$. □

The lift of x has the non-zero entries $(\mathbb{M}_x)_{10} = (\mathbb{M}_x)_{01} = 1$ and $(\mathbb{M}_x)_{k\pm1,k} = 1/\sqrt{2}$ otherwise, from $xT_0 = T_1$ and $xT_k = \frac{1}{2}(T_{k+1}+T_{k-1})$, which is Equation (C.2) at $n = 1$. On a circuit the lift of x is a combination of truncated shifts, built by the post-selected subtraction of Appendix E.2, and a polynomial coefficient is the sum of the powers of that block weighted by its monomial coefficients, so its subnormalisation is $\sum_p |a_p|\,\alpha^p_{\mathbb{M}_x}$. For a coefficient that vanishes where the derivatives are largest, such as the $(1-x^2)$ of R3a, this sum far exceeds what the product $\mathbb{M}_a\mathbb{G}^2$ needs, which is the looseness of the standard form's encoding on R3 (§ 4.6). The folds $\mathbb{n}_1$ and $\mathbb{n}_x$ of the doubled space follow from the same product-to-sum identity applied to a product of two series (Equation (3.7)); their circuits and norms are given in Appendix H.2.

D The Published Circuits as Built

§ 3.3 describes, at the level a reader needs to follow the argument, the circuit Wu et al. use to load $\mathbb{G}$'s column-linear entries as a rotation amplitude [30]. This appendix gives the mechanism in full, together with a rigorous bound on its entrywise error that this thesis derives and verifies – the source states only that the error is "negligible" and decreases "exponentially" with n, without a closed form.

D.1 The circuits as built

The constraint rows (Figs 2 and 8). $\mathbb{B}(x)$ and $\mathbb{D}^{(0)}(x_s)$ are block-encoded by the inverse feature map $U_\tau^\dagger(x)$ followed by $\hat{S}_{n+1} = 2\,|0\rangle\langle 0| - I$ controlled inside a Hadamard sandwich on one ancilla, a two-term LCU on $n+2$ qubits. The encoded block is $\mathbb{B}(x)/\sqrt{2^{n+1}}$, and $\mathbb{D}^{(0)}(x_s)/\big(\sqrt{2^{n+1}}/f(x_s)\big)$ for the collapse row. The global phase of -1 in $\hat{S}_{n+1}$ is a relative phase once the circuit is controlled, and cannot be dropped.

The derivative circuit (Fig. 9). The registers are, in order, a flag qubit acted on by $R_Y(\omega_c)$ with $\omega_c = 2\arccos\sqrt{1/3}$, an X ancilla, an offset register of $m = n$ qubits sandwiched between Hadamards, a value ancilla, and the n-qubit system. Two conditioned subroutines assign the entries on different bases. In $\hat{U}_R$ the offset $|i\rangle$ selects the $(2i+1)$th superdiagonal and the system $|j\rangle$ the jth column; it consists of n singly-controlled $R_Y(\theta_k)$ gates on the value ancilla, controlled by the bits of $|j\rangle$, followed by n cyclic shifts $\hat{R}_n$. In $\hat{U}_C$ the offset selects the $(i+1)$th row; it consists of n multi-controlled $R_Y(\varphi_k)$ gates, one shift $\hat{L}_n$ and $2n$ swaps. With $\mu = 2/\|\mathbb{G}\|_S$ the angles are $\theta_k = 2\arcsin(-2^{k-1}\mu)$ and $\varphi_k = 2\arcsin[-2^{k-1}\mu(1/\sqrt{2}-1)]$. The circuit uses $m+n+3 = 2n+3$ qubits, its gate count is linear in n before the multi-controlled gates are

decomposed and $O(n^3)$ after an $O(n^2)$ decomposition of each, and $\mathbb{G}$ is the top-left $2^n \times 2^n$ block of the unitary multiplied by $(2^{n-1} + 2^n)\|\mathbb{G}\|_S$. These are Wu et al.'s statements; their tests follow.

Costing the rebuild. The rebuild restricts the offsets to odd values. Every non-zero entry of $\mathbb{G}$ has $p-k$ odd, so an even offset, $d = 0$ above all, would load a value onto the diagonal and onto entries that must vanish; the uniform superposition over every offset drawn in Fig. 9 does not exclude them. With the ladder coefficient $\mu = 2/\|\mathbb{G}\|_S$ and the weight $2/N$ that the odd-offset superposition gives each entry, the rebuilt circuit block-encodes $\mathbb{G}$ at $\alpha = N/\mu = N\|\mathbb{G}\|_S/2$.

Lemma 1 (Wu et al.'s norm of $\mathbb{G}$)**.** *For $n \geq 3$, $\|\mathbb{G}\|_S = \|\mathbb{G}^\top\mathbb{G}\|_1 = N(N-1)^2(N+1)/3$.*

Proof. $(\mathbb{G}^\top\mathbb{G})_{pq} = \sum_k \mathbb{G}_{kp}\mathbb{G}_{kq}$ is non-zero only for $p \equiv q \pmod 2$, the sum running over the $k < \min(p, q)$ of the other parity with weight $4pq/c_k$. For odd $p \leq q$ these k are $0, 2, \ldots, p-1$, and $\sum 1/c_k = \frac{1}{2} + \frac{p-1}{2} = p/2$; for even $p \leq q$ they are $1, 3, \ldots, p-1$, again $p/2$. So $(\mathbb{G}^\top\mathbb{G})_{pq} = 2\min(p, q)^2 \max(p, q)$, non-negative and increasing in both indices, and the largest column sum is the last, $q = N-1$, over the odd $p \leq N-1$: $2(N-1)\sum_{i=1}^{N/2}(2i-1)^2 = N(N-1)^2(N+1)/3$. The column sums of $\mathbb{G}\mathbb{G}^\top$ are, to leading order, $N^4/4$, below this from $n = 3$ on. □

The rebuilt circuit's $\alpha = N\|\mathbb{G}\|_S/2$ is therefore a third of the printed prefactor $\frac{3N}{2}\|\mathbb{G}\|_S$ of Equation (3.5), so every cost quoted for Wu et al.'s encoding is on the kinder side of the printed one. With $\|\mathbb{G}\|_S \sim N^4/3$ and $\|\mathbb{G}\|_2 \sim N^2/(2j_{-3/4,1})$ (Theorem 2), the printed prefactor's looseness is $\frac{3N}{2} \cdot \frac{N^4}{3} \cdot \frac{2j_{-3/4,1}}{N^2} = j_{-3/4,1}N^3$, approached from below.

The claims tested. Table D.1 gives the verdict on each claim Wu et al. make for their circuits. The gate count holds as a bound, the error holds to leading order in μ (Proposition 13), and the rows of Figs 2 and 8 hold exactly. The qubit count of Fig. 9 is not testable: the rebuild uses registers of its own, so its $2n + 5$ neither confirms nor refutes the printed $2n + 3$. The prefactor is valid but exceeds $\|\mathbb{G}\|_2$ by a factor growing as N^3 (§ 4.5.1).

Table D.1: Wu et al.'s encodings hold their gate and error claims but not their efficiency. Each claim of Wu et al. [30] for their block encodings (their Figs 2, 8 and 9), as printed and as measured on the rebuilt circuits. $\mathbb{B}(x)$, rank-one zero-constraint row; $\mathbb{D}^{(0)}(x_s)$, rank-one collapse row at the regular point x_s; $\|\mathbb{G}\|_S = \|\mathbb{G}^\top\mathbb{G}\|_1$, Wu et al.'s norm of the differentiation matrix; $j_{-3/4,1}$, first positive zero of $J_{-3/4}$. The error is to leading order in the ladder coefficient $\mu = 2/\|\mathbb{G}\|_S$.

Circuit and claim	Printed	As rebuilt	Verdict
Fig. 2, $\mathbb{B}(x)$	$n+2$ qubits, $\sqrt{2^{n+1}}$	the same, exact	holds
Fig. 8, $\mathbb{D}^{(0)}(x_s)$	$\sqrt{2^{n+1}}/f(x_s)$	the same, exact	holds
Fig. 9, qubits	$2n+3$	$2n+5$ (own registers)	not testable
Fig. 9, gates	$O(n^3)$	$O(n^2)$	holds (as a bound)
Fig. 9, error	decreases exponentially	$O(N^{-5})$, bound derived	holds (leading order)
Fig. 9, prefactor	$(2^{n-1} + 2^n)\|\mathbb{G}\|_S$	valid, $\sim j_{-3/4,1}N^3\|\mathbb{G}\|_2$	loose

D.2 The small-angle ladder and its error

The mechanism. A ladder of n singly-controlled R_y rotations acts on one flag qubit, initialised to $|1\rangle$: bit m of the (register-arithmetic-shifted) column index controls $R_y(\theta_m)$,

$$\theta_m = 2\arcsin\big(-2^m\mu\big), \qquad m = 0, \ldots, n-1, \tag{D.1}$$

for a small real coefficient μ. Because rotations on the same qubit compose by exact angle addition ($R_y(a)R_y(b) = R_y(a+b)$, an $SO(2)$ identity), the net rotation for a column index p with bit set S is $R_y(\Theta(p))$, $\Theta(p) = \sum_{m\in S}\theta_m$, giving the flag qubit's $|0\rangle$-amplitude

$$\mathrm{raw}(p) = -\sin\big(\Theta(p)/2\big) = \sin\Big(\textstyle\sum_{m\in S}\arcsin(2^m\mu)\Big). \tag{D.2}$$

The row-0 correction (entries $\sqrt{2}p$ in place of $2p$) is a second, independently accumulated ladder on the same flag qubit, with per-bit coefficient $\mu(1/\sqrt{2}-1)$, controlled on the same "this is row 0" flag the main construction already computes.

Why this is only approximate. $\mathrm{raw}(p)$ in Equation (D.2) would equal the desired linear value μp exactly only if $\sin\big(\sum_m \arcsin(x_m)\big) = \sum_m x_m$, which does not hold in general – sin of a sum is not the sum of sines. The discrepancy is a genuine, quantifiable small-angle error.

Proposition 13 (Entrywise error of the ladder construction)**.** *Let $J_3(p) := \sum_{m\in S}(2^m)^3$ for column index p with bit set S. Then*

$$\big|\mathrm{raw}(p) - \mu p\big| \;\le\; \frac{\mu^3}{6}\,p^3 \;+\; O(\mu^5 N^5). \tag{D.3}$$

Proof. Expand both non-linearities to leading order in the small parameter μ: $\arcsin(y) = y + y^3/6 + O(y^5)$ and $\sin(x) = x - x^3/6 + O(x^5)$. Summing the first expansion over the fired bits and substituting into the second,

$$\sum_{m\in S}\arcsin(2^m\mu) = \mu p + \tfrac{\mu^3}{6}J_3(p) + O(\mu^5 N^5), \tag{D.4}$$

$$\begin{aligned}\sin\Big(\sum_{m\in S}\arcsin(2^m\mu)\Big) &= \sin(\mu p) + \cos(\mu p)\cdot\tfrac{\mu^3}{6}J_3(p) + O(\mu^5 N^5)\\ &= \mu p + \tfrac{\mu^3}{6}\big(J_3(p) - p^3\big) + O(\mu^5 N^5).\end{aligned} \tag{D.5}$$

Since $p = \sum_{m\in S} 2^m$ and every $2^m \le p$, the multinomial expansion of p^3 term-by-term dominates $J_3(p)$, with equality only when S is a single bit (p itself a power of two): $J_3(p) \le p^3$. Taking the worst case $p = N-1$ gives Equation (D.3). □

This is a leading-order bound, valid once the largest single argument, $2^{n-1}\mu$, is small and well inside arcsin's domain – true from $n = 2$ onward at the default $\mu = 2/\|\mathbb{G}\|_S$ (Equation (3.5)), but not at $n = 1$, where $\mu\cdot 2^0 = \mu = 1$ exactly and the bound is not yet meaningful.

D.3 Measured resources and error

Table D.2: Wu et al.'s encoding of $\mathbb{G}$: qubits, gates and entry error against the printed claims. n, qubits of the system register, $N = 2^n$; $\|\mathbb{G}\|_S = \|\mathbb{G}^\top\mathbb{G}\|_1$, Wu et al.'s norm; $\epsilon_{\mathbb{G}}$, largest entry error of the encoded $\mathbb{G}$, bounded and measured. Exact-mode classical measurement.

n	N	$\|\mathbb{G}\|_S$	Qubits (claim $2n+3$)	Gates	Bound on $\epsilon_{\mathbb{G}}$	Measured $\epsilon_{\mathbb{G}}$
2	4	6.15×10^1	–	–	–	–
3	8	1.18×10^3	–	–	–	–
4	16	2.04×10^4	–	–	–	–
5	32	3.38×10^5	–	–	–	–
6	64	5.50×10^6	–	–	–	–
7	128	8.88×10^7	–	–	–	–
8	256	1.43×10^9	–	–	–	–

E The Structured Exact Encoding of the Differentiation Matrix

E.1 Proof of Proposition 1, and the two diagonal factors

Proof of Proposition 1. $R_0 = \mathrm{diag}(1/\sqrt{2}, 1, \ldots, 1)$ and $\Lambda = \mathrm{diag}(0, 1, \ldots, N-1)$ are diagonal, so $\big(R_0(2U_{\mathrm{odd}})\Lambda\big)_{kp} = (R_0)_{kk} \cdot 2[p > k,\ p - k \text{ odd}] \cdot p$. This is $2p$ for $k \geq 1$ and $\sqrt{2}\,p$ for $k = 0$ when $p > k$ and $p - k$ is odd, and zero otherwise, which is Equation (4.2). $U_{\mathrm{odd}} = \sum_{d\ \mathrm{odd}} S_d$ with $(S_d)_{kp} = [p - k = d]$, the sum running over the $N/2$ odd $d \in [1, N-1]$. □

Lemma 2 (Λ as a sum of Pauli strings)**.**

$$\Lambda = \frac{N-1}{2}\, I - \sum_{j=0}^{n-1} 2^{j-1} Z_j, \tag{E.1}$$

so its LCU has $n+1$ unitary terms, $\lceil \log_2(n+1) \rceil$ selector qubits and $\alpha_\Lambda = N - 1 = \|\Lambda\|_2$.

Proof. On $|p\rangle$ with bits p_j, Z_j has the eigenvalue $z_j = 1 - 2p_j$, so $p = \sum_j 2^j p_j = \sum_j 2^j(1 - z_j)/2 = \frac{N-1}{2} - \sum_j 2^{j-1} z_j$. The weights' magnitudes sum to $\frac{N-1}{2} + \sum_j 2^{j-1} = N - 1$. The negative weights are phases on SELECT's branches, so no value is loaded by rotation. □

Lemma 3 (R_0 as two terms)**.** $R_0 = a\, I + b\,(I - 2\,|0\rangle\langle 0|)$ *with* $a = \frac{1}{2}(1 + 1/\sqrt{2})$ *and* $b = \frac{1}{2}(1 - 1/\sqrt{2})$*, both positive, so* $\alpha_{R_0} = a + b = 1 = \|R_0\|_2$*, with one selector qubit.*

Proof. On $|0\rangle$ the right side is $a - b = 1/\sqrt{2}$; on every other basis state it is $a + b = 1$. □

E.2 The arithmetic SELECT

The ripple-carry adder of Cuccaro et al. [65], on registers a and b of n qubits with a carry-in c and an overflow z, acts as $(c, a, b, z) \mapsto \big(c, a, (a + b + c) \bmod 2^n, z \oplus [a + b + c \geq 2^n]\big)$ on

every input, since its MAJ and UMA identities do not use $c = 0$. Its inverse therefore maps b to $(b - a - c) \bmod 2^n$ and flips z exactly when $b < a + c$: with $c = z = 0$ it is the in-place subtraction $b \leftarrow (b - a) \bmod 2^n$ with the borrow $z = [\, b < a \,]$. The circuit of Figure 4.1 uses an offset register d of n qubits, the carry and the borrow, $n+2$ ancilla in all: PREPARE applies X to the lowest qubit of d and Hadamards to the others, SELECT is the subtraction with $a = d$ and b the system register, and PREPARE† follows.

Lemma 4 (The odd shifts by subtraction)**.** *The circuit block-encodes U_{odd} exactly at $\alpha = N/2$.*

Proof. PREPARE maps $|0\rangle$ to $\sqrt{2/N}\sum_{\ell \text{ odd}} |\ell\rangle$. From $|p\rangle\,|0\rangle$, PREPARE and SELECT give $\sqrt{2/N}\sum_{\ell \text{ odd}} |(p-\ell) \bmod N\rangle\,|\ell\rangle\,|[\, p < \ell \,]\rangle$. Projecting the borrow on zero keeps the terms with $\ell \le p$, for which $(p - \ell) \bmod N = p - \ell$; projecting the offset register on zero after PREPARE† weights each by $\langle 0|\,\text{PREPARE}^\dagger\,|\ell\rangle = \sqrt{2/N}$. The block's entry (k, p) is therefore $\frac{2}{N}\,\#\{\ell \text{ odd} : \ell = p - k \ge 1\} = \frac{2}{N}[\, p > k,\ p - k \text{ odd} \,]$. □

The terms that wrap around carry a borrow of one and vanish on post-selection, and SELECT is a permutation, so one PREPARE serves all $N/2$ terms. The subtractor costs n MAJ and n UMA gates, each two CNOTs and a Toffoli, and one CNOT: $16n + 1$ CNOTs with a six-CNOT Toffoli, which is the transpiled count at every n computed.

E.3 Proof of Proposition 2

Proof of Proposition 2. Subnormalisation. A product of block encodings is a block encoding of the product at the product of the subnormalisations, so $\alpha_{\mathbb{G}} = \alpha_{R_0}\,\alpha_{2U_{\text{odd}}}\,\alpha_\Lambda = 1 \cdot N \cdot (N-1)$ by the two lemmas above and Lemma 4 (a block encoding of U_{odd} at $N/2$ is one of $2U_{\text{odd}}$ at N). Its ratio to $\|\mathbb{G}\|_2$ tends to $2j_{-3/4,1}$ by Theorem 2, since $N(N-1)/N^2 \to 1$.

Ancilla. The encodings are composed as $R_0 \cdot \big((2U_{\text{odd}}) \cdot \Lambda\big)$, Λ applied first, each product sharing one ancilla register between its factors and adding one flag that marks the right factor's failure branch. On the right factor's success branch its ancilla are back at zero, the state the left factor needs, and on its failure branch the flag is raised and the branch never reaches the post-selected output, so the product stays exact. The inner product shares $\max\big(n+2, \lceil \log_2(n+1) \rceil\big) = n+2$ qubits and adds a flag; the outer shares $\max(1, n+3)$ with R_0's selector and adds a flag: $n+4$ ancilla in all. Of these, the $n-1$ qubits that carry the uniform superposition are forced by the $N/2$ terms of U_{odd}'s combination, since an LCU of L distinct terms needs $\lceil \log_2 L \rceil$ selector qubits.

Gates. U_{odd}'s PREPARE and PREPARE† are $2n$ one-qubit gates and its subtractor $O(n)$ gates. Λ's PREPARE is a rotation tree on $\lceil \log_2(n+1) \rceil$ qubits, and its SELECT is n gates Z_j and the negative weights' phases, each controlled on the selector. R_0 is $2n$ CNOTs and one n-controlled Z, and the two flags are an X controlled on Λ's selector and one controlled on the inner product's $n+3$ ancilla. An m-controlled gate costs $O(m^2)$ CNOTs in an ancilla-free

synthesis, so the n-controlled gates of R_0 and the flags give $O(n^2)$; with a linear-cost synthesis that borrows an idle qubit [66] they cost $O(n)$, and Λ's SELECT, $O(n \log n)$, dominates. □

The transpiled two-qubit gate counts of the encoding of $\mathbb{G}$ alone are available only at small n, 109 at $n = 2$ and 471 at $n = 5$: consistent with the bound, but too few to fit it.

Lemma 5 (The norm of the odd shifts)**.** $\|U_{\text{odd}}\|_2 = 1/\big(2 \sin \frac{\pi}{2N+2}\big)$, *so* $\alpha_{U_{\text{odd}}}/\|U_{\text{odd}}\|_2 = N \sin \frac{\pi}{2N+2}$, *which rises to* $\pi/2$.

Proof. Ordering rows and columns by parity, U_{odd} maps odd columns to even rows and even columns to odd rows, and its two blocks are the $M \times M$ upper triangle of ones with its diagonal, $M = N/2$, and the $(M-1) \times (M-1)$ one, padded with zeros. For the $m \times m$ triangle of ones L, $L^{-1} = I - S$ with S the shift above the diagonal, and $(LL^\top)^{-1} = (I - S^\top)(I - S)$ is tridiagonal with $1, 2, \ldots, 2$ on the diagonal and -1 beside it, whose eigenvalues are $4 \sin^2 \frac{(2i-1)\pi}{2(2m+1)}$, $i = 1, \ldots, m$. So $\|L\|_2 = 1/\big(2 \sin \frac{\pi}{4m+2}\big)$, increasing in m, and $m = M$ gives the lemma. □

R_0 and Λ are encoded at their norms, so of the limiting ratio $2j_{-3/4,1}$ the factor $\pi/2$ is U_{odd}'s and the remaining $4j_{-3/4,1}/\pi \approx 1.35$ is the product's: $\|R_0(2U_{\text{odd}})\Lambda\|_2 < \|R_0\|_2 \, \|2U_{\text{odd}}\|_2 \, \|\Lambda\|_2$, because Λ is largest on the last columns while the leading singular vectors of U_{odd} are spread over all of them.

E.4 The norms of the derivative and its powers

Theorem 2 (The norm of $\mathbb{G}$ in the limit)**.** $\|\mathbb{G}\|_2/N^2 \to \|K\| = 1/(2j_{-3/4,1}) \approx 0.4724$ *as* $N \to \infty$, *where* $(Kf)(s) = \int_s^1 t\, f(t)\, dt$ *on* $L^2(0,1)$ *and* $j_{-3/4,1}$ *is the first positive zero of the Bessel function* $J_{-3/4}$.

Proof. Order the rows and columns by parity. $\mathbb{G}$ maps odd columns to even rows and even columns to odd rows, so in that order it is $\left(\begin{smallmatrix} 0 & E \\ O & 0 \end{smallmatrix}\right)$ and $\|\mathbb{G}\|_2 = \max(\|E\|_2, \|O\|_2)$, with, for $M = N/2$ and $0 \le i, j < M$,

$$E_{ij} = \frac{2(2j+1)}{\sqrt{c_{2i}}}\,[\,j \ge i\,], \qquad O_{ij} = 4j\,[\,j > i\,]$$

(row $2i$, column $2j+1$ for E; row $2i+1$, column $2j$ for O; $c_0 = 2$, $c_{k\ge1} = 1$). An $M \times M$ matrix with entries $\frac{1}{M}\kappa(i,j)$ has the operator norm of the integral operator on $L^2(0,1)$ with the step kernel $\kappa(\lfloor Ms \rfloor, \lfloor Mt \rfloor)$, since $J : v \mapsto \sqrt{M} \sum_i v_i \mathbf{1}_{[i/M,(i+1)/M)}$ is an isometry onto the step functions. Here $E/N^2 = \frac{1}{M}\kappa_E$ with $\kappa_E(i,j) = \frac{2j+1}{2M}[\,j \ge i\,]/\sqrt{c_{2i}}$, and $O/N^2 = \frac{1}{M}\kappa_O$ with $\kappa_O(i,j) = \frac{j}{M}[\,j > i\,]$. Both step kernels are bounded and converge to $k(s,t) = t\,[\,t \ge s\,]$ except on a set of measure $O(1/M)$ (the diagonal cells and the first row), and to within $O(1/M)$ elsewhere, so they converge in $L^2([0,1]^2)$. The Hilbert–Schmidt norm bounds the operator norm, so $\|E\|_2/N^2$ and $\|O\|_2/N^2$ both converge to $\|K\|$.

K is Hilbert–Schmidt, so $\|K\|^2$ is the largest eigenvalue λ of K^*K, with $(K^*g)(t) = t\int_0^t g(s)\,ds$. Let $\lambda f = K^*Kf$ and $h(t) = \int_0^t (Kf)(s)\,ds$. Then $\lambda f = t\,h$, $h(0) = 0$, $h'(1) = (Kf)(1) = 0$ and $h'' = (Kf)' = -tf = -t^2h/\lambda$. With $a = \lambda^{-1/2}$, $h'' + a^2t^2h = 0$, whose solutions are $\sqrt{t}\,J_{\pm 1/4}(at^2/2)$, and $h(0) = 0$ selects $J_{1/4}$. With $z = at^2/2$, the identity $zJ'_\nu(z) = zJ_{\nu-1}(z) - \nu J_\nu(z)$ at $\nu = \frac{1}{4}$ gives

$$h'(t) = \frac{1}{2\sqrt{t}}\big(J_{1/4}(z) + 4zJ'_{1/4}(z)\big) = \frac{2z}{\sqrt{t}}\,J_{-3/4}(z),$$

so $h'(1) = 0$ exactly when $J_{-3/4}(a/2) = 0$. The eigenvalues are $\lambda = 1/a^2$ with $a/2$ the positive zeros of $J_{-3/4}$, the largest at the first, and $\|K\| = 1/(2j_{-3/4,1})$. □

Theorem 3 (The powers of $\mathbb{G}$)**.** *For every $k \geq 1$, $\|\mathbb{G}^k\|_2/N^{2k} \to \|K^k\| > 0$.*

Proof. In the parity ordering $\mathbb{G}^k$ is block diagonal or block anti-diagonal, with blocks that are alternating products of k factors E and O. Under the isometry J of the previous proof, with $J^*J = I$, each block of $\mathbb{G}^k/N^{2k}$ is unitarily equivalent to a product of k operators that converge in norm to K, and so converges to K^k. K is injective, since $\int_s^1 tf(t)\,dt = 0$ for all s forces $f = 0$, so $K^k \neq 0$. □

Corollary 1 (The standard form's norm)**.** *For an equation $\sum_{j\leq k} a_j f^{(j)} = r$ of order k with constant coefficients and $a_k \neq 0$, $\|R_{\mathrm{std}}\|_2 = |a_k|\,\|K^k\|\,N^{2k}(1 + o(1)) = \Theta(N^{2k})$ and $\|H_{\mathrm{std}}\| = \Theta(N^{4k})$.*

Proof. The equation block $\sum_j a_j\mathbb{G}^j$ has norm $|a_k|\,\|K^k\|\,N^{2k}(1 + o(1))$ by Theorem 3, its lower orders being $O(N^{2j})$ with $j < k$. A source block $\mathbb{M}_r\mathbb{D}^{(0)}(x_s)$ is $O(\sqrt{N})$, and a constraint row $\mathbb{B}(x)\mathbb{G}^j$ is rank one with norm $\sqrt{N}\|\tau(x)^\top\mathbb{G}^j\| \leq \sqrt{2N}\,\|\mathbb{G}\|_2^j = O(N^{2j+1/2})$ by Proposition 7, with $j \leq k - 1$, so every constraint row is $O(N^{2k-3/2})$. Since $\|R\|_2^2 = \|A^\dagger A + \sum_i C_i^\dagger C_i\|$ lies between $\|A\|^2$ and $\|A\|^2 + \sum_i \|C_i\|^2$, the rows do not move the leading term. □

The same limits fix the looseness of a power encoded as a product: $\alpha_{\mathbb{G}^k} = \alpha_{\mathbb{G}}^k = (N(N-1))^k$, so $\alpha_{\mathbb{G}^k}/\|\mathbb{G}^k\|_2 \to 1/\|K^k\|$, a constant for each k, equal to $2j_{-3/4,1}$ at $k = 1$. For R2a, whose equation block is $\mathbb{G}^2$ to leading order, this is the constant towards which the measured $\alpha_R/\|R_{\mathrm{std}}\|_2$ of § 4.6 levels off; the theorem does not by itself bound the whole residual's ratio.

E.5 Subnormalisation across n

F The Shared Pipeline: Identities and Formulas

Throughout, $R = [R_1; \ldots; R_L]$ is a stacked residual whose block R_j has a block encoding U_j at subnormalisation α_j on one system register with a shared ancilla register a, $H = R^\dagger R$, and $\sigma_0 \leq \sigma_1 \leq \cdots \leq \sigma_{\max}$ are the singular values of R.

Table E.1: The structured encoding rises to $2j_{-3/4,1} \approx 2.117$ times $\|\mathbb{G}\|_2$; Wu et al.'s prefactor grows as $j_{-3/4,1}N^3$. Subnormalisation over $\|\mathbb{G}\|_2$ of the structured encoding (Proposition 2) and of Wu et al.'s prefactor (Equation (3.5)) for the differentiation matrix $\mathbb{G}$, $n = 2$–10, $N = 2^n$. $j_{-3/4,1}$, first positive zero of the Bessel function $J_{-3/4}$. Exact classical computation.

n	$\|\mathbb{G}\|_2$	Structured	Wu et al.'s
2	7.39	1.623	49.9
3	30.1	1.858	468
4	121	1.986	4.05×10^3
5	484	2.051	3.36×10^4
6	1935	2.084	2.73×10^5
7	7739	2.100	2.20×10^6
8	30 957	2.109	1.77×10^7
9	123 827	2.113	1.42×10^8
10	495 308	2.115	1.14×10^9

F.1 The Stacked Residual and Its Reflection

Let a row-select register of $\lceil \log_2 L \rceil$ qubits sit beside a. PREPARE takes it from $|0\rangle$ to $\sum_j (\alpha_j/\alpha_R)\,|j\rangle$ with $\alpha_R = (\sum_j \alpha_j^2)^{1/2}$, and SELECT applies U_j when it holds j. Write Π_a for the projector of a onto $|0_a\rangle$ and Π_{in} for that of a and the row select together onto zero, and $U_R = \text{SELECT} \cdot \text{PREPARE}$.

Lemma 6 (The stack)**.** $\big(\langle j| \otimes \langle 0_a|\big) U_R \big(|0\rangle \otimes |0_a\rangle\big) = R_j/\alpha_R$*; that is,* $\Pi_a U_R \Pi_{\text{in}} = R/\alpha_R$ *with the row select read as output.*

Proof. $\langle j|\,\text{PREPARE}\,|0\rangle = \alpha_j/\alpha_R$, and SELECT on $|j\rangle$ is U_j, whose block is R_j/α_j. □

Unlike a linear combination's selector, the row select is never returned by an inverse preparation: the blocks are stacked, not summed, and their weights α_j^2 add in quadrature.

Proof of Proposition 5. $2\Pi_a - I$ is a Hermitian unitary, so its conjugate V by the unitary U_R is Hermitian and unitary, and $V^2 = I$. Using $\Pi_a^2 = \Pi_a$ and $\Pi_{\text{in}} U_R^\dagger U_R \Pi_{\text{in}} = \Pi_{\text{in}}$,

$$\Pi_{\text{in}} V \Pi_{\text{in}} = 2\big(\Pi_a U_R \Pi_{\text{in}}\big)^\dagger \big(\Pi_a U_R \Pi_{\text{in}}\big) - \Pi_{\text{in}} = \frac{2R^\dagger R}{\alpha_R^2} - I$$

by Lemma 6. For the controlled version: where the controls hold, $U_R^\dagger C(2\Pi_a - I) U_R$ is V; where they fail it is $U_R^\dagger U_R = I$. □

U_R's own gates are never controlled, so controlling V adds a control to one multi-controlled Z and nothing wider than U_R's widest gate. Since $H \succeq 0$ and $\|H\| = \|R\|^2 \le \alpha_R^2$, the spectrum of H' lies in $[-1, 1]$; a singular value s of R sits at $x = -1 + 2s^2/\alpha_R^2$, at the *height* $h = 1 + x = 2s^2/\alpha_R^2$ above the edge. The filters below work with heights, which keep their relative precision where x itself would round to -1. Numerically, V is verified exactly against $2H/\alpha_R^2 - I$ on every exact-mode record, the worst entrywise error 5.9×10^{-15}.

F.2 Qubitisation

Let $\Pi = \Pi_{\text{in}}$ and $W = (2\Pi - I)V$.

Theorem 4 (Qubitisation [45])**.** *For every $k \geq 0$, $\Pi W^k \Pi = T_k(H')$ with $H' = \Pi V \Pi$.*

Proof. Let $|\lambda\rangle$ be an eigenvector of H' in the image of Π with eigenvalue λ. Then $\Pi V |\lambda\rangle = \lambda |\lambda\rangle$ and $\| V |\lambda\rangle \| = 1$, so for $|\lambda| < 1$, $V |\lambda\rangle = \lambda |\lambda\rangle + s |\lambda^\perp\rangle$ with $s = \sqrt{1-\lambda^2}$ and $|\lambda^\perp\rangle$ a unit vector orthogonal to the image of Π. From $V^2 = I$, $V |\lambda^\perp\rangle = s |\lambda\rangle - \lambda |\lambda^\perp\rangle$. On the plane of $|\lambda\rangle$ and $|\lambda^\perp\rangle$, $2\Pi - I = \operatorname{diag}(1, -1)$, so W is the rotation

$$W = \begin{pmatrix} \cos\vartheta & \sin\vartheta \\ -\sin\vartheta & \cos\vartheta \end{pmatrix}, \qquad \vartheta = \arccos\lambda,$$

and $\langle\lambda| W^k |\lambda\rangle = \cos k\vartheta = T_k(\lambda)$. $W^k |\lambda\rangle$ stays in the plane, which meets the image of Π in $|\lambda\rangle$'s line, so $\Pi W^k |\lambda\rangle = T_k(\lambda) |\lambda\rangle$. For $\lambda = \pm 1$, $V |\lambda\rangle = \lambda |\lambda\rangle$ and $W^k |\lambda\rangle = \lambda^k |\lambda\rangle$. The eigenvectors span the image of Π. □

The theorem projects once, after k steps. Applying W and projecting after every step gives other polynomials: the branches with a non-zero ancilla carry $|\lambda^\perp\rangle$'s component, which later steps rotate back into the image of Π. Every verification of a walk or filter circuit therefore restricts once, at the end.

F.3 The Controlled Walk

With $Z_0 = I - 2 |0\rangle \langle 0|$ on a register, $2\Pi - I = -Z_0$, so $W = Z_0^{(\text{in})} U_R^\dagger Z_0^{(a)} U_R$, the two signs cancelling. Controlled on further qubits, W is the same circuit with the two reflections controlled, since where the controls fail both reflections are the identity and $U_R^\dagger U_R = I$. A controlled walk thus holds no gate wider than U_R's widest, and simulating it is a statevector problem of 2^q amplitudes, never an operator problem of 4^q. A circuit's global phase becomes a relative phase once it is controlled: the phases of the linear combination's terms, and GQSP's global phase when a whole preparation is controlled (Appendix F.9), are therefore carried as phase gates on the controls. A dropped phase λ would scale an overlap by $\cos\lambda$.

F.4 The Minimax Edge Filter

Let $x_0 < x_1$ be the ground and first excited values of H', $w(x) = (2x - (x_1 + 1))/(1 - x_1)$ the affine map of $[x_1, 1]$ onto $[-1, 1]$, and $P_d = T_d(w(x))/T_d(w(-1))$ as in Equation (7.6).

Proposition 14 (Suppression and optimality)**.** *$|P_d| \leq 1$ on $[-1, 1]$ with equality at -1, and*

$$\sup_{x \in [x_1, 1]} \frac{|P_d(x)|}{|P_d(x_0)|} = \frac{1}{\cosh(d\theta_0)}, \qquad \theta_0 = \operatorname{arccosh}\big(-w(x_0)\big).$$

No polynomial of degree d achieves a smaller ratio.

Proof. On $[x_1, 1]$, $|T_d(w)| \le 1$. Below x_1, $w < -1$ and $|T_d(w)| = \cosh\big(d \operatorname{arccosh}(-w)\big)$, which increases as x falls to -1; so $|P_d| \le 1$ with equality at -1, and the ratio is $1/|T_d(w(x_0))| = 1/\cosh(d\theta_0)$. Optimality is Chebyshev's extremal property: among polynomials of degree d that take the value 1 at a point z outside $[-1, 1]$, $T_d/T_d(z)$ has the least maximum on $[-1, 1]$, namely $1/|T_d(z)|$, which is Chebyshev's inequality [55]; Lin and Tong use the same property for eigenstate filtering [28]. Apply it in the variable w at $z = w(x_0)$. □

The degree from heights. The degree of Equation (7.7) is the least d with $1/\cosh(d\theta_0) \le \varepsilon$. With the heights $h_j = 1 + x_j = 2\sigma_j^2/\alpha_R^2$,

$$-w(x_0) - 1 = \frac{2(h_1 - h_0)}{2 - h_1},$$

so $\theta_0 = \operatorname{arccosh}(1+e)$ with $e = 2(h_1 - h_0)/(2 - h_1)$, evaluated as $\log\big(1 + e + \sqrt{e(2+e)}\big)$ without forming $1 + e$. For small e, $\operatorname{arccosh}(1+e) \approx \sqrt{2e}$ and $h_1 - h_0 = 2\Delta/\alpha_R^2$, so $\theta_0 \approx 2\sqrt{\Delta}/\alpha_R$ and

$$d \approx \frac{\alpha_R}{2\sqrt{\Delta}} \ln \frac{2}{\varepsilon}. \tag{F.1}$$

The coefficients. P_d is sampled at $2^{\lceil \log_2 2(d+1) \rceil} > d$ Chebyshev–Gauss nodes, and its Chebyshev coefficients are read off by a type-II discrete cosine transform, exact for a polynomial of degree below the number of nodes: interpolation, not a fit. Nothing is truncated.

Proposition 15 (Guarantee)**.** *Let ψ_{init} have overlap $\gamma > 0$ with a one-dimensional target, the ground state. At the degree of Equation* (7.7) *the normalised output $P_d(H')\psi_{\text{init}}/\|P_d(H')\psi_{\text{init}}\|$ has infidelity at most*

$$\Big(\frac{\sqrt{1-\gamma^2}}{\gamma \cosh(d\theta_0)}\Big)^2 \le \frac{1-\gamma^2}{\gamma^2}\,\varepsilon^2.$$

Proof. Write $\psi_{\text{init}} = \gamma u + \sqrt{1-\gamma^2}\,u_\perp$ with u the ground state and $u_\perp$ in the excited span. The filter maps γu to $\gamma P_d(x_0)u$ and $\sqrt{1-\gamma^2}u_\perp$ to a vector in the excited span of norm at most $\sqrt{1-\gamma^2}\,|P_d(x_0)|/\cosh(d\theta_0)$ by Proposition 14. The infidelity is the squared sine of the angle to u, at most its squared tangent, the off-target norm over the on-target one. □

For a target of $t > 1$ singular vectors the same argument holds with the ratios of the angles of the t-th and lower heights. The suppression is measured relative to the ground state's own filtered value, so there is no absolute tail for that value to fall beneath, and the bound is finite whenever $\gamma > 0$; imaginary time's can fail (Appendix F.5). The success probability is $p_G = \|P_d(H')\psi_{\text{init}}\|^2 \ge \gamma^2 P_d(x_0)^2$, with $P_d(x_0) = \cosh(d\theta_0)/\cosh(d\theta_{-1})$, $\theta_{-1} = \operatorname{arccosh}(-w(-1))$: one when the kernel is exact, $x_0 = -1$, and close to one wherever $h_0 \ll h_1$, as for a trusted kernel.

F.5 The Imaginary-Time Filter

On a singular value s of R, $f_\beta(x) = e^{-\beta(1+x)} = e^{-2\beta s^2/\alpha_R^2}$, so $f_\beta(H') = e^{-tH}$ with $t = 2\beta/\alpha_R^2$: imaginary time β in the units of H'. From the generating function of the modified Bessel functions [89], $e^{z\cos\vartheta} = I_0(z) + 2\sum_{k\geq 1} I_k(z)\cos k\vartheta$, at $z = -\beta$ and $x = \cos\vartheta$, with $I_k(-\beta) = (-1)^k I_k(\beta)$,

$$e^{-\beta x} = I_0(\beta) + 2\sum_{k=1}^{\infty}(-1)^k I_k(\beta)\, T_k(x), \tag{F.2}$$

which, multiplied by $e^{-\beta}$, gives the filter and its coefficients,

$$f_\beta(x) = e^{-\beta(1+x)}, \qquad f_\beta(H') = e^{-2\beta H/\alpha_R^2}, \qquad q_k = \big(2 - \delta_{k0}\big)(-1)^k e^{-\beta} I_k(\beta). \tag{F.3}$$

Certified truncation. Truncated at degree d, the series errs on $[-1, 1]$ by at most its tail $\sum_{k>d}|q_k|$, since $|T_k| \leq 1$, and by Proposition 16 the tail is $1 - \sum_{k\leq d}|q_k|$, computed without summing the infinite series. The degree used is the least whose tail is below the tolerance $\varepsilon_t = \varepsilon/10$: a certified degree. For large β and $k = \mathcal{O}(\sqrt{\beta})$, $e^{-\beta} I_k(\beta) \approx e^{-k^2/(2\beta)}/\sqrt{2\pi\beta}$, so the tail is about $\operatorname{erfc}(d/\sqrt{2\beta})$ and $d \approx \sqrt{2\beta\ln(2/\varepsilon_t)}$. Where the library Bessel function overflows, at β of order 10^9, the values come from $e^{-\beta} I_k(\beta) = \frac{1}{\pi}\int_0^\pi e^{-2\beta\sin^2(t/2)}\cos kt\, dt$ by the trapezoid rule, one real fast Fourier transform, spectrally accurate for this analytic periodic integrand.

The time. With $\Delta = \sigma_1^2 - \sigma_0^2$ and

$$\beta = \frac{\alpha_R^2}{2\Delta}\ln\frac{1}{\varepsilon}, \qquad d \approx \frac{\alpha_R}{\sqrt{\Delta}}\Big(\ln\frac{1}{\varepsilon}\,\ln\frac{2}{\varepsilon_t}\Big)^{1/2}, \tag{F.4}$$

the untruncated filter holds every excited state's amplitude to at most ε times the ground state's, since $f_\beta(x_1)/f_\beta(x_0) = e^{-2\beta\Delta/\alpha_R^2} = \varepsilon$ and f_β falls with s. Against Equation (F.1) the factor $\alpha_R/\sqrt{\Delta}$ is the same, and the logarithms make imaginary time's degree approach twice the minimax filter's as $\varepsilon \to 0$ with $\varepsilon_t = \varepsilon/10$.

The a priori bound. For a one-dimensional target and tail T, the argument of Proposition 15, with the truncation an operator of norm at most T, gives the infidelity bound

$$1 - F \leq \Big(\frac{\sqrt{1-\gamma^2}\, f(\sigma_1) + T}{\gamma f(\sigma_0) - T}\Big)^2, \qquad f(s) = e^{-2\beta s^2/\alpha_R^2},$$

wherever the denominator is positive. The tail is absolute while the ground state's filtered value is $f(\sigma_0) = \varepsilon^{\sigma_0^2/\Delta}$ under the time rule, so where the residual has no near-kernel the tail can reach it and the bound certifies nothing. With a trusted kernel, $\sigma_1/\sigma_0 \geq 20$, $\sigma_0^2/\Delta < 1/399$ and $f(\sigma_0) > \varepsilon^{1/399}$.

Comparing the two filters at like error. A degree means something only at a stated error, so the filter study of § 8.1 holds both filters to the same one, at three strengths. *Equal suppres-*

sion η bounds the excited interval relative to the ground state, $\sup_{x\in[x_1,1]} |P(x)|/|P(x_0)| \le \eta$; it bounds every initial state's infidelity by $\eta^2(1-\gamma^2)/\gamma^2$, depends only on x_0 and x_1, and is the strength at which the minimax filter is optimal (Proposition 14). *Equal certified infidelity* ε is the same bound at the problem's own overlap, $\eta = \gamma\sqrt{\varepsilon/(1-\gamma^2)}$. *Equal measured infidelity* ε is reached on the problem's own spectrum and initial state, and a degree counts only if every larger degree up to the certified one also reaches ε, so that neither filter's oscillation lands on ε by coincidence. Imaginary time's β is free: each of its degrees is the least over β, found by a grid search refined about its best point, and so an upper bound on the true least to the refinement's spacing. Both filters are anchored at the spectrum's edge with unit weight (Proposition 16), so GQSP realises either at the same rescaling, and a filter whose success probability falls below 10^{-12} is not counted.

On synthetic spectra with the ground state at the edge, gaps from 10^{-8} to 10^{-1} on the scale of H' and suppressions from 10^{-2} to 10^{-10}, imaginary time needs 1.67–1.92 times the minimax filter's degree at equal suppression, varying with the suppression and hardly with the gap; above the edge the ratio grows, since imaginary time's value on the ground state then falls with its time. Table F.1 gives the study on the problems whose references both forms trust. At equal certified infidelity both filters' degrees come from the same worst-case bound, so the ratio is the equal-suppression ratio at the problem's own γ; every trusted reference has its ground state at the edge, and the ratio lies within the synthetic range in both forms.

Table F.1: The minimax edge filter needs a lower degree than imaginary time at equal error, except at loose ε on some problems. Each cell: imaginary-time (QITE) degree / minimax degree, both forms, at one resolution per problem (n per axis on R5a) whose reference both forms trust. Three equal-error criteria: certified infidelity at $\varepsilon = 10^{-6}$ (both degrees from the worst-case bound at the problem's own γ), and measured infidelity at $\varepsilon = 10^{-6}$ and 10^{-4} on the problem's own spectrum from the geometric initial state. γ^2, that state's overlap with the ground state. Exact emulation (T2).

Problem	Form	γ^2	Certified, 10^{-6}	Measured, 10^{-6}	Measured, 10^{-4}
R1, $n = 4$	standard	0.904	620 / 353	528 / 352	198 / 227
	descriptor	0.904	49 / 28	41 / 27	18 / 18
R2a, $n = 3$	standard	0.499	8603 / 4820	6422 / 4796	5459 / 3317
	descriptor	0.191	636 / 354	533 / 353	400 / 255
R3a, $n = 3$	standard	0.166	13 127 / 7295	9247 / 7135	7236 / 5189
	descriptor	8.0×10^{-3}	5294 / 2904	3072 / 2862	1784 / 2187
R5a, $n = 3$	standard	0.587	23 175 / 13 011	18 055 / 12 611	10 248 / 8650
	descriptor	0.526	8612 / 4829	4776 / 4617	2285 / 3253

An underestimated edge. The degree is set a priori from the spectral edge (§ 7.4), and a study on R2a at $n = 3$ in the standard form measures what an underestimate of it costs, with the imaginary-time filter, whose time the gap sets directly (Equation (F.4)). At the true gap the degree is 17 750 and the infidelity 9.7×10^{-13}. With the gap halved the degree is 25 103, and with it divided by ten 56 131: the factors $\sqrt{2}$ and $\sqrt{10}$ of an inverse square root, with every preparation still within ε. Wu et al.'s own time rule, run in the same study, gives degree 3336 and infidelity 0.24: its time is shorter than the gap rule's and its state does not reach $\varepsilon = 10^{-6}$. The time rule is examined with the rest of the reproduction in Appendix K. Exact emulation

(T2).

F.6 Unit Weight of Both Filters

Proposition 16 (Unit weight of both filters)**.** *For every $\beta > 0$ the Chebyshev coefficients q_k of $e^{-\beta(1+x)}$ (Equation* (F.3)*), and for every d and $x_1 \in (-1, 1)$ those of P_d in Equation* (7.6)*, alternate in sign and satisfy*

$$\sum_k |q_k| = 1, \tag{F.5}$$

so $|P(z)| \le \sum_k |q_k| = 1$ on the unit circle, with equality at $z = -1$.

Each filter is the reflection, about the spectrum's edge, of a function whose Chebyshev coefficients are non-negative and sum to its value of one there.

Proof of Proposition 16. Imaginary time. $I_k(\beta) > 0$ for $\beta > 0$, so the q_k alternate in sign and $\sum_k |q_k| = e^{-\beta}\big(I_0(\beta) + 2\sum_{k\ge1} I_k(\beta)\big)$, the generating function at $\vartheta = 0$ times $e^{-\beta}$, which is $e^{-\beta}e^{\beta} = 1$.

The minimax filter. Let $\tilde{P}(y) = P_d(-y)$. With $a = 2/(1 - x_1)$ and $b = (1 + x_1)/(1 - x_1) \ge 0$, $w(-y) = -(ay+b)$ and $a = 1+b$, so $ay+b = y+b(1+y)$ and $\tilde{P}(y) = T_d\big(y+b(1+y)\big)/T_d(1+2b)$, the factors $(-1)^d$ cancelling. Taylor's theorem, exact for a polynomial, gives

$$T_d\big(y + b(1 + y)\big) = \sum_{m=0}^{d} \frac{b^m}{m!}\,(1 + y)^m\, T_d^{(m)}(y).$$

Every derivative of T_d has non-negative Chebyshev coefficients, by induction from $T_j' = jU_{j-1}$ and $U_{j-1} = 2\sum_i T_i$ over $0 < i < j$ with $i \equiv j - 1 \pmod 2$, plus T_0 when j is odd; $(1 + y)^m = (T_0 + T_1)^m$ has non-negative coefficients; and so does a product of two such series, since $T_iT_j = \frac{1}{2}(T_{i+j} + T_{|i-j|})$. With $b^m/m! \ge 0$ and $T_d(1 + 2b) > 0$, $\tilde{P}$ has non-negative coefficients $\tilde{q}_k$, and $\sum_k \tilde{q}_k = \tilde{P}(1) = P_d(-1) = 1$. Since $T_k(-x) = (-1)^k T_k(x)$, the coefficients of P_d are $(-1)^k\tilde{q}_k$.

In both cases $|\sum_k q_k z^k| \le \sum_k |q_k| = 1$ on the unit circle, with equality at $z = -1$, where the alternating signs align. □

F.7 GQSP and Its Phases

A signal qubit is added to the walk's register. With

$$R(\theta, \phi) = \begin{pmatrix} e^{\mathrm{i}\phi}\cos\theta & e^{\mathrm{i}\phi}\sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

on the signal and $C_0W = |0\rangle\langle 0| \otimes W + |1\rangle\langle 1| \otimes I$, the circuit is

$$U_\Phi = e^{\mathrm{i}\lambda} R(\theta_d, \phi_d)\, C_0W\, R(\theta_{d-1}, \phi_{d-1}) \cdots C_0W\, R(\theta_0, \phi_0).$$

On an eigenvector of W with eigenvalue z, C_0W acts on the signal as $\mathrm{diag}(z,1)$. Motlagh and Wiebe [48] show that for every $P(z)=\sum_{k\le d}q_kz^k$ with $|P|\le 1$ on the unit circle there are phases with $e^{\mathrm{i}\lambda}\langle 0|\,R(\theta_d,\phi_d)\,\mathrm{diag}(z,1)\cdots R(\theta_0,\phi_0)\,|0\rangle=P(z)$. W is unitary, hence normal, so the signal's zero block of U_Φ is $P(W)$ on the walk's register, and restricting once by Π_{in} and applying Theorem 4 gives Equation (7.5). The circuit is d controlled walks and $d+1$ one-qubit rotations on one more qubit than the walk, uses W but never $W^\dagger$, and needs no parity.

The phases are found from the coefficients in four steps.

1. *Margin.* P is divided by $\sup|P|/(1-\mu)$, $\mu=10^{-2}$, so that $1-|P|^2>0$ on the circle and its logarithm is finite. Both filters have $\sup|P|=1$, so the circuit realises $(1-\mu)P$ and succeeds with probability $(1-\mu)^2p_G=0.9801\,p_G$, a constant that the cost estimates leave out.

2. *Weiss's complement.* The unique outer polynomial Q of degree d with $|P|^2+|Q|^2=1$ on the circle is obtained by spectral factorisation: $\frac{1}{2}\log(1-|P|^2)$ is sampled on a fast-Fourier-transform grid, i times its Hilbert transform is added by doubling the causal half of its cepstrum, and the result is exponentiated; the grid grows fourfold until the coefficients beyond degree d fall below 10^{-11} relative [49]. No root-finding is used, which is unstable well below the degrees needed here.

3. *The inverse nonlinear Fourier transform.* (Q,P) is the SU(2) nonlinear Fourier transform of a sequence (F_k), and GQSP is equivalent to it [50]. The sequence is recovered by the fast divide-and-conquer inverse [51], or by layer stripping, both stable because Q is outer.

4. *The angles.* $\theta_k=\arctan|F_k|$ for $k<d$ and $\theta_d=\pi/2-\arctan|F_d|$; $\phi_d=\arg F_d$, $\phi_{d-1}=\arg F_d-\arg F_{d-1}$ and $\phi_k=\arg F_{k+1}-\arg F_k-\pi$ otherwise; λ, a global phase, from the overlap of the $\lambda=0$ polynomial with P at eight roots of unity, weighted by $|P|$.

The polynomial the angles realise is rebuilt from them alone, in coefficient space, and compared with the target on $4(d+1)$ points of the circle. Acceptance is 10^{-12}, in double precision throughout. The realised polynomials meet it to degrees of order 10^6 in the standard form; in the descriptor's filter study two of the filters, at degrees of order 2×10^5, reach just over 10^{-12}.

F.8 Amplification, the Total Count and the Ideal-Encoding Degree

Proposition 17 (Amplification)**.** *With $\vartheta=\arcsin\gamma$ and $k_{\mathrm{aa}}=\lfloor\pi/(4\vartheta)\rfloor$ rounds of amplitude amplification [57], a preparation that succeeds with probability γ^2 succeeds with probability at least $\max(\gamma^2,1-\gamma^2)\ge\frac{1}{2}$, at $2k_{\mathrm{aa}}+1$ applications of the preparation or its inverse.*

Proof. k rounds rotate the success amplitude from $\sin\vartheta$ to $\sin(2k+1)\vartheta$. If $\gamma\ge\sin(\pi/4)$, then $k_{\mathrm{aa}}=0$ and the probability is $\gamma^2\ge\frac{1}{2}$. Otherwise $k_{\mathrm{aa}}\le\pi/(4\vartheta)<k_{\mathrm{aa}}+1$ gives $\pi/2-\vartheta<(2k_{\mathrm{aa}}+1)\vartheta\le\pi/2+\vartheta$, so $\sin^2(2k_{\mathrm{aa}}+1)\vartheta\ge\cos^2\vartheta=1-\gamma^2>\frac{1}{2}$. □

For small γ, $k_{\mathrm{aa}}\approx\pi/(4\gamma)$: a poor initial state costs linearly in $1/\gamma$. Exact amplification needs

γ, which the resource estimate takes from the reference, as the degree takes the spectral edge; without it, fixed-point amplification costs a constant factor more. The geometric initial state is run for both signs of r and the one with the larger success probability kept, which doubles the preparations; the count leaves that factor out, as it leaves out the margin below. The total count of Equation (7.9) takes $p_G = \gamma^2$ and the transpiled two-qubit count $g(W)$ of one walk, and leaves out what grows more slowly: the $d+1$ rotations of GQSP, the initial state's n rotations, the amplification's reflections and the factor $(1-\mu)^{-2}$. A resource estimate computes every quantity at the target resolution itself: α_R and $g(W)$ from the circuit, σ_0, σ_1 and γ from the classical reference, the minimax degree exactly from Equation (7.7).

Proposition 18 (The degree at the ideal-encoding bound)**.** *Every block encoding of R has $\alpha \geq \|R\|_2$, and the degree of Equation* (7.7) *does not increase as α falls. The degree at $\alpha = \|R\|_2$ is therefore the least that the minimax edge filter needs on R through any block encoding of it. The same holds for a block encoding of H itself, at $\alpha_H \geq \|H\| = \|R\|_2^2$.*

Proof. $\alpha \geq \|R\|_2$ is a block encoding's defining condition. The heights are $h_j = 2\sigma_j^2/\alpha^2$; as α falls, h_1 and $h_1 - h_0 = 2\Delta/\alpha^2$ both grow, so $\theta_0 = \arccosh\big(1 + 2(h_1 - h_0)/(2 - h_1)\big)$ grows and $d = \lceil \arccosh(1/\varepsilon)/\theta_0 \rceil$ does not. For an encoding of H the heights are $2\lambda_j/\alpha_H$, the same function of $\alpha_H = \alpha^2$. □

Any difference between a form's degree as built and its ideal degree is its encoder's cost; any difference between two forms at the ideal degree is their Hamiltonians'. With Equation (F.1) and $\alpha_R \geq \|R\|$, $d \gtrsim \frac{1}{2}\sqrt{\|H\|/\Delta}\,\ln(2/\varepsilon)$, a property of H that no encoding removes.

F.9 The Readout Identity for Probabilistic Preparation

Let $\hat{U}_{\text{prep}}$ be the preparation, the initial state's rotations followed by the GQSP circuit, whose output is $|0_a\rangle\, g + |\text{junk}\rangle$, with $g = \sqrt{p_G}\,\psi$ the system part on the all-zero ancilla and $|\text{junk}\rangle$ supported on non-zero ancilla. Let $\hat{U}_\tau(x)$ be Wu et al.'s feature map, one per axis, and $u(x) = \tau(x)/2^{m/2}$ the first column of their product on m axes, $\tau(x)$ the Kronecker product of the axes' feature vectors.

- G: $\hat{U}_{\text{prep}}$, then $\hat{U}_\tau^\dagger(x)$; $P_G(x)$ is the probability that every qubit reads zero.
- C: a selector qubit in $|+\rangle$, $\hat{U}_{\text{prep}}$ controlled on it, a Hadamard on it, then $\hat{U}_\tau^\dagger(x)$; $P_C(x)$ is the all-zero probability, the selector included.

Lemma 7. *The all-zero amplitude of $\hat{U}_\tau^\dagger(x)$ applied to $|v\rangle\,|0\rangle$ is $u(x) \cdot v$.*

Proof. It is $\langle 0|\,\hat{U}_\tau^\dagger\,|v,0\rangle = \overline{\langle v,0|\,\hat{U}_\tau\,|0\rangle}$, and the feature maps' all-zero column restricted to their ancilla's zero is the real vector $u(x)$. □

Proposition 19 (The readout identity)**.** *With N the field's dimension on m axes,*

$$4P_C(x) - P_G(x) - \frac{1}{2^m N} = \frac{2}{2^m}\sqrt{\frac{p_G}{N}}\,\tau(x)\cdot\psi,$$

On one axis, $m = 1$,

$$4P_C(x) - P_G(x) - \frac{1}{2N} = \sqrt{\frac{p_G}{N}}\,\langle\tau(x)|\psi\rangle\,. \tag{F.6}$$

Proof. The junk has a non-zero ancilla and contributes nothing to an all-zero outcome, so by Lemma 7 $P_G = (u\cdot g)^2$. In C, before the closing Hadamard the state is $(|0\rangle\,|0\rangle + |1\rangle\,\hat{U}_{\text{prep}}\,|0\rangle)/\sqrt{2}$; after it the selector's zero carries $(|0\rangle + \hat{U}_{\text{prep}}\,|0\rangle)/2$, whose all-zero-ancilla system part is $(e_0 + g)/2$. So $P_C = (u_0 + u\cdot g)^2/4$ and $4P_C - P_G = u_0^2 + 2u_0\,u\cdot g$. Every feature vector has $\tau_0 = 1/\sqrt{N}$, so $u_0 = 1/\sqrt{2^m N}$ and

$$2u_0\,u\cdot g = \frac{2}{2^m}\sqrt{\frac{p_G}{N}}\,\tau\cdot\psi. \qquad \square$$

The interference with e_0 recovers the overlap's sign, which one probability alone cannot. For one axis and a deterministic preparation, $p_G = 1$, this is Wu et al.'s Eq. 36 [30]. The model is $f_Q(x) = \sqrt{\eta}\,\tau(x)\cdot\psi$, so with f^* the left-hand side the field follows from a ratio against the regular datum and needs no p_G, while the scale does:

$$f_Q(x) = f(x_s)\,\frac{f^*(x)}{f^*(x_s)}, \qquad \eta_e = \Big(\frac{2}{2^m}\Big)^2\,\frac{f(x_s)^2\,p_G}{N\,f^*(x_s)^2}.$$

Taking $p_G = 1$ misstates η_e by the factor p_G. C needs $\hat{U}_{\text{prep}}$ controlled on the selector: the rotations take the control, each controlled walk needs only its two reflections controlled on the selector and the signal together (Appendix F.3), and GQSP's global phase becomes a phase gate on the selector, whose loss would scale f^* by $\cos\lambda$ and η_e by $1/\cos^2\lambda$. Simulated, the two circuits reproduce the exact overlap, the field and η_e on R1, R2a and R5a, the first two on one axis and the third on two.

G The Descriptor Construction in Full

G.1 Block Rows in Full

Table G.1 lists every row block of the chain. On one axis the state is $w = (\hat{f}^{(0)}; \ldots; \hat{f}^{(k)}; 0; \ldots; 0)$, one block of N amplitudes per derivative, padded with zero blocks to a power of two so that the block index is a register of n_f qubits, the most significant index of the state. On several axes the field is block 0 and each axis a owns the chain $\partial_a f, \ldots, \partial_a^{k_a} f$ up to the highest order that the equation or a constraint takes along it; a mixed derivative would need a block of its own and is not built. $\tilde{\mathbb{B}}_a$ and $\tilde{\mathbb{D}}_a$ are the banded factors on axis a, tensored with the identity on the other axes.

A Neumann condition reads the derivative block with the zero row $\mathbb{B}(x)$, the same row that

Table G.1: The chain's row blocks: one equation row, banded recurrences, padding and constraints. Each row block of the descriptor's stacked residual, the state blocks it reads and its operator. $\hat{f}^{(m)}$, coefficients of the m-th derivative; $\mathbb{M}_{a_j}$, multiplication by the coefficient a_j (the standard form's, lifted where it lifts); $\mathbb{M}_r$, multiplication by the source r; $\mathbb{D}^{(0)}(x_s)$, collapse row at the regular point x_s; $\tilde{\mathbb{B}}_a$, $\tilde{\mathbb{D}}_a$, the banded factors of $\mathbb{G} = \tilde{\mathbb{B}}^{-1}\tilde{\mathbb{D}}$ along axis a; c_{pad}, padding weight; $\mathbb{B}(x^*)$, zero row at x^*. Constraint rows are the standard form's (Definition 2) with each power of $\mathbb{G}$ replaced by the block that carries the derivative.

Row block	Reads	Content
The equation	the field and every carried derivative	$\sum_j \mathbb{M}_{a_j} \hat{f}^{(j)} - \mathbb{M}_r \mathbb{D}^{(0)}(x_s) \hat{f}^{(0)}$
Recurrence $m \to m+1$ along axis a	blocks m and $m+1$ of a's chain	$-\tilde{\mathbb{D}}_a \hat{f}^{(m)} + \tilde{\mathbb{B}}_a \hat{f}^{(m+1)}$
Padding block b	block b	$c_{\text{pad}} I$, $c_{\text{pad}} = 1$
A constraint on $\partial_a^m f$	block m of a's chain	a zero row $\mathbb{B}(x^*)$, a collapse row, a datum slice (Equation (4.4)) or a ratio row (§ 6.2)

a Dirichlet condition applies to the field block, and no $\mathbb{G}$. A source enters the equation row as in the standard form, through the collapse row on the field block. A first-order equation or system is not carried as a chain but in the mass form of Equation (5.7), whose rows are the standard form's equation rows multiplied by $I_C \otimes \tilde{\mathbb{B}}$, with the standard form's constraint rows; its state is the standard form's, one block per component. The system rows of several fields and of a Carleman lift, and their ratio rows, are given in Appendix H.1.

Proposition 20 (The padding decouples)**.** *After ordering the columns,*

$$R_{\text{desc}} = \begin{pmatrix} R_{\text{phys}} & 0 \\ 0 & c_{\text{pad}} I \end{pmatrix},$$

so H_{desc} is block diagonal, every polynomial in H_{desc} maps the physical blocks to themselves, and the padding's singular values all equal c_{pad}.

Proof. No equation, recurrence or constraint row reads a padding block, and each padding row block is $c_{\text{pad}} I$ on its own block. □

The reference, the gap, the filter and the initial state are therefore the physical subspace's, and $c_{\text{pad}} = 1$ adds one to α_R^2 and nothing to the gap the preparation meets. Over the whole spectrum the padding's singular value can lie below the physical subspace's first excited one, but an initial state on the field block never meets it.

G.2 Variable Coefficients and the Padded Block Count

A variable coefficient $a_j(x)$ enters the equation row through the standard form's multiplication operator $\mathbb{M}_{a_j}$, the lift of Paine et al. (Proposition 12), on exactly the axes where the standard form lifts it: an axis with a non-constant coefficient or a source. The descriptor's equation row therefore applies the same operators as the standard form's equation block, which Theorem 1 needs. A multiplication by a polynomial is banded, with bandwidth the polynomial's degree,

so it compiles to the same bands as $\tilde{\mathbb{B}}$ and $\tilde{\mathbb{D}}$ (Appendix G.6). The square Galerkin form, which truncates the exact product to the first N modes and never lifts, is used only as a control in the gap study of Appendix G.4.

With $K = 1 + \sum_a k_a$ physical blocks, the padded state has

$$\dim w = 2^{\lceil \log_2 K \rceil} \prod_a N_a, \qquad n_f = \lceil \log_2 K \rceil, \tag{G.1}$$

amplitudes, against the standard form's $\prod_a N_a$. On one axis $K = k + 1$:

Order k	Blocks $k+1$	Field qubits n_f	Padded blocks 2^{n_f}	Padding blocks
2	3	2	4	1
3	4	2	4	0
4	5	3	8	3
7	8	3	8	0

A first-order equation takes the mass form and carries no derivative block. The heat equation has $K = 4$, and Laplace's and the wave equation $K = 5$, padded to eight. The padded count equals the unpadded one exactly when K is a power of two and is less than twice it otherwise; the chain's state is at most $2K$ times the standard form's. This is the qubit cost that accompanies the gate counts of § 5.8.

G.3 Kernel Equivalence on Several Axes and Under Rescaling

The proof of Theorem 1 in the body treats one axis without rescaling. Both extensions follow from one identity. Let E map a field's coefficients p to the state whose block of order m along axis a is $\mathbb{G}_a^m p$, with $\mathbb{G}_a$ the differentiation matrix on axis a tensored with the identity on the others, and whose padding blocks are zero. Let S be the diagonal of block weights of § 5.6, equal to $\prod_a \zeta_a^{m_a}$ on a block of order m_a along each axis and to one on the field and padding blocks.

Proposition 21 (Substitution identity)**.** $R_{\mathrm{desc}} S\,(S^{-1} E p) = \big(R_{\mathrm{std}}\, p;\, 0\big)$*, after ordering the rows: the recurrence and padding rows vanish on* Ep*, and every other row block acts on* Ep *as the standard form's row block of the same name acts on* p*.*

Proof. The recurrence of order m along axis a gives $-\tilde{\mathbb{D}}_a \mathbb{G}_a^m p + \tilde{\mathbb{B}}_a \mathbb{G}_a^{m+1} p = (\tilde{\mathbb{B}}_a \mathbb{G}_a - \tilde{\mathbb{D}}_a)\mathbb{G}_a^m p = 0$, and the padding blocks of Ep are zero. The equation row gives $\sum_j \mathbb{M}_{a_j} \mathbb{G}^j p - \mathbb{M}_r \mathbb{D}^{(0)}(x_s) p$ term by term, each term's derivative being read from the block that holds it, which is the standard form's equation block. A constraint on $\partial_a^m f$ applies its row to $\mathbb{G}_a^m p$, which is the standard form's row. The factor S multiplies the columns of each block by its weight and S^{-1} divides the block of Ep by the same weight, so they cancel. □

Theorem 5 (Kernel equivalence, general form)**.** $\ker(R_{\text{desc}}S) = S^{-1}E \ker R_{\text{std}}$*, and* E *is injective, so the two kernels have the same dimension.*

Proof. If $p \in \ker R_{\text{std}}$, Proposition 21 gives $(R_{\text{desc}}S)(S^{-1}Ep) = 0$. Conversely, let $(R_{\text{desc}}S)v = 0$ and $w = Sv$. The padding rows force the padding blocks of w to zero. Along each axis's chain each recurrence gives $w_{(a,m+1)} = \tilde{\mathbb{B}}_a^{-1}\tilde{\mathbb{D}}_a w_{(a,m)} = \mathbb{G}_a w_{(a,m)}$, since $\tilde{\mathbb{B}}_a$ is invertible, so $w = Ew_0$, and Proposition 21 gives $R_{\text{std}}w_0 = 0$. E is injective because its first block is the identity. □

What it does not say. The theorem says that an exact kernel is present in both forms or in neither, with the same dimension; a one-dimensional kernel is one-dimensional in both. It does not say that near-kernels are equally sharp. The ratio of the first excited to the lowest singular value, which the trust threshold of § 7.2 reads, is each form's own and differs between them. For a Carleman lift the kernel is the truncated lift's in both forms, and its uniqueness is the initial-value problem's, not a property of either form (§ 6.5).

G.4 The Gap Mechanism and the Artefacts Ruled Out

Proposition 22 (The quotient on represented states)**.** *On the represented states the Rayleigh quotient of the rescaled descriptor is*

$$\frac{\|R_{\text{desc}}S\,(S^{-1}Ep)\|^2}{\|S^{-1}Ep\|^2} = \frac{\|R_{\text{std}}\,p\|^2}{\sum_{m=0}^{k}\zeta^{-2m}\|\mathbb{G}^m p\|^2}, \tag{G.2}$$

and the j*-th smallest singular value of* $R_{\text{desc}}S$ *is at most the least, over* $(j+1)$*-dimensional subspaces* V *of coefficient vectors, of the largest square root of this quotient on* V*.*

Proof. Proposition 21 gives the numerator, and the denominator is the norm of the represented state, whose block m is $\zeta^{-m}\mathbb{G}^m p$. By the Courant–Fischer theorem the j-th smallest singular value is the least, over $(j+1)$-dimensional subspaces of the state space, of the largest quotient on the subspace; restricting the subspaces to the range of $S^{-1}E$ can only raise it. □

The bound is not tight. States that break the recurrences cost little where $\tilde{\mathbb{B}}$'s least singular value is small, and it falls with N, as about π/N, with its singular vector at the ends of the interval. Measured against the bound at $n = 8$, the descriptor's lowest non-zero singular value is 0.99–1.00 times it on controlled equations with a single zero or none, 0.04 times it on an equation with zeros at both ends, and 0.35–0.72 times it on the two-axis problems at $n = 4$ per axis.

The concentrated states. Let a_k vanish simply at an endpoint. A polynomial of degree N can concentrate within $\mathcal{O}(N^{-2})$ of the endpoint, where the Chebyshev points cluster, and each of its derivatives then grows by a factor $\mathcal{O}(N^2)$. On such a state $\|a_k p^{(k)}\| = \mathcal{O}(N^{-2})\|p^{(k)}\|$ and the lower orders are of the same size, so the square root of the quotient is $\mathcal{O}(N^{-2}\zeta^k)$ against the top block. Unscaled, the lowest non-zero singular value is $\mathcal{O}(N^{-2})$ and the gap $\mathcal{O}(N^{-4})$.

At an interior zero, where the resolution is $\mathcal{O}(N^{-1})$, the gap is $\mathcal{O}(N^{-2})$. At a double zero at an endpoint, $\|a_k p^{(k)}\| = \mathcal{O}(N^{-4})\|p^{(k)}\|$ and the gap is $\mathcal{O}(N^{-8})$. These are heuristics; the rates are measured. Table G.2 gives the unscaled gap on controlled equations whose solutions are polynomials where the coefficient vanishes, so that their kernels are exact, built with the square multiplication.

Table G.2: A zero of the leading coefficient closes the unscaled descriptor's gap at the rate the Chebyshev resolution at the zero predicts. Unscaled descriptor gap at $n = 5$ and $n = 9$, its factor of decrease per added qubit, and the standard form's gap, on seven equations whose leading coefficient is bounded away from zero, has a simple zero, or vanishes at both ends. Planning probes; exact classical computation.

Equation	Leading coefficient	Descriptor, $n = 5 \to 9$	Per qubit	Standard
$f'' + 4f' + 4f$	1	0.458 → 0.458	flat	7.04
$(2+x)f'' + f' - f$	≥ 1	0.434 → 0.432	flat	0.777
$(2+x)f' + f$	≥ 1	1.01 → 1.00	flat	4.90
$(1-x)f'' - xf' + 3f$	zero at $x = 1$	1.5e-5 → 2.2e-10	÷16	2.02
$(1-x)f' + f$	zero at $x = 1$	2.4e-5 → 3.5e-10	÷16	1.6
$(1-x^2)f'' - 2xf' + 20f$	zeros at $x = \pm 1$	9.1e-7 → 8.0e-13	÷33	20.5
$xf'' + (1-x)f' + 3f$	zero at $x = 0$	9.3e-3 → 3.8e-5	÷4	1.85–2.02

The first excited state lies 96–100 % in the top-derivative block, and at $n = 9$ 86–99 % of that block's function lies within 0.02 of the coefficient's zero. Variable coefficients alone do not close the gap; a zero does.

Two axes. The equation constrains a sum of top derivatives. Its lowest modes are equal and opposite in two top blocks (transport, Laplace's equation) or sit in the block the equation weights least (the heat equation's $\partial_x^2 f$ block, weighted 1/25), and only the slice rows penalise them: by $\mathcal{O}(1)$ against $\|f\|$, which is small against the derivative blocks. With R5b's constraints and resolution, the lowest singular value that is not a near-solution falls from 0.29 to 0.033 for transport, from 0.081 to 0.0018 for the heat equation and from 0.10 to 0.0011 for Laplace's equation as n goes from 2 to 4 per axis, while for $f_t + f$, which has one top block, it stays at 0.87–0.86.

The field block's weight. Let P_0 project onto the field block and let the initial state be supported there. Then $\gamma = |\langle \psi_g | \psi_{\text{init}} \rangle| = |\langle P_0 \psi_g | \psi_{\text{init}} \rangle| \leq \|P_0 \psi_g\| = w_f^{1/2}$ by the Cauchy–Schwarz inequality, with $w_f = \|P_0 \psi_g\|^2$ the field block's weight that the build records hold. For an exact kernel, $\psi_g \propto S^{-1} E b$ with b the solution's coefficients, so $w_f = \|b\|^2 / \sum_m \zeta^{-2m} \|\mathbb{G}^m b\|^2$. Amplitude amplification then costs a factor $1/\gamma \geq w_f^{-1/2}$ (§ 7.4).

What it is not. Each candidate artefact was tested:

Candidate	Test	Outcome
The lift	R3a with the square multiplication	collapses faster: 2.6e-11 at $n = 8$, against 2.7e-8 lifted
The padding	lowest singular value against c_{pad}; the excited state's padding weight	at most 0.74, below $c_{\text{pad}} = 1$, on every padded record; no padding weight
The constraints	controlled equations with none	they collapse too
Precision	lowest singular value against the decomposition's floor $\varepsilon\sigma_{\max}$	six orders of magnitude above it at the smallest
The basis	a descriptor carrying $f^{(j)}$ in the ultraspherical basis $C^{(j)}$	the same rates where a_k vanishes
The reference	dense decomposition against the iterative reference	agree to $1.9\,\varepsilon\sigma_{\max}$

The ultraspherical descriptor keeps a flat gap on regular problems with a norm about four times smaller, and on Legendre's equation gains 160-fold at $n = 9$ over the Chebyshev descriptor, the share of the recurrences, but still collapses.

G.5 Block Rescaling

The rule of § 5.6 returns each axis's weight from the equation's coefficients and the resolution alone:

Problem	ζ
One axis, a_k with no zero on $[-1, 1]$	$2(\bar{a}_j/\bar{a}_k)^{1/(k-j)}$, j the lowest order present, $\bar{a}$ the mean of $\lvert a\rvert$
One axis, a_k with a zero on $[-1, 1]$	$N/2$
Several axes	N_a, times $\sqrt{\max_b \bar{c}_b/\bar{c}_a}$ on an axis of top order two, $\bar{c}_a$ the mean magnitude of its top coefficient

Whether a_k vanishes is decided from its roots, to a tolerance loose enough to find a double root. On R2a the rule gives $\zeta = 2\sqrt{4/1} = 4$, on R2c $\zeta = 2\sqrt{400} = 40$, and on R3a, whose leading coefficient vanishes at both ends, $\zeta = N/2$.

Why each. For a regular equation with constant coefficients, $(\bar{a}_0/\bar{a}_k)^{1/k}$ is the geometric mean of the characteristic roots' magnitudes, the product of the roots being $\pm a_0/a_k$; a solution $e^{\lambda x}$ has $\|f^{(m)}\| \approx |\lambda|^m \|f\|$, so dividing block m by ζ^m balances the blocks at the equation's own scale. The factor two is where the cost's minimum over ζ sits on the regular benchmark problems. Where the leading coefficient vanishes, the concentrated states of Appendix G.4 have quotient $\mathcal{O}(N^{-4}\zeta^{2k})$, so for second-order problems $\zeta \propto N$ makes it of order one; half the resolution is the constant the sweep found. On several axes the resolution is used per axis, as for a singular equation, and an axis of top order two is weighted so that its top-order term $\bar{c}_a\zeta_a^2$ balances the others: the wave equation $f_{tt} = c^2 f_{xx}$, with wave speed c, takes (cN, N), where

$(N/2, N/2)$ costs 10^4 to 10^6 times more.

The calibration and its check. The cost measured is $\sqrt{\|H\|/\Delta}/\gamma$, the walk's queries at an ideal encoding times amplitude amplification, up to the logarithms both forms share. Sweeping ζ over powers of two on the benchmark problems set the two constants. The same problems then checked the rule, so the check is in-sample. Table G.3 gives the window of ζ over which the cost stays within twice its least, and Table G.4 the rule's cost against the grid's best.

Table G.3: The block weight need only be right to about a factor of two. Range of the block weight ζ over which the cost $\sqrt{\|H\|/\Delta}/\gamma$ stays within twice its least, at $n = 7$ on one axis and $n = 4$ per axis on two (one weight shown per two-axis problem). $\|H\|$, Hamiltonian norm; Δ, spectral gap; γ, the initial state's overlap; the cost counts the walk's queries at an ideal encoding times amplitude amplification. Planning probes.

	R2a	R2c	R3a	R4a	R4d	R5a	R5b
Window of ζ	2–16	16–64	32–90	23–90	4–16	8–32	8–16

Table G.4: The a priori rule lands near the best block weight on a grid. Cost $\sqrt{\|H\|/\Delta}/\gamma$ at the rule's ζ, over the least cost on a grid of powers of two; Δ, gap on the physical subspace; γ, overlap of the geometric initial state on the field block. A ratio below one is the grid's spacing. Planning probes, in-sample.

Problems	n	Rule's cost / grid's best
R1, R2a–c, R4b–d (regular)	5, 7	0.99–1.17; R2c at $n = 5$: 1.44
R3a, R3b, R4a (a simple zero)	5, 7	1.00–1.04
R3c, R3d (solved for $v = f/\sqrt{1-x^2}$)	5, 7	1.00–1.18 at $n = 5$; 2.07 and 2.22 at $n = 7$
R5a, R5b, R5c	3, 4 per axis	1.00–1.17

A coefficient that dips. $(\delta + x^2)f'' + f = 0$ has no zero but dips to δ. The rule is within 1.3 times the grid's best for $\delta \geq 0.01$, and 2.3–4.7 times at $\delta = 10^{-3}$, where the resolution sees the dip and it behaves like a zero. None of the benchmark problems dips so.

A double zero. Wu et al.'s R3c and R3d multiply the associated Legendre operator by $1-x^2$, so their leading coefficient $(1-x^2)^2$ vanishes twice at each end. Unscaled, the descriptor's gap there falls as N^{-8}. Balancing the concentrated states needs $\zeta \propto N^2$, by which $\|H_{\text{desc}}\|$ has grown as $\zeta^4 = N^8$. On a grid of ζ the best cost of the descriptor grows as about $N^{3.7}$–$N^{3.8}$ over $n = 5$ to 8, against the standard form's $N^{2.2}$–$N^{2.3}$, and at $n = 8$ it is 760–850 times the standard form's; no block weight brings the two level. Solved for $v = f/\sqrt{1-x^2}$, whose leading coefficient has simple zeros, the two forms' costs grow alike, as $N^{1.8}$–$N^{1.9}$ at the rule's weight against $N^{2.0}$. The ground state stays accurate in both forms at the double zero, its error set by the solution's branch point; it is the cost that the double zero raises.

What it changes in the circuit. Scaling the columns of block m by ζ^m multiplies the coefficient c_t of every term whose column block has order m by ζ^m (Algorithm 1, line 4). The terms, their selectors and their gates are unchanged, and $\alpha_A = \sum_t |c_t|$ changes with the

coefficients. A general diagonal equilibration, a different weight for each amplitude, would need its own diagonal block encoding, with $\Theta(\dim w)$ gates, and is excluded from every circuit.

G.6 The Term IR and Its Compilation

The system rows of the descriptor (the equation, the recurrences and the padding; the rank-one constraint rows are encoded apart) form one square operator on the block register and the axes' registers. Each row block sits at a block index of its own: a recurrence at the index of its lower block, the equation at the lowest index left, and each padding row at its own. The term IR writes this operator as

$$A_{\text{sys}} = \sum_t c_t \, P_t \otimes \bigotimes_a F_{t,a}, \tag{G.3}$$

with P_t a Pauli word $X^x Z^z$ on the block register (or, for a system's coupling, a matching, § 6.6), and each $F_{t,a}$ a *band*: a diagonal D followed by the truncated shift that takes column q to row $q - \ell$ and discards it where that row does not exist, with D the identity, one Z_j or a reflection Q_m.

Lemma 8 (An affine band with exceptions)**.** *A diagonal whose entries are $v_q = a + bq$ except at a few columns m, where $v_m = a + bm + d_m$, is*

$$\operatorname{diag}(v) = \Big(a + b\,\frac{N-1}{2} + \sum_m \frac{d_m}{2}\Big) I - b \sum_{j=0}^{n-1} 2^{j-1} Z_j - \sum_m \frac{d_m}{2}\, Q_m :$$

$n + 1$ terms and one reflection for each exception.

Proof. $\Lambda = \frac{N-1}{2} I - \sum_j 2^{j-1} Z_j$ (Equation (5.12), and Equation (E.1)), and $|m\rangle\langle m| = (I - Q_m)/2$. □

Lemma 9 (The affine part is tight)**.** *If a and $a + b(N-1)$ have one sign, the combination of Lemma 8 without exceptions has $\alpha = \max(|a|, |a + b(N-1)|) = \|aI + b\Lambda\|_2$.*

Proof. $\alpha = |a + b\frac{N-1}{2}| + |b|\frac{N-1}{2}$, the mean of the end values plus half their range, which is the larger end value in magnitude when the two share a sign. □

Every band of the descriptor has this form. $\tilde{\mathbb{D}}$'s diagonal is 2Λ except at column 1, where the normalisation of T_0 makes it $2\sqrt{2}$; $\tilde{\mathbb{B}}$'s and a constant multiplication's are constant except at their ends; a variable coefficient's multiplication bands and their truncation at the register's edge give the other exceptions. The columns a shift discards are free, and are fitted to keep the combination shortest.

The block register. Terms whose axis factors agree are merged across the row and column blocks they join, so that the block register carries their pattern as one small matrix F, paid for once; the recurrences share $\tilde{\mathbb{D}}$ in this way. F is expanded in Pauli words, $F = \sum_{x,z} f_{xz} X^x Z^z$

with $f_{xz} = 2^{-n_f} \sum_b (-1)^{z \cdot b} F_{b \oplus x,\, b}$. Where the merged pattern is a single matrix unit on one block, its 2^{n_f} Pauli words are replaced by one term with a flag qubit post-selected to keep that block, at the same α. A large irregular coupling, a Carleman generator above all, is split into matchings instead (§ 6.6).

Proposition 23 (Compilation)**.** *Let PREPARE$_R$ load* $\operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}$ *and PREPARE$_L$ load* $\sqrt{|c_t|/\alpha_A}$ *at each term's joint index of the factored selectors, with* $\alpha_A = \sum_t |c_t|$*. Then the circuit that applies PREPARE$_R$, then SELECT, then PREPARE$_L^\dagger$, post-selected on the selectors, the overflow qubits and the flag, block-encodes* A_{sys} *at* α_A*.*

Proof. On the joint index of term t, the factored SELECT applies the tensor product of the operations its registers hold: the term's block-register word and, on each axis, its diagonal and then its shift, which is the band. A joint index that no term uses has zero amplitude in both preparations. The post-selected block is $\sum_t \sqrt{|c_t|/\alpha_A} \cdot \operatorname{sgn}(c_t)\sqrt{|c_t|/\alpha_A}\, U_t = \sum_t c_t U_t/\alpha_A$. With one preparation and its own inverse the signs would square away, and the block would be $\sum_t |c_t| U_t/\alpha_A$. □

The truncated shift. On the output register widened by one overflow qubit, initially zero, add $-\ell$ modulo twice the register's size. A row $q - \ell$ inside $[0, N)$ lands in the lower half, and one outside it lands in the upper half, which the projection of the overflow qubit on $|0\rangle$ discards. The addition is an increment of the qubits from j upwards for each set bit j of $|\ell|$, each a ladder of multi-controlled X gates, and the overflow qubit is the shift's only ancilla.

The compilation, line by line. Algorithm 1's lines 1 and 2 are the decomposition of each block into bands and of each diagonal into an affine part and its exceptions (Lemma 8). Line 3 gathers terms whose axis factors agree across the rows and columns they join, so that the block register carries their pattern once, as a small matrix expanded in Pauli words, and the recurrences' common $\tilde{\mathbb{D}}$ is paid for once and not k times. Line 4 applies the block rescaling. In lines 5 and 6 the two preparations differ only in the signs, so that $\langle 0|\, \text{PREPARE}_L^\dagger\, \text{SELECT}\, \text{PREPARE}_R\, |0\rangle = \sum_t c_t U_t/\alpha_A$; one preparation and its own inverse would give $\sum_t |c_t| U_t/\alpha_A$ instead. The selectors are factored, each only as wide as the number of distinct values its terms use, and SELECT applies each distinct operation once, controlled on its own register.

Rank-one terms. A source $|\hat{r}\rangle \langle \tau(x_s)| / f(x_s)$ is compiled apart and summed with the bands by a further combination: a matrix unit on the block register, the feature map reading $\tau(x_s)$, and a rotation tree loading $\hat{r}$, with two ancilla and no arithmetic. The constraint rows are encoded the same way, each as its own rows, and stacked into the reflection of § 7.3.

The IR check. The terms are built from the problem, the operator library, the residual's layout, its lifts and its block weights, never from the assembled matrices. Summing them back and comparing the sum with the independently assembled system rows is the IR mode of

operator verification, possible at any n (Table I.1); the compiled circuit is then compared with the terms exactly, or by probe states where it is too large.

G.7 Proof of the Cost Proposition, and the Norms

Ancilla. The registers are those of Appendix G.6: one overflow qubit for each axis that shifts, at most one flag, and the factored selectors. On axis a the diagonal selector indexes the identity, the n_a gates Z_j and the distinct reflections r_a, so it has $\lceil \log_2(1 + n_a + r_a) \rceil$ qubits. The reflections are bounded in number by the problem's bands, and the block-register words and the shifts depend on the equation's structure, not on n, so every other selector has a width fixed by the problem, and the total is $\mathcal{O}(\log n)$. For R2 the reflections are three ($\tilde{\mathbb{B}}$'s diagonal and second superdiagonal at their first entries and $\tilde{\mathbb{D}}$'s first entry), so the diagonal selector has $\lceil \log_2(n + 4) \rceil$ qubits; the shifts are 0, 1 and 2, two qubits; the block-register selector has four; with the overflow qubit and the flag the count is $\lceil \log_2(n+4) \rceil + 8$. It is verified on the compiled circuit of R2's system rows at $n = 2$, 4, 5 and 12.

Gates. SELECT applies each distinct operation once. A Z_j is one gate controlled on $\mathcal{O}(\log n)$ selector qubits; a reflection is one phase controlled on n_a qubits; a shift by a fixed ℓ is $\mathcal{O}(\log |\ell|)$ increments of at most $n_a + 1$ qubits, each $n_a + 1$ multi-controlled X gates of up to n_a controls, $\mathcal{O}(n^2)$ gates with a linear-cost decomposition of each. The block-register operations act on n_f qubits, fixed by the problem, and the preparations are rotation trees over $\mathcal{O}(n)$ joint indices. There are $\mathcal{O}(n)$ distinct operations, the Z_j costing $\mathcal{O}(\log n)$ each and the rest $\mathcal{O}(n^2)$ in all, so the total is $\mathcal{O}(n^2)$.

Exactness. Proposition 23; every term is a unitary, so no part of the encoding is approximate.

The subnormalisation. Take first a second-order chain with polynomial coefficients whose leading coefficient has no zero, at a fixed $\zeta \geq 1$. The recurrences' $\tilde{\mathbb{D}}$ terms share the block-register pattern $\mathrm{diag}(1, \zeta, 0, 0)$, whose Pauli expansion has $\alpha = \frac{1}{2}(|1+\zeta| + |1-\zeta|) = \max(1, \zeta) = \zeta$, its largest entry. With Lemma 9, $\tilde{\mathbb{D}}$'s affine part is encoded at $2(N-1)$, so these terms contribute $2\zeta(N-1)$ to α_A. Every other term is independent of N: $\tilde{\mathbb{D}}$'s exception at column 1, the $\tilde{\mathbb{B}}$ bands weighted by ζ and ζ^2, the coefficients' multiplication bands weighted by at most ζ^2, and the padding. A source's rank-one term has coefficient $\|\hat{r}\| \, \|\tau(x_s)\| / |f(x_s)|$, and $\|\hat{r}\|^2$ grows as N for a fixed smooth source (Equation (4.1)), while $\|\tau(x_s)\|^2 < 2$ (Proposition 7). On the other side, $\zeta\tilde{\mathbb{D}}$ is a block of A_{desc}, so $\|A_{\mathrm{desc}}\|_2 \geq 2\zeta(N-1)$. Hence $\alpha_A / \|A_{\mathrm{desc}}\|_2 \leq 1 + C/N$ without a source and $1 + C'/\sqrt{N}$ with one, with constants that grow with ζ and the coefficients. For the mass form of a scalar equation $f' = af$, $\alpha_A \leq 2(N-1) + \mathcal{O}(1)$ and $\|\tilde{\mathbb{D}} - a\tilde{\mathbb{B}}\|_2 \geq 2(N-1) - |a|(2 + \sqrt{2})$, with the same conclusion. For a chain of order three or more the recurrences' pattern need not be encoded at its largest entry, and no limit is claimed; no benchmark problem has one. Where the leading coefficient vanishes, $\zeta = N/2$ makes $\zeta\tilde{\mathbb{D}}$,

$\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2 a_2$ all of order N^2 in one row block, and terms of comparable size add in ℓ^1 in α_A, so the ratio tends to a constant, measured near two (§ 5.8).

Optimality. An exact combination whose SELECT indexes L distinct operations needs $\lceil \log_2 L \rceil$ selector qubits [42]. Λ has exactly $n+1$ non-zero Walsh coefficients, so a combination of Pauli-Z strings needs $n+1$ terms for it and $\Omega(\log n)$ selector qubits. The descriptor's count is therefore of the least order among exact combinations that expand their diagonals in Pauli strings, and is not claimed against encodings built otherwise.

The norms of Table 5.1. Each row block of R_{desc} is a finite sum of blocks weighted by powers of ζ: $\zeta^m\tilde{\mathbb{D}}$, of norm $2\zeta^m(N-1)$; $\zeta^{m+1}\tilde{\mathbb{B}}$, of norm at most $(2+\sqrt{2})\zeta^{m+1}$; $\zeta^j\mathbb{M}_{a_j}$, of norm at most $\sqrt{2}\,\zeta^j \sup|a_j|$ (Proposition 12); the padding, of norm one; and the rank-one constraint and source rows, of norm $\mathcal{O}(\sqrt{N})$, since $\mathbb{B}(x) = \sqrt{N}\,|0\rangle\langle\tau(x)|$ with $\|\tau(x)\|^2 < 2$. The norm of R_{desc} is at most the sum of these and at least the largest of them. At fixed ζ the largest is $2\zeta^{k-1}(N-1)$, so $\|R_{\text{desc}}\|_2 = \Theta(N)$ and $\|H_{\text{desc}}\| = \Theta(N^2)$. With $\zeta = N/2$ and $k=2$, $\zeta\tilde{\mathbb{D}}$, $\zeta^2\tilde{\mathbb{B}}$ and $\zeta^2\mathbb{M}_{a_2}$ are each $\Theta(N^2)$, so $\|H_{\text{desc}}\| = \Theta(N^4)$; on two axes with $\zeta_a = N_a$ and $k_a = 2$ the same count gives $\Theta(N^4)$; and with $\zeta \propto N^2$ the block $\zeta^2\tilde{\mathbb{B}}$, of norm at least $2\zeta^2$, gives $\Theta(N^8)$.

G.8 One Query's Success

Proposition 24 (One query on a Haar-random state)**.** *One application of a block encoding of R at α_R, post-selected, succeeds on a Haar-random state of dimension D with mean probability $\|R\|_F^2/(D\alpha_R^2)$.*

Proof. The success probability on ψ is $\|R\psi\|^2/\alpha_R^2$, and $\mathbb{E}\,\psi\psi^\dagger = I/D$ for Haar-random ψ, so its mean is $\operatorname{tr}(R^\dagger R)/(D\alpha_R^2)$. □

In the structured standard form the dense $\mathbb{G}^k$ dominates: for $k=1$, $\|\mathbb{G}\|_F^2 \sim N^4/2$ (Proposition 8) against $\alpha_{\mathbb{G}}^2 \sim N^4$ on $D = N$ amplitudes, so the mean falls as $1/(2N)$. In the descriptor the dominant $\tilde{\mathbb{D}}$ has $\|\tilde{\mathbb{D}}\|_F^2 = \sum_{q<N}(2q)^2 \sim 4N^3/3$, up to its first entry, against $\alpha^2 \sim 4N^2$, so the mean stays $\Theta(1)$. The pipeline post-selects no single query, its walk being unitary, so this compares the encodings and is not a cost of the preparation.

H Systems and the Nonlinear Route in Full

H.1 Systems in Full

The rows of R7. R7 is $u' = v$, $v' = -\pi^2 u$ on $x \in [-1,1]$ with $(u,v)(-1) = (0,\pi)$, so $C = 2$, $\mathbb{A} = \left(\begin{smallmatrix} 0 & 1 \\ -\pi^2 & 0 \end{smallmatrix}\right)$ and the interval needs no rescaling ($T/2 = 1$). With the field index the most

significant, the equation rows of Equation (6.1) are

$$\begin{pmatrix} \mathbb{G} & -I \\ \pi^2 I & \mathbb{G} \end{pmatrix} \text{ (standard form)}, \qquad \begin{pmatrix} \tilde{\mathbb{D}} & -\tilde{\mathbb{B}} \\ \pi^2 \tilde{\mathbb{B}} & \tilde{\mathbb{D}} \end{pmatrix} \text{ (mass form)}, \tag{H.1}$$

the second being the first multiplied on the left by $I_2 \otimes \tilde{\mathbb{B}}$. Two fields fill the one-qubit field register, so there is no padding. In general a padding block $b \geq C$ carries the row $\hat{f}^{(b)} = 0$ of weight one and no other row reads it, so the Hamiltonian is block diagonal between the physical and the padding blocks; the padding's eigenvalue is one, and the gap, the reference and the initial state are those of the physical blocks.

R7's data. The anchor is v, with $y_v = \pi$, and $y_\perp = 0$ because $u(-1) = 0$, so the ratio row divided by y_a is $|u\rangle\langle u| \otimes \mathbb{B}(-1)$: the zero datum holds $u(-1)$ to zero exactly. The regular datum is $v(-1) = \pi$, a collapse row on the v block at $x_s = -1$, which fixes the scale and the readout.

The ratio rows' spectrum. For C components and $\mathbb{F}$ of Equation (6.2):

Proposition 25 (Ratio rows)**.** $\ker \mathbb{F} = \operatorname{span}(y)$*, and the singular values of* $\mathbb{F}$ *are* $\|y\|$ *once,* $|y_a|$ *with multiplicity* $C - 2$*, and* 0 *once.*

Proof. If $\mathbb{F}v = 0$ then $v_c = (v_a/y_a)y_c$ for every $c \neq a$, and trivially for $c = a$, so $v \parallel y$; and $\mathbb{F}y = 0$. The $C - 1$ non-zero rows, indexed by $c \neq a$, are $[\, y_a I \mid -u \,]$ with $u = (y_c)_{c \neq a}$ (columns $c \neq a$, then a), so $\mathbb{F}\mathbb{F}^\dagger$ restricted to them is $|y_a|^2 I + uu^\dagger$, with eigenvalues $|y_a|^2$ on $u^\perp$ (multiplicity $C - 2$) and $|y_a|^2 + \|u\|^2 = \|y\|^2$. □

The rows therefore fix the ratios of the components and leave the scale free, as a zero row does, and their weight against the equation is set by the data: the largest singular value is $\|y\|$ and the smallest non-zero one $|y_a|$, so the anchor with the largest $|y_a|$ conditions them best. For a Carleman lift y is the vector of the data's scaled moments (Appendix H.5).

The ratio rows' circuit. Divided by y_a, the rows are

$$\frac{\mathbb{F}}{y_a} \otimes \mathbb{B}(x^*) = \left(I - |a\rangle\langle a| - \frac{\|y_\perp\|}{y_a} \, |\hat{y}_\perp\rangle\langle a| \right) \otimes \mathbb{B}(x^*), \qquad \hat{y}_\perp = y_\perp / \|y_\perp\|, \tag{H.2}$$

three LCU terms, each a field-register operation times the zero row's encoding: the identity; the projector $|a\rangle\langle a|$, a matrix unit placed by X gates around the zero projector; and the rank-one term, the matrix unit $|0\rangle\langle a|$ followed by a rotation tree that prepares $\hat{y}_\perp$ from $|0\rangle$. The subnormalisation is $(2 + \|y_\perp\|/|y_a|)$ times the zero row's, and a datum with $y_\perp = 0$ drops the third term.

H.1.1 The coupling as a sum of matchings

Construction. A matching $\mathcal{M}$ is a set of entries of $\mathbb{A}$ with at most one in each row and column. It acts as a partial permutation of the field register's basis states followed by a diagonal: column c goes to row $\pi(c)$ with weight $v_{\pi(c)}$. The permutation is completed to a permutation of all basis states and built from transpositions, each a cascade of CNOTs around one multi-controlled X, the only gate that takes the selector's control. The weights, divided by the matching's largest magnitude $\alpha_{\mathcal{M}}$, lie in $[-1, 1]$ and are applied as $R_Y(2 \arccos v_r)$ on a flag qubit, whose zero then carries v_r, with weight zero on a row the matching does not reach. The rotations of every matching form one uniformly controlled R_Y on the field register and the matching selector, 2^{n_f+w} CNOTs for all of them, w the selector's width. A matching whose weights share one value is a bare permutation, its sign moved into the coefficient. The coupling is $\sum_{\mathcal{M}} \alpha_{\mathcal{M}} U_{\mathcal{M}}$, block-encoded at $\alpha = \sum_{\mathcal{M}} \alpha_{\mathcal{M}}$.

How many. The matchings are the colour classes of an edge colouring of the bipartite graph whose two sides are $\mathbb{A}$'s rows and columns and whose edges are its non-zero entries. The greedy colouring takes the entries in row order and gives each the least colour that its row and its column do not yet hold. Let δ be the largest number of entries in a row or column. An entry meets at most $\delta - 1$ others in its row and $\delta - 1$ in its column, so at most $2\delta - 1$ colours are used. δ is the least possible, since the entries of one row need distinct colours, and it suffices for a bipartite graph by König's theorem [77]; on every Carleman lift built, the greedy colouring reached δ. A Pauli-word expansion of the same coupling, $\mathbb{A} = \sum_{x,z} f_{xz} X^x Z^z$, pays for every word with a non-zero coefficient, which for an irregular sparse matrix is most of the 4^{n_f} words.

Table H.1 sets the matchings against the tensor layout of Appendix H.4 on the lifts of Burgers' and Navier–Stokes' couplings, each encoded alone.

Table H.1: The matchings are cheaper only on small lifts; the tensor layout's term count does not grow. The Carleman lift's coupling $\mathbb{A}_K$, encoded alone, for Burgers (R9c, $a/\nu = 1$) and the Navier–Stokes vortex pair (R10c, $a/\nu = 10$): in the symmetric layout as a sum of matchings, in the tensor layout as $2K - 1$ terms on its physical states. M, Galerkin modes; K, Carleman order; $\alpha/\|\mathbb{A}_K\|$, subnormalisation over the coupling's norm; CNOTs, per query; –, symmetric layout not built. Every entry verified by IR to 10^{-10}. Classical construction and verification.

			Symmetric layout (matchings)			Tensor layout		
Problem	M	K	matchings	$\alpha/\|\mathbb{A}_K\|$	CNOTs	terms	$\alpha/\|\mathbb{A}_K\|$	CNOTs
R9c	4	3	7	1.29	4656	5	1.24	7998
R9c	16	3	31	1.34	525 436	5	1.12	17 218
R9c	16	4	45	1.50	6 370 468	7	1.13	26 596
R10c	12	2	5	2.25	4022	3	14.8	23 208
R10c	20	3	25	3.94	636 214	5	11.4	89 408
R10c	48	2	33	5.89	326 444	3	19.9	79 044
R10c	80	2	61	7.34	883 336	3	17.1	78 724
R10c	316	2	–	–	–	3	23.5	288 120

The matchings' count and cost grow with the entries of $\mathbb{A}_K$, the tensor layout's term count

stays at $2K-1$ and its CNOTs grow slowly with M, so the matchings are cheaper per query only on the smallest lifts. The tensor layout's subnormalisation is close to the norm on Burgers and loose in two dimensions, where the Fourier fold's α is N.

H.1.2 Realification and the real embedding

Realification. Let $\iota : \mathbb{C}^m \to \mathbb{R}^{2m}$, $z \mapsto (\operatorname{Re} z, \operatorname{Im} z)$, a real-linear isomorphism, and let a complex residual be $R = R_r + \mathrm{i}R_i$ with R_r, R_i real. Then $\iota(Rz) = R_\iota\, \iota(z)$ with

$$R_\iota = \begin{pmatrix} R_r & -R_i \\ R_i & R_r \end{pmatrix}, \tag{H.3}$$

so $Rz = 0$ if and only if $R_\iota \iota(z) = 0$, and $\ker R_\iota = \iota(\ker R)$, of real dimension twice the complex one: the solution's real scale and its phase. The data close both. Imposed as real ratio rows on the realified components (Equation (6.2)), they fix the direction of $\iota(z(x^*))$ in $\mathbb{R}^{2m}$, which fixes the phase as well as the ratios, so an initial-value problem has a one-dimensional real near-kernel, as a real system does. For one mode, $z' = cz$ realifies to Equation (6.3). R7 is $z' = -\mathrm{i}\pi z$ with $z = \pi u + \mathrm{i}v$: the realified state is $(\pi u, v)$, and R7 carries $(u, v) = D^{-1}(\pi u, v)$ with $D = \operatorname{diag}(\pi, 1)$, so its coupling $\left(\begin{smallmatrix} 0 & 1 \\ -\pi^2 & 0 \end{smallmatrix}\right) = D^{-1}\left(\begin{smallmatrix} 0 & \pi \\ -\pi & 0 \end{smallmatrix}\right) D$ is the realification's under a diagonal change of variables, and its kernel the image of the realification's under $D^{-1} \otimes I$.

The real embedding of the Fourier variables. For the fluids, with $\hat{u}_{\pm\mathbf{k}} = (a \mp \mathrm{i}b)/2$ the complex coefficients of $a\cos(\mathbf{k}\cdot\mathbf{x}) + b\sin(\mathbf{k}\cdot\mathbf{x})$, the variables $v = \sqrt{2}\,\hat{u}$ on the wavevectors of both signs are $v = Ew$, with w the real coefficients and E putting $1/\sqrt{2}$ on $\pm\mathbf{k}$ for a cosine and $\mp\mathrm{i}/\sqrt{2}$ for a sine. E is an isometry, $E^\dagger E = I$, and unitary when the real basis holds every kept wavevector, so the complex Fourier–Galerkin operators act on real data through E, and the real system's coupling is $E^\dagger(\cdot)E$. Burgers' sine modes are the case of one axis and sines only.

H.2 The Doubled Space

The doubled residual. On $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, let a term have coefficient c and $k \le 2$ field factors, the i-th carrying the derivative $\mathbb{G}^{j_i}$, and a multiplier of degree at most one; it becomes $c\,\mathbb{n}_1\big(\mathbb{D}^{(0)}(x_s)^{\otimes(2-k)} \otimes \mathbb{G}^{j_1} \otimes \cdots\big)$ for a constant multiplier and the same with $\mathbb{n}_x$ for the multiplier x. A linear term is thereby carried by one copy of the collapse row, which on ψ is the constant $f_q(x_s)/f(x_s)$, and a source by two. An invariant constraint becomes $\mathbb{n}_1\big(\mathbb{D}^{(0)}(x_s) \otimes \mathbb{B}(x^*)\mathbb{G}^j\big)$, and the readout is $\mathbb{n}_1(\mathbb{D}^{(0)} \otimes I)$, with $f_Q = \eta f_q$ ($p = 1$ in $f_Q = \eta^{1/p} f_q$). The equation block maps the N^2 amplitudes of Ψ to the $2N$ coefficients of the folded register on $n+1$ qubits.

The kernel bound. The equation block has $2N$ rows, so rank at most $2N$, and each constraint block is a fold of a Kronecker product of two rank-one rows, so rank at most one. With m invariant constraints, $\operatorname{rank} R \le 2N + m$ and $\dim\ker H_\otimes = \dim\ker R \ge N^2 - 2N - m$, which

is at least $N^2/2 = 2^{2n-1}$ once $N \geq 8$ for $m \leq 2$. At $n = 2$ the bound is weaker than Wu et al.'s count, which the records meet there by measurement. Table H.2 gives the measured dimensions; at $n = 2$ to 4 the equation blocks have rank $2N - 3$ (R6a) and $2N - 1$ (R6b), each problem with one constraint.

Table H.2: The doubled space's kernel grows as N^2. Kernel dimension of the doubled-space Hamiltonian $H_\otimes$ for R6a ($4f'' + 2(f')^2 + f = 0$) and R6b ($f'' - 2f^2 + x = 0$), against the doubled state's dimension N^2 and Wu et al.'s lower bound 2^{2n-1}; $N = 2^n$. Classical computation (SVD of the stacked residual).

n	N^2	2^{2n-1}	R6a	R6b
2	16	8	10	8
3	64	32	50	48
4	256	128	226	224
5	1024	512	962	960

Higher degree. § 3.4 states the published construction as quadratic only. A term of degree p in the unknown needs p copies, $|\Psi\rangle = |\psi\rangle^{\otimes p}$, of dimension N^p, its factors folded back into one Chebyshev series by the p-fold product of Proposition 31 (a cubic term by a tripled space, a quartic by a quadrupled one). The register therefore grows with the degree, a limitation also noted in the concurrent work of Paine [31]. The argument of Proposition 4 should carry over, since the corresponding kernel is again a subspace and the p-fold product states are not; this thesis does not use the doubled space beyond $p = 2$. Its alternative for every degree is the Carleman lift of § 6.5.

H.3 Proof of Proposition 4

Proof. (1) As a matrix, $u \otimes u$ is $uu^\top$, of rank one. For independent u and v, $uu^\top + vv^\top$ has rank two, so the sum of two product states is not of the form $w \otimes w$, and the product states are not closed under addition. A kernel is a linear subspace, so a kernel containing $u \otimes u$ and $v \otimes v$ contains their sum, and no kernel is exactly the set of product states.

(2) Write $H_\otimes = \sum_\lambda \lambda \Pi_\lambda$ with $\Pi_0 = \Pi$. Then $P_d(H_\otimes)\psi_{\text{init}} = \Pi\psi_{\text{init}} + \sum_{\lambda>0} P_d(\lambda)\Pi_\lambda\psi_{\text{init}}$, and the second term has norm at most $\max_{\lambda>0} |P_d(\lambda)| \, \|\psi_{\text{init}}\| \to 0$. Since $\Pi\psi_{\text{init}} \neq 0$, normalising gives Equation (6.5). The imaginary-time filter has $P(0) = 1$ and $\max_{\lambda>0} e^{-\beta\lambda} = e^{-\beta\lambda_1} \to 0$; the minimax filter takes the value one at the ground state and at most ε on the excited spectrum, and $\varepsilon \to 0$ as its degree grows at a fixed edge.

(3) For $H_\otimes, P \succeq 0$, $\langle\phi, (H_\otimes + P)\phi\rangle = 0$ if and only if $\langle\phi, H_\otimes\phi\rangle = \langle\phi, P\phi\rangle = 0$, so $\ker(H_\otimes + P) = \ker H_\otimes \cap \ker P$. For this to be $\text{span}\{\psi_* \otimes \psi_*\}$, P must vanish on $\psi_* \otimes \psi_*$ and on no other direction of $\ker H_\otimes$, a condition that refers to ψ_*. □

The scope is the one stated: a penalty built without knowledge of the solution. A penalty that knows the solution isolates it, and is answer-dependent.

H.4 The Carleman Lift

The symmetric layout. A polynomial system $u' = \sum_{p\geq 1} F_p u^{\otimes p}$ on M components, F_p of shape $M \times M^p$, is carried on the monomials $z_\alpha = \prod_i u_i^{\alpha_i}$ with $1 \leq |\alpha| \leq K$, once each: $\sum_{k=1}^{K} \binom{M+k-1}{k}$ components, level by level, the fields first. Differentiating a monomial,

$$z'_\alpha = \sum_i \alpha_i u^{\alpha - e_i} u'_i = \sum_i \alpha_i \sum_p \sum_{j_1,\dots,j_p} (F_p)_{i,j_1\dots j_p}\, z_{\alpha - e_i + e_{j_1} + \cdots + e_{j_p}}, \tag{H.4}$$

and $\mathbb{A}_K$ keeps every term whose monomial has degree at most K. A linear system is its own lift, $\mathbb{A}_1 = F_1$. For R8, $y' = -y + y^2$, before the time axis's factor $T/2$,

$$\mathbb{A}_3 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & -3 \end{pmatrix}. \tag{H.5}$$

Proposition 26 (The lift's spectrum)**.** *For a quadratic system with F_1 diagonalisable, $\mathbb{A}_K$ is block upper triangular in its levels, and its eigenvalues are the sums $\lambda_{i_1} + \cdots + \lambda_{i_k}$, $1 \leq i_1 \leq \cdots \leq i_k \leq M$, $1 \leq k \leq K$, of the eigenvalues of F_1.*

Proof. F_1 maps a monomial to monomials of its own degree and F_2 to degree one higher, so level k reads levels k and $k+1$ only. In F_1's eigencoordinates the diagonal block of level k acts on a monomial v^α as $\sum_i \alpha_i \lambda_i v^\alpha$. The change of coordinates is linear and maps the monomials of degree k to polynomials of degree k, so it is a similarity on each level. □

Proposition 27 (The level scale)**.** *Carrying $w = u/s$ multiplies F_p by s^{p-1}, and the lifted generator becomes $D^{-1}\mathbb{A}_K D$, D the diagonal equal to s^k on level k. The truncated solution, the truncation error and ρ are unchanged.*

Proof. $z(u) = Dz(w)$, so $z(w)' = D^{-1}\mathbb{A}_K D z(w)$, whose entry (α, β) is multiplied by $s^{|\beta|-|\alpha|} = s^{p-1}$ for the terms of F_p. A similarity maps solutions to solutions, and ρ gains $1/s$ from the data and s from F_2. □

With $s = \max(\|u_0\|, 1)$, level k of the data's moments has norm $(\|u_0\|/s)^k \leq 1$. Unscaled, data of norm above one grow as $\|u_0\|^k$ with the level, the ground state's weight moves from the fields to the top level, where no initial state independent of the answer reaches it, and the moments inflate the ratio rows' subnormalisation. The readout is scaled back to the physical datum.

The tensor layout. Level k is carried as the whole tensor $u^{\otimes k}$, M^k components in k slots, slot 0 the most significant.

Proposition 28 (The tensor layout's terms)**.** *In the tensor layout the block of $\mathbb{A}_K$ from level*

$k+p-1$ *to level* k *is*

$$\big(\mathbb{A}_K^{\otimes}\big)_{k,\,k+p-1} = \sum_{j=0}^{k-1} I^{\otimes j} \otimes F_p \otimes I^{\otimes(k-1-j)}, \tag{H.6}$$

a sum of terms "F_p *on slots* $j, \dots, j+p-1$*", one per degree* p *and first slot* j *with* $j+p \le K$*:* $\sum_p (K-p+1)$ *terms,* $2K-1$ *for a quadratic field, whatever* M*. With* V *the embedding of the symmetric layout in the tensor layout,* $\mathbb{A}_K^{\otimes} V = V \mathbb{A}_K$*.*

Proof. $\frac{d}{dt} u^{\otimes k} = \sum_j u^{\otimes j} \otimes u' \otimes u^{\otimes(k-1-j)}$ and $u' = \sum_p F_p u^{\otimes p}$, so slot j's factor F_p contracts slots j to $j+p-1$ of level $k+p-1$ into slot j, and one term (p, j) serves every level that occupies its slots. For the embedding, both sides map the moments $z(u)$ to the truncated derivative of $(u, u^{\otimes 2}, \dots, u^{\otimes K})$; the moments over all u span the monomial space, distinct monomials being linearly independent functions, so the identity holds as matrices. □

The individual terms are not equal on symmetric tensors; only their sum is the symmetric layout's. A native complex preparation in this layout would also need a phase on the collapse row and a complex state loader for the data's moments, neither of which is built.

The tensor layout's circuit. It is built for a Fourier–Galerkin equation in structured form, $u' = L_1 u + \sum \mathrm{out}\big((\mathrm{left}\, u) \cdot (\mathrm{right}\, u)\big)$, every operator a Fourier multiplier, in the variables $v = \sqrt{2}\,\hat{u}$ of § H.1.2. The register is K slots of $d\,m$ qubits, each holding a wavevector's components in two's complement with $2^m \ge 3 \max|k_d| + 1$, so that the product of two kept modes never aliases onto a kept mode, and a level register of $\lceil \log_2(K+1) \rceil$ qubits. A level-k state occupies slots 0 to $k-1$ with kept wavevectors and leaves the others at the zero wavevector, which no truncation keeps: these are the physical states. F_1 on slot j is its multiplier's magnitudes on the kept wavevectors, as one uniformly controlled R_Y on a flag (α the largest magnitude), times the projector onto the levels above j; a symbol $c \prod_d k_d^{a_d} |\mathbf{k}|^{2e}$ has the phase $\mathrm{i}^{\sum_d a_d}$, which joins the term's weight. F_2 on slots j and $j+1$ is the projector onto the levels from $j+2$, the Fourier fold of its input multipliers (Appendix H.6), the output multiplier, which is the Galerkin truncation, and a cyclic shift of the slots above j with the level lowered by one. The terms are combined with weights $T/2$ and, for F_2, $s/\sqrt{2}$. The multipliers vanish off the truncation, so the encoding maps physical states to physical states only, with no leakage. It is verified exactly on Burgers' lift on four modes at $K = 2$, to 1.7×10^{-15} with leakage 1.1×10^{-14}, and by IR on every truncation where IR was run, to 316 modes at $K = 2$ on 100 172 physical states; at $K = 3$ on 148 and 316 modes it is a construction record only.

Where the lift is exact.

Proposition 29 (An exact lift)**.** *If the data lie in a subspace* S *that* F_1 *maps into itself and on which* $F_2(u \otimes u) = 0$*, the truncated lift's first level is the solution exactly, at every* K*.*

Proof. The solution is $u(t) = e^{F_1 t} u_0 \in S$, with $F_2(u \otimes u) = 0$ throughout. In the tensor layout, which embeds the symmetric one (Proposition 28), the top level evolves by the Kronecker sum

of F_1 alone, so from $u_0^{\otimes K}$ it is $u(t)^{\otimes K}$. If level $k+1$ is $u(t)^{\otimes(k+1)}$, the F_2 terms feeding level k apply F_2 to $u \otimes u$ in two adjacent slots, which is zero, so level k evolves by F_1's Kronecker sum and is $u(t)^{\otimes k}$. Down to the first level, $z_1 = u(t)$. □

The Taylor–Green vortex is this case (Appendix H.7).

H.5 The Time Axis, Uniqueness and the Error

Rows and data. A system or a lift on $t \in [0, T]$ is carried on $\xi = 2t/T - 1$, and its equation rows are Equation (6.1) with $\mathbb{A}$, or $\mathbb{A}_K$, holding the factor $T/2$. Its data are the ratio rows $(\mathbb{F}/y_a)\otimes\mathbb{B}(-1)$ at $\xi = -1$ with y the scaled moments $z(u_0/s)$, imposing $z_c(-1) = (y_c/y_a)\, z_a(-1)$. On a padding block $\mathbb{F}/y_a$ is the identity, whose rows meet only that block, so the padding still decouples. The two forms share these rows and differ by the invertible $I \otimes \tilde{\mathbb{B}}_\xi$ on the equation rows, so they share their kernel (Theorem 1).

Proposition 30 (A one-dimensional near-kernel)**.** *On $C^1([-1,1])^C$ the solutions of $z' = \mathbb{A}z$ with $z(-1) \in \operatorname{span}(y)$ are $\{c\, e^{\mathbb{A}(\xi+1)} y : c \in \mathbb{R}\}$, a one-dimensional space.*

Proof. A linear initial-value problem has exactly one solution for each initial value, and the ratio rows leave the initial value one degree of freedom. □

The residual is this operator restricted to polynomials of degree below N_t. Its smallest singular value is the polynomial residual of the true solution, which falls spectrally in N_t for an analytic solution, and the rest of its spectrum approximates the continuum operator's, which has no other kernel. The near-kernel is therefore one-dimensional in both forms and for every $\mathbb{A}$, including a lift whose truncation does not converge: the uniqueness is the initial-value problem's, which the lift carries. Every Carleman build in the records has a one-dimensional kernel; the few not trusted lie at $n_t = 3$, where the time axis does not yet resolve the solution.

The error in two parts. With u the reference, $z^{(K)}$ the exact solution of the truncated lift from the data's moments, and $\hat{u}$ the field decoded from the prepared state, the triangle inequality gives Equation (6.8), both terms on the comparison grid. The lift's term is computed by exponentiating $\mathbb{A}_K$ and does not depend on n_t; the time axis's falls spectrally with n_t. The field's error therefore falls with n_t until the lift's floor and then holds.

H.6 Product Folds

A product of p fields on n qubits has degree at most $p(N-1) < 2^\ell N$, $\ell = \lceil \log_2 p \rceil$, so it is represented exactly on $n + \ell$ qubits, where it is the pointwise product of the fields' values at the nodes. Let $\mathbb{U}$ be the Chebyshev–Gauss transform on $n + \ell$ qubits (coefficients to node values, orthogonal), E the zero-padding from n to $n + \ell$ qubits, and $\mathcal{D}_p = \sum_i \langle i|^{\otimes p}$ the map that keeps the diagonal.

Proposition 31 (The nodal fold is exact and tight)**.** *The fold*

$$\mathbb{m}_1^{(p)} = 2^{p\ell/2}\,\mathbb{U}^\top \mathcal{D}_p \left(\mathbb{U}E\right)^{\otimes p} \tag{H.7}$$

satisfies $\langle\tau(x)|_n^{\otimes p} = \langle\tau(x)|_{n+\ell}\ \mathbb{m}_1^{(p)}$ *exactly. For* $p = 2$ *it is block-encoded at* $\alpha = 2$*, and* $\|\mathbb{m}_1^{(2)}\|_2 = 2\sqrt{\lambda_{\max}(Q \circ Q)} \le 2$*, with* $Q = \mathbb{U}E(\mathbb{U}E)^\top$ *the projector onto the padded fields' node values and* $\circ$ *the entrywise product.*

Proof. The normalised bases on n and $n+\ell$ qubits differ by $\varphi_k^{(n)} = 2^{\ell/2}\varphi_k^{(n+\ell)}$, so a field with coefficients b on n qubits has the values $2^{\ell/2}\,\mathbb{U}Eb$ at the $2^\ell N$ nodes. A product of p such fields is represented exactly on $n+\ell$ qubits and its node values are the pointwise product $2^{p\ell/2}\mathcal{D}_p \bigotimes_i (\mathbb{U}Eb_i)$; $\mathbb{U}$ is orthogonal, so $\mathbb{U}^\top$ returns the coefficients, which is the first claim. (The same map follows from repeated product-to-sum, $T_jT_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$.) For $p = 2$, $\ell = 1$, the circuit applies $\mathbb{U}$ to both copies, each padded by one qubit, copies the first into the second with $n+1$ CNOTs, so that the second is zero exactly on the diagonal, post-selects it on zero, and applies $\mathbb{U}^\dagger$ to the first: a block encoding of $\mathbb{m}_1/2$, so $\alpha = 2$. For the norm, $\mathbb{U}^\top$ is orthogonal, so $\|\mathbb{m}_1\| = 2\|Y\|$ with $Y = \mathcal{D}_2(\mathbb{U}E \otimes \mathbb{U}E)$, and $YY^\top = \mathcal{D}_2(Q \otimes Q)\mathcal{D}_2^\top$ has entry (i,j) equal to Q_{ij}^2. $Q \circ Q$ has non-negative entries and row sums $\sum_j Q_{ij}^2 = Q_{ii} \le 1$, so its largest eigenvalue is at most one. □

The norm rises with n, from 1.58 at $n = 2$ towards about 1.665 (computed to $n = 12$), so $\alpha = 2$ is within 1.21 of it at every n: the encoding is tight to a constant and not optimal. $\mathbb{m}_x$, the fold with the multiplier x, multiplies by the node values $\bar{x}_m = \cos\theta_m$ before $\mathbb{U}^\top$; since θ_m is linear in m, $\mathrm{diag}(\bar{x}) = (D + D^\dagger)/2$ for the diagonal phase $D = \mathrm{diag}(e^{\mathrm{i}\theta_m})$, one phase gate per qubit combined on one ancilla, and α stays 2. The circuit's ancillas are the two padding qubits and $n-1$ spare qubits that return the register to zero for stacking. $\mathbb{U}$ is the discrete cosine transform, a dense gate counted apart and budgeted as a cited primitive [78]; it is not built gate by gate.

The Fourier fold. The tensor layout's fold is the circular convolution $(u{*}v)_{\mathbf{k}} = \sum_{\mathbf{p}+\mathbf{q}\equiv\mathbf{k}} u_{\mathbf{p}}v_{\mathbf{q}}$, which is $\mathrm{QFT}^\dagger\mathcal{D}_2(\mathrm{QFT} \otimes \mathrm{QFT})$ times $N^{1/2}$ for each axis: the QFT takes both copies to grid values, a CNOT from each qubit of the first copy to the second leaves the second at zero exactly where the grid points agree, the second copy is post-selected, and the inverse QFT returns the first to coefficients. Its α, $N^{1/2}$ per axis, is the convolution's own norm on the register, since nothing restricts its input. The QFT is built.

H.7 Fourier–Galerkin Operators

Burgers. On $x \in [0, 2\pi)$, Equation (6.9) from $u(x,0) = a\sin x$ keeps an odd solution odd (u_x is even, uu_x and u_{xx} odd), so $u = \sum_k b_k(t)\sin kx$.

Proposition 32 (Burgers' Galerkin system)**.** *Truncated to $k \le M$,*

$$b_k' = -\nu k^2 b_k - \frac{k}{4}\Big[\sum_{j+l=k} b_j b_l - \sum_{|j-l|=k} b_j b_l\Big], \qquad j,l \le M, \tag{H.8}$$

the sums over ordered pairs.

Proof. $u^2 = \frac{1}{2}\sum_{j,l} b_j b_l\big(\cos(j-l)x - \cos(j+l)x\big)$, so $-(u^2/2)_x = \frac{1}{4}\sum_{j,l} b_j b_l\big((j-l)\sin(j-l)x - (j+l)\sin(j+l)x\big)$. Its coefficient of $\sin kx$ takes $-k/4$ from each pair with $j+l=k$ and $+k/4$ from each pair with $j-l=\pm k$, since $(-k)\sin(-kx) = k\sin kx$. With $\nu u_{xx} \mapsto -\nu k^2 b_k$ and the projection onto $k \le M$, this is the system. □

So $F_1 = \mathrm{diag}(-\nu k^2)$ and F_2 is the bracket, each times $T/2$ on the time axis. The data $b(0) = (a, 0, \dots, 0)$ enter as ratio rows anchored on b_1, which also carries the regular datum. The least damped mode is $k=1$, so $\rho = a\|F_2\|/\nu$, the factors $T/2$ cancelling, and $\|F_2\|$ grows with M. The reference for the lift and the time axis is the Galerkin system's IVP, integrated with its exact sparse Jacobian, which is stiff at sixteen modes (νk^2 up to 256ν). The truncation's own error is measured against the Cole–Hopf solution [79, 80]: with $\beta_B = a/(2\nu)$,

$$u(x,t) = \frac{4\nu \sum_{m\ge1} m\, I_m(\beta_B)\, e^{-\nu m^2 t} \sin mx}{I_0(\beta_B) + 2\sum_{m\ge1} I_m(\beta_B)\, e^{-\nu m^2 t}\cos mx}, \tag{H.9}$$

from $u = -2\nu\,\partial_x \ln\varphi$, which turns Burgers' equation into the heat equation with $\varphi(x,0) = e^{-\beta_B} e^{\beta_B \cos x}$. The runs, all over $T=1$, are R9a to R9e at $a/\nu = 0.03$, 0.3, 1, 3 and 10, each on $M = 4$, 8 and 16 modes.

Navier–Stokes. Equation (6.10) is carried on $\cos(\mathbf{k}\cdot\mathbf{x})$ and $\sin(\mathbf{k}\cdot\mathbf{x})$ over the half-plane $k_x > 0$, or $k_x = 0 < k_y$, with $0 < |\mathbf{k}|^2 \le k_c^2$: $k_c^2 = 4$, 5, 9, 16 and 25 keep $M = 12$, 20, 28, 48 and 80 modes, and 49 and 100, used for the tensor layout's coupling alone, keep 148 and 316. The mean vorticity is invariant and zero, and the stream function is eliminated, $\hat\psi_{\mathbf{k}} = \hat\omega_{\mathbf{k}}/|\mathbf{k}|^2$. With φ_a a basis function of wavevector $\mathbf{p}$, φ_b one of $\mathbf{q}$, and $\mathbf{p}\times\mathbf{q} = p_x q_y - p_y q_x$,

$$J\big(\varphi_a/|\mathbf{p}|^2, \varphi_b\big) = \sigma_a\sigma_b\,\frac{\mathbf{p}\times\mathbf{q}}{|\mathbf{p}|^2}\, h_a(\mathbf{p}\cdot\mathbf{x})\, h_b(\mathbf{q}\cdot\mathbf{x}), \tag{H.10}$$

with $\sigma = -1$, $h = \sin$ for a cosine and $\sigma = 1$, $h = \cos$ for a sine, since $\nabla\varphi_a = \sigma_a \mathbf{p}\, h_a$ and $J(f,g) = \nabla f \times \nabla g$. The product $h_a h_b$ is a sum of the modes $\mathbf{p}\pm\mathbf{q}$, each folded into the half-plane, and the truncation keeps those in the set; every coefficient is rational and computed in exact arithmetic. Mode c's equation is $w_c' = -\nu|\mathbf{k}_c|^2 w_c + \sum_{a\le b} c_{cab} w_a w_b$, each coefficient times $T/2$, with $\nu = 1$; the data enter as ratio rows anchored on the mode with the largest datum, which carries the regular datum.

Two properties follow. *Modes of equal $|\mathbf{k}|$ do not interact.* In the complex basis the Jacobian's

coefficients are $\widehat{J}_{\mathbf{k}} = -\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p}\times\mathbf{q})\,\hat{\psi}_{\mathbf{p}}\hat{\omega}_{\mathbf{q}}$, and with $\hat{\psi}_{\mathbf{p}} = \hat{\omega}_{\mathbf{p}}/|\mathbf{p}|^2$, symmetrised over $\mathbf{p}\leftrightarrow\mathbf{q}$,

$$\widehat{J}_{\mathbf{k}} = -\frac{1}{2}\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p}\times\mathbf{q})\big(|\mathbf{p}|^{-2} - |\mathbf{q}|^{-2}\big)\hat{\omega}_{\mathbf{p}}\hat{\omega}_{\mathbf{q}}, \tag{H.11}$$

which vanishes for $|\mathbf{p}| = |\mathbf{q}|$; the real basis is a unitary change of the complex one. *Without viscosity the truncation conserves the energy and the enstrophy exactly,*

$$E = \tfrac{1}{4}\textstyle\sum_m w_m^2/|\mathbf{k}_m|^2, \qquad Z = \tfrac{1}{4}\textstyle\sum_m w_m^2. \tag{H.12}$$

With P the projection onto the kept modes, $Z' \propto \langle\omega, PJ(\psi,\omega)\rangle = \int J(\psi,\omega^2/2) = 0$ and $E' \propto \langle\psi, PJ(\psi,\omega)\rangle = \int J(\psi^2/2,\omega) = 0$, because $J(f,g) = \partial_x(fg_y) - \partial_y(fg_x)$ is a divergence, whose integral over the periodic square vanishes.

Initial conditions. The Taylor–Green vortex $\omega_0 = 2\sin x\sin y = \cos(x-y) - \cos(x+y)$ occupies two modes with $|\mathbf{k}|^2 = 2$, whose coupling vanishes, so $\omega = \omega_0 e^{-2\nu t}$ exactly in any truncation that holds them, and in the equation itself. Its data lie in a subspace that F_1 preserves and on which F_2 vanishes, so its lift is exact at every order (Proposition 29): R10a validates the lift, the time axis and the preparation at a known answer. The counter-rotating pair is $\pm e^{-|\mathbf{x}-\mathbf{c}|^2/(2\sigma^2)}$ with $\sigma^2 = \frac{1}{4}$ at $\mathbf{c} = (3\pi/4,\pi)$ and $(5\pi/4,\pi)$, periodised and projected exactly onto the modes: it has no cosine part, and its sine coefficients are proportional to $(-1)^{k_x+k_y+1}\sin(\pi k_x/4)\,e^{-|\mathbf{k}|^2/8}$, so modes with $k_x \equiv 0 \pmod 4$ have zero datum. It spans several $|\mathbf{k}|$, so its triads are active. It is scaled to $\|w_0\| = a$ and run at $a/\nu = 3$, 10 and 30 (R10b to R10d) over $T = 1$, and at $a/\nu = 30$ also over $T = 1/8$. The reference for the lift and the time axis is the Galerkin system's IVP, with its exact sparse Jacobian; the truncation's own error is measured against a pseudo-spectral DNS of Equation (6.10) from the same data, whose slope inside the truncation is the Galerkin system's.

I Verification Protocol

I.1 Modes, Sizes and Tolerances

A block encoding (U,α) of an operator X is exact when $\alpha\,\Pi_{\text{out}}U\Pi_{\text{in}} = X$, the projectors fixing the ancilla at zero. Every mode of Table I.1 compares that block, or its action on vectors, with a target built from the problem by the operator library and the residual's assembly, never from the compiler that produced the circuit; a target rebuilt from the circuit's own parts would check the compiler against itself. The pipeline's usual target is the stacked reflection's block $2H/\alpha_R^2 - I$ with H formed from the assembled residual; the atoms (the derivative, a lift, a feature-map row) are compared with their operators in the same way.

Exact. The whole block is reconstructed, one statevector evolution per input basis vector, and its largest entrywise error $\|E\|_{\max}$ is compared with $10^{-10}\|X\|_2$. An $r\times c$ block within that

Table I.1: Every circuit is compared with an independently assembled operator. Each verification mode by what it compares, where it is used and its tolerance. X, the target: the reflection's block $2H/\alpha_R^2 - I$ or an atom's operator; α, the circuit's subnormalisation; ψ, an input state; IR, the compiled term list. The approximation mode applies only to Wu et al.'s approximate encoding of $\mathbb{G}$, which enters resource estimates alone.

Mode	Comparison	Used on	Tolerance
Exact	the whole block, one statevector evolution per input column, entry by entry	circuits whose input columns it can afford, to 25 qubits	max-abs $\leq$ $10^{-10}\|X\|_2$
Probe	4 structured and 16 Haar-random inputs against sparse products $X\psi/\alpha$	every other circuit of at most 24 qubits	relative $\leq 10^{-9}$ per input
IR	the compiled terms, summed to a sparse matrix, against the assembled operator	lifts and couplings at any size	relative max-abs $\leq 10^{-10}$
Approx.	Wu et al.'s $\mathbb{G}$: entry error against its bound	the sizes the exact mode reaches	reported, not gated

tolerance entry by entry has $\|E\|_2 \leq \sqrt{rc}\,\|E\|_{\max}$, so the mode certifies the circuit completely at that size. Its cost is c evolutions of 2^q amplitudes.

Probe. The circuit is applied to 4 structured vectors and 16 Haar-random ones, each one evolution at any size, and compared with a sparse product of the target, which is never formed densely; a probe v passes when $\|Ev\| \leq 10^{-9}\max(1, \|Xv\|)$, the random probes seeded by the run.

Proposition 33 (What random probes see)**.** *For ψ Haar-distributed on dimension D,*

$$\mathbb{E}\,\|E\psi\|^2 = \frac{\|E\|_F^2}{D}.$$

If the error lies in one column, $E = \delta\, w\, e_j^\dagger$ with $\|w\| = 1$, then $\|E\psi\| = \delta|\psi_j|$ and $D|\psi_j|^2$ has mean one.

Proof. $\mathbb{E}\,\psi\psi^\dagger = I/D$, so $\mathbb{E}\|E\psi\|^2 = \mathrm{tr}(E^\dagger E)/D$; for the column error $\|E\psi\|^2 = \delta^2|\psi_j|^2$. □

Random probes therefore certify the root-mean-square error $\|E\|_F/\sqrt{D}$, and an error of size δ confined to one column is seen at about $\delta/\sqrt{D}$: caught with certainty only when $\delta \gtrsim 10^{-9}\sqrt{D}$. The probe mode is evidence of correctness, not a proof of it. Unlike the exact mode, whose cost grows with the number of input columns, it costs twenty evolutions at any size, so it runs on every circuit of at most 24 qubits that the exact mode does not cover.

IR. The compiled terms of a descriptor residual, a coupling or a lift are summed back into a sparse matrix and compared with the operator assembled from the problem, entry by entry relative to its norm, at tolerance 10^{-10}, at any size and with no simulation (Appendix G.6). The two sides are independent constructions, so the comparison is not a tautology. What it leaves open is the step from terms to gates, which the exact and probe modes check where a circuit

can be simulated. Together they chain: the assembled operator equals the terms at every n, and the terms equal the circuit at every simulable size. The compiler is structural, one atom per term, so the circuit at a larger size is the same construction, but it is not independently verified there.

Approximation. Wu et al.'s approximate encoding of $\mathbb{G}$ (Appendix D.1) is compared with $\mathbb{G}$ and its entrywise error reported beside the bound of Proposition 13; the mode never fails, since the encoding is approximate by design and enters resource estimates only.

Circuits wider than the probe ceiling are construction-only records: their subnormalisation, register split and gate counts are recorded, with no verdict.

I.2 Structured Probes

The four structured probes target errors that random vectors dilute. A defect that lives in one column is seen at full strength by the basis vector that selects it and at about $\delta/\sqrt{D}$ by any other. The first basis vector is the T_0 mode and the start of every chain of shifts; the last is where a truncated shift must not wrap round to the start of the register; the middle one, index $D/2$, sets only the most significant qubit, the edge between the two halves of a register and between the blocks of a padded state, where block offsets and axis orders go wrong; the uniform vector tests the sum of all columns at once. A defect confined to columns that no structured probe selects is caught only by the random probes, with the sensitivity of Proposition 33. The unit tests of each atom complement them by checking it exactly at small n.

I.3 Per-Atom Checks

Every atom is verified on its own before it is composed: the structured factorisation of $\mathbb{G}$ (Appendix E), the lifts and folds (Appendix C.7), the feature-map rows and the collapse row, each band of the descriptor (Appendix G.6), the matchings and the tensor layout (§ H.1.1), each exactly at small n and by probe or IR beyond. The banded factorisation $\tilde{\mathbb{B}}\mathbb{G} = \tilde{\mathbb{D}}$ holds to rounding at $n = 1$ to 10. The composed reflection is then verified whole, so an error in the composition, a block offset or a mis-ordered register, is not hidden by correct atoms.

I.4 The Classical Reference: Eigensolver Against SVD

The reference is the smallest right singular vector of R, computed by a backward-stable SVD (LAPACK's divide-and-conquer driver, with the QR-based driver where it fails to converge), in double precision only; H is never formed.

Proposition 34 (The two bounds)**.** *A backward-stable SVD returns the exact singular vectors of $R + \delta R$ with $\|\delta R\|_2 \le c\,\epsilon_{\mathrm{mach}}\sigma_{\max}$, and the computed ground state makes with the exact one an angle with*

$$\sin\vartheta_{\mathrm{SVD}} \le \frac{\|\delta R\|_2}{\sigma_1 - \hat{\sigma}_0} \approx \frac{c\,\epsilon_{\mathrm{mach}}\sigma_{\max}}{\sigma_1 - \sigma_0}, \tag{I.1}$$

$\hat{\sigma}_0$ *the perturbed value. A backward-stable symmetric eigensolver applied to the formed* H *returns the eigenvectors of* $H + \delta H$ *with* $\|\delta H\|_2 \approx c\,\epsilon_{\text{mach}}\sigma_{\max}^2$, *and*

$$\sin\vartheta_{\text{eig}} \lesssim \frac{c\,\epsilon_{\text{mach}}\sigma_{\max}^2}{\sigma_1^2 - \sigma_0^2}.$$

Proof. The first is Wedin's $\sin\Theta$ theorem for singular subspaces [82] applied to the perturbation δR; the second is the Davis–Kahan theorem [83] applied to δH, with the gap $\lambda_1 - \lambda_0 = \sigma_1^2 - \sigma_0^2$. □

The ratio of the two is $\sigma_{\max}/(\sigma_1 + \sigma_0)$, the residual's condition number on the complement of its near-kernel: the eigensolver loses that factor. When $\sigma_0 \ll \sigma_1$ the SVD's bound is about $\epsilon_{\text{mach}}\sqrt{\|H\|/\Delta}$, the square root of the eigensolver's. Both require σ_0 to be isolated, the trusted-kernel condition of Appendix I.6. On the standard form the eigensolver's angle grows with n by the factor that $\sigma_{\max}/\sigma_1$ grows, 2^{2k} per qubit at order k, until it saturates; Wedin's, smaller by that factor, stays below one (Table I.2). The bounds are what a backward-stable algorithm can promise: on the standard form the smallest singular value is returned at the floor $\epsilon_{\text{mach}}\sigma_{\max}$ rather than at its true, smaller value, which is why the kernel is detected by a ratio against that floor. A single-precision residual has $\epsilon_{32}\sigma_{\max}/(\sigma_1 - \sigma_0) > 10^{-3}$ on R2a from $n = 5$, and is refused.

Table I.2: The eigensolver loses the residual's condition number; the SVD does not. Error $1 - |\cos|$ of the classical reference against the exact coefficients (cos, cosine of the angle between them) on $f'' + 4f' + 4f = 0$, $f(-\frac{1}{2}) = 0$: the standard form by a symmetric eigensolver on $R_{\text{std}}^{\top}R_{\text{std}}$ and by SVD of R_{std}, and the descriptor. Classical computation.

n	Standard, eigensolver	Standard, SVD	Descriptor
2	1.7×10^{-1}	1.7×10^{-1}	1.6×10^{-2}
3	8.6×10^{-8}	8.6×10^{-8}	5.8×10^{-9}
4	0	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
5	3.7×10^{-15}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
6	1.7×10^{-10}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
7	1.2×10^{-4}	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$
8	0.73	$\leq 2 \times 10^{-16}$	$\leq 2 \times 10^{-16}$

I.5 The Iterative Reference

Beyond 8,192 physical columns the dense SVD takes too long and too much memory, and the smallest singular triplets are found iteratively. Lanczos iteration finds the largest eigenvalues of a symmetric operator, and the smallest singular values of R are the largest eigenvalues of $(R^{\dagger}R + \mu I)^{-1}$, which is applied without forming $R^{\dagger}R$.

Proposition 35 (The augmented system)**.** *The solution of* $\left(\begin{smallmatrix} I & R \\ R^{\dagger} & -\mu I \end{smallmatrix}\right)\left(\begin{smallmatrix} r \\ x \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 \\ c \end{smallmatrix}\right)$ *has* $x = -(R^{\dagger}R + \mu I)^{-1}c$.

Proof. The first row gives $r = -Rx$; substituting, $-R^{\dagger}Rx - \mu x = c$. □

One sparse LU factorisation of the augmented matrix serves every Lanczos step. It does not fill as the normal matrix does: a dense constraint row of R, with N entries, appears once in R and once in $R^\dagger$, where it would put a dense $N \times N$ block into $R^\dagger R$. The shift $\mu = (10^{-9}\sigma_{\max})^2$ keeps the augmented matrix non-singular where R has an exact kernel and orders the iteration; it does not enter the values, since a Rayleigh–Ritz step on R itself follows. The Lanczos window holds 34 eigenvalues, twice the seventeen singular values kept, so that a degenerate cluster cut by the window's edge, which a single start vector meets only through rounding, is still found whole. $\sigma_{\max}$ comes from Lanczos iteration on $R^\dagger R$, where squaring loses nothing at the top of the spectrum.

Upper bounds and the a posteriori estimate. For V with orthonormal columns, the j-th smallest singular value of RV is at least the j-th smallest of R (Cauchy interlacing), so the computed $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ never fall below the true ones. For the returned v, with $r = \|Rv\| \geq \sigma_0$, the infidelity against the true ground state $v_\star$ satisfies $1 - |\langle v|v_\star\rangle|^2 \leq (r^2 - \sigma_0^2)/(\sigma_1^2 - \sigma_0^2) \leq r^2/\sigma_1^2$, and the reported bound is $r^2/(\tilde{\sigma}_1^2 - r^2)$, which is at least r^2/σ_1^2 when no singular value below $\tilde{\sigma}_1$ is missed. It is therefore an estimate, exact to rounding once the iteration has converged and nothing below $\tilde{\sigma}_1$ is missing; it floors at about $(\sigma_0/\sigma_1)^2$ where R has no kernel, and is sharp where it has one.

The start vector. Both Lanczos runs start from a Gaussian vector of fixed seed: deterministic, answer-independent and with a component on every singular vector. A uniform start would not do: if a permutation P of the columns commutes with $R^\dagger R$ and $Pu = u$, every Krylov vector is P-fixed, so the iteration never meets a singular vector of another symmetry sector. On the Taylor–Green vortex, whose residual has such a symmetry, a uniform start misses most of the lowest singular values.

Checks. Where both run, the iterative reference reproduces the dense one's ground state to the dense one's own error. Past the standard form's moderate n, where the dense SVD's backward error $\epsilon_{\text{mach}}\sigma_{\max}$ is no longer small against σ_1, the iterative reference is the more accurate: on R2a from $n = 8$ to 10 its σ_1 is constant to about 10^{-14} relative, where the dense value drifts.

I.6 The Trust Threshold

The ascending singular values are floored at 10^{-300}, so that an exact zero gives a finite ratio, and the ratios σ_k/σ_{k-1} are formed for $k = 1, \ldots, 16$. The kernel's dimension is the k of the largest ratio, the *gap ratio* is that ratio, and the reference is trusted when the dimension is one and the gap ratio at least 20. An exact kernel of dimension k is found because its computed values are at most $c\,\epsilon_{\text{mach}}\sigma_{\max}$ by Weyl's inequality, so its ratio exceeds 20 whenever $\sigma_k/\sigma_{\max}$ exceeds about 4×10^{-15}; beyond that the floor is the gap's own size and nothing in the computed spectrum separates them. For a near-kernel, if the truncated solution $\hat{b}$ nearly minimises the

residual, $\|R\hat{b}\| \approx \sigma_0$, its angle to the ground state is at most about σ_0/σ_1: a gap ratio of 20 puts the ground state within 0.05 of it in sine.

Why 20. A spectrum with no kernel still has ratios, and the detector reports the largest as a kernel's. The threshold must exceed the largest ordinary spacing in the benchmark problems' singular spectra, so that a ratio above it cannot come from spacing alone. Over the classical builds of R1 to R8 the largest such spacing is the stiff R2c's structural σ_2/σ_1, just under 20, with the median far below; the only larger ratios are those of three of the reproduction's panels (Appendix K), whose second near-kernel is not a spacing. The value is empirical, a property of these spectra and not a theorem.

What it does not decide. A kernel of dimension above one is never trusted; the doubled space's is the example (§ 6.4). A second near-kernel higher up, with a larger ratio, moves the reported dimension above one and is untrusted, which is conservative. A one-dimensional kernel with a modest gap ratio, an under-resolved problem found by luck, is not trusted. An iterative reference counts among its smallest seventeen values with the Ritz upper bounds above, and can report a ratio too small by their overestimate. Trust means that a kernel is detected, not that the field has converged; the convergence in n is measured separately.

I.7 The Circuit-Simulation Memory and Time Model

A circuit simulation applies the GQSP circuit one controlled walk at a time on a statevector simulator [84], with gate fusion off, on the walk's qubits and the signal, $q = q_W + 1$; it holds a statevector, never an operator, so its memory grows as 2^q and not 4^q (Appendix F.3).

Memory. A statevector of q qubits is $S = 16{\cdot}2^q$ bytes of complex double-precision amplitudes, and a dense k-qubit gate matrix $16 \cdot 4^k$ bytes, totalling G over a step's dense gates. The peak resident memory is estimated as

$$M_{\text{peak}} = M_0 + 8S + 4G,$$

with M_0 the interpreter and libraries, eight statevector-sized buffers at the simulation's peak and four copies of each gate matrix. The multipliers are measured, not derived, from 20 to 24 qubits, and the model's peak is within 0.85 to 1.05 of the memory resident in a fresh process at 22 qubits. The statevector term alone is 2^{q-23} GiB: 8 GiB at 26 qubits and 128 GiB at 30. Memory is the hard limit of circuit simulation; time is the soft one.

Time. Once the memory fits, one step is simulated untimed to warm the caches, then five times, and the median is t_{step}; a single cold step on a shared node can read many times its steady cost, and the median of warm steps does not move for one slow outlier. The run's estimate is (candidates) $\cdot\, (d+1) \cdot t_{\text{step}}$, $d+1$ steps for the degree and the opening rotation

and one simulation per initial-state candidate. Which resolutions circuit simulation reaches is therefore a joint statement of q and d: the degree is set by the spectrum, and one more qubit roughly doubles both the memory and the step's time.

Refusal. The budget is the job's allocation of memory and time. The estimate is logged against it whatever the verdict, and a run that exceeds either is refused and recorded as beyond the simulation budget, with the estimate and the resource estimate's numbers beside it; if the memory does not fit, the step is never timed.

I.8 Acceptance and the Error Budget

Table I.3: Every preparation is judged at five levels, each against a bar it can be measured to. Applied to every preparation whose reference is trusted; a run that fails a bar is reported as a finding, never re-judged. ψ, ψ_{ref}, prepared and reference states; ε, target suppression; p_G, post-selection success probability; d, filter degree; $\varepsilon_{\mathrm{disc}}$, discretisation error; $B(d_{\mathrm{total}})$, the error budget. The state bar's a priori bound follows from the spectral edge, γ and the filter (Proposition 15); the circuit bar is judged before normalising, which divides rounding by $\sqrt{p_G}$, and on normalised states only where $p_G \geq 10^{-12}$ (Lemma 10).

Level	Measured	Bar
State	infidelity $1 - \lvert\langle\psi_{\mathrm{ref}}\rvert\psi\rangle\rvert^2$ in the form's own space	$\leq \varepsilon = 10^{-6}$, and $\leq 10\times$ its a priori bound
Circuit	circuit simulation against exact emulation from a unit input, before normalising	$\leq 10^{-10}$
Probability	the two tiers' p_G	relative $\leq 10^{-8}$
Phases	realised GQSP polynomial against its target, on $4(d+1)$ points of the unit circle	$\leq 10^{-12}$
Field	decoded field against the reference solution, L^∞	$\leq \varepsilon_{\mathrm{disc}} + B(d_{\mathrm{total}})$, Equation (7.10)

Lemma 10 (Normalising amplifies differences)**.** *For non-zero vectors x and y, $\left\lVert x/\lVert x\rVert - y/\lVert y\rVert\right\rVert \leq 2\lVert x-y\rVert/\lVert x\rVert$, and $\left\lvert\lVert x\rVert^2 - \lVert y\rVert^2\right\rvert \leq \lVert x-y\rVert(\lVert x\rVert + \lVert y\rVert)$.*

Proof. $\left\lVert x/\lVert x\rVert - y/\lVert y\rVert\right\rVert \leq \lVert x-y\rVert/\lVert x\rVert + \lVert y\rVert\left\lvert 1/\lVert x\rVert - 1/\lVert y\rVert\right\rvert = (\lVert x-y\rVert + \lvert\lVert y\rVert - \lVert x\rVert\rvert)/\lVert x\rVert$, and the reverse triangle inequality bounds the second term by $\lVert x-y\rVert$. The second inequality factorises a difference of squares. □

The circuit's output has norm $\sqrt{p_G}$, so a rounding error of about 10^{-16} in the unnormalised output becomes about $10^{-16}/\sqrt{p_G}$ in the normalised state; at $p_G = 10^{-12}$ that is the bar of 10^{-10}. The circuit and its emulation are therefore compared before normalising, from a unit input, and the normalised states only where $p_G \geq 10^{-12}$; below it the comparison would measure rounding, not the circuit. The probability bar follows from the circuit bar by the lemma's second part only for $p_G \gtrsim 4\times 10^{-4}$, and is a separate measurement below. A run that fails a bar is reported as a finding, with its degree and the growth of the difference, and never re-judged against an adjusted bar.

The state metric. For unit vectors the distance between states, $d(a,b) = \min_\phi \|a - e^{\mathrm{i}\phi} b\| = \sqrt{2 - 2|\langle a|b\rangle|}$, is a metric, and it is related to the infidelity $\epsilon = 1 - |\langle a|b\rangle|^2$ exactly by $\epsilon = d^2(1 - d^2/4)$; so $d \approx \sqrt{\epsilon}$, and distances add by the triangle inequality where infidelities do not. Each distance is computed as the norm of a difference of phase-aligned vectors, never by the overlap formula, which cancels and cannot resolve distances below about 10^{-8}, the size of the terms themselves.

The chain. A prepared state passes through three states on its way from the target, the ground state: the untruncated filter's output ψ_0, T2's state ψ_2, and the circuit's state ψ_c, which the realised phases prepare. The filter term $d(\text{target}, \psi_0)$ is computed exactly from the physical subspace's SVD: with $c_i = \langle v_i|\psi_{\text{init}}\rangle$ and w_i the filter's value on the i-th singular value,

$$\epsilon_{\text{filter}} = \frac{\sum_{i\geq 1} w_i^2 |c_i|^2}{\sum_{i\geq 0} w_i^2 |c_i|^2}, \qquad d_{\text{filter}} = \Big(\frac{2\epsilon_{\text{filter}}}{1 + \sqrt{1 - \epsilon_{\text{filter}}}}\Big)^{1/2}.$$

It depends on the edge, the overlaps and the filter, never on the solution. The truncation and phase terms are bounded a priori: if two polynomials of H' differ by at most η in the sum of the absolute differences of their Chebyshev coefficients, their outputs differ by at most η since $\|T_k(H')\| \leq 1$, and by Lemma 10 the normalised states by at most $2\eta/\sqrt{p_G}$. For truncation η is imaginary time's certified tail, and for the minimax filter, which truncates nothing, the rounding of the Chebyshev recurrence, up to about 5×10^{-8} at degrees beyond 10^4; for the phases it is the difference between the realised and target coefficients. Every encoding is exact and the field is decoded classically, so no further term enters.

Proposition 36 (The readout bound)**.** *Write the decoded field as $f = w\,u$ with weight w and $u(x) = u_s\,\ell_x(\psi)/\ell_s(\psi)$, where $\ell_x(\psi) = \langle \tau(x)|D\psi\rangle$ is the field functional at x, $\ell_s = \ell_{x_s}$ and u_s the regular datum. Let ψ_{ref} and ψ be unit vectors, $\delta = \psi - \psi_{\text{ref}}$ with ψ's phase chosen so that $d = \|\delta\|$ is their distance, $s = u_s/\ell_s(\psi_{\text{ref}})$, $u_{\text{ref}} = s\,\ell_x(\psi_{\text{ref}})$, $K = |s| \max_x \|D^\top \tau(x)\|$, $U = \max|u_{\text{ref}}|$ and $W = \max|w|$, the maxima over the comparison grid. Then, wherever $Kd < |u_s|$,*

$$\max_x |f - f_{\text{ref}}| \leq B(d) = \frac{WK\,(1 + U/|u_s|)\,d}{1 - Kd/|u_s|}.$$

Proof. With $a = \ell_x(\psi_{\text{ref}})$ and $a_s = \ell_s(\psi_{\text{ref}})$, $u_{\text{ref}} = sa$, $u_s = sa_s$ and $u = s(a + \ell_x\delta)/(1 + \ell_s\delta/a_s)$, so

$$u - u_{\text{ref}} = s\,\frac{\ell_x\delta - (u_{\text{ref}}/u_s)\,\ell_s\delta}{1 + \ell_s\delta/a_s}.$$

$|s\,\ell_x\delta| \leq Kd$, and $|s\,\ell_s\delta| \leq Kd$ because the regular point's functional is no larger than the grid's largest (on one axis it is a grid point, and on two axes the functional's norm is largest at the corners, which are). The numerator is therefore at most $Kd(1 + U/|u_s|)$ and the denominator at least $1 - Kd/|u_s|$; the weight multiplies by at most W. □

The bound is an inequality, not a first-order estimate, and it holds on states from 10^{-8} to 10^{-1}

from the reference with random phases. The field's error is then at most $\varepsilon_{\text{disc}} + B(d_{\text{filter}} + d_{\text{trunc}} + d_{\text{phase}})$, Equation (7.10), with $\varepsilon_{\text{disc}}$ the classical reference's own field error against the reference solution. The budget is taken on the physical columns, since the descriptor's padding decouples and every state is zero on it.

J Benchmark Catalogue

J.1 Linear problems

Table J.1: The linear benchmark problems R1–R5, as run. Equation, constraint rows, regular point x_s and reference solution. x_z, the root of the reference (or of its derivative) nearest the value Wu et al. print, set up from the reference; P_l, P_l^m, Legendre and associated Legendre functions; $v = f/\sqrt{1-x^2}$, the variable R3c and R3d are solved for. Every source enters by its Chebyshev coefficients; the heat equation's initial function is imposed at $t = -1$. Wu et al.'s printed rows, run only in the reproduction: Appendix K.

	Equation	Constraint rows	x_s	Reference
R1	$f' + 2f = 0$	– (regular datum $f(-1) = 1$)	-1	$e^{-2(x+1)}$
R2a	$f'' + 4f' + 4f = 0$	$f(-1) = 0$	0	$\frac{1}{2}(1+x)e^{-2x}$
R2b	$f'' - 2f' - 3f = 0$	$f(0) = 0$	$\frac{1}{2}$	$\frac{1}{4}\left(e^{3x} - e^{-x}\right)$
R2c	$f'' + 5f' + 400f = 0$	$f'(0) = 0$	0	$e^{-5x/2}\left(\cos\omega x + \frac{\sqrt{7}}{21}\sin\omega x\right)$, $\omega = \frac{15\sqrt{7}}{2}$
R3a	$(1-x^2)f'' - 2xf' + 20f = 0$	$f'(0) = 0$	1	P_4
R3b	$(1-x^2)f'' - 2xf' + 30f = 0$	$f(0) = 0$	1	P_5
R3c	$(1-x^2)v'' - 4xv' + 28v = 0$	$v'(0) = 0$	$\frac{1}{2}$	$f = P_5^1 = -\sqrt{1-x^2}\,P_5'$
R3d	$(1-x^2)v'' - 4xv' + 40v = 0$	$v(0) = 0$	$\frac{1}{2}$	$f = P_6^1 = -\sqrt{1-x^2}\,P_6'$
R4a	$(x-1)f'' - xf' + f = (x-1)^2$	$f'(x_z) = 0$, $x_z = -0.19598$	0	$\frac{3}{2}e^x - \frac{1}{8}x(8x+13) - 1$
R4b	$f'' + 4f' + 4f = e^{-2x}$	$f(x_z) = 0$, $x_z = 1 - \sqrt{3}$	0	$\frac{1}{2}e^{-2x}(x^2 - 2x - 2)$
R4c	$f'' - 5f' + 6f = xe^{2x}$	$f(x_z) = 0$, $x_z = 0.90542$	0	$\frac{1}{6}\left(4e^{3x} - e^{2x}(3x^2 + 6x + 2)\right)$
R4d	$f'' + 4f' + 20f = xe^{-2x}\sin 4x$	$f'(x_z) = 0$, $x_z = -0.74681$	0	$\frac{1}{64}e^{-2x}\left((x-48)\sin 4x - 4(x^2+16)\cos 4x\right)$
R5a	Laplace, $f_{xx} + f_{yy} = 0$	$f(\pm 1, y) = f(x, -1) = 0$; datum slice $f(x, 1) = \sin\frac{\pi(x+1)}{2}$	$(0, 1)$	$\cos\frac{\pi x}{2}\,\sinh\frac{\pi(y+1)}{2}/\sinh\pi$
R5b	heat, $f_t = \frac{1}{25}f_{xx}$	$f(t, \pm 1) = 0$; datum slice $f(-1, x) = e^{4\pi^2/25}\sin 2\pi x$	$(-1, \frac{1}{4})$	$e^{-4\pi^2 t/25}\sin 2\pi x$
R5c	wave, $f_{tt} = 4f_{xx}$	$f(t, \pm 1) = 0$; datum slice $f(0, x) = \sin 2\pi x$; $\partial_t f(0, x) = 0$	$(0, \frac{1}{4})$	$\cos 4\pi t\,\sin 2\pi x$

J.2 Nonlinear and coupled problems

J.3 Run Options

A problem's data enter as the four kinds of row of Definition 2, listed with the problems that use them in Table J.3.

Every run is a problem of Table 4.1 at a resolution, with the options of Table J.4. The defaults are the configuration every comparison of § 8 uses; the alternatives are controls, labelled ablations, or the printed choices kept to test the published method.

Table J.2: The nonlinear and coupled benchmark problems R6–R10. Equation, data and reference. x_s, regular point; x_z, exact root nearest the printed value; ρ, Carleman ratio; a, initial amplitude; ν, viscosity; $k \le M$, wavenumbers of the M sine modes; ω_0, initial vorticity; J, the advective nonlinearity; w_0, initial mode vector; IVP, initial-value problem; DNS, direct numerical simulation; (truncation), the reference for the Galerkin truncation's error.

	Equation	Data	Reference
R6a	$4f'' + 2f'^2 + f = 0$ (Wu et al. Eq. 29)	$f'(0) = 0$, $x_s = 0$	$1 - x^2/8$
R6b	$f''-2f^2+x = 0$ (Wu et al. Eq. 34)	$f(x_z) = 0$, $x_z = 0.026147$ (printed 0.02615)	boundary-value solve to 10^{-12}
R7	$u' = v$, $v' = -\omega^2 u$, $\omega = \pi$	$u(-1) = 0$, $v(-1) = \omega$	$u = \sin \omega(x + 1)$
R8	$y' = -y + y^2$, $t \in [0, 3]$	$y(0) = \frac{1}{2}$ ($\rho = 0.5$)	$1/(1 + e^t)$
R9	viscous Burgers, periodic, $u_0 = a \sin x$	sine modes $k \le M$, $M = 4, 8, 16$; $a/\nu = 0.03$–10	Galerkin IVP; Cole–Hopf (truncation); exact lift
R10a	2-D vorticity Navier–Stokes, $\omega_0 = 2 \sin x \sin y$	$J \equiv 0$ (Taylor–Green)	exact viscous decay
R10b–d	2-D vorticity Navier–Stokes, counter-rotating vortex pair	$\|w_0\| = a$, $a/\nu = 3, 10, 30$; active triads	Galerkin IVP; DNS (truncation); exact lift

Table J.3: Every constraint row either leaves the solution's scale free or ties data to it. The four rows that carry a problem's data, grouped by that role. Each enters the stacked residual at weight one, is the same in both forms, and acts on the block holding the quantity it constrains: a derivative's zero row is $\mathbb{B}(x^*)\mathbb{G}^j$ in the standard form and $\mathbb{B}(x^*)$ on the j-th derivative block of the descriptor's chain. x^*, the point or slice of the data; x_s, a regular point; $\tau(x)$, the Chebyshev polynomials evaluated at x; $\mathbb{S}_a$, slice along axis a; $\hat{g}$, coefficients of g; a, the anchor component of a vector datum y and $y_\perp$ the rest; $\mathbb{M}_r$, multiplication by the source r; $\mathbb{G}$, differentiation matrix. Definitions: §§ 4.3 and 6.2.

Row	Data it carries	Operator	Problems
Vanishes on the solution and its multiples: fixes the shape, leaves the scale free			
Zero	$f^{(j)}(x^*) = 0$ at a point: Dirichlet ($j = 0$), Neumann ($j = 1$), or the same on a boundary of a PDE	$\mathbb{B}(x^*)\mathbb{G}^j$, with $\mathbb{B}(x) = \sqrt{N}\,\|0\rangle\langle\tau(x)\|$	R2–R6
Datum slice	an initial or boundary *function* g on the slice $x_a = x^*$ of a PDE	$\mathbb{S}_a(x^*) - \hat{g}\,\langle\tau(x_s)\|\,/f(x_s)$	R5
Ratio	a vector datum y at x^*: the initial values of a coupled system or of a Carleman lift	$\big(\mathbb{F}/y_a\big) \otimes \mathbb{B}(x^*)$, with $\mathbb{F} = y_a(I - \|a\rangle\langle a\|) - y_\perp\langle a\|$ fixing the direction of y	R7–R10
Maps the solution to a constant, whatever its scale: ties data to the scale			
Collapse	the regular datum $f(x_s)$, which fixes the scale after preparation; through it a source r, a non-zero point datum y^*, and the linear terms of the doubled space	$\mathbb{D}^{(0)}(x_s) = \mathbb{B}(x_s)/f(x_s)$, entering as $-\mathbb{M}_r\mathbb{D}^{(0)}(x_s)$ for a source and as $\mathbb{B}(x^*)\mathbb{G}^j - y^*\mathbb{D}^{(0)}(x_s)$ for a datum	every problem (the scale); R4 (sources); R6

Table J.4: The model's input options and the values used. Every run is a problem of Table 4.1 at a resolution with these options, at the default unless a result states otherwise. n, qubits per Chebyshev axis ($N = 2^n$); n_t, qubits on a time axis; M, Galerkin modes; ζ, block weight; K, Carleman order; ε, target suppression; r, ratio of the geometric product state.

Option	Values	Default and use
Resolution	n qubits per Chebyshev axis, $N = 2^n$; n_t on a time axis; modes M (R9, R10)	swept per problem (Tables J.1 and J.2)
Form	standard; descriptor	both, compared
Descriptor layout	chain (order ≥ 2); mass form (first order)	by the equation's order; the chain on a first-order system as an ablation
Block rescaling	ζ a priori (Equation (5.11)); none	ζ; unscaled as an ablation
Encoding of $\mathbb{G}$	structured, exact; Wu et al.'s	structured; Wu et al.'s in resource estimates only
Nonlinearity	doubled space; Carleman order K, symmetric or tensor layout	as the problem requires; the doubled space in the standard form only
Filter	minimax edge filter; imaginary time	minimax; imaginary time in the filter study and the reproduction
Target ε	suppression of the excited spectrum relative to the ground state	10^{-6}; also 10^{-4} and 10^{-8} in the filter study
Initial state	geometric product state, $r = \pm\frac{1}{2}$; uniform; random	geometric; uniform and random as controls
Tier	circuit simulation (T1); exact emulation (T2); resource estimate (T3)	every tier the run's size allows (§ 7.5)

J.4 Results by problem

The linear benchmark problems and the coupled oscillator R7 in both forms, with the minimax edge filter at $\varepsilon = 10^{-6}$ and the geometric initial state, under the conventions of § 7.7. Table J.5 gives the largest resolution at which each form's reference is trusted and its preparation was emulated, with the prepared state's infidelity against the reference and the filter's degree. Every one of these preparations meets ε, its a priori bound and its field budget (§ 8.1). The standard form's emulations stop at smaller n than the descriptor's because its degree grows as N^{2k}; a dash marks a problem with no trusted reference at the resolutions emulated, R2b and R2c in the standard form and the wave equation R5c in both, whose references are trusted only at larger n (§ 8.3). Table J.6 gives the resource estimates at the largest resolution estimated for every problem, as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$, and Table J.7 the growth of these degrees with N, of which Table 8.2 carries a representative of each class.

Table J.5: Every trusted preparation meets its target, in both forms. Per problem and form: the largest resolution n (per axis on R5) with a trusted reference and an exact emulation, the prepared state's infidelity against the reference, and the minimax filter's degree. –, no trusted reference at the resolutions emulated. Exact emulation (T2).

	Standard			Descriptor		
Problem	n	Infidelity	Degree	n	Infidelity	Degree
R1	5	1.0×10^{-13}	3240	8	8.0×10^{-14}	860
R2a	3	9.9×10^{-13}	9198	8	3.8×10^{-12}	9128
R2b		–		8	2.2×10^{-12}	23 279
R2c		–		8	2.2×10^{-12}	240 658
R3a	4	1.2×10^{-12}	227 610	7	3.4×10^{-12}	146 002
R3b	4	5.2×10^{-12}	190 516	7	1.5×10^{-11}	149 490
R3c	4	1.2×10^{-12}	493 783	7	8.2×10^{-13}	377 948
R3d	4	5.8×10^{-12}	493 555	6	5.8×10^{-12}	119 478
R4a	3	2.1×10^{-13}	41 951	7	2.6×10^{-13}	226 204
R4b	4	3.0×10^{-12}	247 819	8	2.3×10^{-11}	8892
R4c	4	2.6×10^{-12}	232 606	8	2.8×10^{-12}	18 331
R4d	4	1.0×10^{-12}	54 837	8	1.7×10^{-12}	9412
R5a	3	3.5×10^{-13}	25 421	4	3.5×10^{-13}	84 913
R5b	4	3.2×10^{-11}	27 692	5	2.8×10^{-11}	136 658
R5c		–			–	
R7	7	1.4×10^{-12}	356 379	9	2.5×10^{-12}	21 143

K Reproduction Record

K.1 All panels

K.2 The Fig. 5 triage

R4a's source is a polynomial, represented exactly, so its rounded point acts alone: the printed $f'(-0.2) = 0$ stands for an extremum at $x \approx -0.196$, and the error holds at 3.1×10^{-3} for

Table J.6: As built the descriptor's degree is the lower on every problem; at the ideal-encoding bound the standard form's is the lower on R3. Minimax edge filter degree at $\varepsilon = 10^{-6}$, $n = 8$ on one axis and $n = 5$ per axis on R5, as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$ (α_R, the residual's subnormalisation; $\|R\|_2$, its norm). At the ideal bound the standard form's degree is the lower on R3a–R3d and the two are close on R5b. Resource estimates (T3).

		As built		$\alpha_R = \|R\|_2$	
Problem	n	Standard	Descriptor	Standard	Descriptor
R1	8	2.1×10^5	8.6×10^2	1.0×10^5	8.5×10^2
R2a	8	1.2×10^{10}	9.1×10^3	1.5×10^9	8.7×10^3
R2b	8	4.8×10^{10}	2.3×10^4	6.2×10^9	2.2×10^4
R2c	8	6.0×10^9	2.4×10^5	7.8×10^8	1.8×10^5
R3a	8	1.7×10^{10}	5.7×10^5	1.1×10^5	2.8×10^5
R3b	8	1.4×10^{10}	5.8×10^5	9.3×10^4	2.8×10^5
R3c	8	3.6×10^{10}	1.5×10^6	4.1×10^5	7.1×10^5
R3d	8	3.6×10^{10}	1.7×10^6	4.1×10^5	8.4×10^5
R4a	8	5.8×10^{10}	9.0×10^5	6.7×10^9	4.0×10^5
R4b	8	1.8×10^{10}	8.9×10^3	2.3×10^9	8.1×10^3
R4c	8	1.7×10^{10}	1.8×10^4	2.2×10^9	1.6×10^4
R4d	8	4.0×10^9	9.4×10^3	5.2×10^8	8.4×10^3
R5a	5	8.0×10^6	8.4×10^5	8.1×10^5	2.1×10^5
R5b	5	4.5×10^5	1.4×10^5	6.0×10^4	6.3×10^4
R5c	5	4.7×10^7	1.6×10^6	5.4×10^6	5.8×10^5
R7	8	1.4×10^6	1.1×10^4	6.8×10^5	1.0×10^4

Table J.7: The degree's growth with N on every linear problem and on R7. Exponent of $N = 2^n$ in the minimax filter's degree between the two largest resolutions estimated ($n = 7 \to 8$ on one axis, $n = 4 \to 5$ per axis on two), as built and at the ideal-encoding bound $\alpha_R = \|R\|_2$, grouped by class: k, equation order; a_2, leading coefficient. †Reference not trusted at $n = 4$ per axis; slope indicative only. Two-point slopes of a degree with logarithmic factors (§ 8.2). Resource estimates (T3).

		Standard		Descriptor	
Class	Problem	As built	Ideal	As built	Ideal
First order, $k = 1$	R1	2.01	2.00	0.983	1.01
	R7	2.00	2.00	0.954	0.994
Second order, $a_2 \neq 0$	R2a	4.01	4.00	0.945	1.01
	R2b	4.01	4.00	0.951	1.01
	R2c	4.01	4.00	0.656	0.975
	R4b	4.01	4.00	0.916	1.01
	R4c	4.01	4.00	0.882	1.01
	R4d	4.01	4.00	0.872	1.00
Second order, a_2 vanishes	R3a	4.01	2.00	1.97	2.00
	R3b	4.01	2.00	1.95	1.98
	R3c	4.01	2.00	1.98	2.01
	R3d	4.01	2.00	1.97	2.00
	R4a	4.01	4.01	1.99	2.02
Two axes, per axis	R5a	4.09	4.03	3.31	3.32
	R5b	4.01	3.90	2.23	2.30
	R5c†	2.93	2.86	2.75	2.76

Table K.1: The exact solution reproduces the printed scale factor of every ordinary panel, to within one unit of the last printed digit, and of no partial differential or nonlinear panel. Per panel of Wu et al. at their printed n: the printed scale factor η, the exact solution's $\eta_e = \|b\|^2$ (b, its Chebyshev coefficients; Equation (4.1)) and the paper-faithful ground state's η. †Ground state not trusted (degenerate kernel, or gap ratio below 20). Classical computation.

Fig.	Problem	n	Printed η	Ideal $\eta_e = \|b\|^2$	Ground state	Agreement
3b	R2a	3	1.29	1.29009	1.28731	✓
3c	R2b	3	32.47	32.47741	32.22636†	✓(one unit low)
3d	R2c	5	451.71	451.70962	451.52312	✓
4a	R3a	3	1.75	1.74512	1.74512	✓
4b	R3b	3	1.49	1.48779	1.48779	✓
4c	R3c	5	108.51	108.51142	109.70433†	✓
4d	R3d	5	135.38	135.38360	137.46159†	✓
5a	R4a	3	1.35	1.34649	1.34478	✓
5b	R4b	3	14.74	14.74797	14.72531	✓(one unit low)
5c	R4c	4	0.82	0.82116	0.82521	✓
5d	R4d	4	77.51	77.51167	76.93737	✓
6b	R5a	3	5.21559	5.21653	5.24407	×
6d	R5b	4	603.863	598.86606	598.87966	×
6f	R5c	5	216.469	239.82712	239.82712	×
7b	R6a	2	3.75367	3.52344	3.52344†	×
7c	R6b	2	0.07984	0.04383	0.04182†	×

Table K.2: Errors of Wu et al.'s solutions at the printed resolution, as printed and as run. L^∞ error of each panel as printed (the paper-faithful variant) and of the problem as run, at the printed n, and the problem's least error over n; on 1001 points (101×101 on two axes). †Ground state not trusted (degenerate kernel, or gap ratio below 20); –, the doubled space (R6), whose kernel is never trusted. Classical computation.

Fig.	Problem	Printed n	As printed	As run	As run, best over n
3b	R2a	3	1.2×10^{-3}	1.2×10^{-3}	3.2×10^{-13} ($n = 5$)
3c	R2b	3	1.8×10^{-2}†	1.8×10^{-2}†	2.3×10^{-11} ($n = 5$)
3d	R2c	5	2.2×10^{-3}	2.2×10^{-3}	5.6×10^{-12} ($n = 6$)
4a	R3a	3	1.9×10^{-15}	1.9×10^{-15}	1.9×10^{-15} ($n = 3$)
4b	R3b	3	3.0×10^{-15}	3.0×10^{-15}	2.3×10^{-15} ($n = 4$)
4c	R3c	5	8.7×10^{-1}†	4.0×10^{-14}	4.9×10^{-15} ($n = 3$)
4d	R3d	5	1.2†	1.5×10^{-13}	8.0×10^{-15} ($n = 3$)
5a	R4a	3	3.1×10^{-3}	1.8×10^{-6}	3.8×10^{-14} ($n = 4$)
5b	R4b	3	1.9×10^{-2}	7.8×10^{-3}	8.5×10^{-12} ($n = 5$)
5c	R4c	4	7.2×10^{-3}	4.9×10^{-10}	1.6×10^{-13} ($n = 5$)
5d	R4d	4	6.1×10^{-2}	4.9×10^{-7}	5.1×10^{-12} ($n = 5$)
6b	R5a	3	3.9×10^{-3}	1.4×10^{-4}	2.1×10^{-13} ($n = 4$)
6d	R5b	4	1.9×10^{-4}	4.7×10^{-5}	4.6×10^{-12} ($n = 5$)
6f	R5c	5	8.5×10^{-8}	7.3×10^{-8}	6.1×10^{-11} ($n = 6$)
7b	R6a	2	2.2×10^{-16}†	2.2×10^{-16}†	–
7c	R6b	2	6.8×10^{-3}†	6.8×10^{-3}†	–

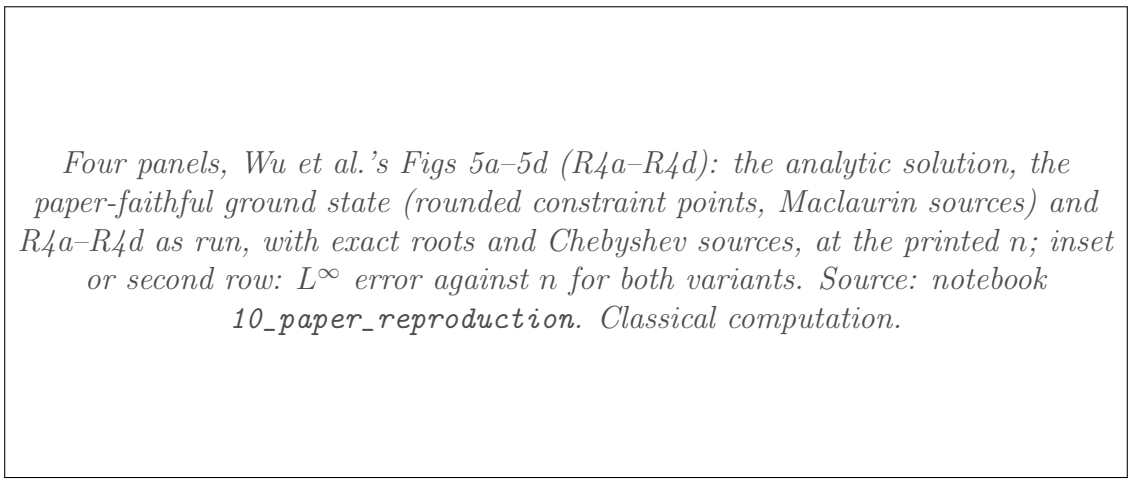


Figure K.1: Printed choices distort Wu et al.'s solutions at the printed resolution.

Table K.3: Each printed choice alone sets an error floor; only both departures together converge. L^∞ error on Wu et al.'s Fig. 5 panels over $n = 4, \ldots, 10$: the paper-faithful variant (rounded constraint point, Maclaurin source), each departure alone (exact root; Chebyshev source coefficients), and both together at their best n. –, R4a's polynomial source is already exact, so its rounded point acts alone. Classical computation.

Fig.	Problem	Faithful	Exact root	Chebyshev source	Both, best
5a	R4a	3.1×10^{-3}	–	–	3.8×10^{-14} $(n = 4)$
5b	R4b	1.6×10^{-2}	4.0×10^{-2}	3.6×10^{-2}	8.5×10^{-12} $(n = 5)$
5c	R4c	7.2×10^{-3}	4.8×10^{-3}	1.0×10^{-2}	1.6×10^{-13} $(n = 5)$
5d	R4d	6.1×10^{-2}	8.1×10^{-2}	1.1×10^{-1}	5.1×10^{-12} $(n = 5)$

every n from 4 to 10, against 3.8×10^{-14} at $n = 4$ at the exact point. On R4b to R4d each printed choice alone sets a floor, and only both departures together converge.

K.3 Panels not reproduced, and printed errors

Printed errors. Wu et al. print the solution of their Laplace example as $\operatorname{csch}\pi \cosh(\pi x/2) \sinh(\pi(y+1)/2)$, which does not satisfy their boundary conditions; the solution, and the reference used throughout, is $\cos(\pi x/2) \sinh(\pi(y+1)/2)/\sinh\pi$. Beyond the printed resolution Wu et al.'s invariant sets for their partial differential examples no longer fix the solution (Appendix C.5), and its heat data at $t = 0$ pose a backward problem on $[-1, 0]$ (Appendix C.3).

K.4 Wu et al.'s claims

K.5 Wu et al.'s filter and its parameters

Wu et al. realise imaginary time by QSVT on a block encoding of $\tilde{H} = H/\|H\|_F$ [30]. The target $p(x) \approx e^{-\beta x}$ has indefinite parity, so Wu et al. find one set of phase angles for their even part and one for their odd part, of maximal degrees d^e and d^o, run the two opposite-parity QSVT sequences and sums them by an LCU: a mixed-parity circuit. Each sequence uses the

Table K.4: The claims of Wu et al. [30] tested in this thesis, and their verdicts. $\mathbb{G}$, differentiation matrix; $\mathbb{M}$, multiplication operator; $\mathbb{n}_1$, product-to-sum fold; $\mathbb{B}$, $\mathbb{D}^{(0)}$, zero and collapse constraint rows; η, scale factor; β, imaginary time; $\lambda_{\max}$, λ_2, largest and second-smallest eigenvalues of H; x_s, regular point.

	Claim	Verdict
A-01	$\mathbb{G}$, $\mathbb{M}$, $\mathbb{n}_1$ and $\mathbb{B}$ as printed at $n = 2$	reproduced
A-02	the printed η of Figs 3–5 (11 panels)	within one unit of the last printed digit; 9/11 equal when rounded
A-03	the printed η of Fig. 6	not reproduced
A-04	the printed η of Fig. 7	not reproduced
A-05	the printed Laplace solution	a typographical error (cosh for cos)
A-06	the $\mathbb{B}$ circuit (Fig. 2): $n + 2$ qubits, prefactor $\sqrt{2^{n+1}}$	confirmed
A-07	the $\mathbb{G}$ circuit (Fig. 9): $2n + 3$ qubits, $O(n^3)$ gates, error decreasing exponentially	–
A-08	the $\mathbb{D}^{(0)}$ circuit (Fig. 8): prefactor $\sqrt{2^{n+1}}/f(x_s)$	confirmed
A-09	the linear H is positive definite with distinct eigenvalues	–
A-10	the doubled-space H has at least 2^{2n-1} zero eigenvalues	–
A-11	the time rule $\beta \geq \lambda_{\max}/(\lambda_2(n-1))$ suffices	–
A-12	the necessary terms scale as $O(\text{poly}(n))$ and are efficient	true for gates; not for the subnormalisation
A-13	the examples are solved accurately at the printed n	distorted where rounded points or Maclaurin sources enter

unitary dilation

$$U(\tilde{H}) = \begin{pmatrix} \tilde{H} & \sqrt{I - \tilde{H}\tilde{H}^\dagger} \\ \sqrt{I - \tilde{H}^\dagger\tilde{H}} & -\tilde{H}^\dagger \end{pmatrix}, \tag{K.1}$$

built from PREPARE and SELECT subroutines over the unitary decomposition of $\tilde{H}$ with $m = \lceil \log_2 L \rceil$ ancilla for L unitaries. The time β is set by Wu et al.'s approximate lower bound, Equation (3.6), and its examples' times and degrees are stated with the method in § 3.3. Wu et al. also note that GQSP, GQET and GQSVT could improve the circuit.

K.6 The tau-truncation ablation

Table K.5: Dropping the top k rows of the equation block (the Chebyshev tau method) gives an exact kernel with accuracy comparable to Wu et al.'s least squares. L^∞ field error and least singular value σ_0 of the residual, least squares against tau truncation, on R2a–R2c at $n = 5$–7; $k = 2$, the equation's order. Classical computation.

Problem	n	Least-squares L^∞	σ_0	Tau-truncated L^∞	σ_0
R2a	5	3.2×10^{-13}	3.0×10^{-11}	1.6×10^{-13}	0
R2a	6	2.2×10^{-12}	4.9×10^{-10}	3.8×10^{-12}	0
R2a	7	2.8×10^{-11}	7.8×10^{-9}	1.0×10^{-10}	0
R2b	5	2.3×10^{-11}	5.3×10^{-11}	3.7×10^{-12}	0
R2b	6	1.8×10^{-10}	8.5×10^{-10}	7.4×10^{-11}	0
R2b	7	9.8×10^{-10}	1.4×10^{-8}	8.3×10^{-10}	0
R2c	5	2.2×10^{-3}	1.6×10^{-2}	1.8×10^{-3}	0
R2c	6	5.6×10^{-12}	5.5×10^{-10}	1.8×10^{-11}	0
R2c	7	2.4×10^{-10}	1.1×10^{-8}	1.9×10^{-10}	0

K.7 The imaginary-time filter's circuit simulations

The reproduction runs Wu et al.'s imaginary-time filter in the standard form, by circuit simulation as well as by exact emulation, at $n = 2$ and 3, the problems as printed and as run. Each circuit simulation is compared with the emulation before normalising, against the bar of 10^{-10} that the minimax records all meet (§ 8.1). All but the three of Table K.6 meet it too. The three are among the highest degrees run, of order 10^5, where the statevector's rounding accumulates over the controlled walks, but the degree alone does not set the difference: on R4a at $n = 3$ a record at a comparable degree passes just under the bar, and on the wave equation R5c records at higher degrees pass by more than an order of magnitude. The difference therefore depends on the problem's spectrum and filter as well as on the degree. The bar is kept at every degree, so these are reported as misses and not re-judged. They concern the circuit simulator's arithmetic, not the filter: the emulation is exact for the circuit as built (§ 7.5), and the minimax filter, whose degrees are smaller at equal suppression, meets the bar on every record.

Table K.6: Three imaginary-time circuit simulations miss the 10^{-10} bar at degrees of order 10^5; others at comparable or higher degrees pass. Largest difference between circuit simulation (T1) and exact emulation (T2), before normalising, from a unit input. Below the rule, passing records for comparison: the largest passing difference (R4a) and the highest degree run (R5c). R6b is the doubled space, whose filter output is a projection.

Problem	n	Qubits	Degree	T1 against T2
R6b	2	17	211 721	2.6×10^{-10}
R6b as printed	2	17	211 722	2.1×10^{-10}
R4a as printed	3	20	81 195	1.5×10^{-10}
R4a (passes)	3	20	80 960	9.5×10^{-11}
R5c (passes)	3	19	287 359	4.3×10^{-12}

Layer diagram: `basis` → `operators` → `problems/carleman` → `assemble/ir` → `classical/circuits` → `verify/resources` → `qsp` → `pipeline` → `cli`; *an architecture test fails on any import against the arrow.*

Figure L.1: The layers of the `pihm` package and the one direction in which they may import.

L Software and Reproducibility

L.1 Architecture

L.2 Records and provenance

L.3 Campaigns and computational resources

Table L.1: Resource classes of the Setonix campaigns. Each class's request (partition, CPUs, memory, wall time) and the runs it served.

Class	Request	Typical runs
S	`work`, 1 CPU, 4 GB, 30 min	construction, IR checks, small exact verification
M	`work`, 16 CPUs, 64 GB, 4 h	exact verification to 20 qubits; mid-size exact emulation
L	`work`, 64–128 CPUs, ∼230 GB, 12–24 h	circuit-simulation ceilings; large emulation; probes to 30 qubits
XL	`highmem`, 1 TB	the largest emulation and probe runs

L.4 Tests

L.5 Regenerating the results

L.6 Environment

M Research Proposal

Table M.1: Deviations from the research proposal of May 2025.

Proposed	Outcome	Reason
Extending QSVT to the generalised construction for complex-valued, non-Hermitian operators		
Nonlinear PDEs: Burgers, nonlinear Schrödinger, complex Ginzburg–Landau		
Data-informed Hamiltonians		
(not proposed)		

Masters Research Proposal

Dhyan Patel [23340229]
The University of Western Australia

May 25, 2025

Title: Solving Nonlinear PDEs via Quantum Singular Value Transformation

Supervisor: Professor Jingbo Wang

1 Introduction

1.1 Aims

This thesis aims to develop and evaluate quantum algorithms for solving nonlinear partial differential equations (PDEs) using the Quantum Singular Value Transformation (QSVT) framework. Based on recent work that represents differential equations and their boundary conditions as effective Hamiltonians, this work extends the QSVT to the Generalised Quantum Singular Value Transformation (GQSVT). This extension enables the application of this method to complex-valued differential equations which arise naturally in many areas of physics and engineering. Additionally the project will explore how to incorporate data-informed modifications into these Hamiltonians, with the goal of modelling data-rich regimes.

1.2 Significance

Solving complex nonlinear partial differential equations (PDEs) on quantum hardware is a rapidly emerging prospect with broad applications ranging from fluid dynamics and turbulence modelling to quantum field theory. Established algorithms such as the Harrow-Hassidim-Lloyd (HHL) algorithm and variational methods typically rely on discretisation, which inhibits their scalability to stiff, nonlinear, multiscale or dense systems with fine grid sizes. In contrast, the effective Hamiltonian approach uses spectral methods to encode the solution into a Chebyshev polynomial basis, allowing it to surpass performance when fine-grid encoding is required. The quantum singular value transformation (QSVT) provides a flexible and modular approach to quantum linear algebra, allowing operations like matrix inversion under broader conditions than HHL [1]. However, the current method restricts effective Hamiltonians to real values, thereby limiting the class of operators and solutions to purely real outcomes. Moreover, the HHL paper only addresses linear partial differential equations or nonlinear ordinary differential equations

through a process called linearisation [3] and variational algorithms require classical optimisation [7].

This research is significant because it extends the effective Hamiltonian based framework to complex-valued nonlinear PDEs, which are common in frontier areas of various fields. In domains like geophysical inversion, deep turbulence and real-time weather prediction, incorporating data-informed terms into the Hamiltonian provides a promising path towards optimising quantum solutions to problems where complete physical models are unavailable or difficult to create. By enabling hybrid physics and data driven Hamiltonians, this work could expand the applicability of quantum PDE solvers to data-rich, real world problems.

2 Background

2.1 Problem Statement

Many physical systems are described by partial differential equations (PDEs), but solving them remains computationally costly. In the absence of closed-form solutions, many advanced classical algorithms implement numerical methods which are difficult to scale and lose accuracy in systems with a need for more fine grid or time-step discretisations. This is especially true of nonlinear PDEs which are the primary focus of this research.

Quantum algorithms provide a potentially scalable alternative. Quantum Singular Value Transformation (QSVT) allows quantum circuits to encode real-valued polynomials of definite parity. Recent theoretical developments have shown that it is possible to represent differential equations and their boundary conditions as effective Hamiltonians, which are structured matrices that can be processed using QSVT. However, due to QSVT's restriction to real-valued Hermitian Hamiltonians, the range of solvable PDEs is restricted. Systems with damping, gain/loss or complex boundary interactions can lead to complex-valued or non-Hermitian operators.

While novel frameworks such as the Generalised QSVT (GQSVT) have been proposed to handle complex-valued operators and remove parity constraints, they have not been systematically applied to nonlinear PDEs.

The existing effective Hamiltonian approach is largely physics-driven, assuming known governing models. Yet in many high-dimensional, data-rich domains such as geophysical imaging, real-time weather prediction, or turbulence modelling, the governing equations may be incomplete or unavailable. In these settings, data-informed Hamiltonian construction offers a promising path forward towards more flexible and accurate quantum solvers.

This research addresses the gap in current quantum PDE solving frameworks by developing and applying GQSVT based methods for nonlinear, complex-valued PDEs and exploring hybrid physics and data driven constructions for real world data-intensive applications.

2.2 Literature Review

2.2.1 Exploration of Problems

Burgers Equation (Viscous)

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \tag{1}$$

This equation appears in turbulence theory and has a closed form solution in one real spatial dimension with $\nu > 0$. However, complex versions do exist and the equation remains unsolvable analytically outside these circumstances. The nonlinearity term is $u\frac{\partial u}{\partial x}$.

Nonlinear Schrödinger Equation

$$i\frac{\partial \psi}{\partial t} + \nabla^2\psi + |\psi|^2\psi = 0 \tag{2}$$

This equation describes the evolution of complex wave functions in nonlinear media. Its nonlinearity term is $|\psi|^2\psi$.

Complex Ginzburg-Landau Equation

$$\frac{\partial A}{\partial t} = A + (1 + i\alpha)\nabla^2 A - (1 + i\beta)|A|^2 A \tag{3}$$

This equation models phenomena such as superconductivity and fluid convection patterns. The nonlinearity term is $|A|^2A$.

The current physics-informed Hamiltonian method is able to address Burgers equation in its real case, but the Schrödinger and Ginzburg-Landau equations involve are complex-valued fields. This limitation motivates the development of a more robust framework capable of handling a broader class of nonlinear PDEs.

2.2.2 Existing Classical and Quantum Solving Methods

Classical methods for solving PDEs employ mesh based discretisation techniques such as finite difference and finite element methods or spectral methods [11]. These methods approximate the continuous domain onto a discrete grid, converting the PDE into a large system of algebraic equations that can be solved numerically. These methods are well-established and reliable, but for high-resolution or multi-dimensional systems the computational cost increases rapidly. For higher dimensional problems, direct matrix or classical variational methods may be employed [4].

The Harrow-Hassidim-Lloyd (HHL) algorithm was one of the first to demonstrate exponential speedup for solving systems of linear equations, which arise from discretised PDEs [3]. HHL encodes the linear system $A\vec{x} = \vec{b}$ into a quantum state and uses phase estimation to prepare a quantum state proportional to the solution vector $\vec{x}$. However, it is limited to sparse, well-conditioned Hermitian matrices. This makes it impractical for solving PDE problems, particularly nonlinear ones. Importantly, in the 2021 "Grand Unification" paper, it is shown that HHL can be viewed as a special case of QSVT, where

we're looking to find a target polynomial that inverts the singular values of the matrix A.

Variational quantum algorithms use a hybrid quantum-classical loop to optimise a parameterised quantum circuit that approximates the solution to a PDE [7]. This approach also discretises the domains into a grid. While more flexible than HHL, these methods require classical optimisation and suffer from challenges such as barren plateaus and scaling issues for larger systems.

2.2.3 Quantum Singular Value Transformation

The quantum singular value transformation (QSVT) is a powerful framework that generalises quantum linear algebra operations by enabling quantum circuits to apply polynomial transformations to the singular values of block encoded matrices [1]. QSVT enables the application of any parity-constrained polynomial $f(\Sigma)$ to the singular values Σ of a matrix A block encoded within a unitary.

$$U = \begin{bmatrix} A & \cdot \\ \cdot & \cdot \end{bmatrix} \text{ with } \|A\| \leq 1 \tag{4}$$

Assuming we have access to such a unitary U and the ability to apply a sequence of controlled-phase rotations by an angle ϕ, we can construct a unitary $V_{\vec{\phi}}$, dependent on the values of $\vec{\phi}$ we select for our phase rotations, that looks like the following:

$$V_{\vec{\phi}} = \begin{cases} \tilde{\Pi}_{\phi_1} U \left[\prod_{k=1}^{(d-1)/2} \Pi_{\phi_{2k}} U^\dagger \tilde{\Pi}_{\phi_{2k+1}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for odd Poly}^{(\text{SV})}(A) \\ \left[\prod_{k=1}^{d/2} \Pi_{\phi_{2k-1}} U^\dagger \tilde{\Pi}_{\phi_{2k}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for even Poly}^{(\text{SV})}(A) \end{cases} \tag{5}$$

where the sequences of phases $\vec{\phi} = (\phi_0, \phi_1, \ldots, \phi_d)$ is chosen specifically to encode a desired polynomial $f(x)$ of degree d [1]. $\tilde{\Pi}_\phi$ and Π_ϕ represent a controlled phase rotation by ϕ where $\tilde{\Pi}$ and Π are the left and right unitary projectors respectively such that $A := \tilde{\Pi} U \Pi$. By carefully selecting which phase angles to rotate by, we can approximate any real, fixed parity continuous polynomial [6] acting on the singular values of A.

It can be shown that QSVT serves as a unifying framework for many quantum algorithms including quantum search, Hamiltonian simulation, factoring, quantum walks and more [8]. A key limitation of QSVT is that it only supports real-valued polynomials applied to matrices. This restricts its applicability to problems where both the matrix and the function being applied are real-valued [1]. For differential equations, this implies only PDEs with real-valued operators and real-valued solutions can be handled.

For physics-informed partial differential equation (PDE) solvers, QSVT is used in ground-state preparation via Hamiltonian simulation. In this case the target polynomial is $p(x) = e^{-tx}$ [11].

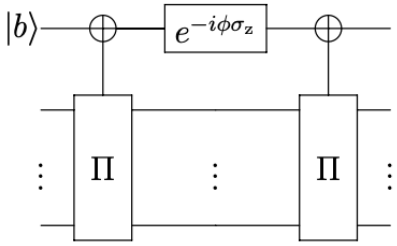


Figure 1: Quantum Circuit Diagram of the implementation of a Projector Controlled Phase Gate, Π_ϕ [2]

2.2.4 Physics-Informed Effective Hamiltonians

This algorithm encodes a differential equation in a latent Chebyshev basis and finds its solution by preparing the ground state of a specifically constructed Hamiltonian [11]. First, consider some differential equation with a differential operator $\mathcal{L}$, solution $u(x)$ and source function $f(x)$:

$$\mathcal{L}[u(x)] = f(x), \qquad x \in [a, b]. \tag{6}$$

We can represent the solution as a state overlap in the Chebyshev basis with a scale factor s, namely

$$u(x) = s\langle T(x)|\psi\rangle \tag{7}$$

Here $|T(x)\rangle$ is known as a "feature map" state, a superposition of computational basis states with Chebyshev coefficients. $|\psi\rangle$ is a parameterised quantum state carrying the unknown amplitudes. Thus, we've effectively represented our solution in a latent Chebyshev basis [11].

To translate differential operators, we use the known differentiation matrices associated with the Chebyshev basis [5]. An example translation is shown below where $\mathbb{G}$ is the generator matrix corresponding to differentiation:

$$\frac{d^2 f(x)}{dx^2} + af(x) = 0 \longrightarrow \mathbb{A} = \mathbb{G}^{\intercal 2} + a\mathbb{I}. \tag{8}$$

Here, $\mathbb{A}$ is the matrix corresponding to the latent space representation of the DE. Crucially, differential operators act simply in the Chebyshev basis, as the following

$$\frac{d}{dx} f(x) = \frac{d}{dx}\langle T(x)|\psi\rangle = \langle T(x)|D^\intercal|\psi\rangle. \tag{9}$$

As a result, all terms in the DE can be rewritten as linear operators acting on $|\psi\rangle$.

To implement constraints, we use $\mathbb{B}(x)$, a data transformation operator dependent on x. For example, for specific constraints such as $f(x_0) = 0$ or $f'(x_1) = 0$, we can use $\mathbb{B}(x_0)$ or $\mathbb{B}(x_1)\mathbb{G}^\intercal$ [11]. From these, we can create our effective Hamiltonian:

$$\mathbb{H} = \mathbb{A}^\intercal \mathbb{A} + \mathbb{B}^\intercal \mathbb{B}. \tag{10}$$

We may add additional boundary constraints as required, which would effectively be more $\mathbb{B}$ matrices. Therefore, by construction, solving the original DE is equivalent to solving the eigen-problem:

$$\mathbb{H}_{\text{eff}}|\psi\rangle = 0, \tag{11}$$

and the desired solution corresponds to the ground state of the effective Hamiltonian. H is a real symmetric positive-definite matrix with distinct eigenvalues and mutually orthonormal eigenvectors [11].

We then block encode the effective Hamiltonian, and use a ground state preparation routine, QSVT, to amplify the amplitude of the ground state. Specifically, we use quantum imaginary time evolution, constructing the polynomial approximation of $e^{-t\mathbb{H}}$. This exponentially suppresses the excited eigen-components, leaving a higher probability of measuring the ground state. Importantly, the algorithm directly prepares the lowest energy solution without classical optimisation [11].

Finally, to measure the quantum state, we use a Chebyshev transformation and Hadamard test to measure the state we have prepared. This controlled overlap test effectively measures the overlap $\langle T(x)|\psi\rangle$. In practice this is done via interferometric measurement [11].

This same framework is extendable to multi-variable and nonlinear DEs. For a function of two variables $f(x, y)$, we use a tensor product Chebyshev basis with two registers, preparing a joint feature state $|T(x)\rangle \otimes |T(y)\rangle$. Take Laplace's equation for example:

$$\frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = 0 \longrightarrow \mathbb{A} = \mathbb{G}_x^{\intercal 2} \otimes \mathbb{G}_y^{\intercal 2}. \tag{12}$$

For nonlinear terms, we add extra registers. For example, a term like $f'(x)^2$ leads to a product of two copies of $|\psi\rangle$ in latent space. We create a new, higher dimensional latent Chebyshev space where $|\Psi\rangle \equiv |\psi\rangle \otimes |\psi\rangle$, so that the nonlinearity term becomes linear in the larger space. This naturally handles all nonlinearity. Inhomogeneous terms are handled via Maclaurin series expansion. For all these cases, we then construct the effective Hamiltonian as usual and solve [11]. This approach allows us to encode the solution over the entire domain, rather than using grid discretisation. Therefore, it is possible to evaluate the solution function at any arbitrary input values. However, as discussed above, QSVT can only handle real values. This restricts the class of differential equations we're able to solve.

Example Solutions:

Below are results from the application of this algorithm on example differential equations [11]. The paper did not attempt to solve nonlinear PDEs, but there are examples of both linear PDEs and nonlinear DEs. $f(x)$ is the known solution and $f_Q^*(x)$ is the quantum solution.

1. Heat Equation

$$\frac{\partial f(t,x)}{\partial t} - k\frac{\partial^2 f(t,x)}{\partial x^2} = 0, k = \frac{1}{25}, \mathcal{BC} = \{f(0,x) = \sin(2\pi x), f(t,\pm 1) = 0\} \tag{13}$$



Figure 2: Solution for the Heat Equation Example. [11]

2. Nonlinear Example 1

$$4\frac{d^2 f(x)}{dx^2} + 2\left(\frac{df(x)}{dx}\right)^2 + f(x) = 0, \mathcal{BC} = \{f(0) = 1, f'(0) = 0\} \tag{14}$$

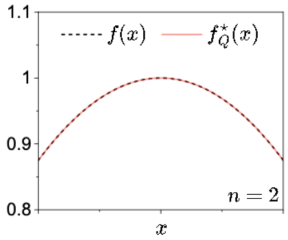


Figure 3: Solution of Nonlinear Example 1 [11]

3. Nonlinear Example 2

$$\frac{d^2 f(x)}{fx^2} - 2f(x)^2 + x = 0, \mathcal{BC} = \{f(-1) = -0.1, f(1) = 0.1\} \tag{15}$$

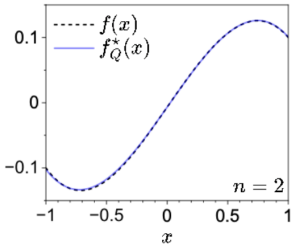


Figure 4: Solution of Nonlinear Example 2 [11]

The model performs well for all examples shown. The errors are almost negligible and decrease as the number of qubits used (shown as n on the figures) increase. The experiments were conducted on classical simulations of quantum hardware and are therefore currently computationally costly, however based on these results it is projected this algorithm will become useful in the early fault-tolerant era [11].

2.2.5 Generalised Quantum Singular Value Transformation

The generalised QSVT (GQSVT) framework extends the standard QSVT algorithm by lifting the restrictions on coefficient phases and polynomial parity [9]. The standard QSVT restricts the target polynomial $p(x)$ to have real coefficients and definite parity. However, the generalised version allows complex polynomials and indefinite parity. These generalisations are inherited from developments in generalised quantum signal processing (GQSP) which is itself an extension of the quantum signal processing framework that QSVT was built on [10]. Eliminating the restrictions on real polynomials and Hamiltonians imposed by QSVT will allow us to solve complex-valued differential equations. GQSVT also promises a simplification in complexity for the crucial and time-consuming step of calculating phase factors. The task was of complexity $\mathcal{O}(d^2)$ for QSVT, GQSVT promises a complexity of $\mathcal{O}(d)$ [9]. This should speed up calculations even for problems which are currently possible via the existing method.

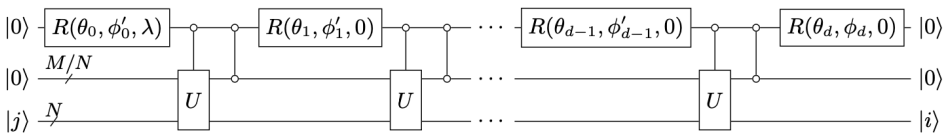


Figure 5: Example circuit for the generalised quantum singular value transformation for a Hermitian matrix A, block encoded as U. R is a rotation matrix analogous to Π for QSVT [9].

3 Research Plan

3.1 Method

This research will be primarily theoretical and computational. The goal is to design, simulate and evaluate novel quantum algorithms for solving nonlinear, complex-valued partial differential equations (PDEs). Development will be conducted using using *Mathematica* for symbolic manipulation and proof-of-concept derivations. Simulations will be created using *Python* based simulators such as *Qiskit* and *Pennylane* and results will be compared to classical solving algorithms. Efforts wil be split into two phases. First, extending the current QSVT algorithm to GQSVT and second, integrating data-informed terms into effective Hamiltonians to enable solving PDEs in data-rich regimes. This research represents a new research direction within QUISA.

Proposed Plan:

1. Reproduce the Current Physics-Informed QSVT Method:
   (a) Implement the standard QSVT framework in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
2. Extend to GQSVT:
   (a) Create the GQSVT algorithm in *Python*.
   (b) Choose an appropriate problem.
   (c) Implement a new physics-informed Hamiltonian PDE solver in *Python*.
   (d) Compare quantum solution against classical benchmarks
3. Experiment with more data-informed terms in the Hamiltonian

Timeline:

Semester 1 2025: Research Proposal.

Semester 2 2025: Recreation of QSVT Physics-informed method. Begin extension to GQSVT.

Semester 1 2026: Completion of a working GQSVT enhanced algorithm, including simulation and evaluation in comparison to classical algorithms.

Semester 2 2026: Experimentation with data-informed terms.

3.2 Preliminary Work

In semester 1 2025, work consisted of a thorough literature review, achieving an in-depth understanding of QSVT and the physics-informed Hamiltonian approach for PDEs. This was pivotal in developing a thesis that could both build on the newest developments in the field and be achievable within the time-frame of a Master's course. A working *Python* implementation of QSVT has been created and work in underway on implementing the current physics-informed Hamiltonian approach using QSVT.

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Table L.2: The test tiers of the `pihm` package. What each tier tests and when it runs.

Tier	Content	Run
Unit and property	operator identities over random n and x; primitives; arithmetic; IR; filters; phases	every commit
Golden	fixtures from the frozen legacy code and Wu et al.'s printed $n = 2$ matrices	every commit
Reproduction	Wu et al.'s targets, both variants	every commit
Integration	every problem, both forms, smallest n, all stages	every commit
Documentation	Sphinx build, doctests, spelling	every commit
Claims	one test per claim; writes the claims report	at gates
Slow	larger-n verification; circuit-simulation ceilings	at gates, on Setonix